

PrimeWave™ Design Environment WaveView Tool User Guide

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About the PrimeWave Design Environment

WaveView Tool User Guide

The *PrimeWave Design Environment WaveView Tool User Guide* describes the PrimeWave™ Design Environment waveform viewer, the WaveView tool. The WaveView tool is a graphical waveform viewing and analysis program for analog/mixed-signal IC design simulations, which can help you to use simulation tools more effectively.

This preface includes the following sections:

- [New in This Release](#)
- [Related Products, Publications, and Trademarks](#)
- [Conventions](#)
- [Customer Support](#)
- [Statement on Inclusivity and Diversity](#)

New in This Release

Information about new features, enhancements, and changes, known limitations, and resolved Synopsys Technical Action Requests (STARs) is available in the PrimeWave Design Environment Release Notes on the SolvNetPlus site.

Related Products, Publications, and Trademarks

For additional information about the PrimeWave™ Design Environment tool, see the documentation on the Synopsys® SolvNetPlus support site at the following address:

<https://solvnetplus.synopsys.com>

You might also want to see the documentation for the following related products:

Synopsys® PrimeWave Design Environment

Synopsys® Custom Compiler™

Synopsys® PrimeSim™ Continuum™

Synopsys® PrimeSim™

Synopsys® PrimeSim™ Pro

- Synopsys® PrimeSim™ SPICE
- Synopsys® PrimeSim™ HSPICE®
- Synopsys® PrimeSim™ XA
- Synopsys® PrimeSim™ CCK
- Synopsys® PrimeSim™ MOSRA
- Synopsys® VCS® PrimeSim™ AMS
- Synopsys® Synopsys FineSim®
- Synopsys® Verdi®
- Cadence® Spectre® Circuit Simulator
- MathWorks® MATLAB

Conventions

The following conventions are used in Synopsys documentation.

Convention	Description
Courier	Indicates syntax, such as <code>write_file</code> .
<i>Courier italic</i>	Indicates a user-defined value in syntax, such as <code>write_file design_list</code>
Courier bold	Indicates user input—text you type verbatim—in examples, such as <code>prompt> write_file top</code>
Purple	• Within an example, indicates information of special interest. • Within a command-syntax section, indicates a default, such as <code>include_enclosing = true false</code>
[]	Denotes optional arguments in syntax, such as <code>write_file [-format fmt]</code>
...	Indicates that arguments can be repeated as many times as needed, such as <code>pin1 pin2 ... pinN</code> .
	Indicates a choice among alternatives, such as <code>low medium high</code>
\	Indicates a continuation of a command line.

Convention	Description
/	Indicates levels of directory structure.
Bold	Indicates a graphical user interface (GUI) element that has an action associated with it.
Edit > Copy	Indicates a path to a menu command, such as opening the Edit menu and choosing Copy .
Ctrl+C	Indicates a keyboard combination, such as holding down the Ctrl key and pressing C.

Customer Support

Customer support is available through SolvNetPlus.

Accessing SolvNetPlus

The SolvNetPlus site includes a knowledge base of technical articles and answers to frequently asked questions about Synopsys tools. The SolvNetPlus site also gives you access to a wide range of Synopsys online services including software downloads, documentation, and technical support.

To access the SolvNetPlus site, go to the following address:

<https://solvnetplus.synopsys.com>

If prompted, enter your user name and password. If you do not have a Synopsys user name and password, follow the instructions to sign up for an account.

If you need help using the SolvNetPlus site, click REGISTRATION HELP in the top-right menu bar.

Contacting Customer Support

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Statement on Inclusivity and Diversity

Synopsys is committed to creating an inclusive environment where every employee, customer, and partner feels welcomed. We are reviewing and removing exclusionary language from our products and supporting customer-facing collateral. Our effort also includes internal initiatives to remove biased language from our engineering and working

environment, including terms that are embedded in our software and IPs. At the same time, we are working to ensure that our web content and software applications are usable to people of varying abilities. You may still find examples of non-inclusive language in our software or documentation as our IPs implement industry-standard specifications that are currently under review to remove exclusionary language.

1

Getting Started with the PrimeWave Design Environment Waveform Viewer

This chapter details getting started with the PrimeWave Design Environment's waveform viewer, the WaveView tool.

The PrimeWave Design Environment's waveform viewer, the WaveView tool, is a graphical waveform viewing and analysis program for analog/mixed-signal IC design simulations, which can help you to use simulation tools more effectively with:

- High-performance waveforms database I/O to access large amount of simulation data efficiently.
- Extensive mixed-signal display functions and analysis capabilities to extract design parameters from simulation result.

The WaveView tool opens automatically when you run a simulation from the PrimeWave Design Environment with **Plot Waveforms** set.

This chapter contains information on the following topics:

- [Basic GUI Usage](#)
- [Basic WaveView Tool Usage](#)
- [Running ACE Scripts](#)

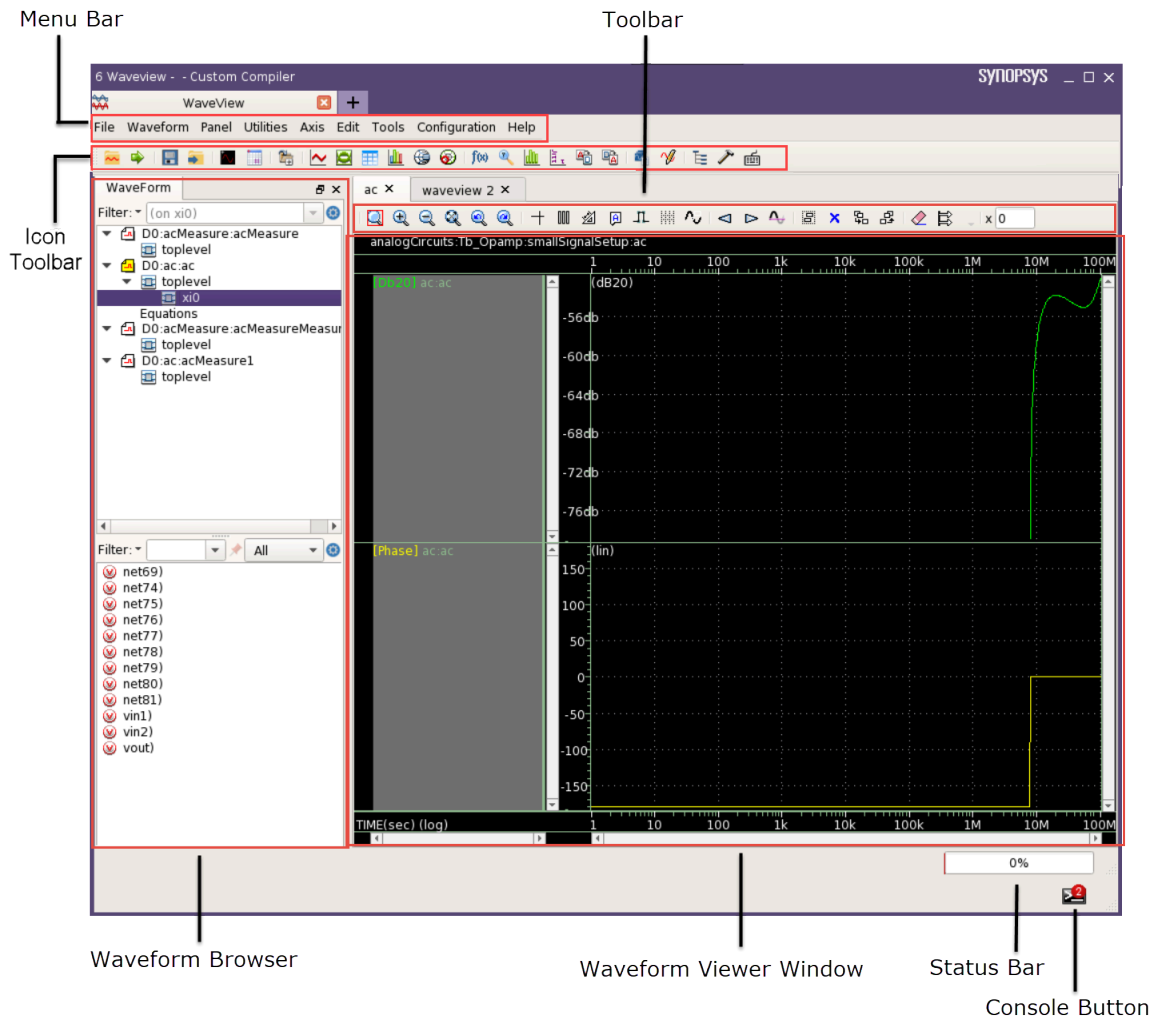
Basic GUI Usage

The following topics describe basic GUI usage:

- [Viewing the GUI](#)
- [Supported GUI Values](#)
- [Viewing the Waveform Browser](#)

Viewing the GUI

The following figure shows the WaveView GUI:



The main window is comprised of the following parts and associated functions:

- Menu Bar — the top application menu bar.
- Icon Toolbar— shortcuts for frequently used menu options.
- Toolbar — shortcuts for the waveform viewer window.
- Waveform Browser — contains the Waveform Browser including the top hierarchy browser and the bottom signal list window. Choose **Configuration > Show/Hide Browser** to open the Waveform Browser.

- Waveform Viewer Window — the waveform viewing area. Multiple waveform viewers can be opened inside the waveform viewer window.

The widths of the WaveForm Browser area and the Waveform Viewer window can be adjusted using the vertical pane bar in between the Browser and the Waveform Viewer area.

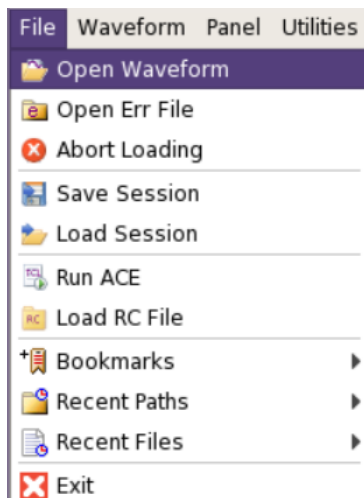
- Status Bar — reports progress of operations such as waveform loading and signal searching.
- Console button — opens/closes the Console.

The following sections describe the waveform viewer GUI in detail:

- [File Menu](#)
- [Waveform Menu](#)
- [Panel Menu](#)
- [Utilities Menu](#)
- [Axis Menu](#)
- [Edit Menu](#)
- [Tools Menu](#)
- [Configuration Menu](#)

File Menu

The **File** menu appears as follows:

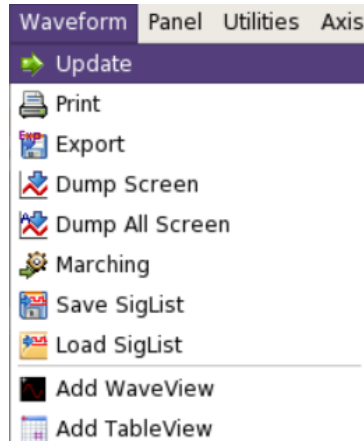


The following options are available:

- **Open Waveform** — imports files into the tool. See [Opening a Waveform File](#).
- **Open Err File** — open an .err file. See [Opening Err Files](#).
- **Abort Loading** — abort the loading of a file.
- **Save Session** — saves the current session. See [Save Session](#).
- **Load Session** — loads a saved session. See [Load Session](#).
- **Run ACE** — runs an ACE script. See [Running ACE Scripts](#).
- **Load RC File** — loads an RC file. See [Opening RC Files](#).
- **Bookmarks** — manage waveform bookmarks. See [Bookmarking Waveform Files](#).
- **Recent Paths** — view recently used file paths.
- **Recent Files** — view recently used files.
- **Exit** — exit the waveform viewer.

Waveform Menu

The **Waveform** menu appears as follows:



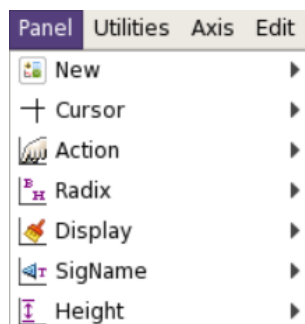
The following options are available:

- **Update** — updates the waveform file. See [Updating Waveform Files](#).
- **Print** — prints the waveform file. See [Printing Waveform Files](#).
- **Export** — exports waveform data. See [Exporting Waveform Data](#).
- **Dump Screen** — dumps the current screen. See [Dumping the Screen](#).

- **Dump All Screen** — dumps all screens. See [Dump All Screens](#).
- **Marching** — automatically updates waveforms from a running simulation job. See [Marching Waveforms](#).
- **Save SigList** — saves the signal list. See [Saving and Loading SigLists](#).
- **Load SigList** — see above.
- **Add WaveView** — adds a new waveform viewer. See [Working With Waveviews](#).
- **Add Tableview** — adds a new tableview. See [Working With Tableviews](#).

Panel Menu

The **Panel** menu appears as follows:

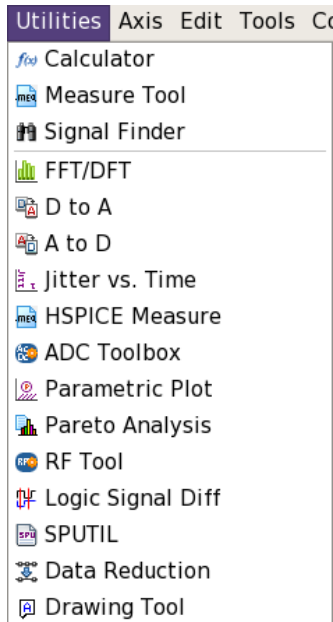


The following options are available:

- **New**: adds a panel type. See [Using Panels](#).
- **Cursor** options: adds or configures cursors. See [Using Cursors](#).
- **Display** options: sets panel display options. See [Setting Panel Display Options](#).
- **Action**: adds a panel type. See [Using Panels](#).
- **Radix** options: sets radix options. See [Setting Radix Options](#).
- **SigName** options: sets signal name options. See [Setting Signal Name Options](#).
- **Height** options: adjusts panel height. See [Adjusting Panel Height](#).

Utilities Menu

The **Utilities** menu appears as follows:

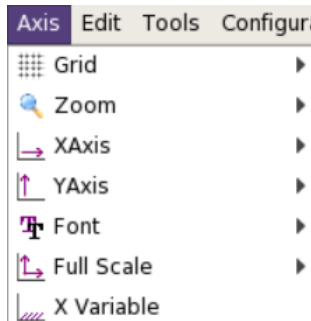


The following options are available:

- **Calculator** — opens the PrimeWave Design Environment Calculator. See Using the PrimeWave Design Environment Calculator in the *Synopsys PrimeWave Design Environment: Calculator Function Reference Manual*.
- **Measure Tool** — opens the Measurement Tool dialog box. See [Using the Measurement Tool](#).
- **Signal Finder** — opens the Signal Finder. See [Searching Signals](#).
- Other utilities — See [Using WaveView Tools](#).

Axis Menu

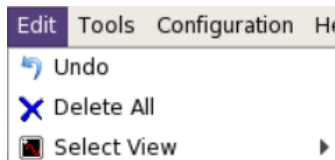
The **Axis** menu appears as follows:



See [Axis Attributes](#) for details.

Edit Menu

The **Edit** menu appears as follows:

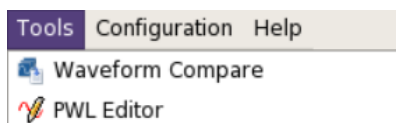


The following options are available:

- **Undo** — undoes an action.
- **Delete All** — deletes all waveform viewer panels.
- **Select View** — selects a waveform viewer panel.

Tools Menu

The **Tools** menu appears as follows:

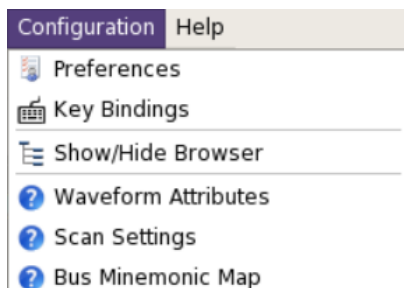


The following options are available:

- **Waveform Compare** — opens the Waveform Compare tool. See [Using the Waveform Compare Tool](#).
- **PWL Editor** — opens the PWL Editor. See [Using the PWL Editor](#).

Configuration Menu

The **Configuration** menu appears as follows:



The following options are available:

- **Preferences** — open Preferences dialog. See [Setting Preferences](#).
- **Key Bindings** — configure key bindings. See [Setting Key Bindings \(Shortcut Keys\)](#).
- **Show/Hide Browser** — show/hide Waveform Browser.
- **Waveform Attributes** — set waveform attributes. See [Setting Waveform Attributes](#).
- **Scan Settings** — set scan settings. See [Scanning Signals](#).
- **Bus Mnemonic Map** — map bus values. See [Mapping Bus Values](#).

Supported GUI Values

All GUI text boxes accept values in the scientific number format (1.00E+02) or numbers with scale unit such as nano (n) and micro (u).

The following scale units are supported:

- T - 1E12
- G - 1E9
- M (upper case) - 1E6
- MEG - 1E6
- x - 1E6
- k - 1E3
- m (lower case) - 1E-3
- u - 1E-6
- n - 1E-9

- p - 1E-12
- f - 1E-15

Viewing the Waveform Browser

See [Using the Waveform Browser](#) for information on viewing and using the Waveform Browser.

Basic WaveView Tool Usage

This section contains information on the following topics:

- [Loading and Saving Sessions](#)
- [Viewing Recent Files and File Paths](#)
- [Log Files and Command History](#)
- [Customizing File Browser Filters](#)

Loading and Saving Sessions

The PrimeWave Design Environment WaveView tool provides save and restore functions for users to save or retrieve a job session to or from a session file. Session files have the `.sx` default file extension.

- [Save Session](#)
- [Load Session](#)

Save Session

To save your current session:

1. Before saving your session, you might consider changing the save directory from the default.
2. Click **Save Session** on the **File** tab.
The **Save Session** dialog box opens.
3. Select or enter the file in which to save your current job session.
4. (Optional) Click the **Use Relative Path in Session File** check box to use a relative path for your session file.

5. (Optional) Click the **Save Current waveform viewer Only** check box to save only the current waveform viewer tab in your saved session file.
6. (Optional) Click the **Save a Copy of Loaded Signals** check box to save loaded signals to a new file named `<session_file_name>.wdf`.

For example, if a session is saved to a file called `foo.sx`, the data file name is `foo.sx.wdf`. When loading the session file, only `foo.sx.wdf` is loaded.
7. Click **OK** to save your job session.

Load Session

The WaveView tool provides save and restore functions for users to save or retrieve a job session to or from a session file. Session files have the `.sx` default file extension.

To load from a session file:

1. Click **Load Session** on the **File** tab.

The **Load Session** dialog box appears.
2. Browse to and select the session file you want to open.

The browser will open to the last location from which a session file was loaded.
3. (Optional) To apply the saved signal selection/layout to other loaded waveform data files in the application, ensure the **Apply to existing waveform files** option is enabled (the default). Select the preferred waveform data file for each restore target using the Waveform Source dialog box.

If the **Apply to existing waveform files** option is not enabled, the session loader restores signal selection/layout using the waveform data files specified in the session file.
4. (Optional) Click the **Suppress popup warning messages** option check box to disable session loading warning messages (missing signals, for example).
5. Click **OK** to load the selected session file.

The session file is loaded into the WaveView tool.

Note:

When loading a session file and applying it to multiple target files, an index number is appended to waveform viewer tab names to avoid duplicate names. For example, for name `waveform viewer1` in session file, the displayed tab names become `waveform viewer 1_1`, `waveform viewer1_2`, and so on. The index starts from 1.

You can also load a session file from the command line with the `-x` option. For example:

```
wv -x setup_file
```

Add the `-y` option to apply the session setup to different waveform data files. For example:

```
wv -y setup_file file1 file2 ... filen
```

Viewing Recent Files and File Paths

Click **Recent Files** in the **File** tab to display a list of recently opened files from the waveform viewer GUI.

Click **Recent Paths** in the **File** tab to display a list of recently used file paths.

Log Files and Command History

To toggle the display of the **Console** window, click the **Console** button at the bottom of the WaveView window.

This section contains information on the following topics:

- [Viewing and Saving Log Files](#)
- [Changing the Default Log File Directory](#)
- [Replaying Log Files](#)
- [Viewing Command History](#)

Viewing and Saving Log Files

Log files are saved for every session. You can view the contents of the log file for your current session by clicking the **Log** tab at the bottom of the Console window.

You can also view log files from previous sessions, which are saved in your working directory as sequentially numbered files (`cx.log.1`, `cx.log.2`, `cx.log.3`, for example).

Changing the Default Log File Directory

By default, the WaveView tool outputs log files in the working directory. Use the `SW_SX_LOG_DIR` environment variable if you want to save the log file to a different directory.

Replaying Log Files

All actions saved in log files can be replayed in the tool. To replay a log file, click **Run ACE** from the **Script** section of the **File** tab, and open a log file.

You can also replay a log file at the command line. To replay a saved log file, enter the following line at the command line:

```
cx -tcl <log_file>
```

As the log file is read, all actions saved in the log file are performed in real time in the GUI.

Viewing Command History

To view a log of the underlying Tcl commands that are called for all actions performed in the tool, click the **History** tab at the bottom of the Console window.

Once you exit the tool, the underlying Tcl commands for all actions you take are saved to a log file. See [Viewing and Saving Log Files](#) for more information.

Customizing File Browser Filters

The file filter option menu in the file loading dialog box can be customized to add your own filter patterns. To customize the menu:

1. Click **Preferences** in the **Configuration** menu.

The **Preference Settings** window opens.

2. Select the **General** category.
3. Specify a waveform filter from the **Waveform File Filter** preference menu.

You can choose to filter **All Files** or **Waveform Files**.

4. (Optional) Click **New** to create a new filter pattern, then enter your own filter pattern using the following format:

```
pattern_label:filter_pattern
```

The `pattern_label` is a character string, and `filter_pattern` is the actual string pattern used for name filtering. Both parts are displayed in the filter option menu. The `pattern_label` can contain spaces.

Multiple filter patterns are delimited by vertical bar (|):

```
Spice3 output:*.raw|*.out
```

Multiple filter option menu pattern entries are delimited by a semicolon (;):

```
Spice3 output:*.raw ; Spice listing:*.out
```

Globbing expressions are supported through wildcards:

```
Hspice:*.tr[0-9]|*.ac[0-9]|*.dc[0-9]
```

If the `pattern_label` field is omitted in an entry, the `filter_pattern` is used as the display string in the filter option menu.

5. Click **OK** or **Apply-save** to save your changes.

To delete a custom filter, select the filter from the **Waveform File Filter** menu and click **Delete**.

Running ACE Scripts

The Analysis Command Environment (ACE) is a Tcl-based programming environment comprised of a set of high-level waveform data operating commands evolved from the old Synopsys APX (Application Programming Extension) API functions. While the old APX API functions are still supported in ACE, the new high-level commands in ACE significantly increase the ease of use to construct scripts for complex analysis needs.

To run an ACE script, click **Run ACE** on the **File** tab.

The ACE commands support:

- Tcl list and Perl array variables for handling a collection of objects.
- Wildcard name pattern for referencing multiple objects.
- Direct use of Tcl/Perl variables in waveform equations. All calculator functions are available for the `sx_equation` command.
- Execution in both GUI mode and non-GUI batch mode.

See Also

- [WaveView™ ACE Command Reference](#)

2

Importing Files

This chapter details how to import files into the PrimeWave Design Environment tool.

This chapter contains information on the following topics:

- [Supported File Formats](#)
- [Opening a Waveform File](#)
- [Opening Multiple Files](#)
- [Opening PWL Files](#)
- [Opening RC Files](#)
- [Reading Compressed Waveform Files \(Linux Only\)](#)
- [Working With WDF Files](#)

Supported File Formats

The following waveform formats are supported:

- Synopsys FSDB files (binary, v6.5 Linux platforms; v6.5 Windows platforms)
- Synopsys HSPICE® tr0, sw0, ac0, mt0, hb0, nw0, mc0, mct0, rlgc, wzo, annotate, listing file (Binary/ASCII)
- Synopsys StarSim™ wdb, wv, xp (binary)
- Synopsys HSIIM®/NanoSim®/CustomSim™ wdb (binary)
- Synopsys HSIIM/NanoSim out ,vector, .mt file (ASCII)
- Synopsys Saber AI/PL (binary/ASCII)
- Synopsys VCS®/Magellan™/DVE VPD (binary)
- Synopsys CustomSim .err, .errz, and .errt files

- Synopsys Galaxy Custom Compiler® Simulation and Analysis Environment (SAE) `resultsMap.xml` and `results.xml` files
- Cadence® Spectre® PSF, PSFXL (support only in Tran section for corners, MC, and sweeps), WSF (binary/ASCII), runObjFile
- Cadence PSPICE® DAT (binary), Allegro® SIM (ASCII)
- Verilog VCD (ASCII), ModelSim® WLF (binary)
- Mentor Graphics® ELDO®/ADMS™ COU, WDB (binary), AdiT TR0/TB0.
- SPICE3/SmartSpice raw format (binary/ASCII)
- Agilent® ADS .ds format (binary), TouchStone® S-parameters (ASCII)
- CSDF (Common Simulation Data Format) (ASCII)
- Text file, CSV (comma separated values) format
- Other in-house simulator formats (SDIF, MICS, TI-punch)
- Synopsys WDF format (binary)
- Binary scope data (Tektronix®, Agilent, Keysight, Teledyne Lecroy®)
- IBIS models, Synopsys Liberty® files

Additional File Format Information

Additional information is available for the following file formats:

- [Special Note for the ELDO COU Format](#)
- [Special Note for the ADMS/ELDO WDB Format](#)
- [Special Note for the PSF Format](#)
- [Special Note for the WDF, FSDB, Binary HSPICE tr0, and PSF Formats](#)

Special Note for the ELDO COU Format

The COU format from ELDO simulator can contain multiple simulation runs. Before loading a COU file, COU reader in the PrimeWave Design Environment tool needs to scan the entire file to determine how many runs the COU file contains.

For very large COU files, the detection process might take a very long time because the reader has to sequentially examine all data in the COU files.

To speed up this COU file loading process, the PrimeWave Design Environment tool has an alternative built-in search algorithm that can cut the file loading time significantly. To

invoke the fast read option, set the `SW_SX_FAST_COU` environment variable to any value in your shell (for example, `setenv SW_SX_FAST_COU`). Due to the nature of the COU format, the fast search algorithm has a trade-off that might potentially determine the number of runs of a COU file incorrectly. Although the possibility is very low, you can always disable the fast search algorithm by unsetting the environment variable (for example, `unsetenv SW_SX_FAST_COU`).

Special Note for the ADMS/ELDO WDB Format

By default, a built-in WDB file reader is used to load data from WDB files generated by ADMS and ELDO. However, for small amount of simulation waveform data, the WDB writer in these simulators might buffer waveform data in the WDB Java server without writing them to a disk file. To work around this problem, you can take one of the following actions:

1. For ELDO, set the `-isaving` option to a smaller value to force more frequent incremental saving to disk files. The file size specified in the `-isaving` option is in MB. For ADMS, set the `JwdbSpillThreshold` option in `$HOME/.vams_setup` to a smaller value to lower the buffer size.
2. Enter `setenv SW_SX_JWDB_API 1` at the command line to use the vendor-provided WDB API to load data directly from the Java WDB server. The API + Java WDB server operation, however, can be very slow for large WDB files.

Special Note for the PSF Format

PSF format assigns a unit for each signal in a simulation output file. For noise analysis, to display different waveforms using proper scales, signals with V^2/Hz unit are automatically converted to $V/\sqrt{Hz}$ by taking the square root of the loaded value during the waveform loading process.

HSPICE splits large output PSF format data (> 2GB) into multiple 2GB PSF files on 32-bit platform. Since 32-bit platform is obsolete and the simulator binary is always 64-bit, there is no 2GB file size limitation for PSF file.

Special Note for the WDF, FSDB, Binary HSPICE tr0, and PSF Formats

Split files are supported for the WDF, FSDB, binary HSPICE tr0, and PSF files. When these files are opened in the PrimeWave Design Environment tool, a search is performed for the subsequent files in the same directory. For example, if the loaded file is `output.wdf`, the PrimeWave Design Environment tool automatically looks for `output.wdf.1`. If the output file is `output.fsdb.1`, the PrimeWave Design Environment tool looks for `output.fsdb.2`. If you do not want to load the subsequent split files, disable the **Automatically connect to subsequent split files** option in the **Loading Waveform Files** dialog box.

Opening a Waveform File

To open a waveform, click **Open Waveform** in the Waveform tab.

To abort loading a waveform file that is taking too long to load, click **Abort Loading** on the **File** tab.

Opening Waveform Files in Case-Insensitive Mode

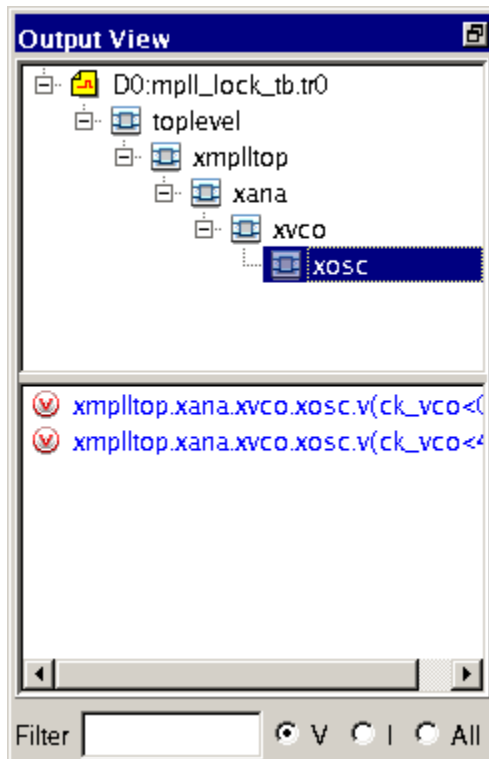
You can open waveform files in case-insensitive mode as follows:

- Using the command-line option `-case_insensitive_mode`
- Using the option `-case_insensitive` with the ACE command `sx_open_sim_file_read`

For FSDB files, the tool does not perform a linear search, which provides significant performance improvement with the ACE command `sx_signal`.

Opening Multiple Files

If multiple files are loaded, to allow simultaneous signal browsing on these multiple target files, the **Open New Browser** context menu function opens multiple signal browsers for each waveform file respectively.



Opening PWL Files

To open textual or PWL data, click **Open Waveform** on the **File** tab and expand the **Additional PWL/Table Format Options** section in the opened dialog box.

The PrimeWave Design Environment tool supports data input in regular text format. Scaling factors (for example, 1ns) and scientific formats (for example, 1.0e-9) are both supported in the textual data files.

To distinguish PrimeWave Design Environment text data files from regular text files, the first line of a text data file is reserved for the format specification and must appear in the following form, where format-type is the format type keyword:

```
#format format-type
```

Lines beginning with a number sign (#) are treated as comments.

Note:

If the file cannot be recognized, use the option **Force Open Waveform as** to open the file as a SPICE PWL or text file.

The following sections describe how to prepare textual input for:

- [Tabulated Data in Real Numbers](#)
- [Tabulated Data in Complex Numbers](#)
- [2-Dimension Sweep Data in Real Numbers](#)
- [2-Dimension Sweep Data in Complex Numbers](#)
- [Measured Data Without Name Header](#)
- [SPICE PWL Sources](#)
- [Opening Err Files](#)

Tabulated Data in Real Numbers

Syntax:

```
#format table
xvar signal1 signal2 ....
value1 value2 value3 ...
...
```

`xvar` and `signaln` are variable names and `valuen` are numeric values. As a special-case exception, the first line can be omitted if the name of `xvar` is `TIME`.

Example 1:

```
#format table
VGATE Ichannel
0.0 0.0
1.0 1u
2.0 5u
3.0 5.2u
...
```

Example 2:

The following data file can be recognized without the `#format` line:

```
TIME V(1) V(2)
0.0 5.0 2.5
1n 4.8 2.6
...
```

If the data file does not contain the x-value column, (commonly seen in measurement data), add keyword `#dataonly` to the `#format` line. The reader automatically assigns a 0-based row index as the x-axis value.

If the signal names are embedded in the last '#' comment line before the actual waveform data values, add keyword `#cmt_name` to the format line.

Tabulated Data in Complex Numbers

Format for complex numbers in (real, imaginary) form.

Syntax:

```
#format complex-ri
xvar signal1 signal2 ....
valuex value1-real value1-imag value2-real value2-imag ...
...
```

Format for complex numbers in (magnitude, angle [0-360]) form.

Syntax:

```
#format complex-ma
xvar signal1 signal2 ....
valuex value1-mag value1-angle value2-mag value2-angle ...
...
```

Format for complex numbers in (magnitude, phase [0-2 π]) form:

Syntax:

```
#format complex-mp
xvar signal1 signal2 ....
valuex value1-mag value1-phase value2-mag value2-phase ...
...
```

Example:

```
#format complex-ri
Freq vout
100meg 0.0 0.0
200meg 1.0 0.1
300meg 1.1 0.2
400meg 1.4 0.4
...
```

2-Dimension Sweep Data in Real Numbers

Syntax:

```
#format 2dsweep xvar2
xvar signal1 signal2 ....
#sweep value1-x2
valuex value1 value2 ...
```

```
...
#sweep value2-x2
valuex value1 value2 ...
...
```

where xvar2 is the name of the second sweep variable.

Example:

```
#format 2dsweep vgs
VDS Ichannel
#sweep 0.0
0.0 0.0
1.0 1u
2.0 5u
3.0 5.2u
#sweep 0.1
0.0 0.0
1.0 2u
2.0 6u
3.0 6.2u
...
```

2-Dimension Sweep Data in Complex Numbers

Format for complex numbers in (real, imaginary) form

Syntax:

```
#format 2dsweep-ri xvar2
#sweep value1-x2
xvar signal1 signal2 ....
valuex
value1-real value1-imag value2-real value2-imag ...
...
#sweep value2-x2
xvar signal1 signal2 ....
valuex value1-real value1-imag value2-real value2-imag ...
...
```

Format for complex numbers in (magnitude, angle [0-360]) form.

Syntax:

```
#format 2dsweep-ma xvar2
#sweep value1-x2
xvar signal1 signal2 ....
valuex value1-real value1-imag value2-real value2-imag ...
...
#sweep value2-x2
xvar signal1 signal2 ....
```

```
valuex value1-real value1-imag value2-real value2-imag ...  
...
```

Format for complex numbers in (magnitude, phase [0-2]) form.

Syntax:

```
#format 2dsweep-mp xvar2  
#sweep value1-x2  
xvar signal1 signal2 ....  
valuex value1-real value1-imag value2-real value2-imag ...  
...  
#sweep value2-x2  
xvar signal1 signal2 ....  
valuex value1-real value1-imag value2-real value2-imag ...  
...
```

Example:

```
#format 2dsweep-ma temp  
freq gain  
#sweep 50  
100meg 1.0 0  
200meg 1.0 90  
300meg 1.1 120  
400meg 1.4 150  
#sweep 75  
100meg 1.0 0  
200meg 1.0 80  
300meg 1.2 110  
400meg 1.3 140  
...
```

Measured Data Without Name Header

To import text measure data that does not have name header line or the x-axis value column, click **Open Waveform** and expand the **Additional PWL/Table Format Options** section.

Select the required options and click **Ok** to load the data file.

SPICE PWL Sources

To import SPICE PWL sources into the PrimeWave Design Environment tool, click **Open Waveform** and expand the **Additional PWL/Table Format Options** section.

Choose the **SPICE PWL Source Data** option and click **Ok** to load the data file. The SPICE PWL sources must not use any variable parameters.

Opening Err Files

To open an .err file, click **Open Err** on the **File** tab. The errors are displayed in the .err File viewer dialog box.

Opening RC Files

Setup files can be loaded from nWave (rc files) and Xelga (swd files). To load a setup file, click **Load RC File** in the **File** tab.

Reading Compressed Waveform Files (Linux Only)

On Linux platforms only, compressed (.Z) or gzipped (.gz) waveform files can be directly opened in the PrimeWave Design Environment tool. The tar.gz waveform files are not readable directly since a tar file can potentially contain multiple files.

By default the current working directory is used as the temporary directory to store the decompressed files. Decompressed files are automatically cleaned up when a the compressed file is closed in the PrimeWave Design Environment tool. Set the `SW_SX_TMP_DIR` environment variable to redirect the temporary directory to a different location.

Working With WDF Files

The WDF (Waveform Data File) format is a Synopsys proprietary waveform storage format. The WDF format not only compresses analog and logic waveform data, it also facilitates fast waveform access for large data files. The compression scheme can be lossy or loss-less (default). Waveform data files in other formats can be converted into WDF files on demand.

The WDF format is directly supported by a number of simulators including HSI, NanoSim, UltraSim, Spectre, and AMS-Designer.

You can extract signal files as a group. For example, you can create the following Tcl script file to extract signal files:

```
set f1 [sx_open_sim_file_read ./group.grp]
sx_current_sim_file $f1
set sig [sx_signal x1]
set sig2 [sx_signal x2]
set f2 [sx_open_sim_file_write ./save2d.wdf]
sx_save_signal $f2 "$sig $sig2"
sx_close_sim_file $f2
sx_close_sim_file $f1
```

Converting Waveform Files to WDF

To convert an open waveform file to WDF format, right-click the name of a waveform file name, and choose **Convert to WDF** from the menu that opens. The whole waveform file is saved into a new WDF file with X range you specify.

Converting Existing Output Files to WDF

One or more output files can be converted (in batch mode) to Synopsys WDF files using the `-c` command-line option.

Lossless Compression

With lossless compression, all original simulation output data points are preserved.

```
wv -c file1 file2 ... filen
```

Lossy Compression

You can use the `-r` command-line option to turn on data reduction for the compression process. Use `-r 1` for moderate reduction or `-r 2` for a more aggressive reduction.

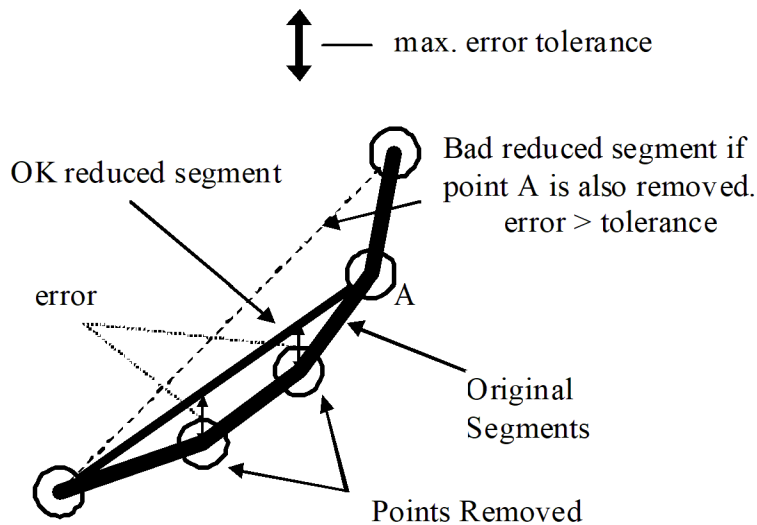
```
wv -c -r (1 or 2) <-rv 0.001> <-ri 10n>
```

Reducing WDF Data

The data reduction option removes redundant data points from straight waveform segments based on user-specified error tolerances. Data points are removed if the error between the original data points and the reduced PWL segments are smaller than the user-specified tolerance.

Figure 1 depicts the scheme used in WDF data reduction.

Figure 1 WDF data reduction scheme



The moderate reduction setting keeps at least 1 point out of four (4) original data points, while the more aggressive reduction mode keeps 1 out of ten (10) points. The `-rv` and `-ri` options are used to specify the reduction tolerance for voltage (or generic) waveforms and current waveforms respectively. The default setting is 5 mV for voltage waveform and 1 nA for current waveforms.

3

Working With Waveforms

This chapter details working with waveforms in the WaveView tool.

The section contains information on the following topics:

- [Using the Waveform Browser](#)
- [Waveform Files and Data](#)
- [Setting Waveform Attributes](#)
- [Working With Waveviews](#)
- [Working With Tableviews](#)
- [Comparing and Checking Waveforms](#)
- [Comparing Logic Signals](#)

Using the Waveform Browser

The following topics are detailed in this section:

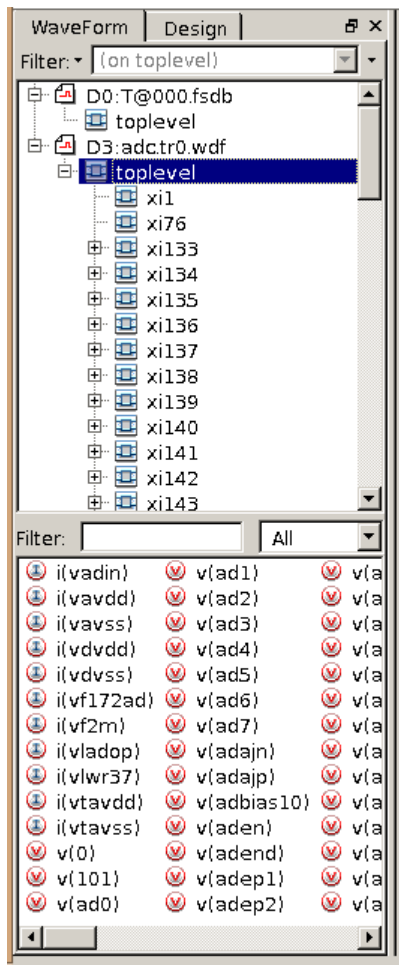
- [Viewing the Waveform Hierarchy Browser](#)
- [Displaying Signals](#)
- [Highlighting Waveforms](#)

Viewing the Waveform Hierarchy Browser

The Waveform Browser consists of an upper hierarchy browser and a lower signal list window. The lower list window displays signals under the selected hierarchy in the upper browser. Only one item can be selected in the upper hierarchy browser, while multiple signals can be selected in the signal windows for drag-and-drop operations.

To distinguish waveform data with the same file names from different directories, in the root entries of the upper hierarchy browser, directory paths of loaded waveform files are displayed using directory prefix `DX` where X is the sequential index of different directories.

To find out the original full paths of directory entries, select from the main menu **File > Show Directory Table** to display the table that maps directory identifiers to full file paths.



Analog Waveform Icon Reference

The following table describes what each analog waveform type icon signifies in the Waveform Browser:

Icon	Analog Waveform Type	Icon	Analog Waveform Type
	Real Voltage		(Real,Imaginary) Complex Generic
	Real Current		(Magnitude,Degree) Complex Generic

Icon	Analog Waveform Type	Icon	Analog Waveform Type
	Real Generic		(Magnitude,Phase) Complex Generic
	(Real,Imaginary) Complex Voltage		Real Voltage Alias
	(Magnitude,Degree) Complex Voltage		Real Current Alias
	(Magnitude,Phase) Complex Voltage		Real Analog Alias
	(Real,Imaginary) Complex Current		Sweeping Parameter
	(Magnitude,Degree) Complex Current		Derived Data Top Level
	(Magnitude,Phase) Complex Current		Wire Type

Digital Waveform Icon Reference

The following table describes what each digital waveform type icon signifies in the Waveform Browser:

Icon	Digital Waveform Type	Icon	Digital Waveform Type
	Logic Integer		Logic Register
	Logic Supply		Logic Alias
	Logic Parameter		Logic Variable
	Logic Wire		(Magnitude,Phase) Complex Generic

Non-Digital/Analog Waveform Icon Reference

The following table describes what each icon signifies besides those used for analog and digital waveform types in the Waveform Browser:

Icon	Icon Type	Icon	Icon Type
	Waveform Data File (no signal displayed, not linked)		Generic signal alias name
	Linked Waveform Data File (linked to netlist)		Voltage signal alias name
	Data file with some of its signals displayed		Current signal alias name
	An old data file that has been overridden by new result		Logic signal alias name
	One or more equations.		Waveform comparison results.

Displaying Signals

Signals from the Waveform Browser can be loaded into a waveview with the drag-and-drop operation. If a hierarchy (or file) entry is dragged from the upper hierarchy browser, all signals in the selected hierarchy level (or a file) are loaded. If a signal (or multiple signals) is dragged from the signal list window, only the selected signals are loaded. You can also double-click in the lower signal window to add an individual signal.

Depending on the signals being dragged, drop hints appear when signals are dragged over a waveview area. If the waveview is empty, hint boxes indicate available initial waveview modes. If the waveview contains existing panels, an insertion hint or drop-in box indicates the drop location.

If any signal from a waveform file is being displayed in a waveview window, the file icon changes, and the signal name is displayed in blue.

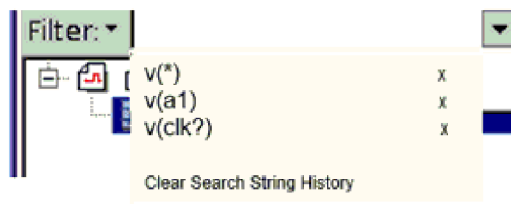
Alternatively, to plot all the signals of an instance, in the Waveform Browser, right-click an instance and select the **Plot Waveform to** option from the context menu.

Pinning a Filter String

The toggle button  pins or unpins a filter string to the current hierarchy. Once pinned (grey), the filter string is restored whenever selecting this hierarchy. The filter string is not editable while pinned. When unpinned (red), the filter string follows the preference of **Keep Search String in the Signal List Filter**.

Viewing Filter History

Filter history allows you to select previous strings entered into the filter line editor. The history dynamically updates as text is entered, and appears as a drop-down when the line editor is focused. For example:



The history list includes the last ten strings. Click **X** to remove an individual string, click **Clear Search String History** to remove all strings, and press Esc to close the history list.

Highlighting Waveforms

To highlight a waveform, click the signal check control button, and click the name of a signal to toggle its highlight state on and off.

You can also toggle a signal highlight by clicking the waveforms, which toggles the highlight states of all signals that intersect the mouse pointer.

To toggle which waveform is highlighted, press J to move down the waveform list and K to scroll up.

Deleting Highlighted Waveforms

Right-click a highlighted signal and choose **Cut** from the signal context menu to delete all highlighted signals.

You can also move highlighted signals in a panel with a drag-and-drop operation. With the signal mode control button armed, follow the steps in [Moving Panels](#) or [Copying Panels](#) to move highlighted signals between panels or waveviews.

Grouping and Ungrouping Highlighted Waveforms

Right-click a highlighted signal, and choose **Group** from the signal context menu to group all highlighted signals into a new panel.

You can also group signals by name by right-clicking signals and choosing **Group By Name** from the menu that opens.

To ungroup highlighted waveforms, right-click a highlighted signal, and choose **Ungroup** from the signal context menu to split each of the highlighted signals into a new panel.

Copying Waveform Names

Right-click a highlighted signal, and choose **Copy Label** from the signal context menu to copy the names of all highlighted signals. This enables you to select and copy multiple signal names at one time. Note that the buffer size is limited to 8192.

Sending a Waveform to Results Analyzer

Right-click a signal in the Waveform Browser or in the WaveView panel and choose **Send to Results Analyzer** to add signal expression/waveform from WaveView to Results Analyzer.

The RA calculator opens where you can construct or modify an expression.

Waveform Files and Data

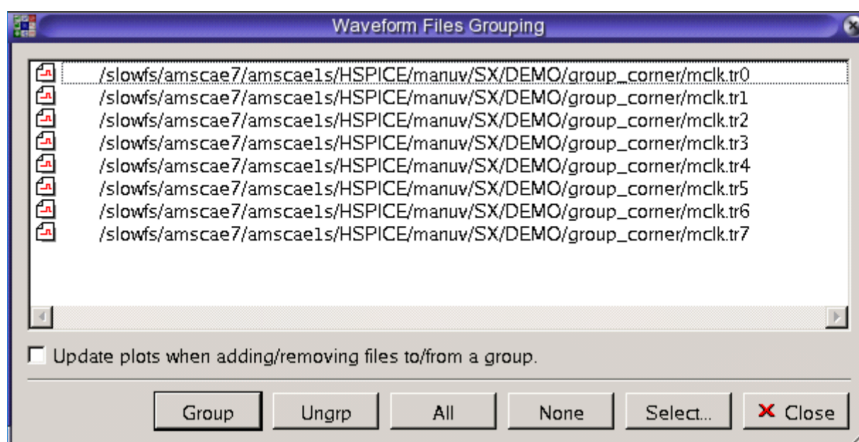
The section contains information on the following topics:

- [Grouping Waveform Files](#)
- [Bookmarking Waveform Files](#)
- [Setting the Active Waveform File](#)
- [Updating Waveform Files](#)
- [Printing Waveform Files](#)
- [Exporting Waveform Data](#)
- [Marching Waveforms](#)
- [Saving and Loading SigLists](#)
- [Dumping the Screen](#)

Grouping Waveform Files

Multiple waveform files can be collected into a group. When signals are loaded from a member in the grouped waveform files, the PrimeWave Design Environment tool automatically loads the same signals from the other waveform files in the same group.

The grouping function is useful for loading signals from corner/alter simulations that usually involve multiple output files. To group multiple waveform files, right-click a waveform file, and choose **File Grouping** from the menu that opens. The **Waveform Files Grouping** window appears.



Alternatively, click the **Group/Ungroup All Files**  icon in the menu toolbar to group or ungroup all files in this browser.

Select waveform files in the list to group/ungroup them. For waveform formats that have associated sweeping parameters (such as the COU and the PSF format), click **Select** to filter files based on their parameter value conditions or by name of the .fsdb files.

The **File Grouping** function only ties multiple files into a group. It does not assign any parameter condition to each individual file. As a result, parametric analysis is not possible on these grouped files. To add corner parameter conditions to files, the files must be grouped into a file set using a link file (similar to the runObjFile approach). The link file not only specifies which files to be grouped, it also defines the associated parameter conditions for each file in the group. Grouped files using the link file are processed as a single file as if the result is from a sweep analysis. Since the link file provides sweeping parameter conditions, parametric analysis is also supported.

To create the link file, right-click a file name, and choose **Create File Set** from the menu that opens. To add parameter conditions to a file, enter the value in the (name=value) format, multiple values are delimited by comma (for example, temp=0,vdd=3.3). Click **Add** to add the parameter values to a file and the new values are indicated in the file list. Note that each of the files in a file set must have the same number of the same parameters. Finally, enter the path to the link file and click **Ok** to create the link file. Once a link file is generated, it can be opened directly as a regular waveform file.

If you want to update the plotted waveforms after files are added to or removed from a group, click the **Update plots when adding/removing files to/from a group** check box before clicking the **Group** or **Ungrp** buttons.

Bindkey for the **File Grouping** is available in the menu command.

Bookmarking Waveform Files

A waveform file can be added to the bookmark list. Click **Bookmarks** and choose **Manage Bookmarks** in the **File** tab.

The **Add/Manage Bookmarks** window opens.

You can also reload a waveform file from the bookmark list from this window.

Setting the Active Waveform File

To have all plotted signals come from a specified waveform file, right-click a waveform file name in the WaveForm Browser and select **Set as Source of Plotted Signals**.

Updating Waveform Files

The WaveView tool can read growing waveform data files during a simulation job. Choose **Update** from the menu in the **File** section of the **Waveform** tab to update all displayed and derived signals.

If you restart a new simulation to override the old waveform data file, the update function automatically detects re-created data files and prompts you to select between keeping the old result or updating to new result.

If you keep the old data file, only the signals that are loaded are available from the old data file (because the old file is overwritten). An old data file is indicated by its blue file icon. To update an old data file to the new result, right-click the file in the Waveform Browser and choose **Update WDF** from the context menu.

To display signals from both the new and the old data file, choose to keep the old result. The old result file is saved and automatically opens the new data file.

If you need to update the WDF signal list, you can right-click the name of a file in the Waveform Browser and choose **Reload WDF Signal List** from the menu that appears.

Printing Waveform Files

On Linux platforms, the WaveView tool supports printout in the PostScript format only. To submit a print job, choose **Print** from the menu in the **File** section of the **Waveform** tab open the **Print Setup** window.

The print target is defaulted to the current active waveview, which can be changed by clicking **Change**. Select your preferred settings and click **Print** to submit the print job.

PostScript Print Layout

Multiple print options are supported for the PostScript printout:

- Print only the active waveview to a single page as shown on the screen.
- Print all signals in the active waveview using the display height. Multiple pages might be needed if the waveview has signals that are scrolled out of the view.
- Print all waveviews in a single page, multi-column layout. The user can choose the number of columns preferred.
- Print all waveviews, each in a single page, as shown on the screen display.

Select the **Color PostScript** option if your printer supports color PostScript.

The default print setup is black/white in landscape orientation. The default printer device and user-defined print command can be predefined in the .spxrc file.

Printing on Linux Platforms

On Linux platforms, the Print Setup dialog box supports print out to different paper sizes and printers. Printer names can be predefined in the Printer Devices field of the Preference Settings dialog box. Multiple printer names are delimited by semicolon (;). Selected paper size and printer device are saved in `$HOME/.spxlast` when the application is closed. The stored values are automatically used as the default when the application starts again. If the `PRINTER` environment variable is defined in your Linux shell, it is always used as the default printer.

Printing on Windows Platforms

On MS Windows platforms, in addition to the PostScript print-to-file option, the WaveView tool also supports printout to all installed printers except line printers. PostScript format is supported as a file output option only. Choose **Print** from the menu in the **File** section of the **Waveform** tab to open the MS Windows version of the Print Setup dialog box. Select preferred options and click **Continue** to select a printer device, or select the path for the PostScript output file.

Exporting Waveform Data

To export waveform data, choose **Export** from the menu in the **File** section of the **Waveform** tab. Enter your output path and other settings in the Waveform Export Parameters dialog.

Confirm your output file path to open the **Waveform Export Parameters** dialog box.

Choose the panels or waveviews you want to export in the **Target** section: **Selected Panels**, **All Panels**, **Selected Waveviews**, or **All Waveviews**. If you choose a selected

panel or waveform option, you are prompted to choose those panels or waveviews once you click the OK button.

Also, choose which signal names you want to export in the **Exported Signal Names** section: **Full Hierarchy** or **Net Name Only**. Click **Append File Name** if you want to append the file name to the associated exported signals.

Waveform data in the active waveview can be saved using several different formats. The following formats are supported:

Format	Description
Text Table	Tabulated format. You can specify output options in the Text Table Output Option section of the Waveform Export Parameters dialog box. Selecting the Suppress X-axis column option hides the x-axis in the table output. Selecting the Hex Logic Values option changes the binary logic output values to hexadecimal values. Selecting the Do not extend waveforms over the longer x-range of other waveforms option prevents waveforms from being extended over an x-range of another waveform that is longer.
PWL Source	SPICE PWL (piecewise linear) format. You can specify PWL logic levels and slew values in the Logic Waveforms to PWL Sources section, as well as any netlist syntax options in the PWL Source Netlist Syntax Options section of the Waveform Export Parameters dialog box.
WDF	Synopsys WDF compressed format. You can specify the Set Start X and Set Stop X values in the Text Table Output Option section of the Waveform Export Parameters dialog box.
VCD	VCD file with analog support. You can specify the VCD Timescale in the lower right-hand corner of the Waveform Export Parameters dialog box.
M-File	file. Choose the panels or waveviews you want to export in the Target section: Selected Panels, All Panels, Selected Waveviews, or All Waveviews. If you choose a selected panel or waveform option, you are prompted to choose those panels or waveviews once you click the OK button. Also, choose which signal names you want to export in the Exported Signal Names section: Full Hierarchy or Net Name Only. Click Append File Name if you want to append the file name to the associated exported signals. You can specify the Number of Valid Digits in the lower-left corner of the Waveform Export Parameters dialog box.

Format	Description
FSDB	FSDB file. Choose the panels or waveviews you want to export in the Target section: Selected Panels, All Panels, Selected Waveviews, or All Waveviews. If you choose a selected panel or waveform option, you are prompted to choose those panels or waveviews once you click the OK button. Also, choose which signal names you want to export in the Exported Signal Names section: Full Hierarchy or Net Name Only. Click Append File Name if you want to append the file name to the associated exported signals. You can specify the starting and stopping X values by clicking the Set X Start and Set Stop X options in the Text Table Output Options section.
OUT	.OUT file. Choose the panels or waveviews you want to export in the Target section: Selected Panels, All Panels, Selected Waveviews, or All Waveviews. If you choose a selected panel or waveform option, you are prompted to choose those panels or waveviews once you click the OK button. Also, choose which signal names you want to export in the Exported Signal Names section: Full Hierarchy or Net Name Only. Click Append File Name if you want to append the file name to the associated exported signals. You can specify the starting and stopping X values by clicking the Set X Start and Set Stop X options in the Text Table Output Options section.

By default, the WaveView tool only exports waveforms in selected panels using the x-axis step of the very first signal. Click **All Panels** to export waveforms in all panels.

Marching Waveforms

To automatically update waveforms from a running simulation job, choose **Marching** from the menu in the **File** section of the **Waveform** tab.

Enter the required update interval and the abort trial limit and click the **Start** button to start waveform marching. Automatic update is stopped if the waveform file is not growing after the retrial limit. The automatic update function can be manually stopped at any time by clicking the **Stop** button in the same dialog box.

If the automatic waveform marching is enabled, it is indicated with an icon in the main application toolbar.

Saving and Loading SigLists

Signal list files include all signal names that are currently plotted.

To save a signal list file, click **Save SigList** from the menu in the **File** section of the **Waveform** tab.

To load a signal list file, click **Load SigList** from the menu in the **File** section of the **Waveform** tab. Click the **Plot in current waveview check** box to plot all signals in the current waveview; otherwise, all signals are plotted in a new waveview.

Dumping the Screen

All waveforms in a waveview can be included when performing a screen dump, even if the waveforms are scrolled above or below the currently visible portion of the waveview window.

To include all off-screen waveforms, choose **Dump Screen** from the menu located in the **File** section of the **Waveform** tab. The **Save Screen Dump** dialog box opens. Click the **Use Maximum WaveView Height** check box, and click **OK**.

On Microsoft platforms, you can copy and paste the waveview contents to any application that has access to the clipboard. To copy the screen bitmap of a waveview window to the clipboard, choose **Copy to Clipboard** from the waveview context menu. To export a waveview in the vector-based Windows EMF (Enhanced Meta File) format, choose **Screen Dump to EMF**.

On Linux platforms, you can dump the waveview contents to the JPEG, PNG, or EMF formats. Choose **Dump Screen** from the waveview context menu to dump the display content.

Dump All Screens

To dump all screens, choose **Dump All Screens** from the menu located in the **File** section of the **Waveform** tab. The **Save Screen Dump** dialog box opens.

Setting Waveform Attributes

To modify waveform attributes for a waveform:

1. Double-click the signal name of the waveform, or click **Settings** on the **Waveform** tab, and choose **Waveform Attributes** from the menu that opens.

The **Waveform Attributes** dialog box opens with the attributes for the chosen signal.

2. Change any of the following waveform display attributes:
 - **Line Color**
 - **Line Width**

- **Dash Type**
 - **Marker Shape**
3. For analog signals, change the following attributes as necessary:
 - **Display Preference**
 - **Hide Signal**
 - **Axis** (left or right)
 4. For digital signals, change the following attributes as necessary:
 - **Display Preference** and type
 - **Mnemonic On**
 - **Radix**
 5. For analog or digital signals, enter the marker frequency, which changes the display frequency of the data point markers.
 6. (Optional) Click the **Turn on Alias** check box to change the alias of a signal.
 7. Click **Apply** to save your changes.

Working With Waveviews

The waveview window is the waveform display area in the WaveView tool. Click over the corresponding tab area of a waveview to bring it to the foreground.



The following topics are covered in this section:

- [The Active Waveview](#)
- [Adding a Waveview](#)
- [Deleting Waveviews](#)
- [Displaying WaveView Windows](#)
- [Displaying All Open Waveviews](#)
- [Displaying Waveview Titles](#)
- [Displaying the Hierarchy and Signal Browsers](#)
- [Displaying the Console Window](#)
- [Undoing and Redoing a Waveview Operation](#)
- [Refreshing Waveviews](#)
- [Renaming Waveviews](#)
- [Clearing Waveview Contents](#)

- [Dumping the Waveview Contents](#)
- [Docking and Undocking Waveviews](#)
- [Toggling the Tab Background Color](#)
- [Changing the Order of Tabbed Waveviews](#)
- [Synchronizing Waveviews](#)
- [Deleting All Open Waveviews](#)
- [Using the Drawing Tool](#)

The Active Waveview

In the dock mode, the top waveview is considered the active waveview. In the undock mode, the top-most waveview or the selected waveview is the active waveview. Clicking the tab of a waveview window makes that waveview the active waveview.

Adding a Waveview

An empty waveview is initially opened when the WaveView tool starts. To add more waveviews, click **WaveView** in the **Add** section of the **Waveform** tab.

The newly created waveview is placed on the top of the waveview stack.

Deleting Waveviews

To delete the active waveview, click the X button of a waveview tab.

Displaying WaveView Windows

A waveview window can have one or more non-overlapping Panels. Panels in a waveview can be arranged in two modes: vertical stack or independent row/column. In the vertical stack mode, panels stack top down and share a common horizontal axis. In the independent row/column mode, panels are arranged from left to right in a row. Panels in the row/column mode are independent of each other. A small icon at the upper-right corner of the waveview indicates the waveview orientation.

The following display modes are available:

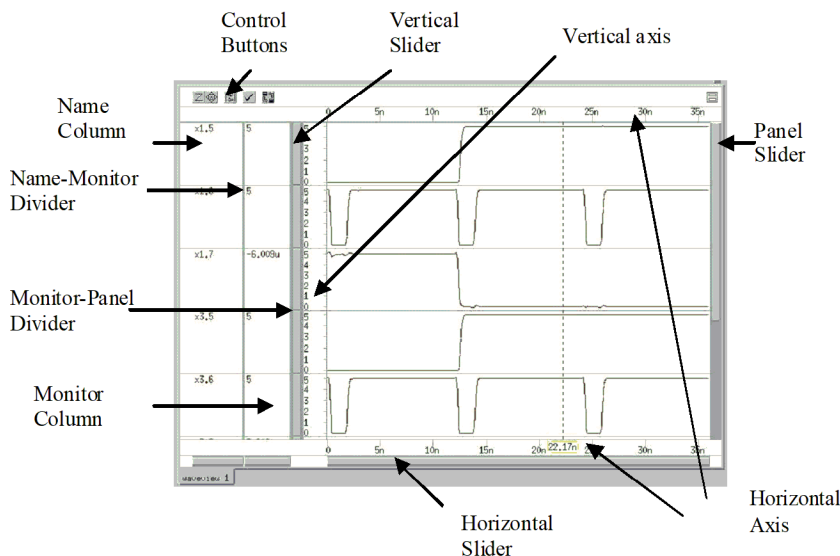
- [Stack Mode](#)
- [Row and Column Mode](#)
- [Vertical Row and Column Mode](#)

- [Horizontal Row and Column Mode](#)
- [Tiled Row and Column Mode](#)
- [TableView Mode](#)

Stack Mode

To view waveviews in vertical stack mode, right-click a waveview tab and choose **Stack Layout** from the menu that opens. Waveviews are displayed in this mode by default.

The following figure shows the vertical layout display.



The vertical layout has the following display components:

- **Control Buttons:** Provides shortcuts to some of the commonly used waveview functions.
- **Vertical Slider:** Panes zoomed waveforms vertically. It is not displayed for logic panels.
- **Vertical Axis:** The vertical axis of a panel. For a logic panel, it displays the vector width and radix.
- **Panel Slider:** Panes the entire panel stack vertically when the waveview height is too small to show all panels. Mouse wheel can be used to scroll this sidebar.
- **Horizontal Axis:** The common x-axis of all panels, which appears at both the top and the bottom of a vertical waveview.
- **Horizontal Slider:** This slide bar is used to pane zoomed waves horizontally.

- **Monitor Column:** Displays waveform or cursor-related values. Multiple monitors can be added. A horizontal slide bar at the bottom scrolls long value strings horizontally.
- **Monitor-Panel Divider:** Defines the left boundary of the waveform plotting area. Click and drag over the divider to resize the width of monitor columns.
- **Name-Monitor Divider:** Defines the border between the name and monitor area. Click and drag the divider to resize the width of name and monitor columns.
- **Name Column:** Displays waveform names. A vertical slide bar scrolls name list vertically. A slide bar at the bottom scrolls long name strings horizontally.

Row and Column Mode

To view waveviews in row and column mode, right-click a waveview tab and choose **Row/Column Layout** from the menu that opens.

Vertical Row and Column Mode

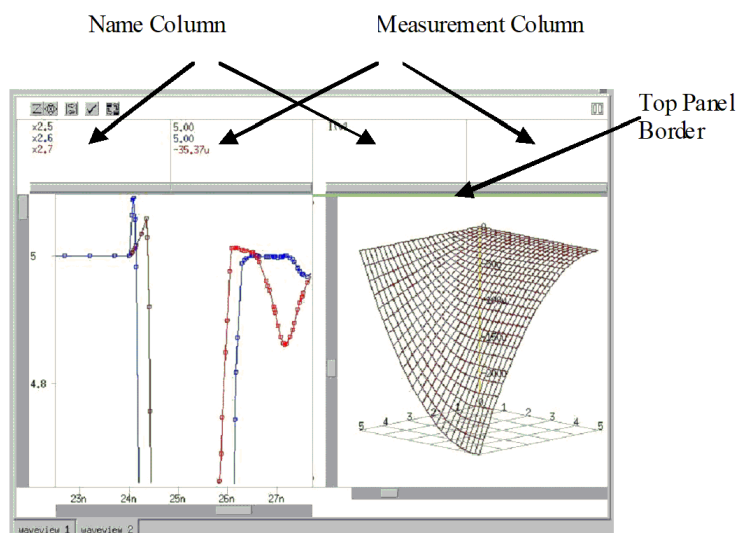
To view waveviews in vertical row and column mode, right-click a waveview tab and choose **Single Column Layout** from the menu that opens.

Horizontal Row and Column Mode

To view waveviews in horizontal row and column mode, right-click a waveview tab and choose **Single Row Layout** from the menu that opens.

You can use the mouse to change the height and width of the Signal Name panel in **Single Row Layout** by hovering the mouse over the edges of the panel until the cursor becomes a double arrow, then clicking and dragging to adjust the size.

The following figure shows the independent row and column display.



Panels in the independent row/column layout are arranged in a similar way to the Vertical Stack mode, except for the following differences and additional display components:

- **Name column:** The name column is located at the top of each panel. The vertical and horizontal slide bars can be used to scroll long list or long name strings.
- **Monitor column:** The monitor column is at the top of each panel. A horizontal slide bar at the bottom scrolls long value strings horizontally.
- **Top Panel border:** The top border can be moved to redefine the top boundary of the waveform plotting area. Drag the divider and move it vertically to resize name/monitor height.

Each waveview window is associated with a waveview context menu that can be invoked from the upper right corner of the window, or from the waveview tab area. Choose an item from the waveview context menu to rename, delete or refresh a waveview, or edit the waveview title.

Tiled Row and Column Mode

To view waveviews in tiled row and column mode, right-click a waveview tab and choose **Tile Row/Column Layout** from the menu that opens.

TableView Mode

To view tabular data, click the  button in the button toolbar. Only tabular views are allowed in tableviews.

See [Working With Tableviews](#) for more information on tabular panels.

Displaying All Open Waveviews

To display a list of currently open waveviews from which you can choose, click **Select View** in the **Edit** section of the **Waveform** tab.

Displaying Waveview Titles

To hide or display the title of a waveview, right-click a waveview tab and choose either **Show Title** or **Hide Title** from the menu that opens.

Displaying the Hierarchy and Signal Browsers

To toggle the Hierarchy and Signal Browser displays, click **Browser** from the **Configuration** tab or press **Ctrl-H** on your keyboard.

Displaying the Console Window

To toggle the display of the Console window, click **Console** on the **Configuration** tab. The Console window is displayed at the bottom of the WaveView tool main window.

Undoing and Redoing a Waveview Operation

To undo a waveview operation, click **Undo** in the **Edit** section of the **Waveform** tab. To redo a waveview operation, click **Redo**.

You can undo or redo panel, line, axis, cursor, monitor, measurement meter, drawing tool, and tabular operations.

Some waveview operations, such as adding or deleting a waveview, cannot be undone. The maximum undo and redo steps is limited to 100.

Refreshing Waveviews

Right-click the tab of a waveview, and choose **Refresh Waveview** from the menu that opens to refresh the display content of the active waveview.

Renaming Waveviews

Right-click a waveview tab, and choose **Rename Waveview** from the menu that opens to rename the active waveview.

Clearing Waveview Contents

To clear the contents of a waveview window, right-click inside a waveview and choose **Clear Waveview** from the menu that opens.

Dumping the Waveview Contents

All waveforms in a waveview can be included when performing a screen dump, even if the waveforms are scrolled above or below the currently visible portion of the waveview window.

To include all off-screen waveforms, right-click a waveview tab, and choose **Dump Screen** from the menu that opens. Click the **Use Maximum WaveView Height** check box, and click **OK**.

On Microsoft platforms, you can copy and paste the waveview contents to any application that has access to the clipboard. To copy the screen bitmap of a waveview window to the clipboard, right-click a waveview tab, and choose **Copy to Clipboard** from the menu that

opens. To export a waveview in the vector-based Windows EMF (Enhanced Meta File) format, choose **Screen Dump to EMF**.

On Linux platforms, you can dump the waveview contents to the JPEG, PNG, or EMF formats. Right-click a waveview tab, and choose **Dump Screen** from the menu that opens to dump the display content.

Docking and Undocking Waveviews

Overlapping (docked) waveviews in the waveview window can be undocked into individual pop-up windows. Right-click a waveview tab, and choose **Dock/Undock** from the menu that opens to undock the waveview. To dock and undocked waveview, click the dock button .

Toggling the Tab Background Color

To toggle the background color of individual waveform tabs between black and white, right-click on a tab and select **Black/White Background**.

Changing the Order of Tabbed Waveviews

If you have multiple waveviews open at the same time, you can change the order of those waveviews by dragging the tabs left or right.

Synchronizing Waveviews

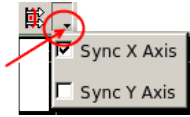
A stack-mode waveview can be synchronized with other stack-mode waveviews. Cursors and the x-axis of a synchronized waveview are updated automatically when the x-axis display range or the active cursor location changes in other synchronized waveviews. Right-click a waveview and choose **Sync/Unsync** from the menu that opens to toggle the sync state of the active waveview, or choose **Sync/Unsync All** to toggle the sync states of all waveviews.

Toggling Sync Mode

The  toggle button in the WaveView toolbar switches the synchronization status of the current waveview. When in sync mode () , clicking the button removes all panels in the current waveview from the sync group. When in unsync () , clicking the button adds all panels to the sync group.

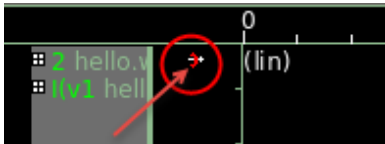
Setting Sync Axes

The option Sync Axes in the WaveView toolbar sets which axes to sync when in sync mode. This option is disabled in unsync mode. For example:



Adding Panels to a Sync Group

To add a panel to a sync group, **right-click** the panel and select **Add to Sync Group**. If the target panel is selected, all selected panels are added to the sync group. When in a sync group, the panel displays the following icon:



To remove a panel from a sync group, **right-click** and select **Remove from Sync Group**. If the target panel is selected, all selected panels are unsynced.

Deleting All Open Waveviews

To delete all currently open waveview windows, click **Delete All** in the **Edit** section of the **Waveform** tab.

Using the Drawing Tool

To open the Drawing Tool, choose **Drawing Tool** from the menu located in the **Utilities** section of the **Waveform** tab.

The following options are available:

Option Category	Option	Description
Remark Text	Remark Text	The remark text to add to your panel.
	Anchored to Waveform	Anchors the remark text to the waveform you choose with a visible line.
	Vertical	Orients remark text vertically on a panel.
	Border	Places the remark text on the border of a panel.

Option Category	Option	Description
Line	Font	Specifies the font for a remark. The default is Normal .
	Text Color	Specifies the color for a remark. The default is Auto .
	Line Color	Specifies the line color. The default is white .
	Line Width	Specifies the line width. The default is 1 .
	Dash Type	Specifies the line type. The default is a solid line .
	Position (X1, X2, Y1, Y2)	Specifies the X and Y coordinate position of a line, which depends on the kind of line you want to draw (Horizontal, Vertical, or Arbitrary)
	Position (Horizontal, Vertical, Arbitrary)	Specifies the type of line you want to draw.
	Fill X Range	Draws a line that fills the entire range of the X-axis when you draw a Horizontal line.
	Fill Y Range	Draws a line that fills the entire range of the Y-axis when you draw a Vertical line.

Working With Tableviews

The Tabular panel displays data such as waveform results and equations in a spreadsheet format. To create a Tabular panel, choose **Tabular** from the menu in the **Panel** section of the **Waveform** tab.

A new tabular panel is created on the current tableview. If the current view is not a tableview, a message is displayed that asks if you want to create a tableview.

Equation								
File	Equation	Specification		Result				Pass/Fail
		Min	Max	Value	Min	Max	Mean	
D0.yu.tr0	y _{max} (lv(1))	0	6	5				OK
D0.tran_sweep.tr0	y _{max} (lv(1))	2	7		1	8	4.5	25% Fail
D0.tran_sweep.tr0	mean(lv(1))	2	4		0.6	4.8	2.7	25% Fail

Waveform Compare						
Target	Check	Rule	Master File	Time Range	Tolerance	Pass/Fail
/remote/sandwor...	check1	Rule-1	/remote/sandwor...	0~20n	V[abs:0.0001 rel:0.0001] I[abs:1e-09 rel:0.01]	25% Fail
tran_sweep.tr0	check1(bound_chk)	bound_chk		10n~20n	[0 4]	Fail
tran_sweep.tr0	check2(env_chk)	env_chk	yu.tr0		v(1) v(2)	Fail

The Tabular panel can display the following types of information in separate table sections:

- equations
- signals
- waveform comparison results
- monitors

By default, only signals with scalar values are displayed in the tabular window (for example, when you double-click a signal from signal list). If you drag and drop a waveform result to tabular window, the signal is also displayed in the spreadsheet.

Note:

You can plot the value pairs from an X-Y panel into tabular format by right-clicking anywhere in an X-Y panel, and choose **Plot in Tabular** from the menu that opens. A new tableview is created with all the plotted value pairs from the X-Y panel.

Working With All Types of Tabular Data

This section contains the following information on how to sort and group all types of data, as well as how to hide and show table rows.

- [Hiding and Showing Rows](#)
- [Sorting Tabular Column Data by Ascending or Descending Order](#)
- [Filtering Tabular Data](#)

Hiding and Showing Rows

To hide a row in a Tabular window, right-click any row, and choose **Hide Row** from the menu that opens.

To show all rows that are hidden, right-click anywhere in the table, and choose **Unhide All Rows** from the menu that opens.

Sorting Tabular Column Data by Ascending or Descending Order

To sort column data by ascending or descending order, right-click a column title in the Tabular panel, and choose **Sort(Asc)** or **Sort(Desc)** to sort the data in ascending or descending order, respectively.

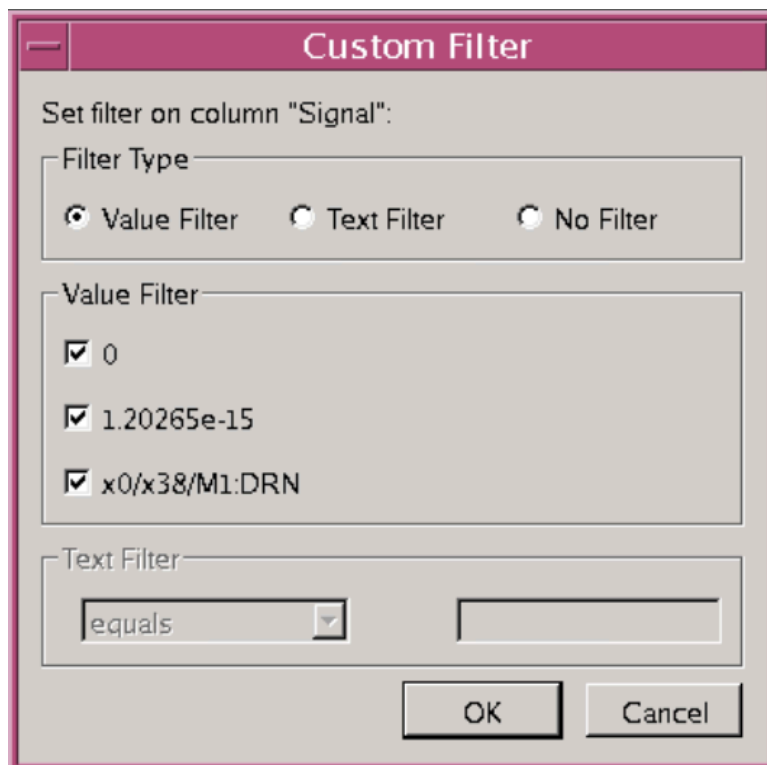
To sort the equations by signal names used in the equation strings, right-click the Equation column and select **Sort by Signal**.

Filtering Tabular Data

To filter tabular data:

1. Right-click a column heading in the Tabular panel that contains the data you want to filter, and choose **Value Filter** from the menu that opens.

The **Custom Filter** dialog box opens.



2. Choose a Filter type: **Value Filter**, **Text Filter**, or **No Filter**.

If you want to filter data by value, click **Value Filter**, and move on to the next step.

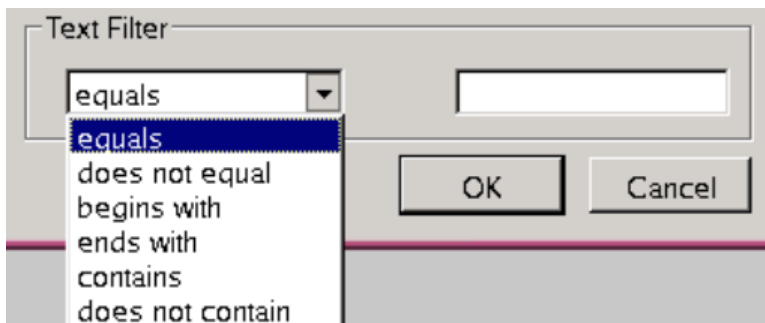
If you want to filter data by a text string, click **Text Filter**, skip to [Step 4](#).

If you click **No Filter**, skip to [Step 6](#).

3. Click the check boxes next to each value you want to filter from the table.

Skip to [Step 6](#).

4. Choose a filter qualifier from the menu in the **Text Filter** section.



The following qualifiers are available:

- **equals**
- **does not equal**
- **begins with**
- **ends with**
- **contains**
- **does not contain**

5. Enter a text string in the text box next to the filter qualifier menu to include in your filter.

6. Click **OK** to save your filter settings.

The tabular data is filtered in the chosen column.

Working With Tabular Equation Data

The Equation section of a Tabular panel contains the following equation data:

Column Title	Description
File	File name.

Column Title	Description
Equation	Equation name (or equation string if no name is specified). If no corner or sweep results are available, the signal waveform is displayed instead of the equation.
Specification	Equation specifications.
Result	Equation results.
Pass/Fail	Pass or fail status. If no specifications exist, pass or fail result are not available.

This section contains information on the following topics:

- [Modifying Tabular Equations](#)
- [Adding or Removing Tabular Statistics Results](#)
- [Marking Annotating Equation Information From Equation Results](#)
- [Exporting Tables](#)

Modifying Tabular Equations

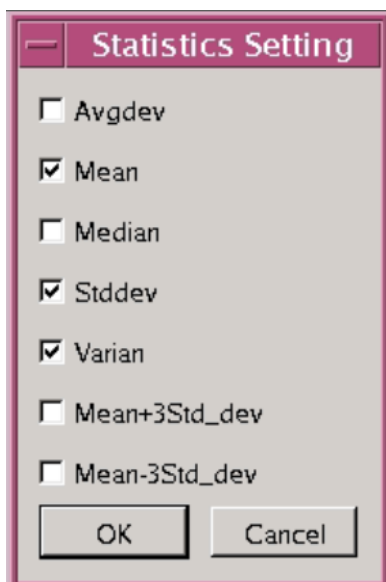
To modify an equation, right-click the name of an equation in the **Equation** column, and choose **Modify** from the menu that opens.

See PrimeWave Design Environment Calculator Overview for information on the Calculator.

Adding or Removing Tabular Statistics Results

To add or remove statistics in the Equation section of a Tabular panel:

1. Right-click the **Result** column, and choose **Statistics Setting** from the menu that opens.



The **Statistics Setting** dialog box opens.

2. Click the check box next to each result type you want to display in the table.
3. Click **OK** to save your changes.

The chosen result types are displayed in the table.

Marking Annotating Equation Information From Equation Results

If there is annotation information, left-click the link in the Equation column or right-click and select **Show in WaveView** to annotate the information in the waveview. For sweep scalar results, first expand the details.



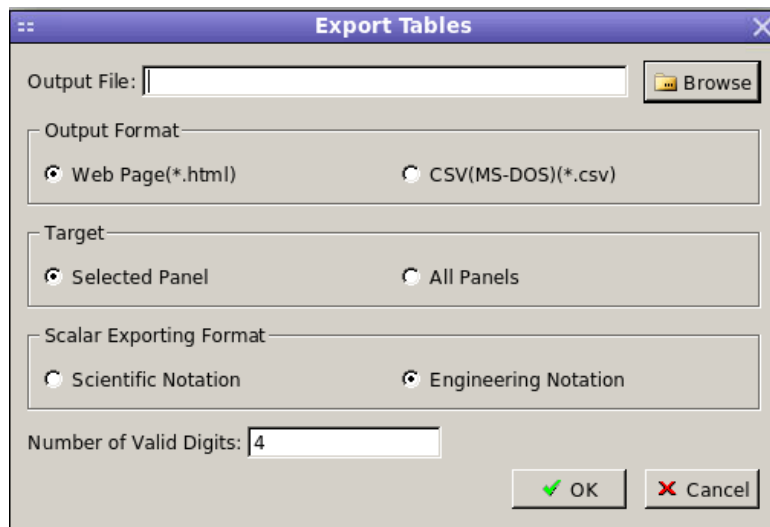
For example:

Exporting Tables

To export a table to an HTML or .csv file:

1. Click the  button on the toolbar.

The **Export Tables** dialog box opens.



2. Enter an **Output File** name, or click **Browse** to browse to and select one.
3. Choose an Output Format: **Web Page (*.html)** or **CSV (MS-DOS)(*.csv)**.
4. Choose a Target you want to export: **Selected Panel** or **All Panels**.
5. (Optional) Click the **Export Data With Unit Scale Factor** option to export the tabular data with the unit scale factor.

This option is enabled by default.

6. Enter the number of valid digits in the **Valid Digits** text box.
7. Click **OK** to export your table data.

Working With Tabular Signal Data

The Equation section of a Tabular panel contains the following equation data:

Column Title	Description
File	File name.

Column Title	Description
Signal	Signal name.
Hierarchy	Hierarchy location of the signal.
Value	Value of the signal or waveform data.
Unit	Unit of measure.

The Signal section of a Tabular panel contains the following signal data:

- File
- Signal
- Hierarchy
- Value
- Unit

Viewing Waveform Data

To view waveform data in the Signal section of a Tabular panel, click the  icon in the Value column. A waveview opens with the waveform data displayed.

Working With Tabular Waveform Comparison Data

The Waveform Compare section of a Tabular panel contains the following waveform comparison data:

Column Title	Description
Target	Target file.
Check	Waveform check.
Rule	Rule name.
Master File	Master file name.
Time Range	Time range values.
Tolerance	Tolerance value.
Pass/Fail	Pass or fail status.

This section contains information on the following topics:

- [Viewing Tabular Waveform Check Details](#)
- [Viewing Tabular Waveform Check Signal Details](#)

Viewing Tabular Waveform Check Details

To view waveform check details, click the name of a waveform check in the Check column in the Waveform Compare section. A new table opens with Signal and Difference Points/Rate information for that waveform check.

Click the Waveform Compare link to return to the Waveform Compare section, or you can click a signal in the Signal column to display signal details. See [Viewing Tabular Waveform Check Signal Details](#) for more information.

Viewing Tabular Waveform Check Signal Details

To view waveform check signal details, click the name of a signal in the waveform check details table (see [Viewing Tabular Waveform Check Details](#) for more information). The check signal details table is displayed, which contains x-axis, Master (value), (Target) Value, Diff, Diff(%) information. Double-clicking on a row in the details table plots the related signals on a waveview displays a vertical line showing where the difference occurs.

Comparing and Checking Waveforms

This section contains information on the following topics:

- [Sampling and Converting Waveform Data](#)
- [Creating a Waveform Compare Control File](#)
- [Defining Aliases](#)
- [Defining Rules](#)
- [Defining Parameters for Comparison and Checks](#)
- [Controlling X- and Z-Level Constraints](#)
- [Checking Waveforms](#)
- [Saving Waveform Comparison Results](#)
- [Using the Waveform Compare Tool](#)

Sampling and Converting Waveform Data

The following general rules apply when comparing waveforms:

- If a logic bus signal has different width (number of bits) between the master and the target file, only the less significant bits of the longer bus value are compared with the shorter bus value.
- Multiple signals are compared sequentially during the comparison process.
- Analog waveforms are compared based on the final V/I tolerance that is evaluated from user-defined tolerance values as:

```
tolerance = MAX(abstol, reltol*master_waveform_value)
```

- A signal must exist in both the master and one of the target files for the comparison to take place.
- If one of the signals from the master file and the target file is a logic signal, logic comparison is used to compare the two signals.
- For analog waveforms, voltage tolerance is used if master and target signals are not both current type.

Creating a Waveform Compare Control File

Before the waveforms are compared, create a check rule file to compare signals and generate reports of any differences that exist.

The check rule file must include the following items:

- Path to the master file and the target files.
- Names of signals to compare and name mapping information if the signal names are different between the master file and the target files.
- Start, stop, delay, and step values for the comparison to take place.

Note:

When adding comments, ensure each line begins with a semicolon (;).

Defining Aliases

Aliases define values that can be later referenced in the Rule and the Check sections of the control file. For example, you can define the following aliases to substitute a reference string in a control file:

```
alias small_step "step=200ps"
alias medium_step "step=.5ns"
alias initial_period "start=0ns, stop=5ns"
alias stable_period "start=5ns, stop=16n"
alias end_period "start=16ns, stop=20n"
```

Defining Rules

The Rule section of the Control File defines rules that you can use in the Check section to perform the actual checks. Multiple rules can be defined in the Rule section, and each rule must be defined using the following syntax format:

```
rule rule_name begin
  parm1=parm1_value
  parm2=parm2_value
end
```

The waveform compare function samples and compares waveform data at the steps you define. Each comparison rule can contain one master file and comparison parameters. You can specify the absolute and relative tolerances for voltage or current signals, respectively. Voltage tolerances are used if the signal type is unknown. Warnings are issued if the difference between the master signal and the target signal exceeds the following condition:

$$\text{abstol} + \text{reltol} * \text{master_signal}$$

To use a “one-direction” relative tolerance for comparison, add a plus (+) or minus (-) sign in front of the relative tolerance value.

You can define the following parameters in the Rule section:

- Type of check represented by a keyword. The following checks are supported:
 - Sample-based waveform compare function.
See [Checking Waveforms](#) for more information.
 - Waveform monotonicity check.
See [Checking Waveform Monotonicity](#) for more information.
 - Waveform absolute bound check.
See [Checking Waveform Bounds](#) for more information.

- Waveform envelop check.

See [Checking the Waveform Envelope](#) for more information.

- Waveform Pattern Check

See [Checking the Waveform Pattern](#) for more information.

- Comparison sampling step (overwrites global definition)
- Master file definition
- Check parameters such as tolerances and boundaries.

For example, `v_abstol`, `v_reltol`, `i_abstol`, and `i_reltol` overwrites the defaults and Global definitions.

Note:

Aliases can be used in comparison sampling steps and tolerance or boundary check parameters.

Defining Parameters for Comparison and Checks

You can define parameters to control the waveform compare and checks. Any parameters defined outside of the Rule and Check sections are treated as global parameters. Any locally defined parameters inside of the Rule and Check sections supersede defined global parameters.

The following parameters are available:

- [a2d_threshold](#)
- [batchsize](#)
- [bus](#)
- [delay](#)
- [full_compare_log](#)
- [ignorex](#)
- [ignorez](#)
- [len_reltol](#)
- [length_check](#)
- [map](#)

- `mapfile, map_hier`
- `master`
- `match_type`
- `mean_range`
- `mean_step`
- `mean_tol`
- `pattern_check`
- `ref_clock`
- `report_style`
- `report_type`
- `report_vio_sig_only`
- `show_diff`
- `show_maxmin`
- `signal`
- `start, stop, step`
- `t_tol`
- `use_a2d_always`
- `use_sim_steps`
- `v_abstol, i_abstol, v_reltol, i_reltol`
- `when`
- `x_absmgn`
- `x_chkrange`
- `x_values`
- `xval_precision`

a2d_threshold

The `a2d_threshold` statement applies to both the master and the target files. It describes the voltage threshold value for converting an analog signal to a digital signal or the

threshold range. This statement can be redefined in the check section to specify different thresholds for target signals. Default is 1.5. The syntax for the `a2d_threshold` value is:

```
a2d_threshold value or
a2d_threshold low_value high_value
```

Note:

You can also use the `a2d_threshold` statement in the check section for target file signals using the same syntax. For example:

```
alias num1 2
alias num2 3

rule rule1 begin
    master yu.tr0
    step 1n
    use_a2d_always
    a2d_threshold num1 num2
    xval_precision 16
end

check begin
    target yu.tr0
    signal v(2)
    a2d_threshold 1 4
    rule rule1
end
```

batchsize

The `batchsize` statement controls the number of signals that are compared at a time, which can save memory and improve performance. The syntax for the `batchsize` statement is:

```
batchsize value
```

For FSDB and WDF files, set `value` to 1. For PSF or *.tr0 files, use the following formula to calculate the appropriate `value`:

```
value <= host_RAM / (filesize * 2 / num_of_signals)
```

For example:

```
batchsize 200
```

bus

The bus statement describes how a master logic bus value is interpreted when compared to an analog target signal. The following keywords are used with the bus statement:

- scale

Specifies the scaling factor from the decimal bus value.

- signed/unsigned

Specifies if the bus value is signed or unsigned.

- pwc/pwl

Specifies if PWC or PWL interpolation is used for value in between two bus value transitions. The syntax of the bus statement is:

```
bus bus_signal_name scale=value unsigned pwc
```

delay

The delay value applies to both the master and the target files. It describes the shift amount of x-axis value the user prefers to add to the original waveforms. The value can be redefined in the master or target file section. Positive delay value implies shifting the waveform toward the positive x-axis direction. Default is zero. The syntax for the delay parameter is:

```
delay value
```

Note:

If you use delay with use_sim_step, the following occurs:

- The waveform shifts by the specified delay value.
- The time points from sim_steps (in the [start stop] time range when specifying time_range) is compared.

full_compare_log

The full_compare_log statement creates a log file using the <rule_file_name>.log file name syntax to record the comparison details for all points.

This log file records the following information:

- master values
- target values

- difference values (target minus master values)
- tolerance

ignorex

The ignorex statement ignores x states during comparison.

ignorez

The ignorez statement ignores z states during comparison.

len_reltol

The len_reltol statement sets the relative length tolerance. The default is 0.01.

length_check

The length_check statement checks the length of a curve.

map

The map command is required when the compared signal names differ between the master and target files. One map statement is required for each mapped signal. The syntax of the map statement is:

```
map master_signal_name target_signal_name
```

mapfile, map_hier

The mapfile statement specifies a separate map file, which you can create to define master and target signals for comparison. The syntax of the mapfile statement is:

```
mapfile <mapfile_name>
```

The following example is a sample map file:

```
;mapfile1
;Master_SigName      Target_SigName
node_1      node_2
node_3
node_5      node_5
node_6      node_7
```

In this example, `node_1` and `node_2`, as well as `node_6` and `node_7`, are compared as the Master Signal and Target Signal for each pair, respectively. The `node_3` and `node_5` are compared as the same name.

The `map_hier` statement maps the hierarchy of the defined signals. The syntax of the `map_hier` statement is:

```
map_hier <master_signal_hier> <target_signal_hier>  
<master_signal_leaf_name> [<target_signal_leaf_name>]
```

The following example is a sample map file that uses the `map_hier` statement:

```
;commands inside the map file  
map_hier testbench testbench.ifourphase  
v(ck*)  
end_hier
```

Note:

The `mapfile` statement can only be specified in the Check section, and `map_hier` can only be specified within a `mapfile` statement.

master

The `master` statement declares the file path to be used as the golden file when comparing waveforms.

match_type

The `match_type` statement forces target signals to be the same type as the master signals.

mean_range

The `mean_range` statement performs a mean comparison for a derived set of waveforms based on the defined starting and stopping range values. The syntax for the `mean_range` statement is:

```
mean_range start stop
```

When a mean check is enabled, the mean version waveforms are first created for the master and target signals. Using the mean check does not affect the time points used for the check. If the time value falls in one of the `mean_range` periods, the mean waveforms are used instead of the original waveforms. Multiple `mean_range` statements can exist in a single check section.

mean_step

The `mean_step` statement samples waveforms and performs mean comparisons based on the defined step value. The syntax for the `mean_step` statement is:

```
mean_step step_value
```

Default is 1ns. Mean comparisons are enabled as long as the `mean_tol` statement appears in the rule section.

mean_tol

The `mean_tol` statement performs a mean comparison based on the defined tolerance value. The syntax for the `mean_tol` statement is:

```
mean_tol value
```

Default is 0.01.

pattern_check

The `pattern_check` statement searches for a pattern in logic signals. The syntax for the `pattern_check` statement is:

```
pattern_check <pattern>
```

For example:

```
pattern_check 101
```

ref_clock

A reference trigger clock signal can be used to determine the x-axis locations at which the comparison takes place. Two statements are required to define the reference trigger clock. Their syntax are:

```
ref_clock signal_name rise|fall|crossref_threshold value
```

The reference clock signal can be a logic or analog signal. For an analog reference signal, the `ref_threshold` value also needs to be defined. If the `ref_threshold` statement is omitted, the `a2d_threshold` value is used as the default. If `ref_clock` is defined in the rule file, the `stop/start/step` values are ignored.

report_style

The `report_style` statement defines the type of comparison report to produce. The syntax for the `report_style` statement is:

```
report_style value
```

For example, including the following line creates a report you can use with NanoSim:

```
report_style nanosim
```

report_type

The `report_type` statement defines what information is included in a comparison report. The syntax for the `report_type` statement is:

```
report_type value
```

You can specify the `value` as either `point`, `range`, or `.err` (`point` is the default value).

report_vio_sig_only

When the `report_vio_sig_only` statement is specified, only signals with violations are reported in the output file. The `report_vio_sig_only` statement can be in the global or rule section, and it applies to all the following comparison types:

- waveform comparison
- monotonicity check
- length check
- bound check
- envelope check
- pattern searching

The syntax for the `report_vio_sig_only` statement is:

```
report_vio_sig_only
```

show_diff

When the `show_diff` statement is specified, the analog value differences in percentage are reported in the output file.

For example:

```
rule v_check begin
    master inv.mt0
    use_sim_steps
    v_abstol 0.0001
    show_diff
    show_maxmin
end

check begin
    target inv.mt1
    signal *
    rule "v_check"
end
```

show_maxmin

When the `show_maxmin` statement is specified, the max, min, and mean of the analog value differences are reported in the output file.

For example:

```
rule v_check begin
    master inv.mt0
    use_sim_steps
```

```
v_abstol 0.0001
show_diff
show_maxmin
end

check begin
  target inv.mtl
  signal *
  rule "v_check"
end
```

signal

The signal statement defines the signal to be compared and can be added in the Check section or used globally. Wildcards (* and ?) can be used in the name pattern. If the signal statement is omitted, all signals are compared. The syntax for the signal statement is:

```
signal name1 name2 ... type=analog|logic|all -exclude
```

The `type` specifies the type of signal, and `-exclude` excludes the specified values from wildcard (* or ?) matching.

For example:

```
signal * -exclude i*
```

or

```
signal v* -exclude v(a1) v(a2)
```

start, stop, step

These values apply to the comparison process. The start and stop statements can be used globally, and the step statement can be defined in the Rule section or globally. All three can be redefined in the master file section. Default is 0. Syntax for these values are:

```
start | stop | step value
```

t_tol

The `t_tol` value specifies the point just before the continuous violation time range is exceeded. If the continuous violation time range is larger than `t_tol`, then the violation is reported; otherwise, it is considered as a passing result.

The `t_tol` setting is only used when comparing one master versus one target, analog to analog comparison, and analog to digital comparison only. Setting `t_tol` to zero disables glitch filtering. The syntax for the `t_tol` statement is:

```
t_tol value
```

For example:

```
rule rule1 begin
    master target.tr0
    step 0.2n
    a2d_threshold 1.5
    t_tol 0.9n
    xval_precision 16
end

check begin
    target output_tb065.evcd
    signal v(a1)
    map v(a1) TESTBENCH.INST1.MEM1/MS_1T1C
    time_range 0 10n
    rule rule1
end
```

use_a2d_always

The `use_a2d_always` statement forces a comparison between two analog signals with converted logic values used for `a2d_threshold`.

use_sim_steps

The `use_sim_steps` statement uses the time steps saved in waveform files to determine where waveform comparisons occur on the x-axis.

v_abstol, i_abstol, v_reltol, i_reltol

The `v_abstol`, `i_abstol`, `v_reltol`, `i_reltol` values define the error tolerances for comparing analog waveforms. The values can be redefined in the target file section. Defaults are:

```
v_abstol=50uv, i_abstol=1nA, v_reltol=0.001, i_reltol=0.01
```

The syntax for setting the tolerance values is:

```
v_abstol | i_abstol | v_reltol | i_reltol value
```

The final error tolerance used internally for waveform comparison is evaluated using the following equation

$$\text{MAX}(\text{ABS}(\text{abstol}), \text{reltol} * \text{ABS}(\text{master_v}))$$

Note:

`v_abstol` and `v_tolerance` are equivalent statements. `i_abstol` and `i_tolerance` are also equivalent statements.

when

The when statement executes when the specified condition is met. If when is used in the global or rule section, it applies to the master file. If the specified signal is not located in the master file, the tool tries to find the signal in the target file. If when is used in the check section, it applies to the target file.

Only simple conditions ($\neq$ and $=$) are supported, and only one signal can be specified. For example:

```
when "v(b) != 1"
```

Arbitrary when statements are supported for logic signals.

x_absmgn

The x_absmgn value specifies the x-axis tolerance margin and is used to compare analog signals. The syntax for the x_absmgn statement is:

```
x_absmgn value
```

The value without a plus (+) or minus (-) sign specifies bidirectional tolerance. Adding a plus or minus sign specifies a unidirectional tolerance (x_absmgn +5n or x_absmgn -3n, for example).

x_chkrange

The x_chkrange statement performs logic transition waveform comparisons based on the value, which searches for a signal transition within the defined window. The syntax for the x_chkrange statement is:

```
x_chkrange value
```

The value without a plus (+) or minus (-) sign specifies bidirectional checking. Adding a plus or minus sign specifies unidirectional checking (x_chkrange +5n or x_chkrange -5n, for example).

Note:

The x_chkrange statement can be used in both rule and check sections when comparing digital signal transitions.

x_values

The x_values statement specifies the time points to compare.

For example:

```
rule rule1 begin
    master comp_r.tr0
end
```

```
check check1 begin
    target comp_r.tr0
    signal v(rpin1)
    map v(rpin1) v(rpin2)
    x_values 1n 20n 45n 100n
    rule rule1
end
```

xval_precision

The xval_precision statement controls the x-value precision. The syntax for the xval_precision statement is:

```
xval_precision n
```

The *n* is the number of digits past the decimal point.

Controlling X- and Z-Level Constraints

X- and Z-level constraints change depending on the way you use some waveform comparison statements.

For example, when you use use_sim_steps and x_chkrange, the following X- and Z-level constraints are applied:

- The X and Z states remain as is and are compared. This is the default behavior.
- The default X and Z states change in the following manner if you use the following commands as described:
 - setx 0: Sets X-states to “Low” for comparison.
 - setx -1: Sets X-states to the previously known state for comparison.
 - setx 1: Sets X-states to “HIGH” for comparison.
 - setz 0: Sets Z-states to “Low” for comparison.
 - setz -1: Sets Z-states to the previously known state for comparison.
 - setz 1: Sets Z-states to “HIGH” for comparison.

Note:

Using the ignorex or ignorez statements with the x_chkrange statement might produce unwanted results.

Checking Waveforms

This section contains information in the following topics:

- [Waveform Compare Utility](#)
- [Waveform Check Example](#)
- [Checking Waveform Monotonicity](#)
- [Checking Waveform Bounds](#)
- [Checking the Waveform Envelope](#)
- [Checking the Waveform Pattern](#)
- [Rules Section Example](#)

Waveform Compare Utility

The Waveform Compare Utility checks the rules specified in the check section sequentially on the specified targets, time range, and signal list. Checking parameters can be redefined to and apply to the subsequent checks in the check section. The check is performed sequentially on all rules specified in the Check section and reports the violations for each check.

You can specify the following parameters in the Check section:

- Target files.

Multiple target files can be delimited by spaces or commas. For example:

```
target "t1.dat t2.dat"
```

- Check start/stop time (x-axis) range.

The `time_range` statement defines the check start and stop times.

For example:

```
time_range 10n 20n
```

- Target signals.

Signal names can contain the asterisk (*) or question mark (?) wildcards. For example:

```
signal "*"
```

- The rule to use.

A predefined rule must be specified for each check. For example:

```
rule "initial_v_check"
```

Waveform Check Example

The following is a sample waveform check:

```
check begin
;; value can be a pre-defined alias
time_range "0ns, 6ns"
;; the rule must have been defined in the rule section
rule "initial_v_check"

;; missing target files are automatically removed from the

;; target file list
target "t1.dat, t2.dat, t3.dat, t4.dat"
signal "s1,s2,s3"
time_range stable_period
rule "v_check"

;; check different time range
target "t1.dat, t2.dat, t3.dat"
signal "s1,s2,s3"
time_range end_period
rule "last_master"

target "t1.dat"
signal "s1"
time_range "5n,15n"
rule "mono_chk"

;rule "bound_chk"

target "t1.dat"
signal "s1"
; varcond_target "VDD=5:TEMP=100"
rule "env_chk"
end
```

Checking Waveform Monotonicity

The monotonicity check function screens each of the specified signals in the target file for Y-value magnitude inversion. The monotonicity check does not require any input parameter. A warning is issued when the monotonicity is reversed and restored.

```
rule mono_chk begin
monotonicity_check
end
```

Checking Waveform Bounds

The bound check function screens every data point of the specified signals in the target file against the absolute lower and upper bounds defined in the `upper_bound` and

`lower_bound` statements. A warning message is issued for each data point that exists outside of the bounds.

```
rule bound_chk begin
  upper_bound 6.2
  lower_bound 0.5
end
```

Checking the Waveform Envelope

The envelope check function screens every data point of the specified signals in the target file against the two 'bounding' signals defined in the `bound_sig1` and `bound_sig2` statements. A warning message is issued for each data point that exists outside of the bounds.

```
rule env_chk begin
  master gold1.dat
  bound_sig1 s1
  bound_sig2 s2
  ; master_sig1  sweep1.dat
  ; master_sig2  sweep2.dat
  ; varcond_sig1 "VDD=5:TEMP=0"
  ; varcond_sig2 "VDD=5:TEMP=100"
end
```

Checking the Waveform Pattern

The waveform pattern check searches for a specified pattern in logic signals. If the pattern is found, a message is issued with the x value where the pattern starts.

```
rule pattern_chk begin
  pattern_check 101011
end

check begin
  target foo.vcd
  signal *
  time_range "0ns 40ns"
  rule "pattern_chk"
end
```

Rules Section Example

The following is a sample rules section:

```
rule initial_v_check begin
  master gold1.dat
  step    small_step
  v_tolerance 1mv
  v_reltol   +0.01
end
```

```
rule v_check begin
    master gold2.dat
    step .1ns
    v_tolerance 1mv
;    delay 1ns
    v_reltol -0.01
end
```

If a waveform compare rule contains no master file, the previous master file in the check sequence is used. For example:

```
rule last_master begin
    step .2ns
    v_tolerance 2mv
end
```

Saving Waveform Comparison Results

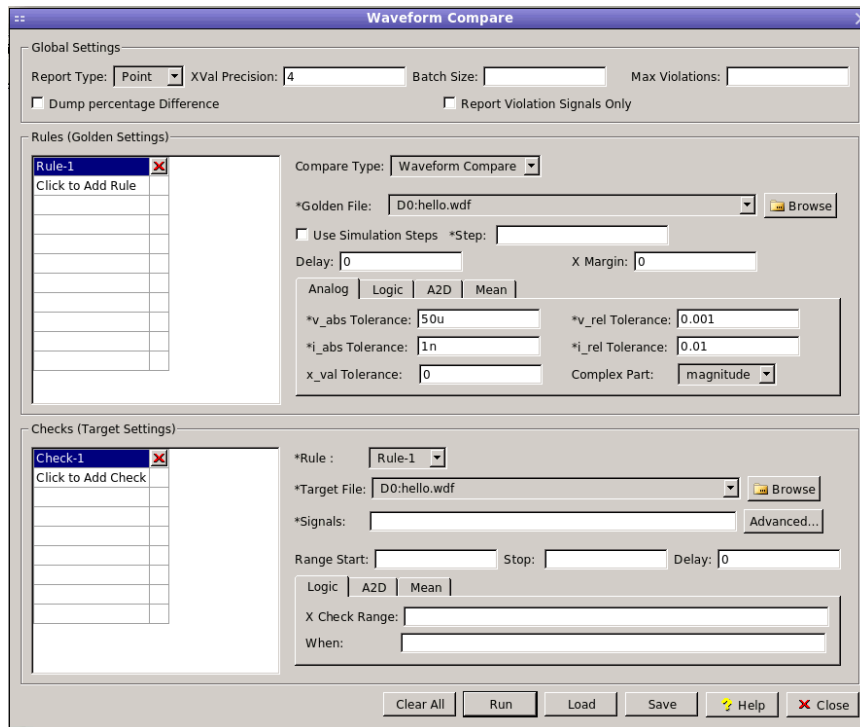
Waveform comparison results can be saved in a session file with the violated intervals highlighted by adding the `-x session_file` option to the `wv -compare` command. The operating system and version information is also included in this file. You can load the session file into the WaveView tool for further analysis.

Using the Waveform Compare Tool

To compare waveforms in the GUI:

1. Click **Waveform Compare** in the **Tools** tab.

The **Waveform Compare** window opens.

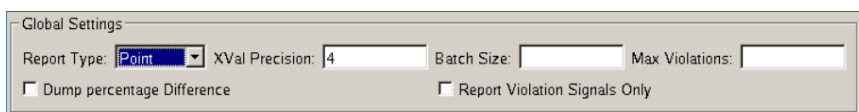


2. Specify Global Settings.
See [Specifying Global Settings](#).
3. Add one or more rules.
See [Adding Rules \(Golden Settings\)](#).
4. Add one or more checks.
See [Adding Checks \(Target Settings\)](#)
5. Run your comparison.
See [Running Comparisons](#).

Specifying Global Settings

To specify Global Settings for a waveform comparison:

1. Ensure you have the **Waveform Compare** window open (click **Waveform Compare** in the **Tools** tab).
2. Choose the type of report you want to generate from the **Report Type** menu in the Global Settings.



Global Settings

Report Type: **Point** XVal Precision: Batch Size: Max Violations:

☐ Dump percentage Difference ☐ Report Violation Signals Only

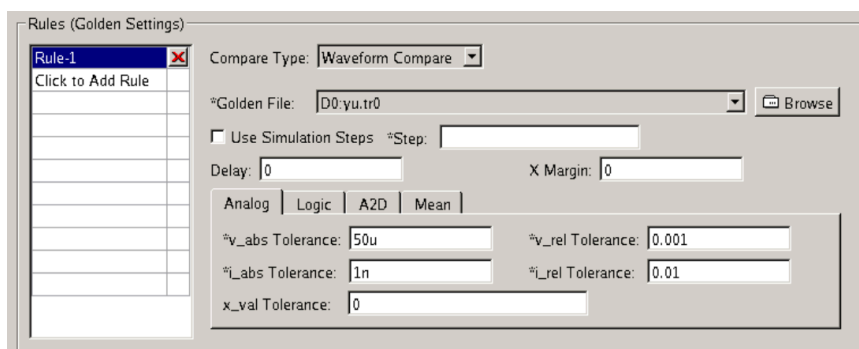
3. (Optional) Enter values for the **XVal Precision**, **Batch Size**, and **Max Violations**.
4. (Optional) Click the **Dump Percentage Difference** check box to include the percentage difference in the comparison report.
5. (Optional) Click the **Report Violation Signals Only** check box to include only the signals that have violations in your comparison report.
6. To add rules to your waveform comparison, continue to the next section, [Adding Rules \(Golden Settings\)](#).

Adding Rules (Golden Settings)

Note:

If you already have a rule file set up, click the **Load** button to load your rule file.

To add rules to your waveform comparison, choose what kind of comparison you want to run from the **Compare Type** menu in the **Waveform Compare** window. Double-clicking on rule name in the rule table enters editing mode where you can modify the rule name. No spaces are allowed in rule names.



Rules (Golden Settings)

Rule-1

Click to Add Rule

Compare Type: **Waveform Compare**

*Golden File:

☐ Use Simulation Steps *Step:

Delay: X Margin:

Analog | Logic | A2D | Mean

*v_abs Tolerance: *v_rel Tolerance:

*i_abs Tolerance: *i_rel Tolerance:

x_val Tolerance:

The following comparison types are available. Any settings for each comparison type that has an asterisk (*) next to it are required:

Comparison Type	Description
Waveform Compare	Compares waveforms. Continue to the Adding Waveform Compare Rules section.
Envelope Check	Checks the specified envelope. Continue to the Adding Envelope Check Rules section.

Comparison Type	Description
Monotonicity Check	Checks for monotonicity. No further rule setup is necessary—continue to the Adding Checks (Target Settings) section.
Bound Check	Checks the specified bounds. Continue to the Adding Bound Check Rules section.
Length Check	Checks the specified length. Continue to the Adding Length Check Rules section.
Pattern Search	Searches for the specified pattern. Continue to the Adding Pattern Search Rules section.

Adding Waveform Compare Rules

To add waveform compare rules:

1. Choose a golden file from the **Golden File** menu in the **Waveform Compare** window, or browse to and select the golden file you want to use.
2. Enter the step value you want to use for the waveform comparison in the **Step Value** text box, or click the **Use Simulation Steps** check box to use the steps from your simulation instead.
3. (Optional) Enter delay and X margin values into the **Delay** and **X Margin** text boxes.
4. Enter values for the following Analog tolerance values on the Analog tab:
 - **v_abs Tolerance**
 - **i_abs Tolerance**
 - **v_rel Tolerance**
 - **i_rel Tolerance**
 - **x_val Tolerance**
5. (Optional) Click the **Always Convert Analog signals to Logic then Compare** check box to always compare analog to digital.
6. Enter values for the **High** and **Low** A2D Threshold.
7. Click the Logic tab, and choose values from the **Treat as X** and **Treat as Z** menus.
8. Enter a value into the **X Check Range** text box for the X shift you allow when comparing a transition.

This is enabled only when **Use Simulation Steps** is selected.

9. Enter a value into the **When** text box to specify a logical condition.

The data is compared when the condition is true (for example, when "xclk/clkin==0") .

10. Click on **Mean** tab.
11. Click the **Enable Mean Check** check box to check the mean values of each step instead of comparing the original values.
12. Click **Mean Steps** to specify the steps used in a mean check.
13. Click **Mean Tolerance** specify the absolute tolerance for mean value comparisons.
14. To add checks to your waveform comparison, continue to the next part of the waveform setup, [Adding Checks \(Target Settings\)](#).

Adding Envelope Check Rules

To add envelope check rules:

1. Choose a signal file from the **Bound Signals File** menu in the **Waveform Compare** window, or browse to and select a file.
2. Enter signal names in the **Bound 1** and **Bound 2** text boxes.
3. To add checks to your waveform comparison, continue to the next part of the waveform setup, [Adding Checks \(Target Settings\)](#).

Adding Bound Check Rules

To add bound check rules, click the **Upper Bound** or **Lower Bound** check boxes in the **Waveform Compare** window, and enter values for the upper bound or lower bound limit, respectively.

To add checks to your waveform comparison, continue to the next part of the waveform setup, [Adding Checks \(Target Settings\)](#).

Adding Length Check Rules

To add length check rules:

1. Choose a golden file from the **Golden File** menu in the **Waveform Compare** window, or browse to and select the golden file you want to use.
2. Enter values for the **Absolute Tolerance** and **Relative Tolerance**.
3. To add checks to your waveform comparison, continue to the next part of the waveform setup, [Adding Checks \(Target Settings\)](#).

Adding Pattern Search Rules

To add pattern search rules, enter a pattern in the **Pattern to Use** text box in the **Waveform Compare** window.

To add checks to your waveform comparison, continue to the next part of the waveform setup, [Adding Checks \(Target Settings\)](#).

Adding Checks (Target Settings)

You can add multiple checks for each rule you add. To add checks to your waveform comparison setup:

1. Choose a rule from the **Rule** menu in the **Waveform Compare** window, which you already set up in the [Adding Rules \(Golden Settings\)](#) section.

The displayed check settings change depending on the type of rule you choose.

2. Choose a target file from the **Target File** menu, or browse to and select the target file you want to use.
3. Enter the names of one or more signals you want to check into the **Signals** text box.
4. Click **Advanced** to specify more signal related settings.
5. Enter a value for the **Max Hierarchy Level**.
0 is the default.
6. Choose a type of signal from the **Signal Type** menu.
7. Click the **Match Type** check box to match the type of signal.

If you are setting up a check for a rule that is Length Check comparison, no further setup for this check is available. Skip back to [Step 1](#) to add more checks for rules, or if you are finished, skip to [Step 16](#). Otherwise, continue to the next step.

8. Enter values for the **Range Start** and **Stop**.

If you are setting up a check for a rule that is Monotonicity or Bound Check comparison, no further setup for this check is available. Skip back to [Step 1](#) to add more checks for rules, or if you are finished, skip to [Step 16](#). Otherwise, continue to the next step.

9. Enter a value for the **Delay**.
10. Specify the master to target signal name mappings in the Signal Mapping Table.
11. Click the **Logic** tab.
12. Enter a value into the **X Check Range** text box for the X shift on a target signal if different from the X Check Range value in Rule section.

This works only when you enable **Use Simulation Steps**.

13. Enter a value into the **When** text box to specify a logical condition using target file signals.

The data is compared when the condition is true (for example, when `xclk/clkin_b==0`).

Skip back to [Step 1](#) to add more checks for rules, or if you are finished, continue to the next step.

14. Click on the **A2D** tab, and enter the **Low** and **High** A2D thresholds if they differ from the thresholds you set in Rule section.
15. Click the **Mean** tab, and enter the Mean check start and end values.
16. Your waveform comparison setup is complete. Continue to the final part of the waveform comparison, [Running Comparisons](#).

Running Comparisons

To run a waveform comparison:

1. Ensure you have the waveform comparison setup complete.

See [Using the Waveform Compare Tool](#) for more information.

2. Click the **Run** button.

The Save Waveform Compare Result File dialog box opens.

3. Enter a name for your result file, and click **OK** to save it in the required location.

The waveform comparison runs, and the results are saved in your result file. The result file is automatically opened in the WaveView tool, and the result is plotted to a tableview.

See [Working With Tableviews](#) for more information on working with tabular waveform comparison data.

Comparing Logic Signals

The batch mode sample-based waveform compare function compares mixed-signal waveforms between multiple target waveform files and a golden master waveform file. The comparison rules are defined in a check rule file that is used to control the comparison process.

To compare waveforms in batch mode using the Waveform Compare utility:

1. Review the general rules that apply when comparing waveforms.

See [Sampling and Converting Waveform Data](#) for more information.

2. Create a check rule file if you want to compare signals and generate reports of any differences that exist.

See [Creating a Waveform Compare Control File](#) for more information.

3. Use the following syntax at the command line to start a comparison:

```
wv -compare rule_file [output_file] [-x session_file]
```

The `output_file` is the name of the output file where the comparison results are saved. If you do not specify a name, the comparison result is displayed directly in the standard output. If you specify `-x session_file`, a session file is created, which saves the comparison files, signals, and differences information. You can load this file into the PrimeWave Design Environment tool to view the comparison results.

During comparison, if any errors occur due to unlocated or mismatched signals, a list of the mismatched signals are saved in a `mismatch.txt` file.

Setting User-Specified Colors for Diff Markers

You can specify colors for diff markers in the session file created by waveform compare, as follows:

```
wv -compare <rulefile> -x <session_file> -diff_color  
    <color_or_color_list>
```

or:

```
wv -x <session_file> -diff_color <color_or_color_list>
```

where `-diff_color` specifies a color or list of colors from these supported color keywords: black, white, grey, red, magenta, brown, blue, cyan, green, gold, darkgreen, yellow, orange, purple, **or** lightpurple.

4

Plotting Signals

This chapter details plotting signals in the WaveView tool.

This section contains information on the following topics:

- [Using Panels](#)
- [Panel Types](#)
- [Searching Signals](#)
- [Scanning Signals](#)
- [Data Reduction](#)
- [Direct Plot Mode](#)

Using Panels

This section contains information on the following topics:

- [Panel Actions](#)
- [Specifying Panel Attributes](#)
- [Setting Panel Display Options](#)
- [Setting Signal Display Options](#)
- [Setting Radix Options](#)
- [Setting Signal Name Options](#)
- [Adjusting Panel Height](#)

Panel Actions

The following panel actions are available:

- [Adding Panels](#)
- [Deleting Panels](#)
- [Grouping Panels](#)
- [Creating a Panel Set](#)
- [Selecting Panels](#)
- [Displaying Previous and Next Loaded Signals](#)
- [Adding Labels to a Panel](#)
- [Moving Panels](#)
- [Copying Panels](#)

Adding Panels

Panels are created automatically when signals are dropped into a waveview window. The default panel type is selected based on the signal type.

To manually add a specific type of panel into the active waveview window, choose one of the panel types from the menu in the **Panel** section of the **Waveform** tab.

See [Panel Types](#) for information on the different panel types.

Deleting Panels

To delete a panel, right-click a panel, and choose **Delete Pane** from the menu that opens. If the target panel is selected, all selected panels are deleted.

You can also select a panel and click **Action** in the **Panel** section of the **Waveform** tab, and choose **Delete** from the menu that opens.

Pasting Deleted Panels

Deleted panels can be pasted back to any waveviews by choosing **Paste** from the panel context menu. The deleted panels are always pasted after the current panel in which the mouse pointer resides.

Grouping Panels

Selected panels can be grouped into one panel. To group panels, select the panels you want to group, right-click one of the selected panels, and choose **Group** from the menu that opens.

You can also select a panel and click **Action** in the **Panel** section of the **Waveform** tab, and choose **Group** from the menu that opens.

If the grouped panels are in a logic panel type, a new bus signal is formed. The Bus Configuration dialog box allows you to configure the bus signal.

Panel grouping action can also be invoked from the Panel Group control button.

Ungrouping Panels

Signals inside a panel can be split so that each signal is placed into a new panel. To ungroup panels, select the panels you want to ungroup, right-click one of the selected panels, and choose **Ungroup** from the menu that opens.

You can also select a panel and click **Action** in the **Panel** section of the **Waveform** tab, and choose **Ungroup** from the menu that opens.

Creating a Panel Set

See [Panel Set](#) for information on using a panel set to group signals.

Selecting Panels

To select all panels in a waveview, click **Action** in the **Panel** section of the **Waveform** tab, and choose **Select All** from the menu that opens.

You can also select one or more individual panels by clicking the name or monitor area of a single panel or clicking multiple panels as you hold the **Shift** or **Ctrl** keys.

Deselecting All Panels

To deselect all panels, click **Action** in the **Panel** section of the **Waveform** tab, and choose **Select None** from the menu that opens.

You can also use the Panel Select control button to toggle between the Select-All and the Unselect-All operations.

Displaying Previous and Next Loaded Signals

To display the previously loaded signal from the Signal Browser, click **Action** in the **Panel** section of the **Waveform** tab, and choose **Scan Previous** from the menu that opens.

To display the next loaded signal from the Signal Browser, click **Action** in the **Panel** section of the **Waveform** tab, and choose **Scan Next** from the menu that opens.

Adding Labels to a Panel

You can use the Drawing Tool to add labels to a panel, as well as control the line color and width on a panel.

See [Using the Drawing Tool](#) for more information on the Drawing Tool.

Moving Panels

To move a panel, click and hold a name or value area of a panel. Drag the panel to a new location and release the mouse button. If the source panel is selected, all selected panels are moved.

During the dragging process, a drop hint appears to indicate the drop location.

Copying Panels

To copy a panel, click and hold a name or value area of a panel. Drag the panel to a new location and release the mouse button. If the source panel is selected, all selected panels are copied.

During the dragging process, a drop hint appears to indicate the drop location.

Panels can be copied from a waveview window to another waveview window. Use the preference settings to select between the **Move** and the **Copy** operations. Panels can also be copied and pasted using the corresponding menu entries in the panel context menu.

Specifying Panel Attributes

To change the attributes of a panel, right-click anywhere inside a panel, and choose **Attributes** from the menu that opens.

The following attributes are available depending on the type of panel:

Panel Type	Attribute	Description
All Panels	Panel menu	Specifies the panel to which the attributes are applied.
X-Y, Histogram, Polar Plot, and 2D Sweep	Fit to WaveView	Displays a waveview to fit the height.
	Default Height	Displays a waveview at the default height.
	Signal Name Display menu	Specifies which signal name to display.
	Signal Name Type menu	Specifies the displayed signal name type.
	Sort Names	Sorts the names of signals in alphabetical order.

Panel Type	Attribute	Description
	Display Preference menu	Specifies the display preference of a panel.
	Show Right Axis	Displays the right-hand y-axis.
	I-Signals to Right Axis	Uses the right-hand y-axis for the current signal.
StatEye		
	Zero Threshold	The zero threshold value (the default is $1.00001\text{E}-20$).
	Full Waveform Range	When enabled, the full waveform range of the Z display range is used. When disabled, you can enter Min and Max range values for the Z display range.
	Use Waveform Range	When enabled, the waveform range of the Z axis range is used. When disabled, you can enter Min and Max range values for the Z axis range.
	Z Scale Type	Specifies the type of Z scale: Linear or Log scale (the default scale is Log). Note: When the scale type is Log , the Min and Max values for range setting need to be in the log10 scale.
	Set User Colors	Specifies colors for BER values. BER values can be a single value (for example, 1m), or a range (for example, $2\text{m}-3\text{m}$).
Smith Chart		
	Impedance Chart or Admittance Chart	Specifies a Smith chart as an Impedance Chart or Admittance Chart .
	Impedance Data or Admittance Data	Specifies displayed data as Impedance Data or Admittance Data .
	Characteristic Z	Default value is $5.0000\text{E}+01$.

Setting Panel Display Options

This section contains information on the following topics:

- [Grid](#)
- [Zoom](#)
- [XAxis](#)
- [YAxis](#)
- [Font](#)
- [Full Scale](#)
- [X Variable](#)
- [Axis Attributes](#)

Grid

You can toggle the grids of selected panels on or off with the Grid control button.

Default bindkey: **G**.

The following grid settings are available from the Grid menu:

- [All Grid On](#)
- [H-Grid Only](#)
- [V-Grid Only](#)
- [All Grid Off](#)
- [User Settings](#)

All Grid On

To display the full grid, click **Axis** on the **Waveform** tab, then click **Grid**, and choose **All Grid On** from the menu that opens.

H-Grid Only

To display only the horizontal grid, click **Axis** on the **Waveform** tab, then click **Grid**, and choose **H-Grid Only** from the menu that opens.

V-Grid Only

To display only the vertical grid, click **Axis** on the **Waveform** tab, then click **Grid**, and choose **V-Grid Only** from the menu that opens.

All Grid Off

To hide the grid display, click **Axis** on the **Waveform** tab, then click **Grid**, and choose **All Grid Off** from the menu that opens.

User Settings

To define the grid parameters of XY panels, click **Axis** on the **Waveform** tab, then click **Grid**, and choose **User Settings** from the menu that opens.

User-defined grids start from the Start value and repeat over the Step interval in both directions away from the startpoint.

Zoom

Zoom operations are supported in the following panel types:

- X/Y zoom in X-Y, 2D-Sweep and Eye-diagram panels
- X zoom in logic panels
- Radial zoom in Polar Coordinate panels

Depending on the default action type of the left mouse button in a waveview (the default left mouse button action can be specified in **Left Button Default** in the **Preference Settings** window), the zoom operation is invoked with:

- Cursor mode: You can invoke the zoom operation by first arming a waveview into the zoom mode. Click the **zoom** control button to activate the zoom mode.
- Zoom mode: The zoom operation automatically starts when you hold down and drag the left mouse button inside a waveform.
- Default Bindkey: Press the **Z** key to arm the zoom mode and release it to unarm.

If the left mouse button is defaulted to cursor operations (see [Setting Key Bindings \(Shortcut Keys\)](#)), continue clicking the zoom control button, which cycles the zoom mode in the following order: X/Y zoom (), X zoom (), Y zoom (), no zoom ().

The current zoom mode is indicated with the following icons:



where:

- **Zoom Mode**  — Click this zoom mode icon to cycle through the three zoom modes:
 -  X/Y zoom

Default Bindkey: Press the **Z** key to arm the X/Y zoom operation. X/Y zoom is indicated by the red square around the zoom mode icon.

If you arm the X/Y-zoom operation of a waveview with the **Z** bindkey, releasing the **Z** bindkey without any zoom action results in a 2x zoom out. The same 2x zoom out action can also be invoked using the **Zoom Out** control button.

-  X zoom

Default Bindkey: Press the **H** key to arm the horizontal zoom operation. X zoom is indicated by red vertical lines around the zoom mode, zoom out, zoom in, and zoom to fit icons.

If you arm the X-zoom operation of a waveview with the **H** bindkey, releasing the **H** bindkey without any zoom action results in a 2x x-axis zoom out.

-  Y zoom

Default Bindkey: Press the **V** key to arm the vertical zoom operation. Y zoom is indicated by red horizontal lines around the zoom mode, zoom out, zoom in, and zoom to fit icons.

If you arm the Y-zoom operation of a waveview with the **V** bindkey, releasing the **V** bindkey without any zoom action results in a 2x y-axis zoom out.

- **Zoom In**  — Click to zoom in on all selected panels (or all panels, if none are selected). Default zoom in amount is 50%.
- **Zoom Out**  — Click to zoom out on all selected panels (or all panels, if none are selected). Default zoom out amount is 2x.
- **Zoom to fit**  — Click to restore full page plotting on selected panels (or all panels, if none are selected). The zoom to fit is applied to the selected panel axes. If no panel is selected, the zoom to fit is applied to all panels.
- **Undo Zoom**  and **Redo Zoom**  — Click to traverse up and down a zoom sequence, undoing or redoing a zoom step, respectively. You can **right-click** and select **Remove Icon** to remove these icons from the toolbar.

Use the undo-zoom function to undo a zoom operation. Default Bindkey: **U**

Use the redo zoom function to redo a zoom operation. Default Bindkey: **R**

Setting Zoom Ratio

The preference **Zoom In Ratio %** allows you to select the % for zooming in and out (out being the inverse).

Set Zoom Area

To specify the the x-axis and y-axis zoom range operations manually, click **Axis** on the **Waveform** tab, then click **Zoom**, and choose **Set Zoom Area** from the menu that opens. Enter the zoom range parameters, and click **Ok** to set the zoom level on all panels or selected panels of the active waveview.

Note:

Y ranges are in db (or db10) units if the target panel y-axis is in the db (or db10) scale

Full UnZoom

To unzoom the display, click **Axis** on the **Waveform** tab, then click **Zoom**, and choose **Full Unzoom** from the menu that opens.

Default bindkey: **X**

Using Sliders in Zoomed Panels

In a zoomed panel (X-Y, logic or 2D-Sweep), the zoomed area can be interactively shifted using the horizontal and vertical slide bars. The waveview is redrawn at real time when the slide bar is moved. This graphic intensive operation can become noticeably slow especially when you work from a remote host.

The real-time screen redraw can be disabled for the slide bar to avoid the slow graphic operation. To disable real-time redraw, disable the **Real-time waveview scroll** option in the **Preference Settings** window (click **Settings** on the **Configuration** tab). With the real-time waveview scroll disabled, only the grid rulers are interactively redrawn during slide bar movement.

XAxis

The following x-axis types are available from the **XAxis** menu:

- [Linear](#)
- [Log10](#)
- [dB20](#)

Linear

To set a linear x-axis for a plotted signal, click **Axis** on the **Waveform** tab, then click **XAxis**, and choose **Linear** from the menu that opens.

Log10

To set a log10 x-axis for a plotted signal, click **Axis** on the **Waveform** tab, then click **XAxis**, and choose **Log10** from the menu that opens.

dB20

To set a dB20 x-axis for a plotted signal, click **Axis** on the **Waveform** tab, then click **XAxis**, and choose **dB20** from the menu that opens.

YAxis

The following y-axis types are available from the YAxis menu:

- [Linear](#)
- [Log10](#)
- [dB20](#)
- [dB10](#)
- [Y2 - Linear](#)
- [Y2 - Log10](#)
- [Y2 - dB20](#)
- [Y2 - dB10](#)
- [Y2 - Phase](#)

Linear

To set a linear y-axis for a plotted signal, click **Axis** on the **Waveform** tab, then click **YAxis**, and choose **Linear** from the menu that opens.

Log10

To set a log10 y-axis for a plotted signal, click **Axis** on the **Waveform** tab, then click **YAxis**, and choose **Log10** from the menu that opens.

dB20

To set a dB20 y-axis for a plotted signal, click **Axis** on the **Waveform** tab, then click **YAxis**, and choose **dB20** from the menu that opens.

dB10

To set a dB10 y-axis for a plotted signal, click **Axis** on the **Waveform** tab, then click **YAxis**, and choose **dB10** from the menu that opens.

Y2 - Linear

To set a Y2 - linear y-axis for a plotted signal, click **Axis** on the **Waveform** tab, then click **YAxis**, and choose **Y2 - Linear** from the menu that opens.

Y2 - Log10

To set a Y2 - log10 y-axis for a plotted signal, click **Axis** on the **Waveform** tab, then click **YAxis**, and choose **Y2 - Log10** from the menu that opens.

Y2 - dB20

To set a Y2 - dB20 y-axis for a plotted signal, click **Axis** on the **Waveform** tab, then click **YAxis**, and choose **Y2 - dB20** from the menu that opens.

Y2 - dB10

To set a Y2 - dB10 y-axis for a plotted signal, click **Axis** on the **Waveform** tab, then click **YAxis**, and choose **Y2 - dB10** from the menu that opens.

Y2 - Phase

To set a Y2 - Phase y-axis for a plotted signal, click **Axis** on the **Waveform** tab, then click **YAxis**, and choose **Y2 - Phase** from the menu that opens.

Font

The following font sizes are available from the Font menu:

- [Normal](#)
- [Large](#)

Normal

To set the axis font size to the default normal size, click **Axis** on the **Waveform** tab, then click **Font**, and choose **Normal** from the menu that opens.

Large

To set the axis font size to a larger size, click **Axis** on the **Waveform** tab, then click **Font**, and choose **Large** from the menu that opens.

Full Scale

The following axis scales are available from the Full Scale menu:

- [Set X Full Scale](#)
- [Set Y Full Scale](#)
- [Default Y Full Scale](#)

Set X Full Scale

To set a fixed plotting range for the x-axis, click **Axis** on the **Waveform** tab, then click **Full Scale**, and choose **Set X Full Scale** from the menu that opens.

Set Y Full Scale

To set a fixed plotting range for the y-axis, click **Axis** on the **Waveform** tab, then click **Full Scale**, and choose **Set Y Full Scale** from the menu that opens.

Default Y Full Scale

Fixed plotting ranges can be assigned to the x-axis of X-Y panels. Click **Axis** on the **Waveform** tab, then click **Full Scale**, and choose **Default Y Full Scale** from the menu that opens. The **Default Y Axis Range** dialog box opens. Click **Ok** to apply the change to selected panels of the active waveview.

X Variable

To set an X variable:

1. Click **Axis** on the **Waveform** tab, and click **X Variable**.

The **Setting X Variable** dialog box opens with the default X variable is enabled by default.

2. With the dialog box open, right-click a signal in the **Waveform Browser** signal list window, and choose **Use as X Variable**. In case you do not want to use default X variable, deselect the **Use Default X Variable** option.

Dragging a signal to the **X Variable** field, resets the **Parameter Variable** field to empty. Selecting a **Parameter Variable** from the menu, resets the **X Variable** field to empty. So, you can set only one field.

When the **Change X Variable** dialog box is opened, right-click on signal name in the signal browser, the **Use a X Variable** menu appears.

If you drag a new signal, the **X Variable** shows the new signal name replacing the old one.

Note:

The field is not cleared if **Use Default X Variable** is selected. This is so that if you switch back you do not have to set it again.

3. Click **Apply** to apply the change to an active vertical waveview or all the selected panels in an active horizontal waveview.

By default, signals in X-Y panels are plotted with respect to their internal X variables defined in the original waveform data file. The X variable can be replaced with a signal so that other signals are plotted as functions of the selected signal.

In a vertical waveview, if the waveview contains no logic panel, you can assign a signal as the common X variable so that all signals are plotted against the selected X variable. In a horizontal waveview, the same operation can be applied individually to all selected X-Y panels.

You can set the sweep parameter as the X variable on the X-Y panel. Envelopes are enabled on the X-Y panel for sweep signals.

To change the X variable, from the signal filter of the waveform browser (**Configuration > Browser**), select any signal. Right-click the mouse button and select **Use as X Variable** from the displayed options. This changes the X variable of the plotted waveform to the selected signal. Alternatively, the X variable can be changed by dragging and dropping the required signal to the x-axis of the waveview window.

Axis Attributes

To set axis attributes:

1. Double-click on an axis.

The **Axis Attributes** window opens.

2. Choose an axis from the **Axis** menu.

All axes in the active waveview window are available.

3. Click one of the zoom level buttons to change the display:

- **Zoom In** increases magnification to show detail.
- **Zoom to Fit** displays the maximum number of data points to show the entire range of a signal. The entire range is determined by the minimum and maximum values entered in the **Full Scale Range** text boxes at the bottom of the **Axis Attributes** window.

4. Enter values for the zoom level in the **Min** and **Max** text boxes.

These values specify the range you want to display.

5. (Optional) Click the **Use Waveform Range** check box to specify the range of a signal, and enter the Min and Max values for that range.
6. (Optional) Click the **Grid On** check box to display the grid for the selected axes.

You can click the **Auto Grid** check box and enter **Start** and **Stop** values for the gridding. You can only turn this option off for linear axes.

7. Choose the axis label font size from the **Font** menu.

The following font sizes and styles are available:

- **Normal**
- **Large**
- **Gothic**

- **Times**
- **Roman**
- **Courier**

8. Click **Apply** to save your changes.

Setting Signal Display Options

This section contains information on signal display settings for viewing panels:

- [Full Logic Value](#)
- [Short Logic Value](#)
- [VCD Logic](#)
- [RTF Logic](#)
- [PWL Waveform](#)
- [PWC Waveform](#)
- [Bar Chart](#)
- [Data Point Only](#)

Full Logic Value

Logic vector values by default are displayed in their shortest possible form. To display vector values in full length, click **Display** in the **Panel** section of the **Waveform** tab, and choose **Full Logic Value** from the menu that opens.

Short Logic Value

To display vector values in short length, click **Display** in the **Panel** section of the **Waveform** tab, and choose **Short Logic Value** from the menu that opens.

VCD Logic

To plot a signal as a VCD logic waveform, click **Display** in the **Panel** section of the **Waveform** tab, and choose **VCD Logic** from the menu that opens.

RTF Logic

To plot a signal as an RTF logic waveform, click **Display** in the **Panel** section of the **Waveform** tab, and choose **RTF Logic** from the menu that opens.

PWL Waveform

To plot a signal in piecewise linear format, click **Display** in the **Panel** section of the **Waveform** tab, and choose **PWL Waveform** from the menu that opens.

PWC Waveform

To plot a signal in piecewise-constant format, click **Display** in the **Panel** section of the **Waveform** tab, and choose **PWC Waveform** from the menu that opens.

Bar Chart

To plot a signal as a bar chart, click **Display** in the **Panel** section of the **Waveform** tab, and choose **Bar Chart** from the menu that opens.

Data Point Only

Waveforms are displayed by plotting piece-wise-linear (PWL) lines between data points. To display the actual data points, click **Display** in the **Panel** section of the **Waveform** tab, and choose **Data Point Only** from the menu that opens to toggle the data point markers on or off.

Default bindkey: **D**.

Setting Radix Options

To select a radix setting for all panels, click **Radix** in the **Panel** section of the **Waveform** tab, and choose one of the following supported radix settings:

- **Binary**
- **Octal**
- **Decimal**
- **Signed**
- **Hexadecimal**
- **ASCII**
- **Mnemonic on/off**

Logic vector values by default are displayed in binary. You can also choose **Radix** from the panel context menu to select a radix setting for the panel in action.

Setting Signal Name Options

This section contains information on items listed in the **SigName** menu, which is located in the **Panel** section on the **Waveform** tab:

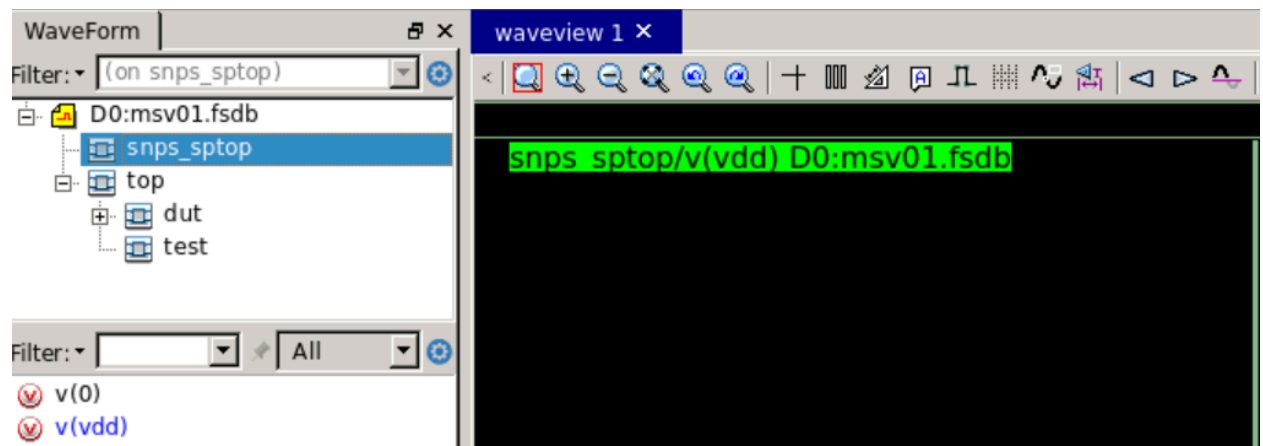
- [Sort](#)
- [Leafname](#)
- [Fullname](#)
- [Fullname+file](#)
- [Fullname+path+file](#)
- [Netname](#)
- [Netname+file](#)
- [Netname+path+file](#)
- [Alias](#)

Sort

To sort signal names alphabetically, choose **Sort** from the **SigName** menu in the **Panel** section of the **Waveform** tab.

Leafname

Leafname is the netname along with the signal type. For example, in the following image, if the full name of a signal is `snps_sptop/v(vdd)`, its leaf name is `v(vdd)`.



To display the leaf names for signals, choose **Leafname** from the **SigName** menu in the **Panel** section of the **Waveform** tab.

Fullname

To display the full names for signals, choose **Fullname** from the **SigName** menu in the **Panel** section of the **Waveform** tab.

For example, snps_stop/v(vdd).

Fullname+file

To display the full names and file names for signals, choose **Fullname+file** from the **SigName** menu in the **Panel** section of the **Waveform** tab.

For example, snps_stop/v(vdd) mvs01.fsdb.

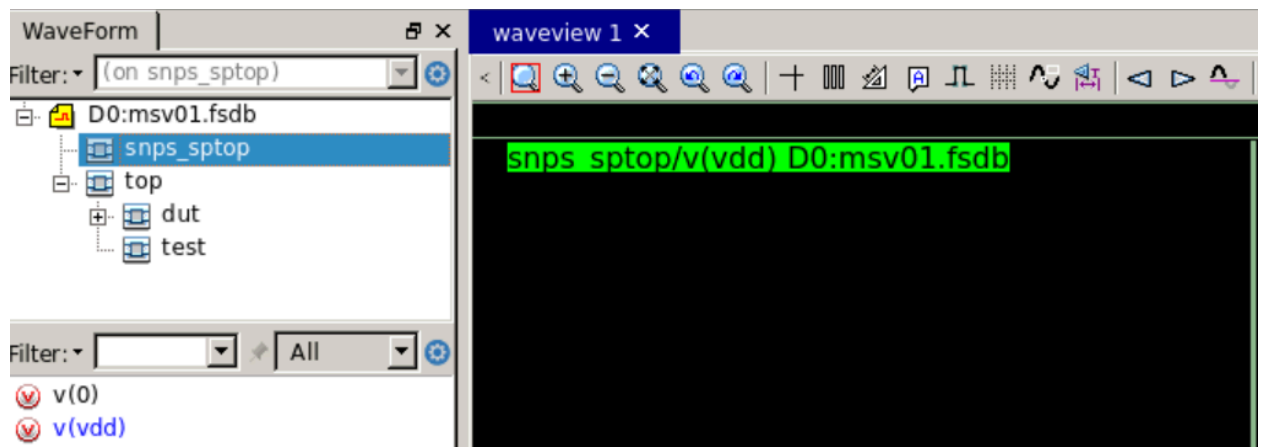
Fullname+path+file

To display the full names, file names, and paths for signals, choose **Fullname+path+file** from the **SigName** menu in the **Panel** section of the **Waveform** tab.

For example, snps_stop/v(vdd) D0:mvs01.fsdb.

Netname

Netname is the name of the wire or net in the schematic. For example, in the following image, if the full name of a signal is snps_sptop/v(vdd), its net name is vdd.



To display the net names for signals, choose **Netname** from the **SigName** menu in the **Panel** section of the **Waveform** tab.

Netname+file

To display the nets names and file names for signals, choose **Netname+file** from the **SigName** menu in the **Panel** section of the **Waveform** tab.

For example, vdd mvs01.fsdb.

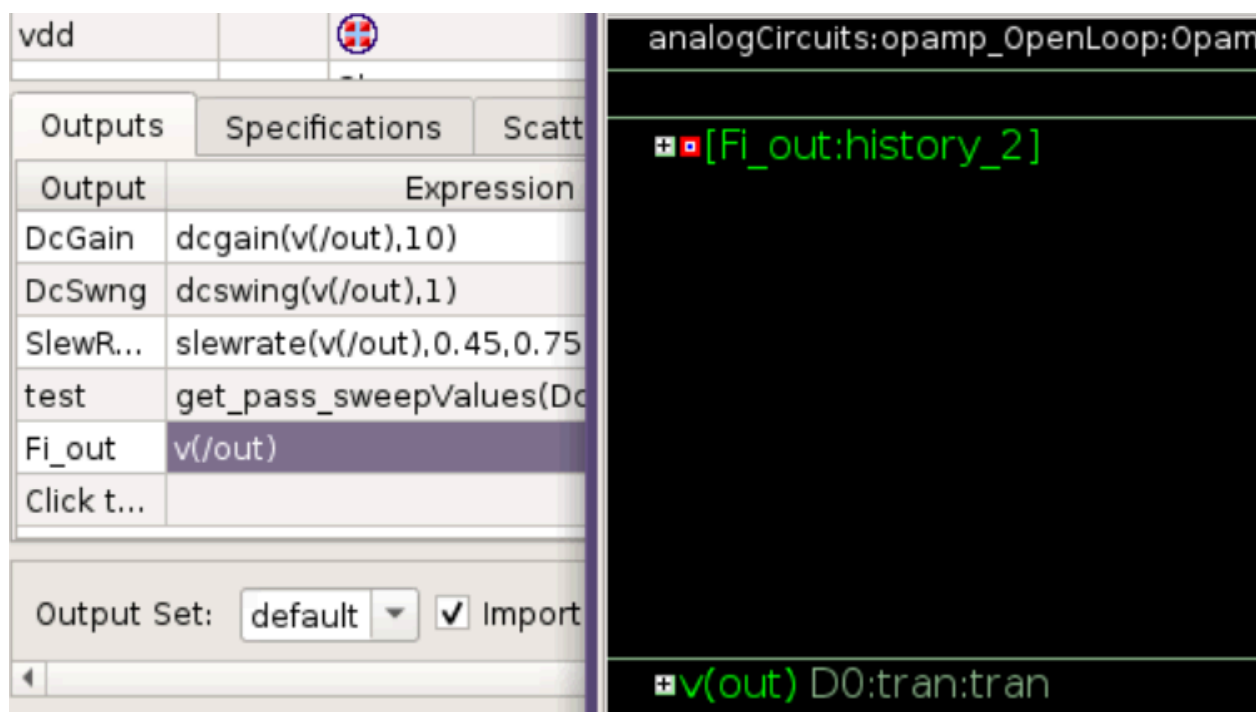
Netname+path+file

To display the net names, file names, and paths for signals, choose **Netname+path+file** from the **SigName** menu in the **Panel** section of the **Waveform** tab.

For example, vdd D0:mvs01.fsdb.

Alias

In PrimeWave, the expression name in the **Outputs** tab is taken as the signal alias name. For example, in the following image, if Fi_out is the expression name entered for a signal v(/out) in the PrimeWave **Outputs** tab, then the alias name for the v(/out) signal is taken as Fi_out. The alias name can also be edited and changed as required.



To display the aliases for signals, choose **Alias** from the **SigName** menu in the **Panel** section of the **Waveform** tab.

Fit Longest Label

To fit the signal name width to the longest signal label, right-click on the canvas, choose **Signal Name > Fit Longest Label**. A shortcut icon for this is available in the toolbar



Adjusting Panel Height

This section contains information on items listed in the **Height** menu, which is located in the **Panel** section on the **Waveform** tab:

- [Fit](#)
- [Default](#)
- [Fit All](#)
- [Default All](#)
- [Manual Adjust](#)

Fit

In a vertical stack waveview, the height of selected panels can be automatically adjusted to fit to the full window height. To fit the full window height, click **Height** in the **Panel** section of the **Waveform** tab, and choose **Fit** from the menu that opens.

You can also use the default bindkey **T** to toggle selected panels between fit and unfit modes.

If the selected panels are separated into multiple groups, each group is fitted respectively. If there are too many panels in a group and the panel height becomes too small, the height is limited to the user-defined minimum height.

Default

To restore the default window height, click **Height** in the **Panel** section of the **Waveform** tab, and choose **Default** from the menu that opens.

Panel heights in a vertical stack waveview are calculated automatically based on the following preference settings:

- Maximum number of visible panels
- Minimum X-Y panel height

You can adjust the height of a panel by dragging the bottom edge of the left y-axis area. If the panel in action is a selected panel, the height of all selected panels is changed.

Individual panel height adjustment is available for analog panels only. To adjust the height of all logic panels, change the value of Logic Panel Height in the preference settings.

Fit All

In a vertical stack waveview, the height of selected panels can be automatically adjusted to fit to the full window height. To fit the full window height for all windows, click **Height** in the **Panel** section of the **Waveform** tab, and choose **Fit All** from the menu that opens.

You can also use the default bindkey **T** to toggle selected panels between fit and unfit modes.

If the selected panels are separated into multiple groups, each group is fitted respectively. If there are too many panels in a group and the panel height becomes too small, the height is limited to the user-defined minimum height.

Default All

To restore the default window height for all windows, click **Height** in the **Panel** section of the **Waveform** tab, and choose **Default All** from the menu that opens.

Panel heights in a vertical stack waveview are calculated automatically based on the following preference settings:

- Maximum number of visible panels
- Minimum X-Y panel height

You can adjust the height of a panel by dragging the bottom edge of the left y-axis area. If the panel in action is a selected panel, the height of all selected panels is changed.

Individual panel height adjustment is available for analog panels only. To adjust the height of all logic panels, change the value of Logic Panel Height in the preference settings.

Manual Adjust

To manually adjust the height for a windows, click **Height** in the **Panel** section of the **Waveform** tab, and choose **Manual Adjust** from the menu that opens.

Panel Types

This section contains information on the following topics:

- [X-Y Panels](#)
- [Eye Diagrams](#)
- [Logic Waveforms](#)
- [Tabular Panels](#)
- [Histogram Panels](#)

- [Smith Chart](#)
 - [Polar Plot Panels](#)
 - [2-D Sweeps](#)
 - [3-D Sweeps](#)
 - [Pair Plot](#)
 - [Constell](#)
 - [Panel Set](#)
 - [Parametric Plots](#)
 - [Plotting Clock Jitter](#)
 - [HSPICE QQ Plot](#)
-

X-Y Panels

This section contains information on the following topics:

- [Creating an X-Y Panel](#)
- [Plotting X-Y Panel Values in Tabular Format](#)
- [FFT/DFT Conversion](#)
- [A to D Conversion](#)

Creating an X-Y Panel

To create a new X-Y panel, choose **X-Y** from the menu in the **Panel** section of the **Waveform** tab.

An X-Y panel is designed to display analog waveforms. It is the only panel type that is allowed in both vertical stack and row/column waveviews. The X-Y panel accepts signals that are represented in (valuex,valuey) pairs, ((x,y) data points, for example), where valuex can be the default sweeping variable of a signal, or the signal value of a regular signal; valuey can be either the value of a real- signal or the magnitude/phase/real/imaginary part of a complex signal. The zoom operation is supported along both axes, and the logarithmic scale is also supported along both axes.

Waveforms in linear X-Y panels can be displayed with different plotting options including PWL (piece-wise-linear) line-type, PWC (piece-wise-constant) line-type, bar chart and data-point only mode. The default line type is determined by the type of waveform.

Plotting X-Y Panel Values in Tabular Format

To plot the value pairs from an X-Y panel into tabular format, right-click anywhere in an X-Y panel, and choose **Plot in Tabular** from the menu that opens. A new tableview is created with all the plotted value pairs from the X-Y panel.

FFT/DFT Conversion

To open the FFT/DFT Parameters window, choose **FFT/DFT** from the menu located in the **Utilities** section of the **Waveform** tab.

FFT and DFT operations are used to convert time-domain signals into their frequency-domain spectrum representation. The FFT/DFT operations can be performed on waveforms in the X-Y panels only.

FFT/DFT operations are supported when using various windowing functions. The FFT/DFT dialog box accepts input parameters and applies DFT/FFT to all signals in selected panels of the active waveview. If no panel is selected in the active waveview, all signals in the active waveview become the FFT/DFT target. If the active waveview has only one panel, the FFT/DFT operation is by default applied to all waveforms in the panel. If the active waveview has more than one panel, you need to select the FFT target panels.

The FFT/DFT algorithm requires the following parameters:

- The number of sample points: Determines the number of time-domain sample points used for FFT/DFT operation. This number is internally rounded to the next power of two that is greater than the user input for FFT. A slow run time warning is displayed for DFT operations over 1024 points.

OR

The sampling frequency: The frequency of the sampling signal for the FFT/DFT operation. The sampling rate must be greater than 1 hz.

Note:

If you have a defined time range, you need to specify either the number points or sampling rate. If you do not have a time range specified, the number of points and the sampling rate are used to determine the start and stop times.

- The sampling window start time: Determines the start time of the time-domain window for FFT/DFT conversion.
- The sampling window end time: Determines the end time of the time-domain window for FFT/DFT conversion. The specified window end time must be greater than its start time.

You must choose which three input parameters are in use. Deselect an input parameter using the toggle switch in front of the parameter. If the **Enable zero-padding after stop**

time option is selected, all four parameters need to be entered, the sampling stop time is adjusted based on the other three input parameters. For example:

```
window_end_time < window_start_time + number_of_samples *  
(1/sampling_frequency)
```

Evenly spaced zeros are padded between the user specified end time and the one calculated by the waveview.

Click the **Check** button to check the input parameters before a FFT/DFT operation. It automatically fills the unspecific value and modifies the end time if necessary when four sampling parameters are specified. Use the **Save** and **Load** buttons to save and load FFT input parameters respectively.

To enter the start and stop time based on the frequency of the input signal, enter the input signal frequency in the Input Signal Frequency field and click **Apply** to calculate the start/stop time.

Window functions can be applied for the FFT/DFT operation. If a window function is selected, the waveform data is multiplied with the selected windowing function before the FFT/DFT operation. No window function is used if the **Rectangle** window type is selected. The gaussian and the kaiser-bessel window types require the additional **Alpha** parameter.

The FFT/DFT output can be normalized against the waveform DC component or a user-defined value in Magnitude Normalization. All harmonic components are displayed in reference to the reference.

If the **Suppress DC before FFT/DFT** option is selected, the dc component is excluded from the target waveforms in the FFT/DFT operation. This is to minimize the error introduced by those signals with large DC component and very small AC magnitudes.

The Output Frequency range settings specify the range of FFT/DFT spectrums for display or text output. The fmin and fmax value define the output spectrum range of interest. Specifying **frequency of interest** only displays or outputs the multiple of this frequency harmonic; all other harmonics are omitted.

The FFT/DFT results can be displayed in a regular X-Y panel or in a spectrum panel. User can also send the results to either a text window or text file for further processing. Y-axis scale is automatically set as db for the FFT/DFT plot.

Select the **Display Sampled Waveform** option to display the time-domain sampled waveform, including the effect of the windowing function.

The FFT result can be displayed in a regular X-Y panel, or a special spectrum panel, or into text window or file. See [Viewing the Spectrum Panel](#) for more information.

This section contains information on the following topics:

- [Viewing the Spectrum Panel](#)
- [Calculating SNR/THD Using FFT](#)
- [Calculating SNR/ENOB With Respect to Full-Scale](#)
- [Calculating the FFT of Complex Signals](#)
- [Changing the FFT Axes Scale](#)

Viewing the Spectrum Panel

The spectrum panel displays spectrum magnitude of harmonic components at different frequencies. Choose **Plot Complex** from the context menu to change the value to the phase, real, or imaginary part of the complex data.

The Spectrum panel uses bar chart by default to display waveforms. You can change it to a line chart by choosing **Configure Spectrum** from the panel context menu.

Cursors in the Spectrum panel report waveform values based on the current complex plot mode. Instead of using piece-wise-linear (PWL) interpolation, the piece-wise-constant (PWC) value is used within each spectrum interval.

Calculating SNR/THD Using FFT

To calculate and display the total harmonic distortion (THD), signal to noise ratio (SNR), signal to noise and distortion ratio (SNDR), and effective number of bits (ENB) of the target signals, select the **Evaluate SNR/THD/SNDR/ENB** option. The result is displayed in the text window or the text output file.

For SNR/THD/SNDR/ENOB calculation, windowing function is usually used if the sample range does not cover integer cycles of the input signal. However, the time-domain windowing functions also spread the original harmonics of the input signal spectrum into multiple spectrum components. To accommodate the spreading effect, an additional parameter (**Bin Size**) is required to describe how many spectral components on each side of a harmonic frequency need to be included as the power of the harmonic.

Specify the cut-off frequency for the noise components in the Band Limit field. This value affects the SNR/SNDR/ENOB calculation. Set to 0 if no band limit is used.

Note:

To include counting floor components when calculating SNR/SNDR, enable the **Count floor noise components in SNR/SNDR calculation** option, which is available in the Preferences (**Settings** on the **Configuration** tab).

To calculate SNR/THD/ENOB with respect to the ADC's full-scale range(dBFS), enable the **Calculate SNR/THD/ENOB with Respect to Full-Scale**,

Compatible with Analog Devices option. See [Calculating SNR/ENOB With Respect to Full-Scale](#) for more information.

Calculating SNR/ENOB With Respect to Full-Scale

To calculate SNR/ENOB/SFDR/THD compatible with Analog Devices, harmonics of input signal f_a fall at frequencies equal to $|\pm Kf_s \pm n f_a|$, where n is the order of the harmonic,

$K = 0, 1, 2, 3, \dots$, and max harmonic index is set to 5 by default.

The magnitude of f_a is scaled to ADC full scale (1.0) when calculating SNR/SFDR, if not normalized. The formula of ENOB is changed to $(\text{SNR}-1.76)/6.02$. And SFDR is calculated with largest noise or harmonic distortion component.

Calculating the FFT of Complex Signals

If the input signal of FFT/DFT is a complex signal, a non-symmetric spectrum is generated. The **Suppress DC before FFT/DFT** option is ignored if the input signal is in complex values. To prepare a complex input signal from transient analysis (for RF-mixer applications), use the `complex()` equation builder function to combine two real signals into a complex signal.

Changing the FFT Axes Scale

To change the scale of the x- and y-axis:

1. Click **Settings** on the **Configuration** tab.
The **Preference Settings** window opens.
2. Click the **Signals** category, and then select the desired scales in the **Default FFT Signal** for the **X-Scale** and **Y-Scale**
3. Click **OK** to save the preferences.

These preferences are saved in the `.spxrc` preference file.

A to D Conversion

Analog waveforms can be converted to logic waveforms based on user-specified threshold values. The conversion procedure can be applied only to waveforms in the X-Y panels.

If the active waveview has only one panel, the A/D conversion operation is by default applied to all waveforms in this panel. If the active waveview has more than one panel, you need to select panels for the A/D conversion operation.

To open the **A/D Conversion Parameters** window, choose **A to D** from the menu located in the **Utilities** section of the **Waveform** tab.

You can enable the **Suffix of Result Signal Name** option, which adds a suffix to the name of the converted signal name for any time of A to D conversion.

This section contains information on the following topics:

- [Single-Bit A/D Conversion](#)
- [Multi-Bit A/D Conversion](#)
- [Creating Bus Signals for Analog Signals](#)

Single-Bit A/D Conversion

Analog waveforms can be converted to single-bit logic waveforms in one of two ways:

- With a single threshold defined, data points above the threshold level are converted to logic H, and those below to logic L.
- With two thresholds defined, all points above the high threshold level are converted to logic H, while all points below the low threshold level are converted to logic L. All other points in between are converted to logic X.

The converted waveforms are displayed in logic panels. Since logic panels are allowed only in vertical waveviews, a new vertical waveview is automatically created if the current active waveview is in horizontal mode.

The two-threshold conversion requires that a signal stays long enough in the X region (region between the high and low thresholds) to generate an X state in the converted logic waveform. The minimum duration requirement is defined by the rise (fall) slew if the signal level moves from below (above) the low (high) threshold level. The rise and fall slews are defined using the `d2a_rise_slew` and `d2a_fall_slew` configuration options.

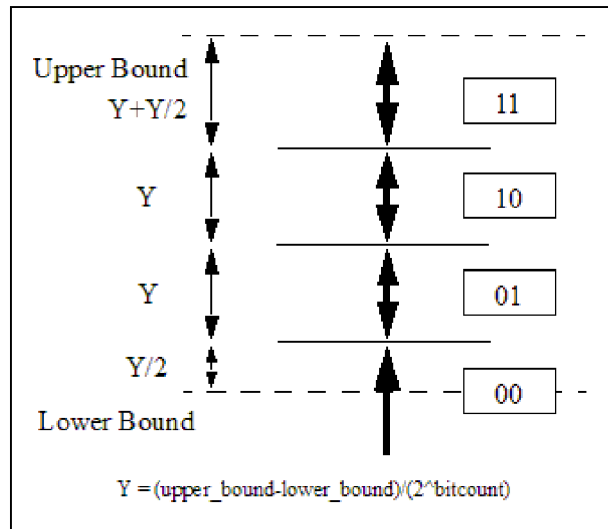
You can specify the default settings for A to D conversions in the **Threshold** category of the **Setting Preferences** window (click **Settings** in the **Configuration** tab).

Multi-Bit A/D Conversion

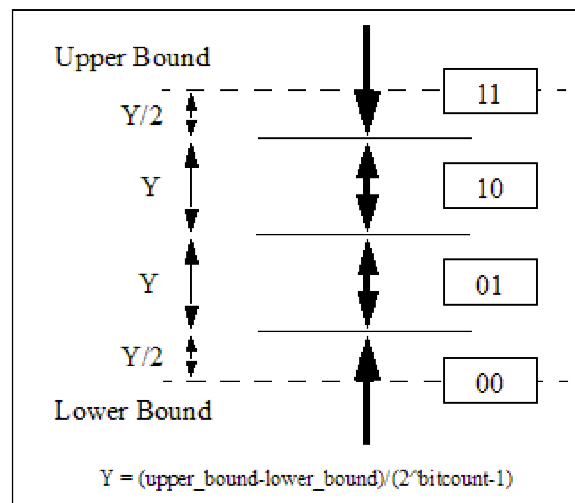
With multi-bit A/D conversion, analog waveforms are converted to multi-bit logic signals. The quantization thresholds are determined automatically based on the number of bits and user-specified lower/upper bounds.

Two quantization threshold schemes are available:

- $Y = (\text{upper_bound} - \text{lower_bound}) / (2^{\text{bit_count}})$: Maintains asymmetric partitions as $(Y+Y/2)$ for the largest codeword and $Y/2$ for the smallest codeword.



- $Y = (\text{upper_bound} - \text{lower_bound}) / (2^{\text{bit_count}} - 1)$: Maintains a symmetric partition as $Y/2$ for both the largest and smallest codewords. This option is selected by default.



Creating Bus Signals for Analog Signals

To create bus signals for analog signals when converting from analog to digital, click the **Create Bus with Selected Analog Signals** check box at the bottom of the A/D Conversion Parameters dialog box as part of your conversion setup.

Eye Diagrams

To create an empty Eye diagram panel, choose **Eye Diagram** from the menu in the **Panel** section of the **Waveform** tab.

Signals are then dragged from the Waveform Browser and dropped into the Eye diagram. The Eye diagram by default displays signals in the folded mode.

A folded Eye diagram partitions signals into continuous intervals that are then overlapped to form the Eye diagram. The interval width can be user-specified or automatically determined.

Zoom operations are supported in both folded and unfolded eye diagrams.

The shift parameter is used to control the alignment of a folded Eye diagram. A slider at the lower-left corner of the Eye diagram panel allows users to interactively change the shift setting.

This section contains information on the following topics:

- [Showing Eye Diagram Data Points](#)
- [Displaying the Envelope Center](#)
- [Unfolding Eye Diagrams](#)
- [Tracing Waveform Points in Eye Diagrams](#)
- [Configuring Eye Diagrams](#)
- [Generating Jitter Histograms](#)
- [Displaying the Eye Height Distribution](#)
- [Adding User-Defined Masks](#)

Showing Eye Diagram Data Points

To display the data points of an eye diagram, right-click an eye diagram, and choose **Show Data Points** from the menu that opens.

Displaying the Envelope Center

To display the envelope center of an eye diagram, right-click an eye diagram, and choose **Show Envelope Center** from the menu that opens.

After it is selected, the center trace is displayed along with the eye diagram. To disable this option, choose it again.

Unfolding Eye Diagrams

An Eye diagram can be unfolded by right-clicking an eye and choosing **Unfold Eye** from the menu that opens. The unfolded Eye diagram displays the original signals and the corresponding intervals highlighted in the red box at the top.

Similar to an oscilloscope, the Eye diagram also accepts a signal as an external trigger to construct the folded Eye diagram. To specify the external trigger signal, drag the signal from the Waveform Browser and drop it into the lower-right corner of an Eye diagram panel. A hint box highlights the drop location for the external trigger signal.

Tracing Waveform Points in Eye Diagrams

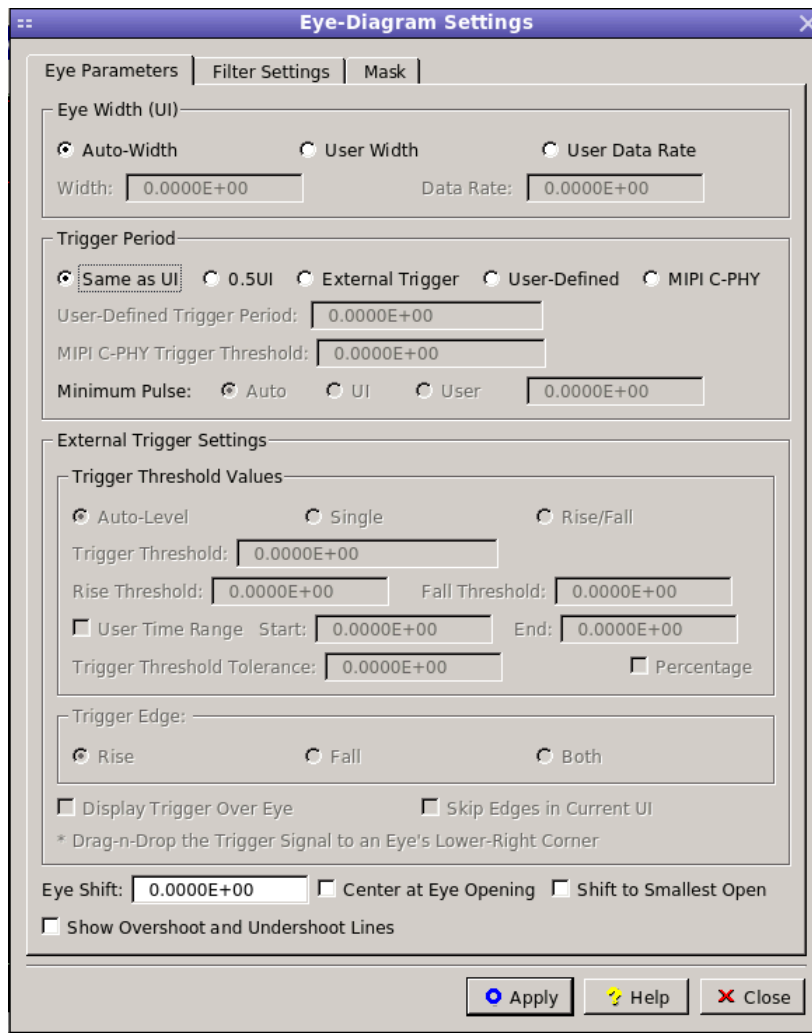
You can use cursors in an eye diagram to trace a waveform point location in the unfolded eye. Right-click an unfolded eye, and choose **Trace** from the menu that opens for the corresponding eye diagram location. You can trace **Eye Fuss**, **upper outer**, **upper inner**, **lower outer**, **lower**, **inner**, **left outer**, **left inner**, **right outer**, **right inner**, **upper margin inner**, **lower margin inner**, **upper margin outer**, or and **lower margin outer**. For an eye diagram that shows an eye mask, additional trace options are available. Cursors are jumped to the corresponding location when the eye is unfolded.

Configuring Eye Diagrams

You can change eye diagram parameters from the **Eye Diagram Settings** dialog box. Right-click an eye diagram, and choose **Configure Eye** from the menu that opens. The **Eye Diagram Settings** window opens.

Note:

If you are configuring eye diagram settings for a state eye, only the settings on the Mask tab are visible. See [Setting Mask Parameters](#) for more information.



Eye diagram settings can be copied and pasted between multiple eye diagrams. Right-click an eye diagram, and choose either **Copy Settings** or **Paste Settings** from the menu that opens.

The following sections explain the eye diagram settings that are available:

- [Setting Eye Parameters](#)
- [Setting Filter Parameters](#)
- [Setting Mask Parameters](#)
- [Setting External Trigger Parameters](#)

Setting Eye Parameters

The following parameters are set from the **Eye Parameters** tab in the **Eye Diagram Settings** window:

Parameter Name	Description
Auto-Width	Automatically determines the eye width or frequency. In Auto-Width mode, an eye diagram always uses the first signal in the panel to determine the eye width.
User Width	Specifies the eye width or frequency using the value you enter in the Width text box.
User Data Rate	Specifies the data rate using the value you enter in the User Data Rate text box.
Same as UI	Sets the eye trigger period to the same as the eye diagram UI width.
0.5UI	Sets the eye trigger period to 0.5UI.
External Trigger	Sets an external trigger using the settings in the External Trigger Settings section of the Eye Diagram Settings window. See the following table for the External Trigger Settings.
User-defined	Sets the eye trigger period to the value your enter in the User-Defined Trigger Period text box.
MIPI C-PHY	Sets the MIPI C-PHY trigger period to the value your enter in MIPI C-PHY Trigger Threshold text box. Default: 0 Minimum Pulse: Select Auto, UI (default), or User. If Auto is selected, it calculate minimum pulse among all three bits. Searching of the next trigger will skip 0.5*min_pulse after each trigger is found. The minimum pulse setting is saved in session file. The ACE command for the minimum pulse: - min_pulse_mode=[AUTO UI USER] and - min_pulse=[user_min_pulse_value] For example: <pre>set_eye_obj [sx_create_eye \$eye_sigs ui=2.22222e-10 phase=-1.11111e-10 trigger=CPHY cphy_vref=2.22222e-10 min_pulse_mode=USER min_pulse=2e-10 trig_level=auto trig_tolerance=0 ttpercentage=1 trig_edge=both start_t=4e-09 stop_t=7.6e-08]</pre>
Eye Shift	Sets the eye shift value
Center at Eye Opening	Centers the eye diagram at the eye opening.
Show Overshoot and Undershoot Lines	Displays the overshoot and undershoot lines with eye diagrams.

The following Eye Parameter settings are available in the External Trigger Settings section when you choose the **External Trigger** option for the trigger period:

Parameter Name	Description
Auto-Level	Uses the average magnitude of the external trigger signal as the trigger level. The value in Trigger Threshold Tolerance field specifies the edge height tolerance (the default is 0.0000E+00). Click the Percentage check box to change the tolerance to a percentage value. When enabled, this option calculates the percentage as $\text{tolerance} = \text{value} * 0.01 * (\text{ymax} - \text{ymin})$. Edges with heights smaller than the tolerance are eliminated.
Single	Uses a single trigger value you enter in the Trigger Threshold text box. The value in Trigger Threshold Tolerance field specifies the edge height tolerance (the default is 0.0000E+00). Click the Percentage check box to change the tolerance to a percentage value. When enabled, this option calculates the percentage as $\text{tolerance} = \text{value} * 0.01 * (\text{ymax} - \text{ymin})$. Edges with heights smaller than the tolerance are eliminated.
Rise/Fall	Uses the rising and falling trigger values you enter in the Rise Threshold and Fall Threshold text boxes, respectively. The value in Trigger Threshold Tolerance field specifies the edge height tolerance (the default is 0.0000E+00). Click the Percentage check box to change the tolerance to a percentage value. When enabled, this option calculates the percentage as $\text{tolerance} = \text{value} * 0.01 * (\text{ymax} - \text{ymin})$. Edges with heights smaller than the tolerance are eliminated.
Rise	Sets the trigger at the rising edge.
Fall	Sets the trigger at the falling edge.
Both	Sets the trigger at both the rising and falling edge.

Setting Filter Parameters

The following parameters are set from the Filter Settings tab in the Eye Diagram Settings window, which is used to exclude unwanted time periods when forming an eye diagram:

Parameter Name	Description
No Filter	Specifies that no filter is enabled.
Index Range	Enables a filter for a specified index range. Enter values for the Starting Window Index and Number of Windows to select the range.

Parameter Name	Description
Time Range	Enables a filter for a specified time range. Enter values for the Range Starting Time and Range Stopping Time .
Skip Frequency	Specifies the number of next trigger edges that are skipped. For example, if Skip Frequency is set to 2, the next 2 trigger edges will be skipped for a valid edge.
DDR 2/3T Mode	Enables or disables the DDR 2/3T mode. When enabled, if the trigger edge is bigger clock+0.5*period, the edge is skipped. For example: <code>set eye_obj [sx_create_eye \$signal ui=auto trigger=ext trigsig=\$strobe "option=ddr2t"]</code>
All Bit Patterns	Shows all bit patterns.
Remove 0->X	Filters out 0 to x transitions.
Remove 1->X	Filters out 1 to x transitions.

Setting Mask Parameters

User-defined masks can be added using mask files.

The following parameters are set from the Mask tab in the Eye Diagram Settings window, which specifies a mask that can be superimposed over a folded eye diagram as a timing specification:

Parameter Name	Description
Calculate Mask	Select to calculate mask on eye diagram waveforms. Choose the mask type and color from the Mask Type and Color menus, respectively.
Mask Type	The mask type: Standard or MIPI C-PHY .
Color	The mask color.
Absolute Size Values	Enables absolute size values for the mask.
User-Defined Vertical Boundary Values	Uses the vertical boundary values you define
Show Mask on Top of Waveforms	Displays the mask on top of eye diagram waveforms.
Center Mask to Eye Opening	Centers a hexagon mask to the eye opening.
Show Worst Incursion	Displays the worst violation incursion on a stateye.

Parameter Name	Description
Show Voltage and Timing Margin	Displays the voltage and timing margins at the mask corners.

Cursors in a folded eye diagram trace the eye fuss (eye height/width) or the envelopes of the eye pattern. Select the required **Eye fuss** values and envelop measurements from the cursor context menu. If a cursor is selected to trace the eye envelope, the cursor is automatically moved to its corresponding unfolded position when the eye diagram is unfolded. For more details, see [Tracing Waveform Points in Eye Diagrams](#).

Setting External Trigger Parameters

The **External Trigger** tab in the **Eye Diagram Settings** window specifies external trigger parameters only when external signals are set in the eye diagram panel.

All signals and associated trigger values are displayed in the **Trigger Signals** table. You can change the trigger signal by dragging and dropping a new trigger into the table and clicking **Apply**.

Generating Jitter Histograms

Cursors are normally used in Eye Diagrams to trace the eye envelope or fuss. They can be also used to produce jitter histograms from the eye envelope pattern. To create the jitter histograms, move a cursor and position its horizontal hair at a preferred level. Right-click a cursor, and choose **View Histogram** from the menu that opens.

The time (the offset into a folded eye window) is automatically sampled for each cross-point between the horizontal hair and the eye waveform, and creates histograms for the two areas.

The generated histograms display the jitter distributions of the two cross section areas.

Displaying the Eye Height Distribution

To display the vertical distribution values of the height of an eye as a histogram over the eye diagram:

1. Right-click an eye diagram, and choose **Eye Height Distribution** from the menu that opens.

The **Eye Height Distribution** dialog box opens.

2. Enter the **X Value**.

The eye height distribution begins with this x-axis value.

3. Click **OK**.

A histogram of vertical distribution values is displayed over the eye diagram.

Adding User-Defined Masks

A mask is an abstract reference overlaid in eye diagrams to qualify signals timing integrity. Two built-in masks are provided: a standard mask for common near-end timing specifications and the IEEE802.3 mask. You can add your own mask definitions using mask files as well.

Saving and Loading Masks

Use the **Save Mask** and **Load Mask** buttons of the **Mask** tab of the **Eye-Diagram Settings** dialog to save and load masks. This feature is supported for both standard and C-PHY masks.

Saved mask settings include all mask options shown on the **Mask** tab dialog. When saving a mask spec, you can choose to append the current mask setting to an existing file.

Mask File Syntax

A mask file can contain one or multiple mask definitions. A mask consists of multiple segments that are lines plotted between adjacent points from a node list. Each mask definition is organized in the following format:

```
mask "maskname" direct|normalized
"nodename", X, Y
<newpath>
"nodename", X, Y
...
```

The `mask` keyword starts a new mask definition, which must be followed by a quoted mask name and then a type keyword (`direct` or `normalized`) that specifies the mask type. The words `mask`, `direct`, `normalized`, and `newpath` are reserved keywords.

Each node in the list consists of a quoted node name, the X value of the node, and the Y value of the node. For a `direct` type mask, `x` is the X value into an eye window, and `y` is the Y value of the node. For a `normalized` type mask, the X and Y values denote a node position as fractions into the eye window. `X=0` (`Y=0`) is on the left (bottom) window edge and `X=1` (`Y=1`) is on the right (top) window edge.

A `newpath` keyword breaks line strokes and starts a new drawing stroke.

Incorporating Mask Files

The `SW_SX_MASKFILE` environment variable is used as a pointer to the user mask files.

To incorporate a mask file:

- On Linux Platforms

```
setenv SW_SX_MASKFILE file1:file2:filen
```

`file1...filen` are the path to the mask files. A column (:) is placed between file paths.

- On Windows 95/98 Platforms

Open a MSDOS window and edit AUTOEXEC.BAT in the top directory of your boot hard-drive (C:\, for example). Add the following line to the AUTOEXEC.BAT file:

```
SET SW_SX_MASKFILE=file1;file2;filen
```

The file names are separated by semicolon (;) instead of column (:) because Windows uses column (:) as a drive delimiter.

- On Windows NT Platforms

Open the **Control Panel** and double-click the **System** icon. After you see the **System Properties** dialog box, select the **Environment Variables** tab. Add the `SW_SX_MASKFILE` environment variable as described in [#unique_327_Connect_42_CEGBDFHC](#).

- On Windows XP/2000 Platforms

Open the **Control Panel** and double-click the **System** icon. After your see the **System Properties** dialog box, click the **Advanced** tab and click the **Environment Variables** button. Add the `SW_SX_MASKFILE` environment variable as described in [#unique_327_Connect_42_CEGBDFHC](#).

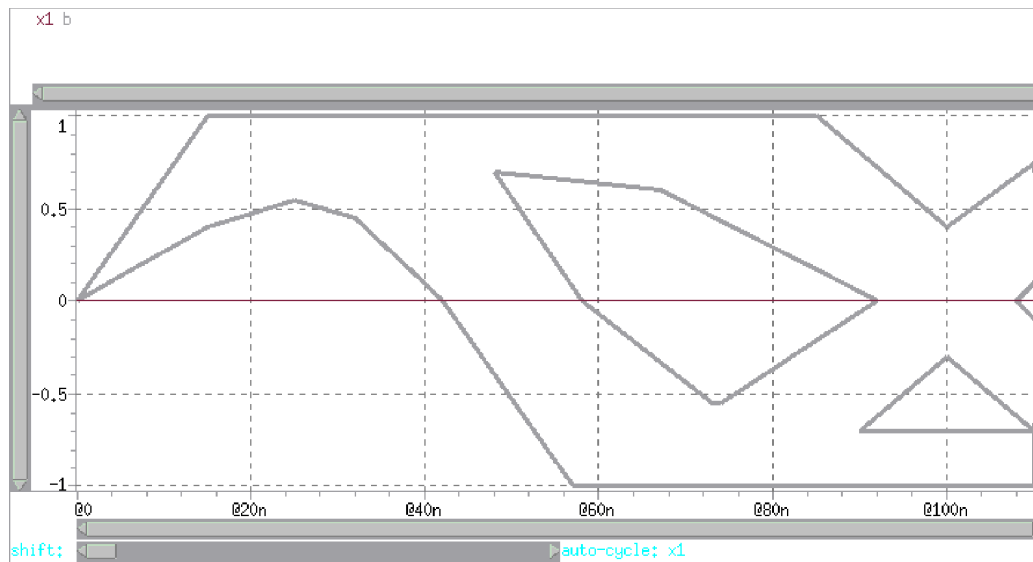
- On Windows Me Platforms

Click **Start**, choose **Run**, then enter `msconfig.exe` and click **OK** to run the `msconfig.exe` program. Click the **Environment tab** and enter `SW_SX_MASKFILE` as a new environment variable. Make sure that the box before `SW_SX_MASKFILE` is checked to enable the environment variable.

These mask files are read at initialization time and the mask menu is expanded in the Eye Configuration dialog box with the user-defined masks.

Mask File Example

The following mask example defines an IEEE802.3 mask as shown in the following figure.



```
mask "IEEE802.3" direct
"A", 0.0n, 0.0
"C", 15.0n, 0.4
"D", 25.0n, 0.55
"E", 32.0n, 0.45
"F", 42.0n, 0.0
"G", 57.0n, -1.0
"T", 110.0n, -1.0
"V", 110.0n, -0.7
"S", 111.0n, -0.15
"R", 108.0n, 0.0
"Q", 111.0n, 0.15
"P", 110.0n, 0.75
"O", 100.0n, 0.4
"N", 85.0n, 1.0
"B", 15.0n, 1.0
"A", 0.0n, 0.0
newpath
"H", 48.0n, 0.7
"I", 67.0n, 0.6
"J", 92.0n, 0.0
"K", 74.0n, -0.55
"L", 73.0n, -0.55
"M", 58.0n, 0.0
"H", 48.0n, 0.7
newpath
"U", 100.0n, -0.3
"W", 90.0n, -0.7
"V", 110.0n, -0.7
"U", 100.0n, -0.3
```

Logic Waveforms

A logic panel displays signals using digital timing diagram. It is permitted only in vertical stack waveviews. Logic panels accept signals that are represented in (time, logic) pairs. Various logic states including logic strength levels are supported. Zoom operation is supported along the x-axis direction.

Logic panels cannot be manually added from the application top menu. They are automatically created when logic signals are dropped into a vertical waveview.

Current vector width (number of bits) and radix setting of a logic panel are indicated in the left column of each panel (B: binary, O: octal, D: decimal, H: Hex-decimal, A: ASCII). The width digits are red color-coded for bus signals formed from originally separated bits, or yellow color-coded for a single bit extracted from an originally grouped bus signal.

Logic panels can be arbitrarily mixed with linear X-Y panels in a vertical stack waveview.

This section contains information on the following topics:

- [Mnemonic Mapping for Bus Signals](#)
- [D to A Conversion](#)

Mnemonic Mapping for Bus Signals

Bus values can be mapped to predefined text strings. You can provide a translation table in an external file.

Syntax:

```
# comment lines start with '#'
value1 name1
value2 name2
...
```

The values are the bus values to be mapped.

The following formats are supported:

- 2400
- 0xAF32
- 'hAF32
- 'b1001_0100_0101
- 'd2400

- 'b110110111100
- 'o371

The names are the text strings to be mapped to. A text string must be quoted if the name contains spaces.

Wildcards are supported in Bus Mnemonic expressions. The ? character is a single bit for wildcards. It works only on 'b, 'o, and 'h.

For example:

```
2'b1?  
3'o1?1  
4'h1?1?
```

You can load the translation table by clicking **Settings** in the **Waveform** tab and then choosing **Bus Mnemonic Map** from the menu that opens. Once a translation table is loaded, click Radix in the Panel section of the Waveform tab, and choose **Mnemonic On/Off** from the menu that opens to toggle the mapping settings.

The mapped values displayed in digital panel area is filled with color. This color is based on color information in map file.

D to A Conversion

Logic waveforms can be converted into analog waveforms based on user-specified analog levels. D/A conversion can be applied only to waveforms in the logic panels.

If the active waveview has only one panel, the D/A conversion operation is by default applied to all waveforms in this panel. If the active waveview has more than one panel, you need to select panels for the D/A conversion operation.

To open the **D/A Conversion Parameters** window, choose **D to A** from the menu located in the **Utilities** section of the **Waveform** tab.

Both single-bit and multi-bit D/A conversions are supported. Analog levels for multi-bit D/A conversion are determined based on rules. Enter the analog voltage levels for the low and high logic states.

There are two modes for D2A sampling: fixed rate sampling and value transition sampling. With the value transition sampling, the D-to-A conversion is completed only at the time when the input digital signal changes its value.

X and Z states can be converted to low or high voltage levels based on user preferences.

Click the **Extend to Full Display Range** button to extend the x-axis range to the current full x-axis display range for each converted analog waveforms, even if the original logic data has an x-axis range that is shorter than the full range.

You can also enable the **Suffix of Result Signal Name** option, which adds a suffix to the name of the converted signal name.

Rising and falling slews of the converted analog waveforms can be specified in the **Preference Settings** window (click **Settings** in the **Configuration** tab), or using the **d2a_rise_slew** and **d2a_fall_slew** configuration options.

Tabular Panels

To create a new Tabular panel, choose **Tabular** from the menu in the **Panel** section of the **Waveform** tab.

See [Working With Tableviews](#) for more information on how to add a tableview and work with tabular panels.

Histogram Panels

To create a histogram panel, an empty histogram panel must be created first by choosing **Histogram** from the menu in the **Panel** section of the **Waveform** tab.

Signals can then be dragged from the **Waveform Browser** and dropped into the histogram. Histogram panels can be placed only in horizontal waveviews.

A histogram plots an individual histogram for each signal in the panel. The average value and standard deviation can be annotated on top of each histogram and the number of bins and data bounds can be reconfigured. The WaveView tool can also calculate the distribution percentages of regions divided by user-defined markers from the histogram mean center.

Configuring Histograms

To configure a histogram, right-click a histogram, and choose **Configure Histogram** from the menu that opens. The **Histogram Settings** window opens.

Right-click on a histogram panel and select **Export Histogram Measures** to bring up a save file dialog for exporting the histogram measures. The m, s, RANGE, and BIN-Width values for the panel are saved in the specified file.

The following settings are available:

Setting	Description
Number of Bins	The number of bins to display in the histogram.
Show Average Line	Enables the average bar. The histogram is plotted by default.

Setting	Description
Show a Single Histogram of All Signals	Plots a single histogram of all signals in a panel.
Y Value Normalized by Runs	Normalizes the Y-axis and sets it to the relative frequency (the number of occurrences divided by the number of Monte Carlo runs, for example).
Show Normal Distribution Curve	Displays the normal distribution curve over the histogram.
Find Distribution	Displays the data distributions for the following specified range in percentage units: • [beginning, mean – lower_marker*sigma] • (mean – lower_marker*sigma, mean] • (mean, mean+upper_marker*sigma] • (mean+upper_marker*sigma, ending]
User-Defined Data Bounds	Specifies the data range to be displayed, which is displayed as the X-axis range in a histogram panel.

Smith Chart

Smith charts display complex normalized impedance, admittance, and scattering parameters (S-parameters). To create a Smith chart panel, choose **Smith Chart** from the menu in the **Panel** section of the **Waveform** tab.

The Smith chart accepts signals that are represented in (real, complex) pairs. The normalized impedance is defaulted at 50 ohms and can be modified in Preference Settings (see the `smith_char_impedance` option in the `.spxrc` configuration file).

Smith charts are allowed only in horizontal waveviews.

You can configure a Smith chart by right-clicking a Smith chart and choosing **Attributes** from the menu that opens. The **Smith Chart Settings** dialog box opens.

Enter the required setting preferences and click **OK** to apply the changes.

Cursors (up to two) in a Smith chart can be used to trace the following value types:

- **Gamma** (the reflection coefficient)
- **SWR** (standing wave ratio)
- **Impedance (Z)**

- **Admittance (Y)**
- **Frequency**

You can change cursors by right-clicking on a Smith chart at the location where the cursor hairs cross, and one of these cursor types from the menu that opens

You can use the Data(X,Y) measurement to trace the frequency, magnitude, and phase values of a signal in a Smith chart.

The **Zoom in** and **Zoom out** buttons are used to zoom in and out a Smith chart. The **Full Unzoom** button displays a full-sized Smith chart.

Polar Plot Panels

To create a new polar plot panel, choose **Polar Plot** from the menu in the **Panel** section of the **Waveform** tab.

Polar plots display data on a complex plane. They are allowed only in horizontal waveviews. The Polar plot accepts signals that are represented in (real, complex) pairs. The zoom operation is supported only along the radial direction.

2-D Sweeps

To create a new 2-D sweep panel, choose **2-D Sweep** from the menu in the **Panel** section of the **Waveform** tab.

The 2-D Sweep panel displays 2-dimensional sweeping data on a flat X-Y plane. It is allowed only in horizontal waveviews. Similar to the 3-D Sweep panel, the 2-D Sweep panel accepts signals that are represented in ((x1, x2), value) pairs. Zoom operation is supported along both axes. Logarithmic scale is also supported along both axes (the x1 direction and the y direction, for example).

Both 2-D and 3-D sweep panels support sweep filters that display the selected sweep traces. Right-click a 2-D plot, and choose **Sweep Line Filter** from the menu that opens. You can highlight traces in the **Sweep Line Filter** dialog box that opens.

Enable the **Display Trace Value** option to display value label next to each sweep trace. If a 2D/3D sweep panel contains more than one signal, the first signal or the first highlighted signal is used as the filter target.

In 2-D sweep plot, the min/max Y-value bounds and the average Y-value of a multi-trace signal can be automatically extracted (for example, waveform envelop from statistical simulation result). To invoke this function, right-click a 2-D sweep panel, and choose **Find Envelope Bound** from the menu that opens. All data points are scanned from all traces, and three new waveforms representing the min/max bounds and the average waveform are created.

Displaying Multi-trace Waveforms

A multi-trace signal contains multiple waveform traces. Each trace has a set of associated sweeping parameters (or sub X-variables). For example:

```
Trace 1: Temp=0 Vdd=3.3
Trace 2: Temp=0 Vdd=5
Trace 3: Temp=100 Vdd=3.3
Trace 3: Temp=100 Vdd=5
```

By default, a multi-trace signal is plotted with respect to its main independent sweeping variable (time, for example). In this case, the sweep line filter function in a 2D-sweep panel can be used to selectively display some of the traces.

The 2D-sweep panel also supports plotting a multi-trace signal with respect to one of its sweeping parameters. To change the default X-axis of a 2D-sweep panel to a sub X-variable, right-click a 2-D panel, and choose **Change X Variable** from the menu that opens. Drag and drop a sweeping parameter from the Waveform Browser to the **Setting X Variable** dialog box. Choose **Sweep Line Filter** from the panel context menu to open the Sub X-Variable Plotting Filter dialog box.

Note:

The filter setup is different between using the default x-axis and sub X-Variable x-axis. This is because in the sub X-Variable plot mode, based on the selected sub X-variable, the number of traces might vary depending on the number of unique combinations among other sweeping parameters. The number of traces are automatically detected and sorted in the order of the sub X-Variable values to display waveforms.

A multi-trace signal must have equal number of points in each of its original traces in order to be plotted in the sub X-variable mode.

Plotting a multi-trace signal with respect to another multi-trace signal is not supported.

3-D Sweeps

To create a new 3-D sweep panel, choose **3-D Sweep** from the menu in the **Panel** section of the **Waveform** tab.

The 3-D Sweep panel displays 2-dimensional sweeping data on a 3-dimension coordinate. It is allowed only in horizontal waveviews. The 3-D Sweep panel accepts signals that are represented in ((x1, x2), value) pairs, where x1 and x2 are the first and second independent sweeping variable, respectively, and value can be either the value of a real signal or the magnitude/phase/real/imaginary part of a complex signal. The viewing angles of the 3-dimension coordinate can be interactively adjusted from the two sliders in the panel. The zoom operation is not supported. Logarithmic scale is supported only along the vertical axis.

Pair Plot

To create a new pair plot, choose **Pair Plot** from the menu in the **Panel** section of the **Waveform** tab.

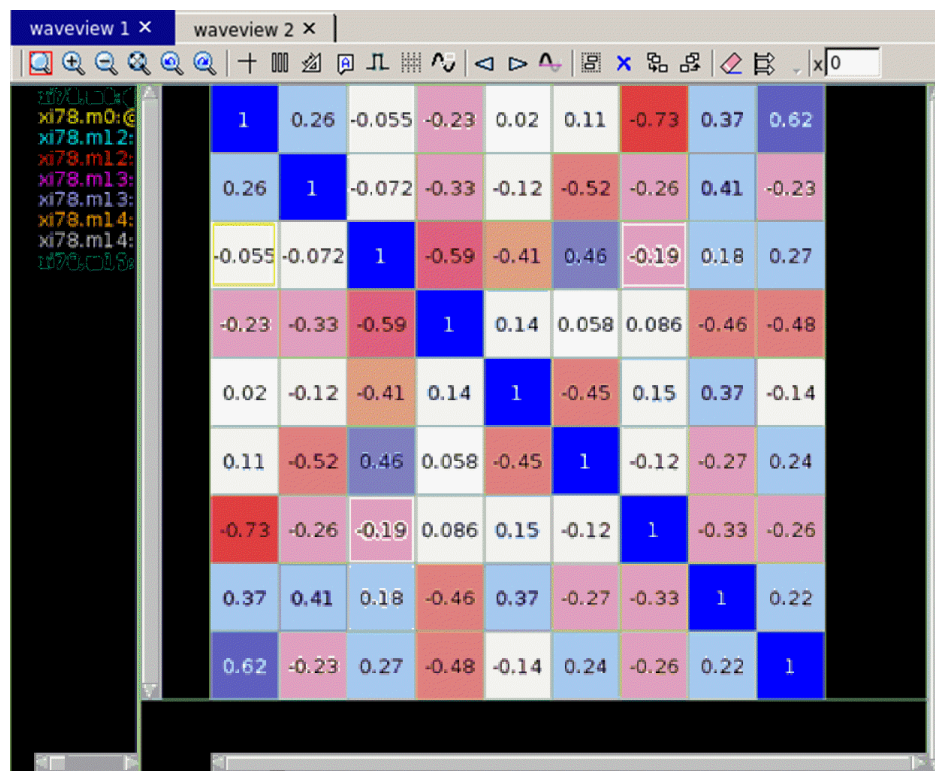
The pair plot displays measurement results in pairs for viewing Monte Carlo data correlation. It is allowed only in row/column waveviews.

The WaveView tool creates a table with a number of rows and columns equal to the number of signals plotted. Each cell/square represents a pair of plotted signals. The signals start from top to bottom, left to right.

Correlation coefficient (R) values are calculated for each pair. Pairs with a value of 0 are shown as white squares. Pairs with a value of 1.0 are shown as blue squares, while values between 0 and 1.0 are a lighter blue, growing darker the closer it is to 1.0. Pairs with a value of -1.0 are shown as red squares, while values between 0 and -1.0 are a lighter red, growing darker the closer it is to -1.0.

If a lot of signals are plotted, the squares are smaller and only the color is shown. The zoom operation is supported.

For example:



When hovering the mouse over signal name, a yellow rectangle highlights the row containing the plotted signal. When hovering over a square cell, the tool shows the R value in a box with canvas background color, as well as the signal names in the y-axis and x-axis areas. When clicking on a square cell, the tool shows a scatter or histogram plot in a separate panel for further analysis. A new panel is created on the first click and reused on each subsequent click.

You can also add a cursor on Pair Plot panel, the R values for each signal is shown in monitor area.

Constell

This panel type supports optical and electrical constellations for OptSim simulations in OptoCompiler.

To create a new constellation, choose **constell** from the **Plot Type** menu.

In this panel, you can plot regular, complex and sweep signals. The x-axis is the real part of the plotted signal and the y-axis is the imaginary part. You can also zoom in and zoom out in this panel.

- Plotting regular signals

For a regular non-complex signal, x-axis values are the signal values, and all y-axis values are equal to zero. The data points are drawn as markers in this panel.

- Plotting complex signals

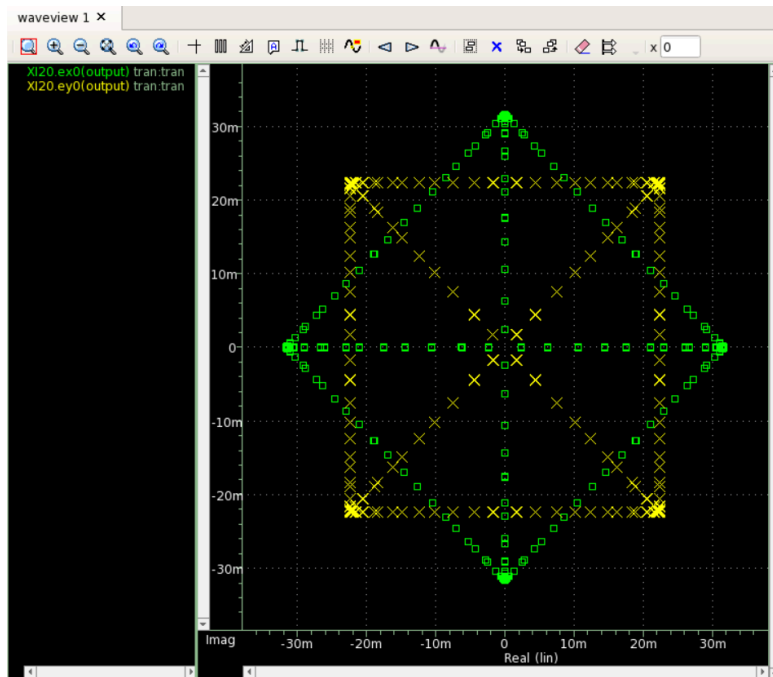
For a complex signal, x-axis values are the real part of the signal and y-axis values are the imaginary part.

- Plotting sweep signals

For a sweep signal, each trace of the sweep signal is plotted as a regular/complex signal in this panel.

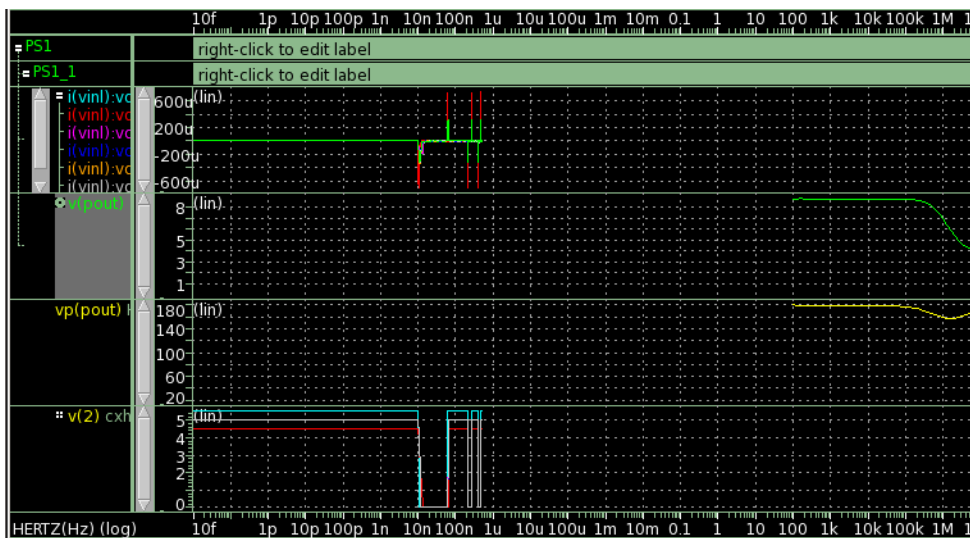
[Figure 2](#) shows an example of a constell plot.

Figure 2 Constell plot



Panel Set

A panel set is a named panel used for grouping signals from different panels. It can contain sublevel panel set panels. To create an empty panel set, choose **Panel Set** from the menu in the **Panel** section of the **Waveform** tab. It appears as follows:



When selecting to **Delete**, **Cut**, or **Copy** a panel set, the operation is applied to all panels contained in the panel set.

Renaming a Panel Set

The name of a panel set is automatically assigned when created. To rename a panel set, right-click the panel and select **Edit Label**.

Expanding a Panel Set

To expand or collapse all panels in a panel set, click the **+/-** icons before the panel set's name. When a panel is collapsed, meters located on that panel are hidden.

Creating a Panel Set From Selected Panels

To create a panel set for selected panels, right-click a panel and select **Create Panel Set**. To ungroup all panels under one single panel set, right-click the panel and choose **Ungroup Panel Set** from the menu it opens.

To insert an empty panel set before a selected panel, right-click the panel and choose Insert **Empty Panel Set**.

Rearranging Panel Set Panels

To move a panel out of a panel set, drag and drop that panel to after the last panel of the panel set, outside of the yellow bounding box. To move a panel to the end of a panel set, drag and drop that panel above the bottom of last panel, inside of the yellow bounding box.

Parametric Plots

The following methods for generating parametric plots are available:

- Using the cursor in a 2D-sweep panel to plot the waveform value versus one of the sweeping parameters. Choose **Plot Y vs. X2** from the cursor context menu. To select a sweeping parameter, right-click the waveform file of the signal in the Waveform Browser and choose **X2 Parameter**. This method is limited to displaying the waveform value only.
- Use the `parametric()` function in the equation builder. The `parametric()` function takes two arguments. The first argument must be a measurement of a multi-trace waveform; the second argument is the sweeping parameter to be used as the x-axis in the parametric plot (for example, `parametric(mean("v(out)"), "TEMP")`). If the second argument is omitted, the default sweeping parameter is used.
- Use the built-in HSPICE parametric plot tool. The parametric tool only accepts HSPICE .MEASURE result files (.mtx/.msx/.max) as the input.

Built-in Parametric Tool for HSPICE .ALTER Simulations

Choose **Parametric Plot** from the menu located in the **Utilities** section of the **Waveform** tab. Parametric plots directly from HSPICE measurement results files (.mt0/.ms0/.ma0 files) are supported.

The Parametric Plot tool can be used to generate scatter plot with measure result annotation. Select the X/Y parameters, the **Scatter Plot** option, and click **Plot** to create the scatter plot.

Generic parametric plot functions can be realized using the parametric() function in the equation builder.

Plotting Clock Jitter

The built-in Jitter vs Time tool can be used to analyze clock jitter between any logic and analog target signal with respect to a reference signal. The reference signal can be an analog or logic signal from simulation output, or an idea signal defined using the width/period/delay parameters. The jitter can be measure against the rise and fall edges of a reference signal.

To invoke the Jitter vs Time tool, choose **Jitter vs. Time** from the menu located in the **Utilities** section of the **Waveform** tab.

This section contains information on the following topics:

- [Selecting Reference Signals](#)
- [Specifying the Target Signal](#)
- [Selecting Reference Edges](#)
- [Target Signal Edges](#)
- [Jitter Types](#)
- [Jitter Output Options](#)

Selecting Reference Signals

The reference signal can be selected from the following sources:

- User-defined Ideal Clock: An ideal clock signal defined by delay, period, and width.
- A Digital Signal: A digital simulation output waveform.
- An Analog Signal: An analog simulation output waveform.

The Digital or Analog Signal field requires an external waveform that can be specified in the External Signal field. Drag and drop the external signal from the Waveform Browser

to the input field. If an analog reference signal is used, threshold value for the rise and fall edges also needs to be specified.

If the **Display Reference Clock Signal** option is selected, the reference signal is shown together with the jitter-versus-time output waveform.

Specifying the Target Signal

The target signal can be either a digital or an analog simulation output waveform. The analog target waveform requires the rise/fall threshold values to be specified.

Selecting Reference Edges

You can select the rise, fall, or both edges of the reference signal as the reference edge for the jitter evaluation.

Target Signal Edges

The following options are available:

- **Same as Reference:** Finds rise-to-rise and fall-to-fall jitters.
- **Rise:** Finds jitter from reference edge to only the rising target edge.
- **Fall:** Finds jitter from reference edge to only the falling target edge.
- **Cross:** Finds jitter from reference edge to the next target switching edge (rise or fall).

The Target Signal Edge options are enabled only if the Jitter Type option is set to **Trigger Time**.

Jitter Types

The Jitter vs. Time tool can evaluate jitter for the following types:

- **Trigger Time:** Measures the edge differences between the trigger edge and the signal edge.
- **Pulse Width:** Measures the pulse differences between the reference signal and the target signal at the reference edge.
- **Period Width:** Measures the period differences between the reference signal and the target signal at the reference edge.

By default, the target signal edges are searched in both directions from a reference switch edge. Select the **Always Search Forward for Target Edge** option to force the target search in the forward direction only.

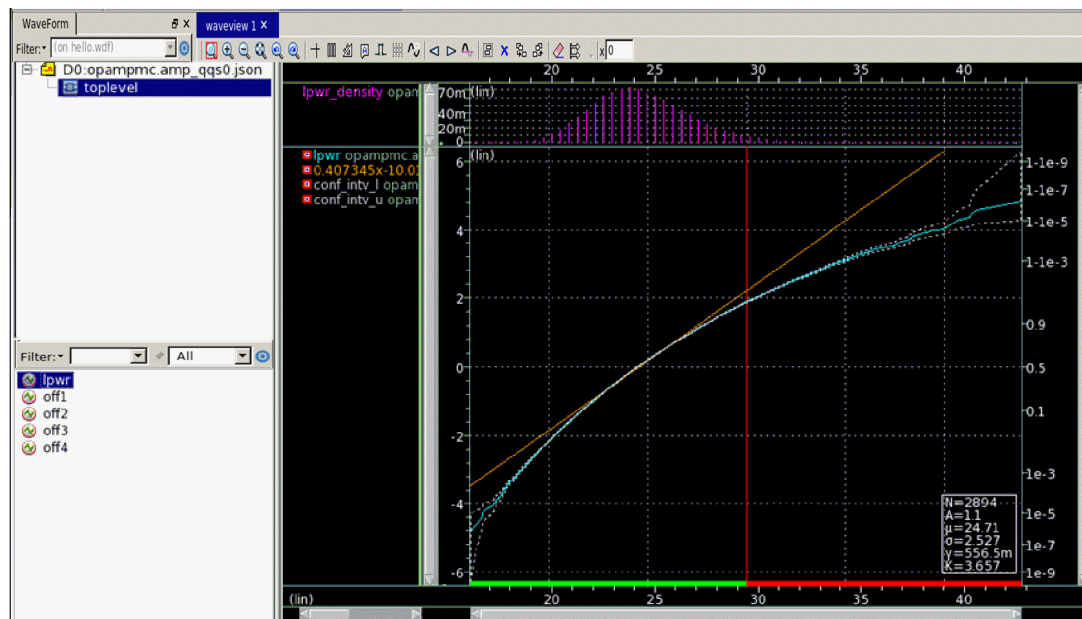
The default target edge search range is half of the clock period at the current reference clock cycle. To limit the maximum search time range, enter the limit in the **Max. Target Edge Search Range** field.

Jitter Output Options

The Jitter vs. Time result is displayed in a linear X-Y panel. PWL waveforms and bar charts are the available plot types. The jitter value sign can be selected between forward-positive or backward-positive. In the forward positive mode, jitter values are positive if the target edge appears later than the reference edge.

HSPICE QQ Plot

The HSPICE QQ statistical plot appears as follows:



After opening files, the measures are listed as signals in the output view.

When plotting QQ measure data, the WaveView tool displays it in an XY panel with Y2 Axis showed. The data array for each measure is used as x-axis and the quantiles array is used as y-axis. The quantiles array is global, which means all measures have the same Y values, but different X values.

The Y2 Axis shows the PNorm values of the standard normal quantile values. The fit line is plotted in the same panel. The Confidence interval lower and upper lines are plotted in the panel as well, as dashed lines.

When plotting QQ measurements, the tool does not share panels with other signals. Each QQ measure is in a new panel, along with fit line and confidence interval lines. Density bars are displayed in a new panel above the QQ measure plot panel.

Statistical summary information is displayed in a box in the bottom right corner of the panel.

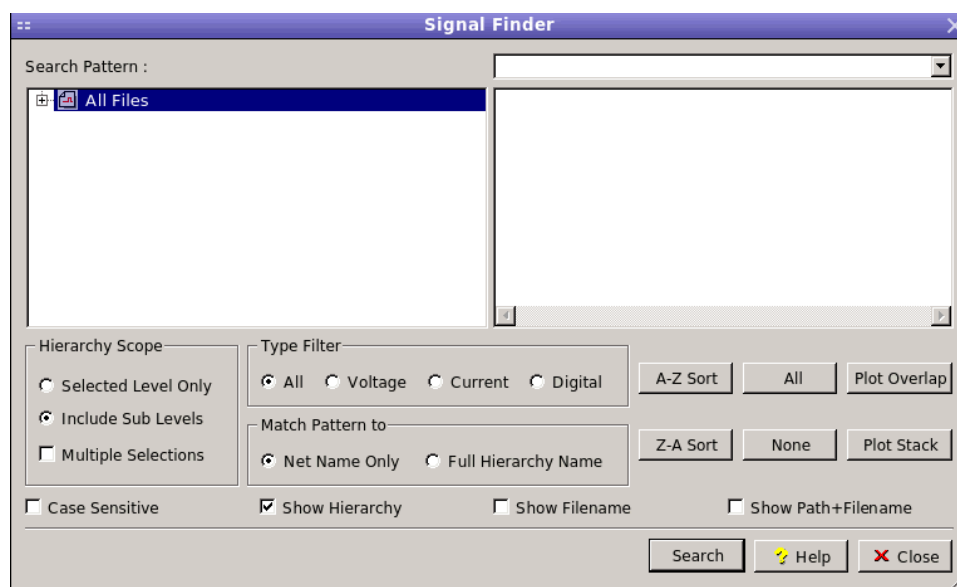
When adding a cursor, the tool shows both left Y and right y-axis values for QQ plot lines, separated by the ; character.

The lower and upper specification limits are plotted as vertical red lines. If both lower and upper specification limits values are -Inf or Inf, there is no specification line displayed. If one of them is not -Inf or Inf, there is one vertical line displayed. If both of them are not -Inf or Inf value, there are two vertical lines displayed.

If lower limit < upper limit, a green bar is shown above the x-axis between lower limit and upper limit, with a red bar out side of [lower, upper] range. If lower limit > upper limit, the tool shows a red bar above the x-axis between lower limit and upper limit, with a green bar out side of [lower, upper] range. If lower limit == upper limit, or both of them are -Inf or Inf value, the tool does not show bars.

Searching Signals

The Signal Finder can be used to search signals from all (or selected) loaded waveform data files and hierarchies. Choose **Signal Finder** from the menu located in the **Utilities** section of the **Waveform** tab to open the **Signal Finder** dialog box.



The signal finder supports name patterns matching with the asterisk (*) and question mark (?) wildcards. Enter a search pattern and hit the return key or click **Search** button to commence a search. The search results are displayed in the right signal list window. Click a signal from the list window to highlight the signal source in the OutputView browser. Drag-and-drop operations are also supported from the signal finder result window into any waveview window. You can sort the results alphabetically by clicking the **A-Z Sort** and **Z-A Sort** buttons.

The search target signals can be refined to Voltage/Current signals only or All Signals by clicking one of the **Type Filter** option radio buttons. To limit the search to a file or a hierarchy level, select the corresponding entry from the left file/hierarchy browser. If the **Selected Level Only** option is selected, the target signals are those in the selected hierarchy/file only. If the **Include Sub Levels** option is selected, the target signals are those in and below the selected hierarchy/file.

On Linux platforms, select the **Multiple Selection** option to select multiple hierarchy entries as the search target.

The **All** and the **None** buttons are used to select/de-select all items in the right signal list window for the drag-and-drop operations.

Click **Plot Overlap** or **Plot Stack** to plot selected signals from the result window in overlap or stack mode.

The **Case Sensitive** option controls the signal name case sensitivity, while the **Net Name Only** and the **Full Hier Name** options control which part of the signal names to be used in the search process.

The **Show Hierarchy**, **Show Filename**, and **Show Path+Filename** options can be used to control the name appearance in the search result browser.

Scanning Signals

Once signals are loaded onto a waveview, you can scan signals from the same files using the **Scan** control buttons. This feature can be used to compare signals from different data files.

Clicking the **right arrow** button advances signals from all selected panels to their corresponding next signals from the same waveform data files. Clicking the **left arrow** scans signals in the reverse direction.

Please note that scanning signals might require reloading data if the next/previous signals are not currently loaded. To speed up the scan process by avoiding data reloading, you might choose to load all waveform data to memory when waveform data files are initially opened (see [Opening a Waveform File](#)).

Default bindkey: **N** (next) or **P** (previous).

Scan Settings

To change the scan settings, click **Settings** on the **Waveform** tab, and choose **Scan Settings** from the menu that opens.

You can configure the scan function into two different modes:

- **Direct Scan:** With the Direct mode, the WaveView tool scans signals according to their order in the waveform data file.
- **Smart Scan:** In the Smart mode, with a master waveform data file selected, the WaveView tool scans signals from the master waveform data file according to their order in the file, while automatically searches other waveform data files for signals with matched names.

Select a scan mode and a master file and click **Apply** to make the change. The **Lock Scope** option controls if the scan is limited within the current hierarchy scope.

Data Reduction

To open the Data Reduction tool, choose **Data Reduction** from the menu located in the **Utilities** section of the **Waveform** tab. The Data Reduction Parameters dialog box appears, and the parameters are acquired for the data reduction operation.

Piecewise-linear waveforms often contain redundant data points. The data point reduction function allows users to remove redundant data points based on user-specified error tolerance. The same data reduction scheme is also used in WDF data reduction.

If the active waveview has only one panel, the reduction operation is by default applied to all waveforms in this panel. If the active waveview has more than one panel, you need to first select panels for the reduction operation.

Click **Ok** to start the data reduction operation. Waveforms with reduced data points are created in new panels in the same waveview.

Direct Plot Mode

To enable direct plot mode, set `wv_dplot=1`, then click the following button at the top of the waveform browser:



Switching to direct plot mode changes the GUI as follows:

- The mini toolbar at the top of the Signal List is hidden.
- There is a new line editor for plotting:



- In the tree browser, the current file is selected and highlighted:



The following features are supported for searching:

- Full scope matching is supported using the delimiter . or /, based on the current file (the end of the delimited part represents the signal).
- Both * and ? are supported for scope and signal (* represents only one level).
- Multiple string searching is supported, separated by whitespace (for example, "x1.x2.v*xx1.xx2.i*").
- Search results are shown in the Signal List with full scope name.
- Currently, search is insensitive and does not support regular expressions.

5

Measuring Waveforms

This chapter details measuring waveforms in the WaveView tool.

This chapter contains information on the following topics:

- [Using Cursors](#)
- [Using Monitors](#)
- [Using the Measurement Tool](#)
- [Supported Measurements](#)
- [Measuring Eye Diagrams](#)
- [RF Measurements](#)
- [PrimeSim HSPICE Measure Tool](#)
- [Pareto Analysis](#)

Using Cursors

This section contains information on the following topics:

- [Adding a Cursor](#)
- [Adding a Horizontal Cursor](#)
- [Adding a Cross-Cursor](#)
- [Jumping Forward and Backward](#)
- [Deleting Cursors](#)
- [Assigning Cursors to Monitors](#)
- [Using Cursor De-Select Mode](#)

Adding a Cursor

Cursors are used to trace waveform-related values. Multiple cursors can be added to a panel. The x-axis position of a cursor is indicated underneath the cursor hair.

To add a cursor, click **Cursor** in the **Panel** section of the **Waveform** tab, and choose **Add Cursor** from the menu that opens.

You can also click the cursor  button in the button toolbar. Cursors can be added for delta measurements by pressing the left- and right-bracket keys ([and]) while moving the cursor.

When multiple cursors are added to a panel, the last moved or added cursor becomes the active cursor.

Waveform values corresponding to the active cursor are displayed in the monitor. In an X-Y panel, distances from all non-active cursors to the active cursor are displayed next to the active cursor.

If you want vertical cursor crossing points with the existing settings, in the **Export V-Cursor Crossings** dialog box, right click a vertical cursor, and choose **Export V-Cursor Crossings With Existed Setting** from the menu that opens. It exports the values without popping up the setup dialog box.

If you want to export Y values at specified X-value crossings to a text file, right-click a vertical cursor, and choose **Export X-Values Crossings** from the menu that opens.

If you want to export some or all vertical cursor crossing points to a text file, right-click a vertical cursor, and choose **Export V-Cursor Crossings** from the menu that opens. In the **Target** field, choose **Selected Cursors** or **All Cursors**.

If you want to export the Y value and data for logic signals in time sequence for all selected cursors, select multiple cursors, right-click on one cursor and select **Export V-Cursor Crossings**. All the values are exported to a file or text window.

This section contains information on the following topics:

- [Selecting With the Cursor](#)
- [Active Cursor and Reference Cursor](#)
- [Moving a Cursor](#)
- [Showing and Hiding Distance Lines](#)
- [Setting Distance Between Two Cursors](#)
- [Locking Pairs of Cursors](#)
- [Locking Cursor Position](#)

- [Snapping the Cursor to an X/Y Value](#)
- [Snapping the Cursor to a Jump Type](#)
- [Setting the Cursor Alias Name](#)
- [Showing and Hiding Cursor Crossing Values](#)
- [Cursors in Smith Charts and Polar Plots](#)
- [Cursors in 2-D Sweep Panels](#)
- [Cursors in 3-D Sweep Panels](#)

Selecting With the Cursor

The cursor is object-based and can select signals or panels, as follows:

- Mousing over a signal or panel highlights the object to be selected with a yellow box.
- **Left-click** selects the object. **Ctrl-click** and **Shift-click** can be used to select multiple objects at once. You cannot select both panels and signals at the same time.
- **Right-click** opens the context menu for the object (and select it if the object is not selected).
- When mousing over the naming area, the cursor appears as follows:
 - For objects that can only be selected, the cursor is an arrow.
 - For objects that can be expanded and collapsed, the cursor is a hand.

Active Cursor and Reference Cursor

Distance lines are shown between a reference cursor and other cursors.

When you add, select, drag, or right-click a cursor, that cursor is set as the active cursor. The active cursor's value is displayed in the monitor.

The first added cursor is set as the reference cursor. The reference cursor can be different than the current cursor and does change when changing the current cursor. There can

only be one reference cursor in an X-Y or logic panel. It is shown with a solid line. For example:



To set a cursor as the reference cursor, **right-click** the cursor and select **Set as Reference Cursor**. Or double-click a cursor and set the cursor as reference cursor in the **Cursor Attributes** dialog.

Moving a Cursor

Depending on the default action of the left mouse button in a waveview (default left mouse button action can be specified in **Left Button Default** preference setting), a cursor can be moved with:

- **Cursor Mode:** If the left mouse button is pressed with mouse pointer near a cursor, the cursor is grabbed. A grabbed cursor can be dragged to a new position. If the mouse pointer is not in the vicinity of any cursor when the left mouse button is pressed, the active cursor is grabbed regardless of its position.
- **Zoom-First Mode:** If the left mouse button is pressed and released without any mouse pointer movement. The active cursor is grabbed to the mouse pointer location when the button is released.

In a zoomed panel, cursors outside the plotting range are placed on the left (or right) boundary of the plotting area. They can be grabbed from the borders of the plotting area. The cursor that is outside but closest to the boundary is grabbed first.

Showing and Hiding Distance Lines

To show or hide the distance lines and delta value of a cursor, **right-click** the cursor and select **Show/Hide Distance**. Hiding distance on a reference cursor hides all distance lines.

When the preference **Hide All Distance between Cursors** is enabled, the tool hides all distance lines and value boxes for all cursors.

Setting Distance Between Two Cursors

To set the distance between cursors, select two cursors, **right-click** on one of them and select **Set Distance**. Enter the required distance in the **Cursor Distance** dialog and click **OK**. The current cursor is moved to achieve the specified distance.

You can also open the **Cursor Distance** dialog by **double-clicking** on the distance value box or **right-clicking** on the distance value box and selecting **Set Distance**.

Locking Pairs of Cursors

To lock a pair of cursors, select two cursors, right-click on one of them and select **Lock distance**.

You can also lock distance by enabling **Lock Distance between Cursors** in the **Cursor Distance** dialog.

To unlock distance, **right-click** on a cursor and select **Unlock Distance**.

To unlock a pair of locked cursors, choose **Release Locked Pair** from the context menu.

Locking Cursor Position

To lock cursor position, enable **Lock Cursor** in the **Set X-value** and **Set Y-value** dialog boxes.

Snapping the Cursor to an X/Y Value

To snap a vertical/horizontal cursor to an X/Y value, respectively, **right-click** on the cursor and select **Set X/Y Value**. Enter the required value in the dialog and click **OK**.

Snapping the Cursor to a Jump Type

To snap the cursor to a jump type, perform the following steps:

1. **Right-click** on the added cursor, select **Attributes**. The **Cursor Attributes** dialog box opens.
2. Choose a **Jump Type**.

If you choose **To**, the cursor jumps to the **X Value** you have set in this field.

3. Select the **Snap Cursor to Jump Type** check box.

This snaps the cursor to the selected jump type.

4. Click **OK**

To enable or disable snapping the cursor to jump type, **right-click** on the added cursor, and select or deselect **Snap Cursor to Jump Type to this position** from the context-

sensitive menu. When enabled, the cursor is snapped to the jump type position set in the earlier procedure.

Setting the Cursor Alias Name

To set an alias name for a cursor, right-click on the cursor value label box and select **Set Alias**. Enter the alias name in the dialog and click **OK**.

Showing and Hiding Cursor Crossing Values

Cursor value display has been enhanced as follows:

- **Showing and hiding V-value crossings:** To show/hide V-value crossing labels along a vertical cursor, **right-click** the cursor and select **Show/Hide V-Value Crossings**.
 - **Show/Hide X Value:** To show/hide X-axis values crossing labels along a vertical cursor, **right-click** the cursor and select **Show/Hide X Value**. To set **Hide X value** as default, choose **Configuration > Preferences > Waveview > Waveview**, select **Hide X Values on the Value Labels at V-Cursor Crossing Points by Default**.
- **Showing and hiding H-value crossings:** To show/hide H-value crossing labels along a horizontal cursor, **right-click** the cursor and select **Show/Hide H-Value Crossings**.

You can also choose to show signal names and file names of all plotted signals or (select signals) as detailed in the **H-Cursor Attributes** dialog.

- **Showing P1/P2 and Delta Values:** The tool shows crossing point values and delta values (P2-P1) of two crossing points between the reference cursor (P1) and the target cursor (P2). By default, the P1 value is shown along with the reference cursor, and the P2 value and delta value are shown along with the target cursor.

When distance lines are hidden, the delta value is not shown.

- When the preference **Display Value Labels at Crossing Points by Default** is enabled, the tool shows crossing labels along the cursor.
- The preference **Show Reference Cursor Crossing Value** can be set to the following:
 - **Individually** — the P1 value is shown along the reference cursor and the P2 value and delta value are shown along the target cursor.
 - **Along with Target Cursor** — the P1 value, P2 value, and delta value are all shown along with the target cursor.

Cursors in Smith Charts and Polar Plots

In Smith charts and Polar plots, a cursor consists of a circle hair and a radial hair. The cross point is the cursor location and its value is displayed. A cursor can be grabbed by clicking the mouse button when the mouse pointer is near any of its two hairs. The cursor values in the monitor column trace the data points that are closest to the hair cross.

Cursors in 2-D Sweep Panels

You can use a cursor in a 2D-sweep panel to create a new 'Y vs. X2' plot. Move the cursor to a preferred position, right-click the cursor, and choose **Plot Y vs X2** from the menu that opens.

Cursors in 3-D Sweep Panels

In a 3D-sweep panel, a cursor consists of a vertical hair, and two crossing hairs lying in the base plane. To grab a cursor in a 3D-sweep panel, move the mouse pointer near the vertical hair base where the two crossing hairs meet.

Adding a Horizontal Cursor

Horizontal cursors are supported in analog X-Y panels. To add a horizontal cursor, click **Cursor** in the **Panel** section of the **Waveform** tab, and choose **Add H-Cursor** from the menu that opens.

You can also switch a vertical cursor to a horizontal cursor by right-clicking a vertical cursor and choosing **Switch to H-Cursor** from the menu that opens. The horizontal cursors trace the x-axis values of the nearest intersect point with waveforms. The x-values are displayed in the monitor column.

You can also click the **Add Cursor** control button to add an H-cursor by setting the **Add H-Cursor with Add Cursor Control Button** option in the preferences. To set this preference, click **Settings** in the Configuration tab, and click the **WaveView** tab. Click the **Add H-Cursor with Add Cursor Control Button** check box, and click **OK**.

This section contains information on the following topics:

- [Annotating Horizontal Cursors](#)
- [Hiding Horizontal Cursors](#)
- [Anchoring Horizontal Cursors](#)
- [Exporting Horizontal Cursor Values](#)

Annotating Horizontal Cursors

In an X-Y panel, labels can be annotated to all the signal crossing points of a horizontal cursor. To enable crossing value labels, right-click an H-cursor, and choose **Crossing Options** from the cursor context menu.

Note:

H-cursor label positions are recorded when the total number of H-crossing labels stay the same during H-cursor movement. The modified cursor label positions are saved in the session file.

Hiding Horizontal Cursors

To hide the H-cursor hair, right-click an H-cursor, click the **Hide Cursor Hair** check box, and click **OK**.

Anchoring Horizontal Cursors

To anchor horizontal cursors to a single panel, right-click an H-cursor, and choose **Crossing Options** from the menu that opens. Click the **Cursor Anchor in Panel** check box, and click **OK**.

Exporting Horizontal Cursor Values

If you want to export all horizontal cursor crossing points to a text file, right-click a vertical cursor, and choose **Export H-Cursor Crossings** from the menu that opens.

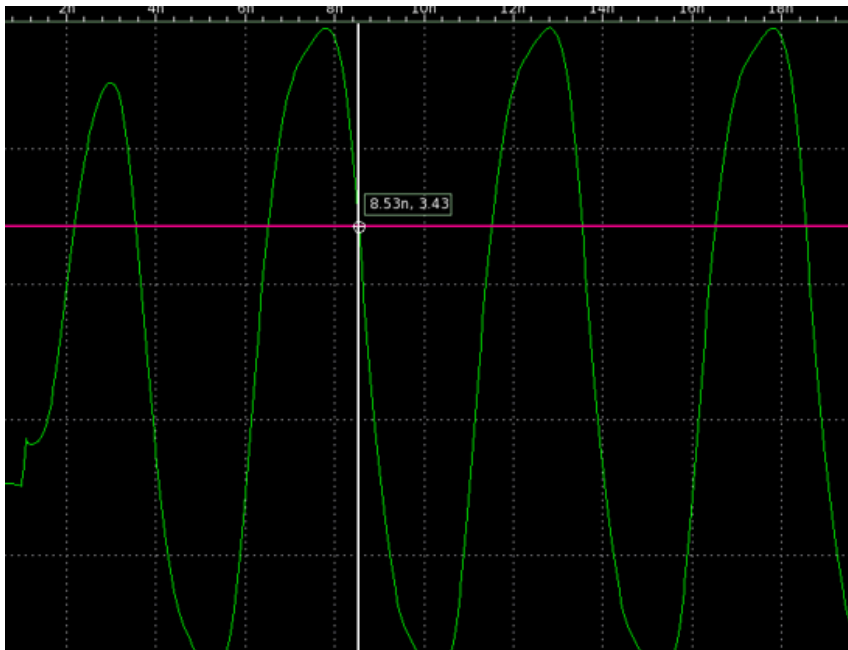
If you want to export X values at specified Y-value crossings to a text file, right-click a vertical cursor, and choose **Export Y-Values Crossings** from the menu that opens.

Adding a Cross-Cursor

Cross-cursors are supported in analog X-Y panels. To add a cross-cursor, click **Cursor** in the **Panel** section of the **Waveform** tab, and choose **Add Cross-Cursor** from the menu that opens.

No monitor is created since there is only one crossing point for all waveforms. The cross-cursor has one vertical line and one horizontal line, showing X and Y values at the crossing point. The crossing point can be in different panels.

It appears as follows:

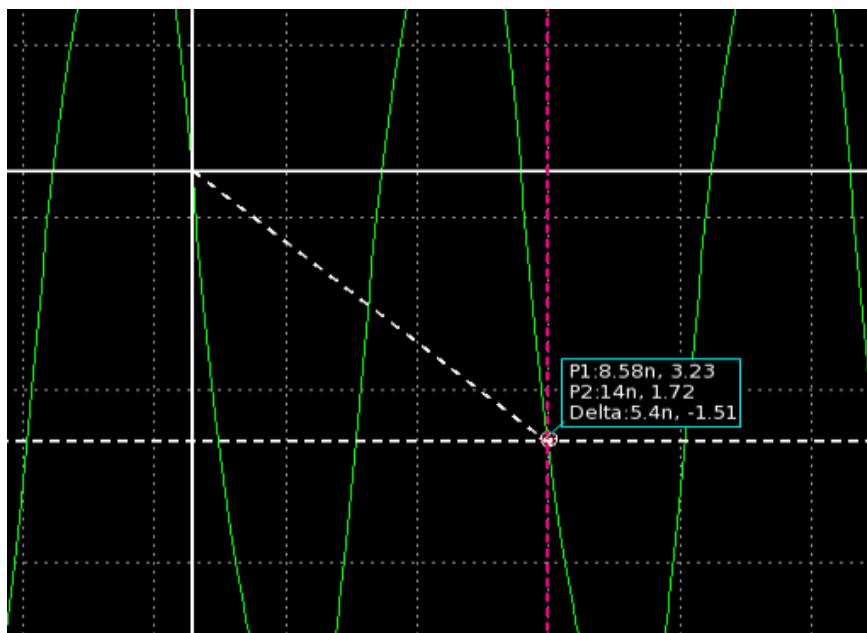


A cross-cursor has two modes:

- Floating (default) — the cursor can be placed at any location, and the vertical and horizontal line can both be moved individually and freely.
- Snap to waveform — when moving the vertical or horizontal line, the other line may be moved to snap to the waveform. It can be snapped to different waveforms. This is enabled by default.

You can set the X or Y value through the right-click Attributes dialog, the icon bar, or using jump forward/backward.

You can pair a cross-cursor with another cross-cursor, which shows the Delta X and Delta Y. For example:



Jumping Forward and Backward

In an X-Y or logic panels, a cursor can be jumped forward to the next data point, peak, switching or crossing point using the cursor context menu.

To jump a cursor forward or backward, right-click near the cursor, and choose **Jump (F)orward "SIGNAL"** or **Jump (B)ackward "SIGNAL"** from the menu that opens to jump the cursor forward or backward on the target signal SIGNAL.

You can also select a cursor, then click **Cursor** in the **Panel** section of the **Waveform** tab, and choose **Jump Forward** or **Jump Backward** from the menu that opens.

The target signal is indicated by a cross marker on the signal waveform at the intercept point with the cursor hair. To select a different target signal, move the cursor near a waveform (along the Y-direction) and the cross marker snaps to the new target waveform.

For digital panels, the cursor can be jumped to the next value change, the next match value, the next rise/fall edge, or the user-defined time.

To jump to a specific X value, right-click a V cursor, and choose **Goto X Value** from the menu that opens.

Default bindkeys: Press the **F** key to jump forward, and the **B** key to jump backward.

Jump Settings

To change the jump parameters, click **Cursor** in the **Panel** section of the **Waveform** tab, and choose **Jump Settings** from the menu that opens to invoke the **Cursor Jump Parameters** dialog box.

Enter a preferred jump type and associated parameters and click **Ok** to make the change. Click **Set as Default** to save the settings in ~/.spxlast as the default setting.

Deleting Cursors

To delete a single cursor, right-click near the cursor you want to delete, and choose **Delete** from the menu that opens to delete the cursor.

You can also select a cursor and click **Cursor** in the **Panel** section of the **Waveform** tab, and choose **Delete Cursor** from the menu that opens.

Deleting All Cursors

To delete all cursors, select a group of cursors, and right-click near the selected group. Choose **Delete All** from the menu that opens to delete all cursors.

You can also select a group of cursors and click **Cursor** in the **Panel** section of the **Waveform** tab, and choose **Delete All** from the menu that opens.

Assigning Cursors to Monitors

When you add a monitor of cursor value, if there is only one monitor, it links to the active cursor. Otherwise, the new added monitor links to a cursor that does not link to any monitor.

Selecting **Auto Link Monitors** from the monitor context menu links all monitors of cursor value to corresponding cursors.

Selecting **Link to Monitor** from the cursor context menu links selected cursors to corresponding monitors. If the tool cannot find a monitor for a selected cursor, it adds a new monitor and link it to this cursor.

Using Cursor De-Select Mode

Disable the option **Move Cursor to Mouse Click Position** and the cursor operates in de-select mode and does not move.

Using Monitors

Monitors are used to trace waveform-related values. Multiple monitors can be added to vertical waveviews, or panels in horizontal. The following monitors are available for analog panels:

Monitor	Description
cursor value	(Y=) Traces waveform values that correspond to the current active cursor position. Logarithmic value is displayed if the corresponding panel has logarithmic Y-axis. To display the logarithmic value of a negative value x , the format $-\log(\text{abs}(x))$ is used.
derivative	(S=) Traces waveform derivative values that correspond to the current active cursor position. Derivative of logarithmic values is displayed if the corresponding panel has logarithmic Y-axis.
1/derivative	(1/S=) Traces the reciprocal of waveform derivative values that correspond to the current active cursor position.
Y delta	(D=) Displays difference between (1) waveform values that correspond to the active cursor position and (2) waveform values that correspond to the other cursor. Delta value is available only when a waveview (or panel) has two cursors.
cursor average	(A=) Displays waveform average between the cursors. Cursor average value is available only when a waveview (or panel) has two cursors.
maximum	(MAX=) Displays waveform maximum values over full or zoomed signal range.
minimum	(MIN=) Displays waveform minimum values over full or zoomed signal range.
peak-to-peak	(PP=) Displays waveform peak-to-peak values over full or zoomed signal range.
average	(AVG=) Displays waveform average values over full or zoomed signal range.
RMS	(RMS=) Displays waveform root mean square values over full or zoomed signal range.

If you choose a **maximum**, **minimum**, **peak-to-peak**, **average**, or **RMS** monitor, you can also specify the following **X-axis Range** settings:

X-Axis Range Option	Description
full signal range	Measures across the full range of a signal.

X-Axis Range Option	Description
zoomed view range	Measures just the zoomed portion of the signal that is currently displayed.
cursor range	Measures between the x-values of two cursors.
user-defined range	Measures between the two specified values.

This section contains information on the following topics:

- [Adding Monitors](#)
- [Copying Monitor Values](#)
- [Reconfiguring Monitors](#)
- [Deleting Monitors](#)
- [Linking Monitors to Cursors](#)
- [Exporting Monitors](#)

Adding Monitors

To add a monitor, click **Cursor** in the **Panel** section of the **Waveform** tab, and choose **Add Monitor** from the menu that opens.

If a vertical waveview (or a panel in a horizontal waveview) has no existing monitor in the cursor value type, a cursor value type monitor is automatically added when a cursor is created.

When you add a monitor of cursor value, if there is only one monitor, it links to the active cursor. Otherwise, the new added monitor links to a cursor that does not link to any monitor.

To add a new monitor, click the control button to open the Monitor Settings dialog box. Select the preferred settings, and click **Ok** to create the monitor.

Copying Monitor Values

To copy the value of a monitor, right-click a monitor, and choose **Copy Value** from the menu that opens. The value is copied to the clipboard.

Reconfiguring Monitors

To reconfigure a monitor, right-click a monitor, and choose **Configure Monitor** from the menu that opens. In the **Monitor Settings** dialog box that opens, reconfigure the monitor, and click **Ok** to make the change.

Deleting Monitors

To delete a monitor, right-click a monitor, and choose **Delete Monitor** from the menu that opens.

Linking Monitors to Cursors

To link a monitor to a selected cursor, right-click a monitor and choose **Link Cursor** from the context menu that appears. In the **Cursor Selection** dialog box that opens, link any desired monitors to cursors.

Exporting Monitors

To export monitors to a text or CSV file, right-click a monitor and choose **Export Monitor** from the context menu that appears. Choose an **Output File** name, **Output Type**, and **Target** in the **Export Columns** dialog box that opens, and click **OK** to export the monitors.

Using the Measurement Tool

To access the Measurement Tool, click **Measure Tool** in the **Utilities** section of the **Waveform** tab.

Once one or more measurements are added to a waveview, you can drag those measurements to take the measurement interactively. If a panel has multiple signals and you want to perform a Width, Slew, Frequency, or Duty-Cycle measurement, the default measurement target is the first selected (highlighted) signal. The first signal is the default if no signal is selected.

Note:

The names of any signals that are currently being measured are displayed at the bottom of measurement configuration dialog box.

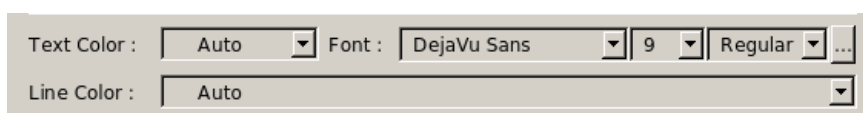
This section contains information on the following topics:

- [Setting the Measurement Font and Color](#)
- [Adding or Removing Measurement Favorites](#)

- [Viewing Measurement Results](#)
- [Setting the Precision of Measurements](#)
- [Exporting Measurements](#)
- [Locking and Unlocking Measurements](#)

Setting the Measurement Font and Color

You can set the font and color in the **Measurement Tool** window to label individual measurements to differentiate between measurements.



Adding or Removing Measurement Favorites

In the Favorites tab of the **Measurement Tool** window, you can choose from several measurement you use most often.

To add a measurement to the list of favorites:

1. Click the **All** tab.

The available measurement categories and corresponding measurement appear.

2. Click a measurement.
3. Click the **Favorite** button at the bottom of the **Measurement Tool** window.

The selected measurement is saved as a favorite and is included as one of the measurements you can choose from the **Favorite** tab.

To remove a measurement from the favorite measurement list:

1. Click the **Favorite** tab.

The current measurement favorites are displayed at the top of the **Measurement Tool** window.

2. Click a measurement.
3. Click the **Remove** button at the bottom of the **Measurement Tool** window.

The selected measurement is removed from the list of favorite measurements you can choose from the **Favorite** tab.

Viewing Measurement Results

To see existing Measurement Tool results from a waveview, right-click a measurement result in a waveview, and choose **Measure Results** from the menu that opens. The **Measure Results** window opens with a table of measurement result values.

Setting the Precision of Measurements

To change the precision of measurements, click **Settings** in the **Configuration** tab, and then click the **General** category in the **Settings Preferences** window that opens. Enter a value in the **Number of Valid Digits** text box.

Exporting Measurements

When you perform a measurement on a waveform, the measurement can be exported to the following equivalent forms:

- Equivalent HSPICE `.MEASURE` statements
- Tcl ACE `sx_equation` statements
- Equation builder equations

To directly export a measurement to the Equation Builder, right-click a measurement and choose **Measurement to Eqn** from the context menu. The equivalent measurement equation and the result are displayed in the Equation Builder.

To export all measurements on a waveview window, right-click a measurement and choose **Measurement Export**. Use the **Dynamic Measurement Export** dialog box to select the measurements to be exported and the preferred export type. Only the following types of measurements can be exported:

- Rise/Fall time
- Width/Frequency
- Difference
- Y-Difference
- Y-Range

In the **Dynamic Measurement Export** dialog box, you can define the export name of individual measurements, enable or disable a measurement in the list, control how to count the number of switch edges, and define the precision for the exported values.

For HSPICE Measure and ACE Tcl export modes, the path of the output file must be specified.

Locking and Unlocking Measurements

To lock or unlock a measurement, right-click the measurement and choose **Lock Measurement** or **Unlock Measurement**, respectively, from the menu that opens.

Supported Measurements

The following measurements are supported:

Measurement	Type	Description
AC Coupled RMS	Level	Calculates the RMS value of the AC component of a waveform.
Amplitude	Level	Calculates the difference between the topline and baseline reference levels.
Average	Level	Calculates the average level of a waveform.
Bandpass	Frequency Domain	Calculates the bandwidth of a bandpass-shaped waveform.
Baseline	Level	Calculates the baseline level of a waveform.
Cpk	Statistics	Calculates an indicator of the process capability for a waveform relative to specified upper and lower limits.
Damping Ratio	S Domain	Calculates the damping ratio of a waveform as -real/mag.
Data(X,Y)	General	Displays the X-axis and Y-axis values of a point on a waveform.
Delay	Time Domain	Calculates the delay between the edges of two waveforms relative to the default or specified topline and baseline levels for both the measured waveform and the reference waveform.
Difference	General	Displays the following information for two points on one or two waveforms: X1, Y1; X2, Y2; and Delta X, Del Y, Slope.
Dpu	Statistics	Calculates the total number of defects per unit based on the points in a waveform.
Duty Cycle	Time Domain	Displays a duty cycle of a periodic waveform.

Measurement	Type	Description
F VS T	Time Domain	Calculates all frequency, period, or duty cycle values for the edges crossing the threshold, and then creates new Frequency vs. time, Period vs. time, or Duty Cycle vs. time waveforms using the specified X range and Y level.
Frequency	Time Domain	Displays the frequency and period of a periodic waveform.
Frequency Value	S Domain	Displays the frequency value at the point of a complex waveform on a Nyquist (or Nichols) plot.
Gain Margin	Frequency Domain	Displays the gain margin of a complex analog waveform in decibels (dB).
Highpass	Frequency Domain	Calculates the corner frequency of a waveform with a highpass shape.
IP2	RF	Calculates the second-order intercept point.
IP3/SFDR	RF	Calculates the third-order intercept point.
Jitter	Time Domain	Displays the histogram of crossing edges with a specified X range or Y range.
Length	General	Calculates the length of a straight line that connects two points on a waveform or two points on two waveforms.
Lowpass	Frequency Domain	Calculates the corner frequency of a waveform with a lowpass shape.
Maximum	Level	Calculates the maximum level of waveform.
Mean	Statistics	Calculates the mean value of a waveform.
Mean+3std_dev	Statistics	Calculates the (mean + 3 standard deviation) value of a waveform.
Mean-3std_dev	Statistics	Calculates the (mean - 3 standard deviation) value of a waveform.
Minimum	Level	Calculates the minimum level of a waveform.
Natural Frequency	S Domain	Calculates the natural frequency of a waveform as the absolute value of an argument.
Overshoot	Time Domain	Calculates the overshoot of a waveform relative to a default or specified topline.
P1dB	RF	Calculates the 1dB compression point.

Measurement	Type	Description
Peak to Peak	Level	Calculates the peak to peak values of a waveform.
Phase Margin	Frequency Domain	Displays the phase margin of a complex analog waveform in degrees.
Quality Factor	S Domain	Calculates the quality factor of a waveform.
Rise/Fall Time	Time Domain	Calculates the risetime and falltime between specified upper and lower levels of a waveform.
RMS	Level	Calculates the RMS (Root Mean Square) value of a waveform.
Settle Time	Time Domain	Calculates the settle time of a waveform with respect to a default or specified settle level and a specified settle band.
Slew Rate	Time Domain	Calculates the slew rate of a waveform relative to the default or specified topline and baseline levels.
Std_dev	Statistics	Calculates the standard deviation of a waveform.
Stopband	Frequency Domain	Calculates the stopband, low, high, or center frequency level or the level at which the measurement is calculated for a stopband-shaped waveform.
Topline	Level	Calculates the topline level of a waveform.
Undershoot	Time Domain	Calculates the undershoot of a waveform relative to a default or specified baseline level.
Width	General	Calculates the pulse width of a waveform.
X at Maximum	Level	Displays the x-value corresponding to the maximum value of a waveform.
X at Minimum	Level	Displays the x-value at the minimum value of a waveform.
Y Diff	General	Displays the Delta Y at a X value between two points on two waveforms.
Y Range	General	Displays the following information for a waveform: Waveform name, peak-to-peak Y range; Minimum Y value, Maximum Y value; and Average Y value, RMS value.
Yield	Statistics	Calculates the ratio of the number of data points between the Y-axis levels Upper and Lower relative to the total number of data points.

General Measurements

The following general measurements are available:

- [Data\(X,Y\)](#)
- [Difference](#)
- [Length](#)
- [Width](#)
- [Y Diff](#)
- [Y Range](#)

Data(X,Y)

Displays the x-axis and y-axis values of a point on a waveform.

Note:

This measurement is supported only in Analog X-Y, 2-D Sweep, and Smith Chart waveview panels.

The following parameters can be set for the Data(X,Y) measurement:

Parameter	Description
Locked Level	Locks the measurement Y level at the specified percentage or level (in Y-axis units).
Target Signal	Uses the min/max Y levels from the signal you are currently measuring. This option is only available when specifying a measurement Y level that is locked at a specified percentage.
All Signals	Uses the min/max Y levels from all signals that are currently open. This option is only available when specifying a measurement Y level that is locked at a specified percentage.
User Specified	Uses the min/max Y values you specify in the Min and Max text fields.
Range options	• Entire Waveform — (default) measures the entire waveform. • Visible Range — measures the visible range. • User Specified — measures the defined range.

Difference

Displays the following information for two points on one or two waveforms: X1, Y1; X2, Y2; and Delta X, Del Y, Slope. The capacitance and inductance values from the X and Y points are also calculated using the following functions:

$$C = \frac{x2-x1}{2\pi \cdot x1x2 \cdot (y1-y2)}$$

$$L = \frac{y1-y2}{2\pi(x1-x2)}$$

Note:

This measurement is supported only in Analog X-Y, Eye Diagram, and 2-D Sweep waveview panels.

The following parameters can be set for the Difference measurement:

Parameter	Description
Locked Y Level	Locks the measurement Y level at the specified percentage or level (in Y-axis units).
Locked X Level	Locks the measurement X level at the specified percentage or level (in X-axis units).
Indep P2 Lock Level	Enables the Lock 2nd Point at independent second Y-level point measurement option.
Lock 2nd Point at	Locks a second measurement Y-level point at the specified percentage or level (in Y-axis units). This option is only available when the Indep. P2 Lock Level option is selected.
Lock P2 X Value	Locks a second measurement X-level point at the specified percentage or level (in X-axis units).
Target Signal	Uses the min/max Y levels from the signal you are currently measuring. This option is only available when specifying a measurement Y level that is locked at a specified percentage.
All Signals	Uses the min/max Y levels from all signals that are currently open. This option is only available when specifying a measurement Y level that is locked at a specified percentage.
User Specified	Uses the min/max Y values you specify in the Min and Max text fields.
Anchor First Point	Anchors the measurement at a single point; further dragging only moves the second point.
Gravity Snap	Snaps anchors to data points when moving.

Parameter	Description
Label Move Independently	Allows label and endpoints to move independently.
Range options	- Entire Waveform — (default) measures the entire waveform. - Visible Range — measures the visible range. - User Specified — measures the defined range.

Length

Calculates the length of a straight line that connects two points on a waveform or two points on two waveforms. If two waveforms are selected, the two waveforms do not need to be the same type.

Width

Calculates the pulse width of a waveform.

Note:

This measurement is supported only in Analog X-Y and Eye Diagram waveview panels.

The following parameters can be set for the Width measurement:

Parameter	Description
Locked Level	Locks the measurement Y level at the specified percentage or level (in Y-axis units).
Target Signal	Uses the min/max Y levels from the signal you are currently measuring. This option is only available when specifying a measurement Y level that is locked at a specified percentage.
All Signals	Uses the min/max Y levels from all signals that are currently open. This option is only available when specifying a measurement Y level that is locked at a specified percentage.
User Specified	Uses the min/max Y values you specify in the Min and Max text fields.
Pulse Type	Specifies the pulse type as Either , Up , or Down .
Width Rate vs. X-axis Plot Option	Calculates the width value of an up or down pulse versus the X-axis.
Range options	- Entire Waveform — (default) measures the entire waveform. - Visible Range — measures the visible range. - User Specified — measures the defined range.

Y Diff

Displays the Delta Y at a X value between two points on two waveforms.

Note:

This measurement is supported only in Analog X-Y waveview panels.

Y Range

Displays the following information for a waveform:

- Waveform name
- Peak-to-Peak Y range
- Minimum Y value
- Maximum Y value
- Average Y value
- RMS value

Note:

This measurement is supported only in Analog X-Y waveview panels.

Specifying the **Measure over Zoomed X-Range** option measures the zoomed area that is currently displayed on screen. If this option is not specified, the whole waveform is measured even if the waveform is zoomed and some of the X-range is not visible.

Time Domain Measurements

The following General measurements are available:

- [Delay](#)
- [Duty Cycle](#)
- [F VS T](#)
- [Frequency](#)
- [Jitter](#)
- [Overshoot](#)
- [Rise/Fall Time](#)
- [Settle Time](#)

- [Slew Rate](#)
- [Undershoot](#)

Delay

Calculates the delay between the edges of two waveforms relative to the default or specified topline and baseline levels for both the measured waveform and the reference waveform. The rising or falling edge on the reference waveform is assumed to cause the corresponding (rising or falling) edge on the measured waveform so that the reference edge occurs before the measured edge.

All rising or falling edges for the measured waveform are determined based on the Trigger setting in the Measure dialog box. From each edge, the corresponding previously occurring edge on the reference waveform is determined. The difference between the two edges on the x-axis is the delay time.

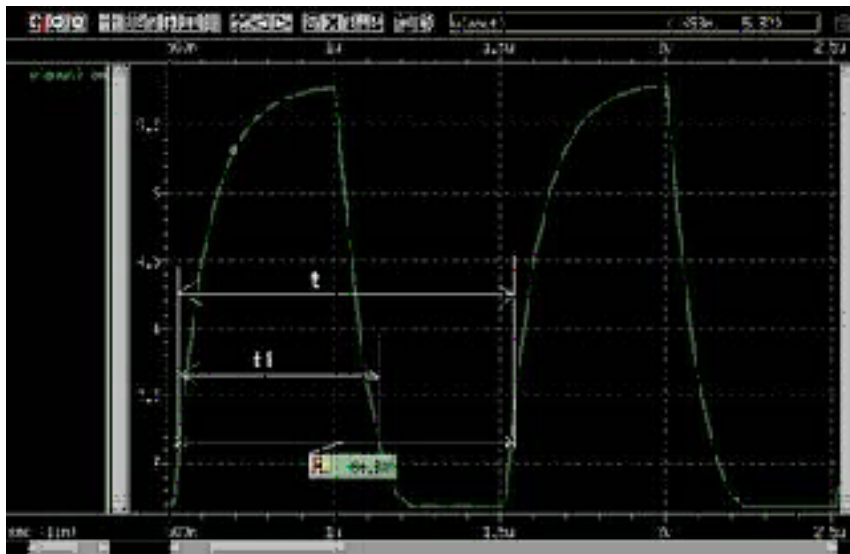
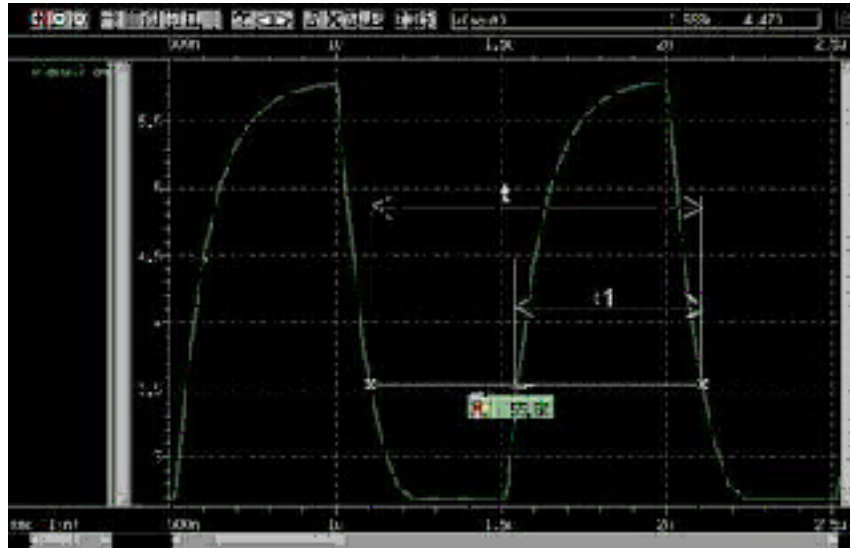
The following parameters can be set for the Delay measurement:

Parameter	Description
Percentage	Locks the delay level at the specified percentages for the signal (L(%)) and reference signal (Ref. L(%)). In signal Y units.
Premise	Specifies the delay value as either (default), positive delay, or negative delay.
Signal Level	Locks the delay level at the specified levels for the signal (L(V)) and reference signal (Ref. L(V)). In signal Y units.
Target Signal	Uses the min/max Y levels from the signal you are currently measuring. This option is only available when specifying a delay level that is locked at a specified percentage range.
All Signals	Uses the min/max Y levels from all signals that are currently open. This option is only available when specifying a delay level that is locked at a specified percentage range.
User Specified	Uses the min/max Y values you specify in the Min and Max text fields.
Trigger	Specifies the trigger type as Either , Rising , or Falling edge(s)
Ref. Trigger	Specifies the reference signal trigger type as Either , Same , or Opposite .
Measure Between Closest Edges by Default	Finds the closest edge between the <code>Signal</code> and <code>Ref. Signal</code> .
Delay Rate vs. X-axis Plot Option	Calculates the delay rate values of the Rise , Fall , or Rise/Fall edges vs. the x-axis.

Parameter	Description
Range options	• Entire Waveform — (default) measures the entire waveform. • Visible Range — measures the visible range. • User Specified — measures the defined range.

Duty Cycle

Displays a duty cycle of a periodic waveform, which is measured in one of the following ways:



The Duty Cycle is calculated as the percentage of:

$$\frac{t1}{t}$$

Note:

This measurement is supported only in Analog X-Y waveview panels.

The following parameters can be set for the Duty Cycle measurement:

Parameter	Description
Locked Level	Locks the measurement Y level at the specified percentage or level (in Y-axis units).
Target Signal	Uses the min/max Y levels from the signal you are currently measuring. This option is only available when specifying a measurement Y level that is locked at a specified percentage.
All Signals	Uses the min/max Y levels from all signals that are currently open. This option is only available when specifying a measurement Y level that is locked at a specified percentage.
User Specified	Uses the min/max Y values you specify in the Min and Max text fields.
Duty Cycle vs. X-axis Plot Option	Calculates the duty cycle value vs. the x-axis.
Range options	- Entire Waveform — (default) measures the entire waveform. - Visible Range — measures the visible range. - User Specified — measures the defined range.

F VS T

Calculates all frequency, period, or duty cycle values for the edges crossing the threshold, and then create new Frequency vs. time, Period vs. time, or Duty Cycle vs. time waveforms using the specified X range and Y level.

Note:

This measurement is supported only in Analog X-Y waveview panels.

The following parameters can be set for the F VS T measurement:

Parameter	Description
Locked Level	Locks the measurement Y level at the specified percentage or level (in Y-axis units).
Target Signal	Uses the min/max Y levels from the signal you are currently measuring. This option is only available when specifying a measurement Y level that is locked at a specified percentage.

Parameter	Description
All Signals	Uses the min/max Y levels from all signals that are currently open. This option is only available when specifying a measurement Y level that is locked at a specified percentage.
User Specified	Uses the min/max Y values you specify in the Min and Max text fields.
Frequency, Period, Delta Period, Duty Cycle	The type of measurement versus time. You can choose one or more types.
Entire Waveform, Visible Range, User Specified	Uses the entire waveform, the visible range, or your specified values for the F vs T measurement.
Delta-P vs T Plot	Plots the delta period versus time. This option is available by right-clicking your mouse and choosing Delta-P vs T Plot from the menu that opens.

Frequency

Displays the frequency and period of a periodic waveform. The period is calculated as the difference in time between two consecutive edges of the same polarity of a waveform. The frequency is calculated as:

$$\frac{1}{\text{period}}$$

You can also measure waveform frequency or parameter versus parameter or index values by right-clicking a Frequency measurement on a multi-member waveform and choosing one the following menu items from the menu that opens:

- **Freq vs Param**
- **Period vs Param**
- **Freq vs Index**
- **Period vs Index**

Note:

This measurement is supported only in Analog X-Y and Logic waveview panels.

The following parameters can be set for the Frequency measurement:

Parameter	Description
Locked Level	Locks the measurement Y level at the specified percentage or level (in Y-axis units).
Target Signal	Uses the min/max Y levels from the signal you are currently measuring. This option is only available when specifying a measurement Y level that is locked at a specified percentage.
All Signals	Uses the min/max Y levels from all signals that are currently open. This option is only available when specifying a measurement Y level that is locked at a specified percentage.
User Specified	Uses the min/max Y values you specify in the Min and Max text fields.
Frequency	Calculates the frequency versus X-axis plot frequency.
Period	Calculates the period versus X-axis plot period.
Range options	- Entire Waveform — (default) measures the entire waveform. - Visible Range — measures the visible range. - User Specified — measures the defined range.
Tolerance Level	Controls the tolerance level for excluding unwanted small rise-fall triggers around the specified ylevel. The tool will ignore the small rise-fall triggers within the specified tolerance range.
Trigger Type	- With Frequency vs x-axis option, controls the versus x-axis plot with Frequency or Period as: - Either — plot with EITHER - Rise — plot with UP - Fall — plot with DOWN.

Jitter

Displays the histogram of crossing edges with a specified X range or Y range. You must click the right mouse button menu on the measurement to complete the measurement.

Note:

This measurement is supported only in Analog X-Y, Eye Diagram, and 2-D Sweep waveview panels.

The following parameters can be set for the Jitter measurement:

Parameter	Description
Locked Level	Locks the measurement Y level at the specified percentage or level (in Y-axis units).
Target Signal	Uses the min/max Y levels from the signal you are currently measuring. This option is only available when specifying a measurement Y level that is locked at a specified percentage.
All Signals	Uses the min/max Y levels from all signals that are currently open. This option is only available when specifying a measurement Y level that is locked at a specified percentage.
User Specified	Uses the min/max Y values you specify in the Min and Max text fields.
Rising	Measures the rising edge jitter.
Falling	Measures the falling edge jitter.
H-Crossing	Measures edges that cross the horizontal range.
V-Crossing	Measures edges that cross the vertical range.
New WaveView	Plots the jitter histogram on a new waveview window.
Active WaveView	Plots the jitter histogram on the active waveview window.

Overshoot

Calculates the overshoot of a waveform relative to a default or specified topline. The overshoot is calculated as the difference between the maximum point on the waveform and the specified (or calculated) Topline value.

The following parameters can be set for the Overshoot measurement:

Parameter	Description
Percentage	Locks the rise/fall margin threshold at the specified high (H(%)) and low (L(%)) percentages. In signal Y units.
Signal Level	Locks the rise/fall margin threshold at the specified high (H(V)) and low (L(V)) signal levels. In signal Y units.
Target Signal	Uses the min/max Y levels from the signal you are currently measuring. This option is only available when specifying a rise/fall margin that is locked at a specified percentage range.

Parameter	Description
All Signals	Uses the min/max Y levels from all signals that are currently open. This option is only available when specifying a rise/fall margin that is locked at a specified percentage range.
User Specified	Uses the min/max Y values you specify in the Min and Max text fields.
Absolute	Specifies the result as "maximum value-high level value."
Percentage	Specifies the result as a percentage of "absolute value/(high level-low level)."
Range options	- Entire Waveform — (default) measures the entire waveform. - Visible Range — measures the visible range. - User Specified — measures the defined range.

Rise/Fall Time

Calculates the risetime and falltime between specified upper and lower levels of a waveform.

Note:

This measurement is only supported in Analog X-Y waveview panels.

The following parameters can be set for the Rise/Fall Time measurement:

Parameter	Description
Percentage	Locks the rise/fall margin threshold at the specified high (H(%)) and low (L(%)) percentages. In signal Y units.
Y Range Options For Percentage Lock Level	- Target Signal Min/Max— Uses the min/max Y levels from the signal you are currently measuring. This option is only available when specifying a rise/fall margin that is locked at a specified percentage range. - Target Signal Topline/Baseline — Uses the topline/baseline from the signal you are measuring. - All Signals — Uses the min/max Y levels from all signals that are currently open. This option is only available when specifying a rise/fall margin that is locked at a specified percentage range. - User-specified — Uses the min/max Y values you specify in the Min and Max text fields. - Each Edge Min/Max — Uses the local edge range to determine the low and high thresholds.
Within X Range	Specifies the X range from which the low and high thresholds are determined.

Parameter	Description
Show Range Levels	The range levels Min/Max or Topline/Baseline will be shown on the graph with the measurement results.
Trigger Type	Specifies trigger type as Rise, Fall, or Either.
Rise/Fall Time vs. X-axis Plot Option	Calculates the Rise, Fall, or Rise/Fall time versus the X-axis
X-Range options	- Entire Waveform — (default) measures the entire waveform. - Visible Range — measures the visible range. - User Specified — measures the defined range.

Settle Time

Calculates the settle time of a waveform with respect to a default or specified settle level and a specified settle band. The settle time is calculated by searching the waveform from right to left to find the first point that is outside of the settle band. The time that the waveform leaves the settle band is used as the settle time.

The following parameters can be set for the Settle Time measurement:

Parameter	Description
Percentage	Locks the settle level at the specified percentage level (Level(%)). In signal Y units.
Signal Level	Locks the settle level at the specified signal level (Level(V)). In signal Y units.
Target Signal	Uses the min/max Y levels from the signal you are currently measuring. This option is only available when specifying a settle level that is locked at a specified percentage range.
All Signals	Uses the min/max Y levels from all signals that are currently open. This option is only available when specifying a settle level that is locked at a specified percentage range.
User Specified	Uses the min/max Y values you specify in the Min and Max text fields.
% of SettleLevel	Calculates the band value as the specified band value percentage multiplied by the settle level (for example, 5*settle_level).
% of PeaktoPeak	Calculates the band value as the specified band value percentage multiplied by the peak to peak range of the signal (for example, 5*peak_to_peak_range).
Absolute	Calculates the band value as the specified value (in signal Y units).

Slew Rate

Calculates the slew rate of a waveform relative to the default or specified topline and baseline levels.

Note:

This measurement is only supported in Analog X-Y waveview panels.

The slew rate is calculated using the difference between the upper and lower levels of a waveform divided by the risetime or falltime of the edge. You can select the upper and lower levels as a percentage of the topline or baseline.

The following parameters can be set for the Slew Rate measurement:

Parameter	Description
Percentage	Locks the rise/fall margin threshold at the specified high (H(%)) and low (L(%)) percentages. In signal Y units.
Signal Level	Locks the rise/fall margin threshold at the specified high (H(V)) and low (L(V)) signal levels. In signal Y units.
Target Signal	Uses the min/max Y levels from the signal you are currently measuring. This option is only available when specifying a rise/fall margin that is locked at a specified percentage range.
All Signals	Uses the min/max Y levels from all signals that are currently open. This option is only available when specifying a rise/fall margin that is locked at a specified percentage range.
User Specified	Uses the min/max Y values you specify in the Min and Max text fields.
Slew Rate vs. X-axis Plot Option	Calculates the slew rate values of the Rise , Fall , or Rise/Fall edges vs. the x-axis.
Range options	- Entire Waveform — (default) measures the entire waveform. - Visible Range — measures the visible range. - User Specified — measures the defined range.

Undershoot

Calculates the undershoot of a waveform relative to a default or specified baseline level. The undershoot is calculated as the difference between the minimum point on the waveform and the specified (or calculated) Baseline value.

The following parameters can be set for the Undershoot measurement:

Parameter	Description
Percentage	Locks the rise/fall margin threshold at the specified high (H(%)) and low (L(%)) percentages. In signal Y units.
Signal Level	Locks the rise/fall margin threshold at the specified high (H(V)) and low (L(V)) signal levels. In signal Y units.
Target Signal	Uses the min/max Y levels from the signal you are currently measuring. This option is only available when specifying a rise/fall margin that is locked at a specified percentage range.
All Signals	Uses the min/max Y levels from all signals that are currently open. This option is only available when specifying a rise/fall margin that is locked at a specified percentage range.
User Specified	Uses the min/max Y values you specify in the Min and Max text fields.
Absolute	Specifies the result as "maximum value-high level value."
Percentage	Specifies the result as a percentage of "absolute value/(high level-low level)."
Range options	• Entire Waveform — (default) measures the entire waveform. • Visible Range — measures the visible range. • User Specified — measures the defined range.

Frequency Domain Measurements

The following Frequency Domain measurements are available:

- [Highpass](#)
- [Lowpass](#)
- [Bandpass](#)
- [Stopband](#)
- [Gain Margin](#)
- [Phase Margin](#)

Highpass

Calculates the corner frequency of a waveform with a highpass shape. The highpass measurement is calculated relative to a default offset or a specified topline and a specified offset.

The corner frequency is calculated by searching from the right to the left until the waveform first falls below the measurement level, which is determined by the specified offset value (from the topline).

The following parameters can be set for the Highpass measurement:

Parameter	Description
Percentage	Locks the topline at the specified percentage (Topline(%)). In signal Y units.
Signal Level	Locks the topline at the specified signal level (Topline). In signal Y units.
Auto Detect	Calculates the topline and uses that value.
Target Signal	Uses the min/max Y levels from the signal you are currently measuring. This option is only available when specifying a topline that is locked at a specified percentage range.
All Signals	Uses the min/max Y levels from all signals that are currently open. This option is only available when specifying a topline that is locked at a specified percentage range.
User Specified	Uses the min/max Y values you specify in the Min and Max text fields.
Offset	Specifies the offset in Y-axis units.
Show Topline	Displays the topline.
Show Topline+Offset Line	Shows the cross line in a panel.

Lowpass

Calculates the corner frequency of a waveform with a lowpass shape. The lowpass measurement is calculated relative to a default or specified topline and a specified offset.

The corner frequency is calculated by searching from the left to the right until the waveform first falls below the measurement level, which is determined by the specified offset value (from the topline).

The following parameters can be set for the Lowpass measurement:

Parameter	Description
Percentage	Locks the topline at the specified percentage (Topline(%)). In signal Y units.

Parameter	Description
Signal Level	Locks the topline at the specified signal level (Topline). In signal Y units.
Auto Detect	Calculates the topline and uses that value.
Target Signal	Uses the min/max Y levels from the signal you are currently measuring. This option is only available when specifying a topline that is locked at a specified percentage range.
All Signals	Uses the min/max Y levels from all signals that are currently open. This option is only available when specifying a topline that is locked at a specified percentage range.
User Specified	Uses the min/max Y values you specify in the Min and Max text fields.
Offset	Specifies the offset in Y-axis units.
Show Topline	Displays the topline.
Show Topline+Offset Line	Shows the cross line in a panel.

Bandpass

Calculates the bandwidth of a bandpass-shaped waveform.

The following parameters can be set for the Bandpass measurement:

Parameter	Description
Percentage	Locks the topline at the specified percentage (Topline(%)). In signal Y units.
Signal Level	Locks the topline at the specified signal level (Topline). In signal Y units.
Auto Detect	Calculates the topline and uses that value.
Target Signal	Uses the min/max Y levels from the signal you are currently measuring. This option is only available when specifying a topline that is locked at a specified percentage range.
All Signals	Uses the min/max Y levels from all signals that are currently open. This option is only available when specifying a topline that is locked at a specified percentage range.
User Specified	Uses the min/max Y values you specify in the Min and Max text fields.

Parameter	Description
Offset	Specifies the offset in Y-axis units.
Show Topline	Displays the topline.
Show Topline+Offset Line	Shows the cross line in a panel.

Stopband

Calculates the stopband, low, high, or center frequency level or the level at which the measurement is calculated for a stopband-shaped waveform. This measurement is calculated relative to a default or specified topline level and a specified offset:

$$f_{\text{center}} = \text{square root of } (f_{\text{low}} * f_{\text{high}})$$

$$\text{quality factor} = f_{\text{center}} / \text{bandwidth}$$

The following parameters can be set for the Stopband measurement:

Parameter	Description
Percentage	Locks the topline at the specified percentage (Topline(%)). In signal Y units.
Signal Level	Locks the topline at the specified signal level (Topline). In signal Y units.
Auto Detect	Calculates the topline and uses that value.
Target Signal	Uses the min/max Y levels from the signal you are currently measuring. This option is only available when specifying a topline that is locked at a specified percentage range.
All Signals	Uses the min/max Y levels from all signals that are currently open. This option is only available when specifying a topline that is locked at a specified percentage range.
User Specified	Uses the min/max Y values you specify in the Min and Max text fields.
Offset	Specifies the offset in Y-axis units.
Show Topline	Displays the topline.
Show Topline+Offset Line	Shows the cross line in a panel.

Gain Margin

Displays the gain margin of a complex analog waveform in decibels (dB). The gain margin is the difference between the gain of the measured waveform and 0 dB at the frequency where the phase shift is -180-degrees.

This measurement might fail if the phase of the measured waveform does not pass through -180-degrees or if the waveform is not complex.

Phase Margin

Displays the phase margin of a complex analog waveform in degrees. The phase margin is the difference between the phase of the measured waveform and the reference degree at the unity gain frequency. The reference can be either -180-degrees or 0-degree.

This measurement might fail if the magnitude of the waveform does not pass through 0 dB or if the waveform is not complex.

The following parameters can be set for the Phase Margin measurement:

Parameter	Description
-180	Specifies the reference as -180 degrees.
0	Specifies the reference as 0 degrees.

Statistical Measurements

The following Statistical measurements are available:

- [Cpk](#)
- [Dpu](#)
- [Mean](#)
- [Mean+3std_dev](#)
- [Mean-3std_dev](#)
- [Std_dev](#)
- [Yield](#)

Cpk

Calculates an indicator of the process capability for a waveform relative to specified upper and lower limits. Cpk is calculated using the following equation:

$$\frac{\text{upper-mean}}{3(\text{std_dev})}$$

or

$$\frac{\text{mean-lower}}{3(\text{std_dev})}$$

In this calculation, `mean` represents the mean value of the scatter plot, `upper` and `lower` represent the specification limits you specify, and `std_dev` represents the standard deviations of the scatter plot. When both the `upper-mean` and `lower-mean` values are provided, the smaller result of these two calculations is displayed as the measurement.

The following parameters can be set for the Cpk measurement:

Parameter	Description
Percentage	Locks the upper/lower threshold at the specified high (H(%)) and low (L(%)) percentages. In signal Y units.
Signal Level	Locks the upper/lower threshold at the specified high (H(V)) and low (L(V)) signal levels. In signal Y units.
Target Signal	Uses the min/max Y levels from the signal you are currently measuring. This option is only available when specifying an upper/lower threshold that is locked at a specified percentage range.
All Signals	Uses the min/max Y levels from all signals that are currently open. This option is only available when specifying an upper/lower threshold that is locked at a specified percentage range.
User Specified	Uses the min/max Y values you specify in the Min and Max text fields.

Dpu

Calculates the total number of defects per unit based on the points in a waveform. Given one or both of an upper and lower specification limit, the Dpu is calculated as the area under a normal distribution that falls outside the specification limits. The mean and standard deviation of the normal distribution are equal to the mean and standard deviation of the points in the waveform.

The following parameters can be set for the Dpu measurement:

Parameter	Description
Percentage	Locks the upper/lower threshold at the specified high (H(%)) and low (L(%)) percentages. In signal Y units.
Signal Level	Locks the upper/lower threshold at the specified high (H(V)) and low (L(V)) signal levels. In signal Y units.
Target Signal	Uses the min/max Y levels from the signal you are currently measuring. This option is only available when specifying an upper/lower threshold that is locked at a specified percentage range.
All Signals	Uses the min/max Y levels from all signals that are currently open. This option is only available when specifying an upper/lower threshold that is locked at a specified percentage range.
User Specified	Uses the min/max Y values you specify in the Min and Max text fields.

Mean

Calculates the mean value of a waveform, which is calculated using the following equation:

$$\frac{1}{N} \sum_{j=1}^N W_j$$

In this calculation, N is the number of points, and array W_j contains the individual points of the waveform.

Mean+3std_dev

Calculates the (mean + 3 standard deviation) value of a waveform.

Mean-3std_dev

Calculates the (mean - 3 standard deviation) value of a waveform.

Std_dev

Calculates the standard deviation of a waveform. This measurement is intended for statistical (discrete) data such as histograms and is calculated using the following equation:

$$\left[\frac{1}{N-1} \sum_{j=1}^N (W_j - \bar{W})^2 \right]^{\frac{1}{2}}$$

Yield

Calculates the ratio of the number of data points between the y-axis levels Upper and Lower relative to the total number of data points.

The following parameters can be set for the Yield measurement:

Parameter	Description
Percentage	Locks the upper/lower threshold at the specified high (H(%)) and low (L(%)) percentages. In signal Y units.
Signal Level	Locks the upper/lower threshold at the specified high (H(V)) and low (L(V)) signal levels. In signal Y units.
Target Signal	Uses the min/max Y levels from the signal you are currently measuring. This option is only available when specifying an upper/lower threshold that is locked at a specified percentage range.
All Signals	Uses the min/max Y levels from all signals that are currently open. This option is only available when specifying an upper/lower threshold that is locked at a specified percentage range.
User Specified	Uses the min/max Y values you specify in the Min and Max text fields.

Level Measurements

Each Level measurement can be performed over one of the following ranges:

- Entire Waveform — (default) measures the entire waveform.
- Visible Range — measures the visible range.
- User Specified — measures the specified range.

The following Level measurements are available:

- [AC Coupled RMS](#)
- [Amplitude](#)
- [Average](#)
- [Baseline](#)
- [Maximum](#)
- [Minimum](#)
- [Peak to Peak](#)

- [RMS](#)
- [Topline](#)
- [X at Maximum](#)
- [X at Minimum](#)

AC Coupled RMS

Calculates the RMS value of the AC component of a waveform. The AC coupled RMS value is calculated using the following equation:

$$\left[\frac{1}{(x_2 - x_1)} \int_{x_1}^{x_2} (w - \bar{w})^2 dx \right]^{\frac{1}{2}}$$

In this calculation, w represents the waveform as well as its average value. The x_1 and x_2 values are the starting and ending points for the waveform. Add WV as an option for the AC Coupled measurement.

Amplitude

Calculates the difference between the topline and baseline reference levels.

Average

Calculates the average level of a waveform.

Baseline

Calculates the baseline level of a waveform.

Note:

This measurement is automatically calculated when enabled.

Maximum

Calculates the maximum level of a waveform.

Minimum

Calculates the minimum level of a waveform.

Peak to Peak

Calculates the peak to peak value of a waveform.

RMS

Calculates the RMS (Root Mean Square) value of a waveform using the following equation:

$$\left[\frac{1}{(x2-x1)} \int_{x1}^{x2} (W^2 dx) \right]^{\frac{1}{2}}$$

In this calculation, W represents the waveform, and $x1$ and $x2$ represent the starting and ending points.

Topline

Calculates the topline level of a waveform. The topline is automatically calculated using the probability density histogram method.

X at Maximum

Displays the x-value corresponding to the maximum value of a waveform.

X at Minimum

Displays the x-value at the minimum value of a waveform.

S Domain Measurements

The following S Domain measurements are available:

- [Damping Ratio](#)
- [Frequency Value](#)
- [Natural Frequency](#)
- [Quality Factor](#)

Damping Ratio

Calculates the damping ratio of a waveform as $-\text{real}/\text{mag}$.

Frequency Value

Displays the frequency value at the point of a complex waveform on a Nyquist (or Nichols) plot.

Natural Frequency

Calculates the natural frequency of a waveform as the absolute value of an argument. The natural frequency is calculated using the following equation:

$$\text{natural frequency} = \sqrt{(\text{real}^2 + \text{imag}^2)}$$

Quality Factor

Calculates the quality factor of a waveform. This measurement is calculated as:

$$\frac{1}{2(\text{damping ratio})}$$

RF Measurements

The following RF measurements are available:

- [P1dB](#)
- [IP2](#)
- [IP3/SFDR](#)

P1dB

Calculates the 1 dB compression point.

IP2

Calculates the second-order intercept point.

IP3/SFDR

Calculates the third order intercept point.

The following parameters can be set for the IP3/SFDR measurement:

Parameter	Description
Calculate SFDR	Specifies that IP3 is calculated as well as SFDR.
Use INTEGRATED_NF	Specifies the Integrated Noise Figure to be used for the SFDR calculation.
Noise Figure (dB)	Specifies the Noise Figure in decibels (dB).
SN ratio (dB)	Specifies the signal-to-noise ratio in decibels (dB).

Parameter	Description
Band width (dB)	Specifies the system Bandwidth in decibels (dB). This parameter is not available when "Use INTEGRATED_NF" is selected.
Integrated NF (dB)	Specifies the Integrated Noise Figure in decibels (dB). This parameter is not available when "Use INTEGRATED_NF" is selected.

Measuring Eye Diagrams

The **Automatic Eye Measurement** tool can take eye opening or aperture measurement on an eye diagram. Right-click an eye diagram, and choose **Measure Eye** from the menu. The **Automatic Eye Measurement** window opens.

Automatic Eye Measurement (on ddcaeh517)

Measure Type

☐ Eye Size
 ☐ Aperture
 ☐ DDR Read
 ☐ DDR Write
 ☐ Overshoot
 ☐ Undershoot
 ☐ Mask
☐ Diamond Opening
☐ Rect. Opening
☐ Hex. Opening
☐ DDR4
☐ DDR5
☐ GDDR5
☒ PAM-N

Input Parameters | **Measure Options** | **Measure Diagram**

Number of Levels:
☒ Eye Open Measurements
☐ Level Measurements
 Amplitude Levels(VA,VB,VC,...): ☒ Even Spaced
☐ Compressed

Timing Reference

☐ Based on Center Eye
☐ Each Eye Independently
☒ Specify References
☒ VREF
☐ X Value
 Eye Center Location
☒ Maximum Eye Width
☐ Maximum Eye Height
☐ Calculate DVDT Min/Max/Avg: Offset from Mid-Point:

Level Measurements Reference

Locate the time at:
☒ Eye Center
☐ Minimum RMS
☒ Time Window from Eye Center: Percent of UI(%)=

Measure Display

Eye Open:
☐ Mid-Point
☐ Skew
☒ Height
☒ Width
 Level:
☒ Mean
☐ Peak to Peak
☐ RMS
☐ Skew

Measure Result

Eye Open Measurements:

	Btm Eye	Mid Eye	Top Eye
Mid-Point			
Skew			
Height			
Width			

Eye Level Measurements:

	Level 0	Level 1	Level 2	Level
Mean				
P2P				
RMS				

☐ Apply to Selected Eye Panels

The following sections contain information on the measurements available in the Automatic Eye Measurement window:

- [Measuring Eye Size](#)
- [Measuring Aperture](#)
- [Measuring DDR Read](#)
- [Measuring DDR Write](#)
- [Measuring Overshoot](#)
- [Measuring Diamond Openings](#)

- [Measuring Rectangle Openings](#)
- [Measuring Hexagon Openings](#)
- [Measuring DDR4](#)
- [Measuring DDR5](#)
- [Measuring GDDR5](#)
- [Measuring PAM-N](#)
- [Measuring Mask](#)
- [Measuring Stateyes](#)
- [Using the Setup/Hold Results Viewer](#)
- [Creating ACE Scripts From DDR Measurements](#)
- [Clearing Eye Measurements](#)
- [Copying Eye Diagram Settings](#)
- [Pasting Eye Diagram Settings](#)
- [Measuring DDR3 Eye Diagrams](#)
- [Measuring DDR4 Eye Diagrams](#)

The same eye measurement functions are also available as ACE functions for batch mode measurements. Refer to the *Analysis Command Environment Reference (ACE) Reference Manual* for details.

Measuring Eye Size

The following eye size measurement options are available:

Option	Description
VREF, AC, and DC Values	The VREF, AC, and DC values use for the eye measurement.
Use DVDT Levels	Defines the lower and upper voltage levels for the DVDT measurement.

Option	Description
Use Time Window from Eye Center	Reports and plots measurement results in the Eye Size measurement naming convention using the specified percentage value, where: • FUSS is TJ_TX • OH is VDIF_AC_TX(max,upper) • MH is VDIF_AC_TX(min,upper) • OL is VDIF_AC_TX(max,lower) • ML is VDIF_AC_TX(min,lower)
Large Font	Sets the font size to large.
Time Unit	Sets the time unit to Auto , ns , or ps .
Apply to Selected Eye Panels	Applies measurements to all selected eye panels.

Eye Measurement Methodology

The permanent default value for **vref**, **vdc**, and **vac** is 0.0. The **vref**, **vdc**, and **vac** values calculate the **MH**, **ML**, **OH**, **OL**, **WH**, and **WL** without checking which one is specified.

You can calculate **MH** and **ML** using the following method:

$MH = (\text{lower limit of the upper eye}) - (vref + vdc)$

$ML = (vref - vdc) - (\text{upper limit of the lower eye})$

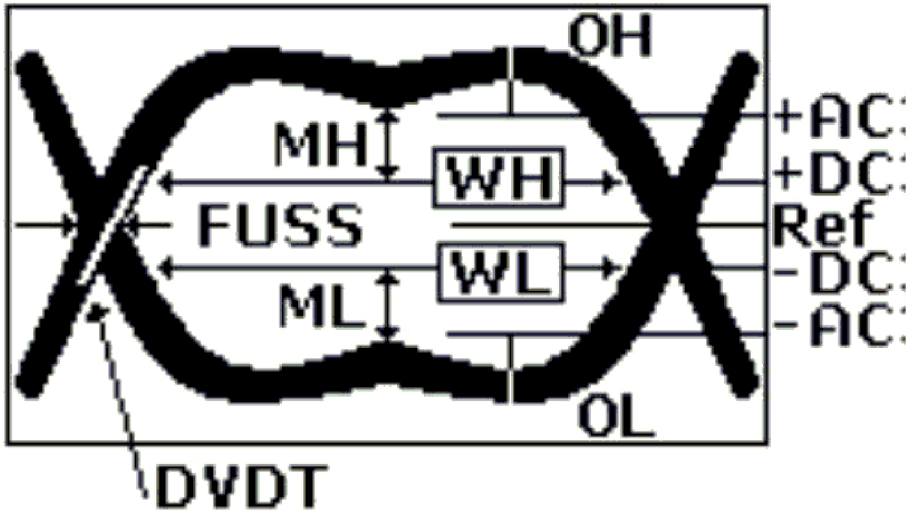
You can calculate **OH** and **OL** using the following method:

$OH = (\text{upper limit of the upper eye}) - (vref + vac)$

$OL = (vref - vac) - (\text{lower limit of the lower eye})$

WH is the eye opening width at level $(vref + vdc)$. **WL** is the eye opening width at level $(vref - vdc)$.

The following figure shows how the expressions mentioned are used to calculate eye diagram measurements:



Measuring Aperture

The following eye aperture measurement options are available:

Option	Description
Use Wider Aperture for VDC-Ringback Divided Eye	Evaluates the width for two openings on two sides of a glitch and uses the wider one as the measurement result. If not enabled, the aperture measurement always starts after any VDC crossing ringback.
Use 2-Box Aperture with no VAC Crossing Before	Sets the value you enter in the Tminac as the time for the width of the first box (the left box). The first box is free of VAC glitches, while the 2nd box (right box) is free of VDC glitches.
Use First Signal as T-Budget Setup/Hold Reference	Sets the first signal in the panel as the reference signal for setup/hold measurement.
VREF, AC, and DC Values	The VREF, AC, and DC values used for the eye measurement. You can enter multiple values for VREF by separating them with colons (:), semicolons (;), or commas (,).
Trigger Clock Delay	Sets the trigger clock delay value you enter in the Trigger Clock Delay text box. This option is disabled if you enable the Delay Clock to Aperture Center option.
Delay Clock to Aperture Center	Sets the trigger clock delay to the aperture center value.

Option	Description
Min/Max Display	Displays the min and max delay from the eye.
Skew	Displays the skew measurement on the eye.
Aperture	Displays the aperture.
Aperture Violation	Displays aperture violations.
Jitter	Displays the point-to-point and RMS jitter as a histogram, which is plotted next to the Vref crossing area.
Timing Budget	Displays the setup and hold measurements.
Guide Grid	Displays the guide grid.
Large Font	Sets the font size to large.
Aper	Sets the line size and color for the aperture.
Time Unit	Sets the time unit to Auto , ns , or ps .
Apply to Selected Eye Panels	Applies measurements to all selected eye panels.

Measuring DDR Read

The following DDR read measurement options are available:

Option	Description
VREF, AC, and DC Values	The VREF, AC, and DC values used for the DDR read measurement.
Guide Grid	Displays the guide grid.
Large Font	Sets the font size to large.
Time Unit	Sets the time unit to Auto , ns , or ps .
Apply to Selected Eye Panels	Applies measurements to all selected eye panels.

Measuring DDR Write

The following DDR write measurement options are available:

Option	Description
VREF, AC, and DC values	The VREF, AC, and DC values used for the DDR write measurement.
Guide Grid	Displays the guide grid.
Large Font	Sets the font size to large.
Time Unit	Sets the time unit to Auto , ns , or ps .
Apply to Selected Eye Panels	Applies measurements to all selected eye panels.

Measuring Overshoot

The following overshoot measurement options are available:

Option	Description
VREF value	The VREF value used for the overshoot measurement.
Large Font	Sets the font size to large.
Time Unit	Sets the time unit to Auto , ns , or ps .
Apply to Selected Eye Panels	Applies measurements to all selected eye panels.

Measuring Diamond Openings

The following diamond opening measurement options are available:

Option	Description
Accuracy(%)	The accuracy percentage of the diamond opening measurement.
Large Font	Sets the font size to large.
Time Unit	Sets the time unit to Auto , ns , or ps .
Apply to Selected Eye Panels	Applies measurements to all selected eye panels.

Measuring Rectangle Openings

The following rectangle opening measurement options are available:

Option	Description
VREF Value	The VREF value used for the rectangle opening measurement.
Large Font	Sets the font size to large.
Time Unit	Sets the time unit to Auto , ns , or ps .
Apply to Selected Eye Panels	Applies measurements to all selected eye panels.

Measuring Hexagon Openings

The following hexagon opening measurement options are available:

Option	Description
VREF Value	The VREF value used for the hexagon opening measurement.
Large Font	Sets the font size to large.
Time Unit	Sets the time unit to Auto , ns , or ps .
Apply to Selected Eye Panels	Applies measurements to all selected eye panels.

Measuring DDR4

The following DDR4 measurement options are available:

Option	Description
Input Parameters	
VREF	The VREF value used for the DDR4 measurement.
AC	The AC value used for the DDR4 measurement.
DC	Sets the DC threshold for eye diagram measurements.
VIHL_AC(min)=	Select to set the VIHL_AC(min) value for calculating tr2 measurements. If a VIHL_AC(min) value is not specified, by default the WaveView tool uses the calculated VIHL_AC value.

Option	Description
Clock Slew	Select to set the clock slew VIH and VIL values.
Enable	Find the Vcent from VRef sweeping using the Start , Stop , and Step values you enter. The VREF value is not used when you enable this option.
Align Bits By	Aligns all bits by Aperture or Eye Open at Vcent .
Report Margins vs. Mask	Measures the margins versus the mask defined by the <code>CX_DRTAB_FILE</code> environment variable.
DDR Spec	The a DDR specification from the drop-down to use for the measurement.
Detailed Results button	See Using the Setup/Hold Results Viewer for information on the Setup/Hold Results Viewer, which you can access via the Detailed Results button.
Measure Options	
Trigger Clock Delay	Sets the trigger clock delay value you enter in the Trigger Clock Delay text box. This option is disabled if you enable the Delay Clock to Aperture Center option.
Delay Clock to Aperture Center	Select to set the trigger clock delay to the aperture center value.
Measure Display	
Skew	This option is enabled by default and cannot be disabled.
Aperture	Displays the aperture.
Guide Grid	Displays the guide grid.
Large Font	Sets the font size to large.
Aper	Sets the line size and color for the aperture.
Time Unit	Sets the time unit to Auto , ns , or ps .
VIHL_AC	Displays vertical bar and value.
Apply to Selected Eye Panels	Applies measurements to all selected eye panels.

Measuring DDR5

The following DDR5 measurement options are available:

Option	Description
Input Parameters	
VREF Value	The VREF value used for the DDR5 measurement.
AC Value	The AC value used for the DDR5 measurement.
Enable	Find the Vcent from VRef sweeping using the Start , Stop , and Step values you enter. The VREF value is not used when you enable this option.
Align Bits By	Aligns all bits by Aperture or Eye Open at Vcent .
Calculate Mask	Calculates mask using the TdIVW1 , TdIVW2 , and VdIVW values you enter.
Display Jitter on New Panel	Displays jitter on a new panel.
Measure Options	
Trigger Clock Delay	Sets the trigger clock delay value you enter in the Trigger Clock Delay text box. This option is disabled if you enable the Delay Clock to Aperture Center option.
Delay Clock to Aperture Center	Sets the trigger clock delay to the aperture center value.
Measure Display	
Min/Max delay	Displays the min and max delay from the eye.
Aperture	Displays the aperture.
Jitter	Displays the jitter.
Guide Grid	Displays the guide grid.
Large Font	Sets the font size to large.
Aper	Sets the line size and color for the aperture.
Time Unit	Sets the time unit to Auto , ns , or ps .
VIHL_AC	Displays vertical bar and value.
Hex. Mask Spec	Displays the hexagonal spec mask.

Option	Description
Max Hex. Mask	Displays the max hex spec mask.
Rect. Mask Spec	Displays the rectangular spec mask.
Max Rect. Mask	Displays the max rectangular spec mask.
Apply to Selected Eye Panels	Applies measurements to all selected eye panels.

Measuring GDDR5

The following GDDR5 measurement options are available:

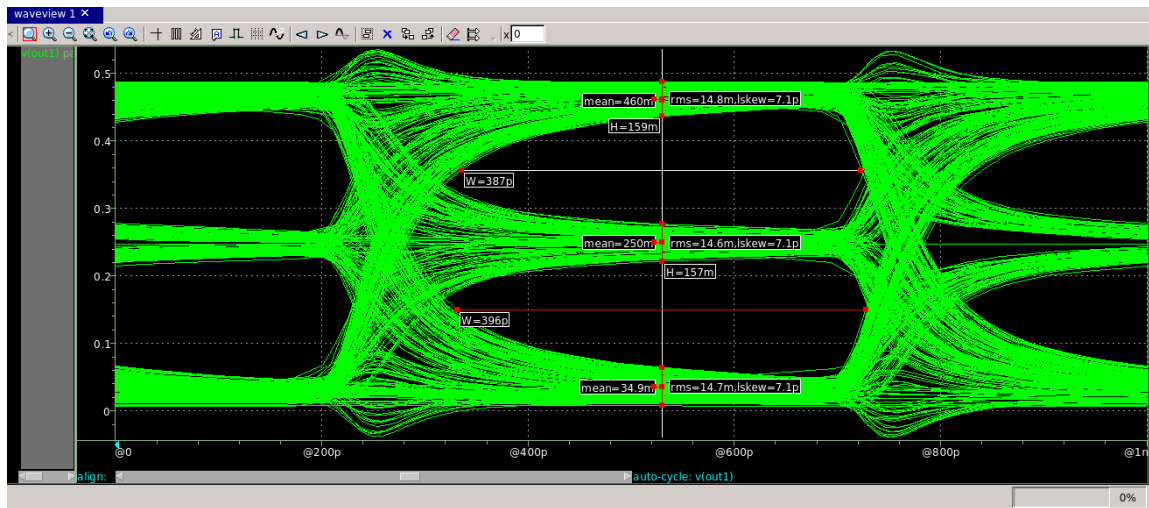
Option	Description
VREF, DC, and AC Values	The VREF, DC, and AC values used for the GDDR5 measurement.
Detailed Results button	See Using the Setup/Hold Results Viewer for information on the Setup/Hold Results Viewer, which you can access via the Detailed Results button.
Trigger Clock Delay	Sets the trigger clock delay value you enter in the Trigger Clock Delay text box. This option is disabled if you enable the Delay Clock to Aperture Center option.
Aperture	Displays the aperture.
Jitter	Displays the jitter.
Guide Grid	Displays the guide grid.
Large Font	Sets the font size to large.
Aper	Sets the line size and color for the aperture.
Time Unit	Sets the time unit to Auto , ns , or ps .
Apply to Selected Eye Panels	Applies measurements to all selected eye panels.

Measuring PAM-N

Select the PAM-N eye measurement in the **Measure Type** section of the **Automatic Eye Measurement** dialog of the GUI.

The PAM-N measurement measures a pulse amplitude modulated signal with N amplitude levels (0, 1, 2, ..., n-1). Currently, PAM3, PAM4, and PAM6 measurements are supported. PAM3 measurement measures a pulse amplitude modulated signal with three amplitude levels (0, 1, 2) and two eye opens— Bottom Eye and Top Eye. PAM4 measurement measures a pulse amplitude modulated signal with four amplitude levels (0, 1, 2, 3) and three eye opens— Bottom Eye, Mid Eye, and Top Eye. PAM6 measurement measures a pulse amplitude modulated signal with six amplitude levels (0, 1, 2, 3, 4, 5) and five eye opens— Eye 0, Eye 1, Eye 2, Eye 3, and Eye 4.

The following figure shows an example for PAM3 measurement:



The options available in the **Automatic Eye Measurement** dialog are as follows:

Input Parameters:

Three types of measurements are supported for PAM-N:

- **Number of Levels** — specify the number of amplitude levels of the measured signal. Currently, measuring signals with 3, 4, and 6 levels are supported.
- **Eye Open Measurements** — for slicer level based receiver, eye height/width are important:
 - Mid-point — time and voltage level of the eye center of each eye
 - Skew — time differences between the individual eye centers and eye center (at maximum height) of the center eye
 - Height — vertical opening of each eye open at certain y level, x value, or eye center
 - Width — horizontal opening of each eye open at certain y level, x value or eye center

- **Eye Level Measurements** — for ADC based receiver, level variation in constellation form is most important
 - Mean — amplitudes at specified x value, eye center or minimum thickness for each level
 - Thickness:
 - Peak-to-peak — Peak-to-peak values at specified x value, eye center, or minimum RMS for each level
 - RMS — RMS values at specified x value, eye center, or minimum RMS from each level
 - Skew — time differences at eye center or minimum thickness to the eye center of the center eye for each level
 - RLM — Level separation mismatch ratio
- **Amplitude Levels (VA,VB,VC,VD):**
 - Even Spaced — voltage amplitude levels are even spaced. By default, amplitude levels are automatically detected assuming the levels are even spaced.
 - Compressed — voltage amplitude for various levels are too high or too low. In this case, you must specify the amplitude levels.

Timing Reference — reference values/measurements are based on:

- Based on Center Eye — time reference for all eye opens is based on eye center of center eye
- Each Eye Independently — time reference for each eye open is based on eye center of each eye itself
- Specify References — time reference for eye open is based on VREF specified. If required, Select X value to specify
- Calculate DVDT Min/Max/Avg — select to automatically measure the DVDT values for minimum, maximum, and average slew rates of PAM4 eyes. Default- Unselected
- Offset from Mid-Point — specify the offset from mid-point to calculate DVDT values of PAM4 eyes.

Eye Center Location — determine the center of each eye:

- Maximum Eye Width — center of the eye is at the mid point of maximum width or width at specified y level
- Maximum Eye Height — center of the eye is at the mid point of maximum height or height at specified x value

Level Measurements Reference — determine the time/x value for level measurements:

- **Eye Center** — time of the level is the average of the centers of the adjacent eyes
- **Minimum RMS** — time of the level is at the minimum level RMS
- **Time window** used to determine the mean of the level and minimum RMS, percentage of UI

Measure Results — results tables for eye open/level measurements, with RLM result shown on the top of the Eye Level Measurements results table. This table extends by 9 rows for Open Eye Measurement if DVDT is calculated.

Apply to Selected Eye Panels — applies the measurement to all selected eye panels.

Measuring Mask

Select the **Mask** eye measurement in the **Measure Type** section of the **Automatic Eye Measurement** dialog of the GUI.

The **Mask** measurement measures the best VREF for the specified mask where the largest vertical margin is found. The measurement returns the best VREF, the largest Vertical Margin, the UI time value, and the voltage value on the eye diagram where the largest Vertical Margin is found.

The following mask measurement options are available:

Option	Description
VREF	The VREF value to place the mask.
Enable	Find the best VREF from vref sweeping using the Start, Stop, and Step values. The VREF setting is not used when you enable this option.
Accuracy(% of UI)	The percentage value used to find the largest Vertical Margin at a certain VREF level.
TdIVW1	Mask specification, hexagon mask center width.
TdIVW2	Mask specification, hexagon mask top/bottom width.

Measuring Stateyes

The following stateye measurement options are available:

Option Category	Option	Description
Bathtub/Jitter Curve	BathtubV/JitterV VREF	When enabled, uses the BathtubV/JitterV VREF value you specify and plots the BathtubV curve.
	BathtubT/JitterT TREF	When enabled, uses the BathtubT/JitterT TREF value you specify.
	Show BathtubT/JitterT next to BER on the right	When enabled, displays the BathtubT/JitterT measurement results to the right of the BER plot.
BER Contour	BER Levels	Performs a BER contour plot based on the BER levels you specify. Note: BERC contour can be displayed on the top of StatEye eye diagram without changing the color grade. Therefore, making the BERC contour visible.
	Show BER Contour on	Displays the BER contour on a BER Eye or in a New Panel .
DDR4/DDR5	VREF and AC	The VREF and AC DDR4 values.
	Find Vcent from VRef Sweeping	When you click the Enable check box, a Vcent measurement from a VRef sweep is performed based on the Start , Stop , and Step values you specify.
	VIHL_AC	Displays the VIHL_AC value in the measurement results, which is the biggest vertical open of the eye. The VIHL_AC measurement scans the eye opening of the average Y value from the left to the right with an accuracy of 1% to find the biggest vertical opening of an eye.
	TdiVW/TclVW_TOTAL	Displays the TdiVW/TclVW_TOTAL value in the measurement results, which is the biggest horizontal opening of an eye. The TdiVW/TclVW_TOTAL measurement scans the eye opening of the average Y value from the top to the bottom with an accuracy of 1% to find the biggest horizontal opening of an eye.

Option Category	Option	Description
	TdiPW/TclPW	Displays the TdiPW/TclPW value in the measurement results, which is the aperture at Vref (where AC=0). You can specify the Vref value directly or use a Vref sweep (in which case Vref is also Vcent).
	Signal Offset	Displays the signal offset value in the measurement results.
	Margins vs Mask	Calculate Mask: The input mask configuration for diamond and hexagonal eye masks. Use Mask Spec File: You can load the mask specification file here. This is the same mask file used for a regular eye. Refer to Sample DDR4 Specification File. Set Mask VREF: This sets the vertical center of the mask at a certain VREF level. Asymmetrical Diamond Mask: Click the check box when using an asymmetrical diamond mask. For more information, refer to Margins vs Mask Measurements.
Jitter	Jitter(P2)	Measures the peak-to-peak jitter.
	Jitter(RMS)	Measures the root mean square jitter.

Margins vs Mask Measurements

Margins vs Mask

☒ Calculate Mask TdiVW1: 40p TdiVW2: 0 VdiVW: 40m

☐ Use Mask Spec File: None ☐ Set Mask VREF: 0

☐ Asymmetrical Diamond Mask

The following measurements are done in the **Margins vs Mask** section:

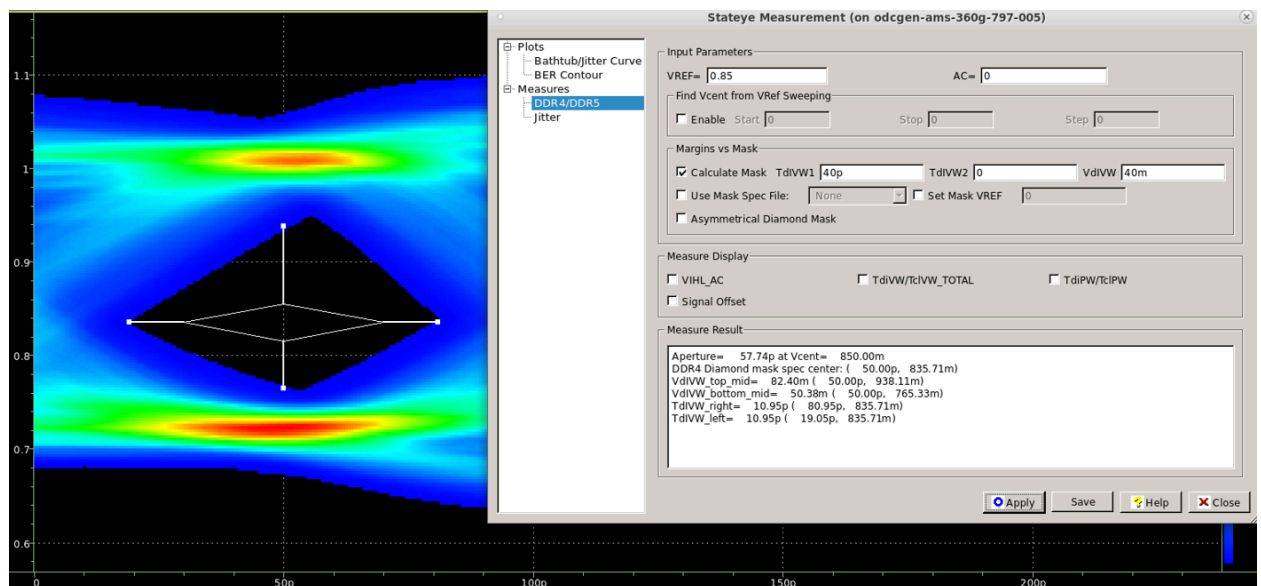
- Margins vs DQ and CA Diamond Mask
- Margins vs DQ Hexagonal Eye Mask

Margins vs DQ and CA Diamond Mask

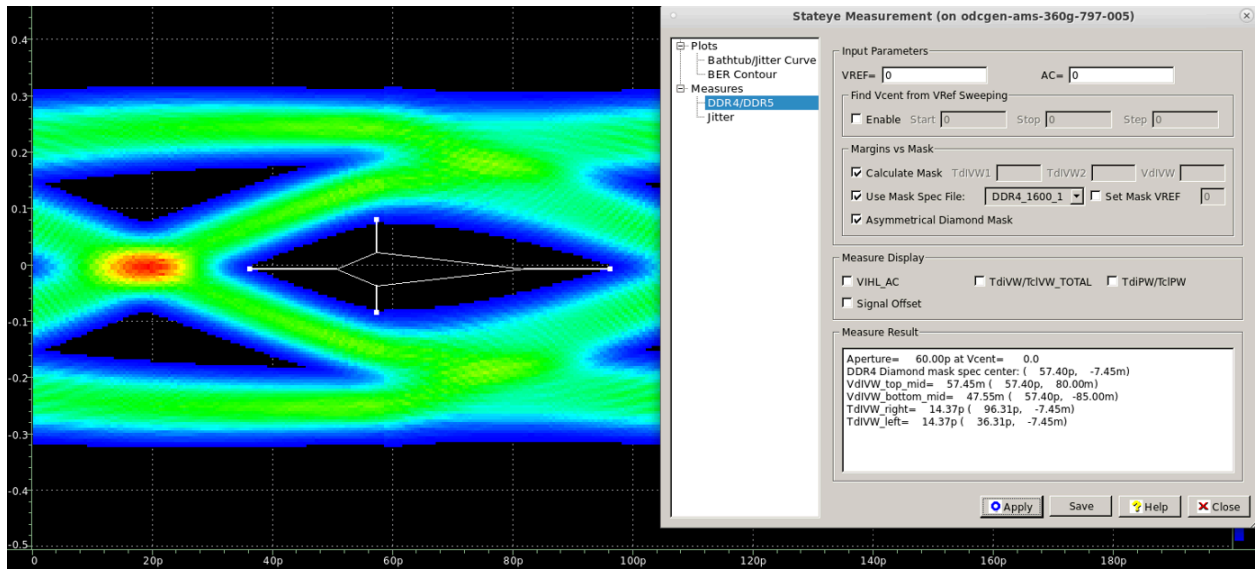
This measurement is done in the following manner:

- Y center is the VREF level of the maximum horizontal eye open, or the specified VREF level (using **Set Mask VREF**).
- X center is:
 - The mid-point of the width at the VREF level for a symmetrical diamond mask.
 - The x location at the maximum vertical eye opening for an asymmetrical diamond mask.

The input mask configuration is as shown in the following image.



You can load a specification file and use asymmetrical mask as shown in the following image.

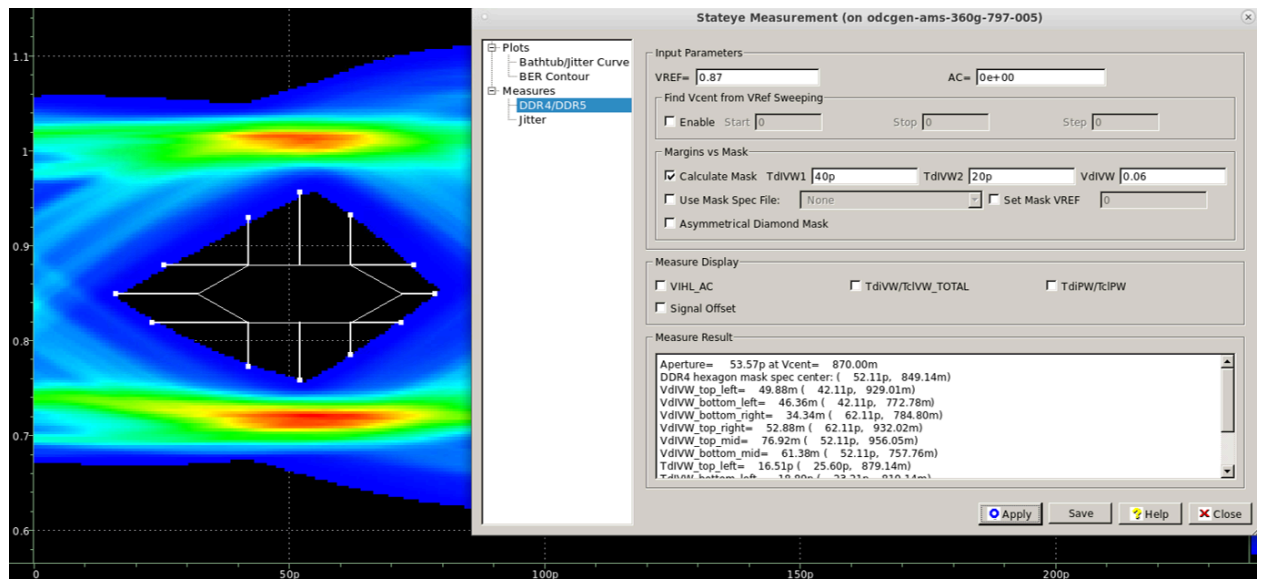


Margins vs DQ Hexagonal Eye Mask

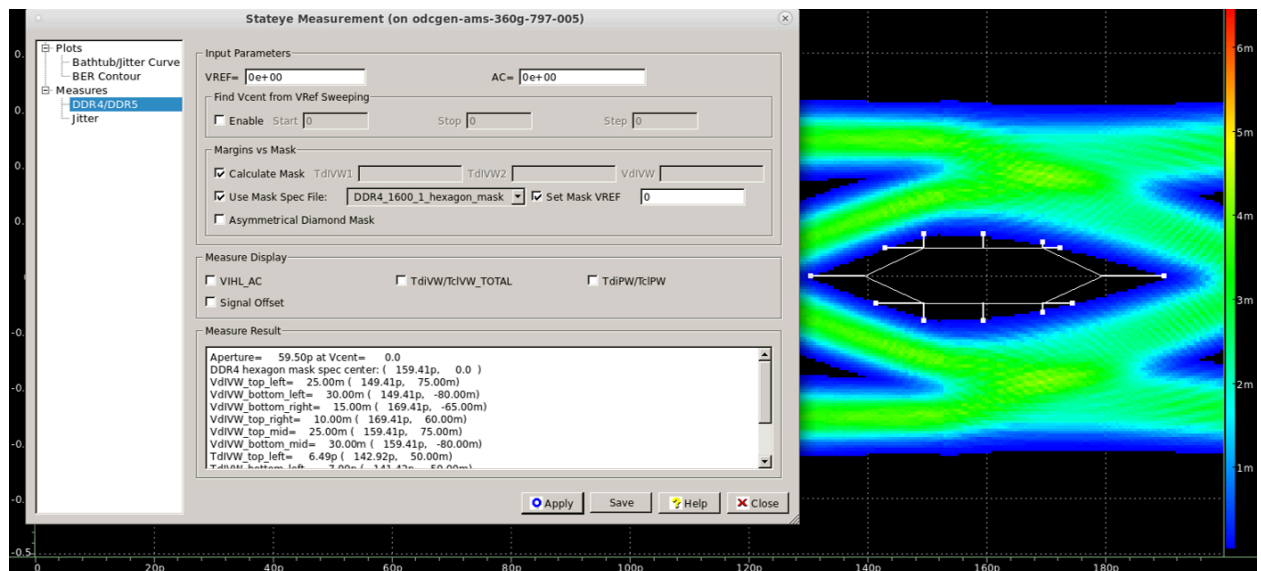
This measurement is done in the following manner by training the Y and X positions to find the maximum summary of all the margin values of the six corners of the mask:

1. Find the maximum horizontal opening and VREF level.
2. Train the X position by placing the mask at the VREF level found earlier and move the mask along VREF horizontally to find the maximum sum of vertical margins.
3. Train the Y position by placing the mask at the X position found in step 2 and move the mask along the X position vertically to find the maximum sum of horizontal margins.

The input mask configuration is as shown in the following image.



You can load a specification file as shown in the following image.



Using the Setup/Hold Results Viewer

To view the Setup/Hold Results Viewer:

1. Click the **Detailed Results** button in the Automatic Eye Measurement window.

The Setup/Hold Results Viewer window opens with the setup and hold results displayed.

2. Click one of the following display category types to display the results you want to see:

- **Setup/Hold**
- **Setup+Hold**
- **Setup**
- **Hold**

If you use a derate table, the margin, measured, and required values are reported, and the two corresponding rows that have the minimum setup and minimum hold margin are highlighted in yellow.

If you do not use a derate table, the minimum setup or hold times are reported, and the two corresponding rows that have the minimum setup or hold times are highlighted in yellow.

3. (Optional) Click the **Data Slew Only** check box to limit the display to data slew values only.

The worst slew values are reported at the bottom of the table. Exported results are also limited when this option is enabled.

4. (Optional) Click the **Export** button to export the selected results to a *.csv file.

Creating ACE Scripts From DDR Measurements

To save an ACE script from the a DDR measurement you currently have set up, click the **Script** button in the Automatic Eye Measurement window.

For each DDR evaluation, the equivalent ACE TCL script is generated in the saved file in lines that are commented out. You can uncomment them to use these lines.

Clearing Eye Measurements

Measurement results are displayed on top of the eye diagrams. Right-click an eye diagram, and choose **Clear Measure** from the menu that opens to clear the measure results.

Copying Eye Diagram Settings

Eye diagram settings can be copied and reused on another eye diagrams. **Right-click** an eye diagram, choose **Copy Settings** from the menu that opens to copy the settings.

Pasting Eye Diagram Settings

Copied eye diagram settings can be applied on top of another eye diagram by performing the following steps:

1. Open the eye diagram you want to apply the copied settings.
2. **Right-click** on the eye diagram, choose **Paste Settings** from the menu. The **Copying Eye Configuration** dialog opens.
3. Choose an eye configuration setting to apply.
4. Select the **Measurement Settings** check box to apply measurements settings along with the selected eye configuration setting.
5. Click OK.

Measuring DDR3 Eye Diagrams

The following DDR3 Eye Diagram panel measurements are available:

- Derate-based setup and hold measurements
- tVAC measurements
- Ringback measurements

You can use these measurements during the design phase to create specifications from a behavioral model or for postsimulation verification.

Measuring Derate-Based Setup and Hold Times

This section contains information on the following topics:

- [Calculating Address, Command, Data Setup, and Hold Derating Values](#)
- [Setting Up DDR3 Measurements](#)
- [Viewing Derating Measurement Output](#)

Calculating Address, Command, Data Setup, and Hold Derating Values

The total setup time (tIS) and hold time (tIH) required is calculated by adding the base value of the setup and hold to the derating value. For example:

```
tIS Total= tIS(Base) + tIS
tIH Total= tIH(Base) + tIH
```

The base value of the setup or hold times is taken from the JEDEC standard. The following table shows sample setup or hold base values of several DDR3 specifications:

Symbol	Reference	DDR3-800	DDR3-1066	DDR3-1333	DDR3-1600	DDR3-1866	DDR3-2133	Units
tIS(base) AC175	V _{IHL(ac)}	200	125	65	45	-	-	ps
tIS(base) AC150	V _{IHL(ac)}	350	275	190	170	-	-	ps
tIS(base) AC135	V _{IHL(ac)}	-	-	-	-	65	60	ps
tIS(base) AC125	V _{IHL(ac)}	-	-	-	-	150	135	ps
tIH(base) DC100	V _{IHL(dc)}	275	200	140	120	100	95	ps

The derating values of setup or hold times are calculated from the slew rate of the differential CLK and DQS signals, as well as the address, command, and data signals. In the JEDEC standard, the derating value lookup table corresponds to different slew rates of CLK or strobe and address, command, or data signals.

ΔtIS, ΔtIH derating in [ps] AC/DC based AC175 Threshold -> VIH(ac)=VREF(dc)+175mV, VIL(ac)=VREF(dc)-175mV																	
		CK,CK# Differential Slew Rate															
		4.0 V/ns		3.0 V/ns		2.0 V/ns		1.8 V/ns		1.6 V/ns		1.4 V/ns		1.2 V/ns		1.0 V/ns	
		ΔtIS	ΔtIH	ΔtIS	ΔtIH	ΔtIS	ΔtIH	ΔtIS	ΔtIH	ΔtIS	ΔtIH	ΔtIS	ΔtIH	ΔtIS	ΔtIH	ΔtIS	ΔtIH
CMD/ ADD Slew rate V/ns	2.0	88	50	88	50	88	50	96	58	104	66	112	74	120	84	128	100
	1.5	59	34	59	34	59	34	67	42	75	50	83	58	91	68	99	84
	1.0	0	0	0	0	0	0	8	8	16	16	24	24	32	34	40	50
	0.9	-2	-4	-2	-4	-2	-4	6	4	14	12	22	20	30	30	38	46
	0.8	-6	-10	-6	-10	-6	-10	2	-2	10	6	18	14	26	24	34	40
	0.7	-11	-16	-11	-16	-11	-16	-3	-8	5	0	13	8	21	18	29	34
	0.6	-17	-26	-17	-26	-17	-26	-9	-18	-1	-10	7	-2	15	8	23	24
	0.5	-35	-40	-35	-40	-35	-40	-27	-32	-19	-24	-11	-16	-2	-6	5	10
	0.4	-62	-60	-62	-60	-62	-60	-54	-52	-46	-44	-38	-36	-30	-26	-22	-10

Setting Up DDR3 Measurements

To set up a DDR3 measurement:

1. Load a “derating table” by setting the following environment variable before invoking the WaveView tool:

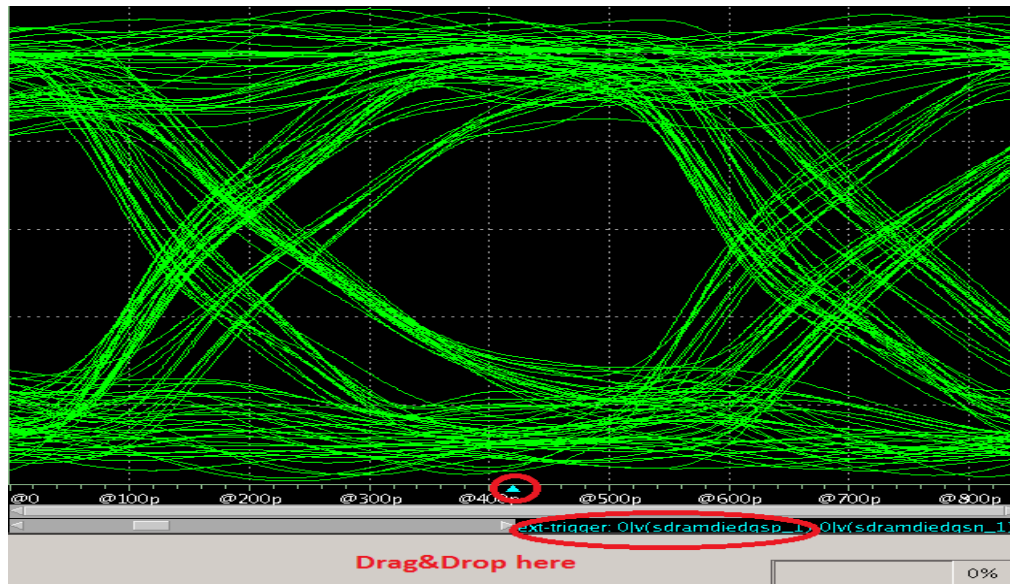
```
setenv CX_DRTAB_FILE <derating_table_file>
```

This file contains the base values of setup and hold, as well as lookup tables for the derating values that are from different DDR3 standards (DDR3-800/1066/1333/1600/1866/2133).

2. Launch the WaveView tool.

3. Plot the eye diagram for the address, command, or data signal and specify the CLK/DQS signal as the external trigger for the eye.

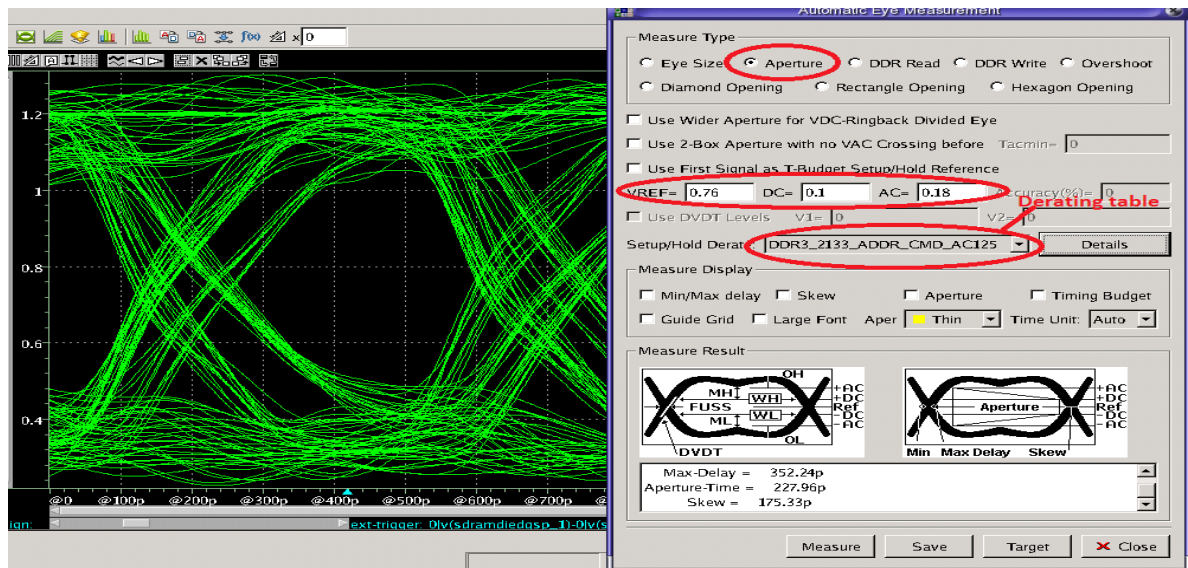
To create an eye diagram with external trigger, plot the eye diagram of the signal, then drag and drop the external trigger signal in to the lower-right corner of the eye diagram panel.



4. Ensure the blue triangle (trigger point) on the x-axis is at the center of the eye.
5. Right-click on the plotted eye diagram, and choose **Measure Eye** from the menu that opens.

The **Automatic Eye Measurement** window opens.

6. Specify the Vref, DC, and AC values, and choose a dithering table from the Setup/Hold Dithering menu.



7. Click **Measure**.

The measurement is performed. Click **Details** to display the measurement report.

Viewing Derating Measurement Output

Two kinds of measurement results are calculated:

- The required setup or hold values, which are calculated by adding the base value to the derating value. The derating values are obtained after measuring the slew values.
- The actual setup or hold values from the waveform.

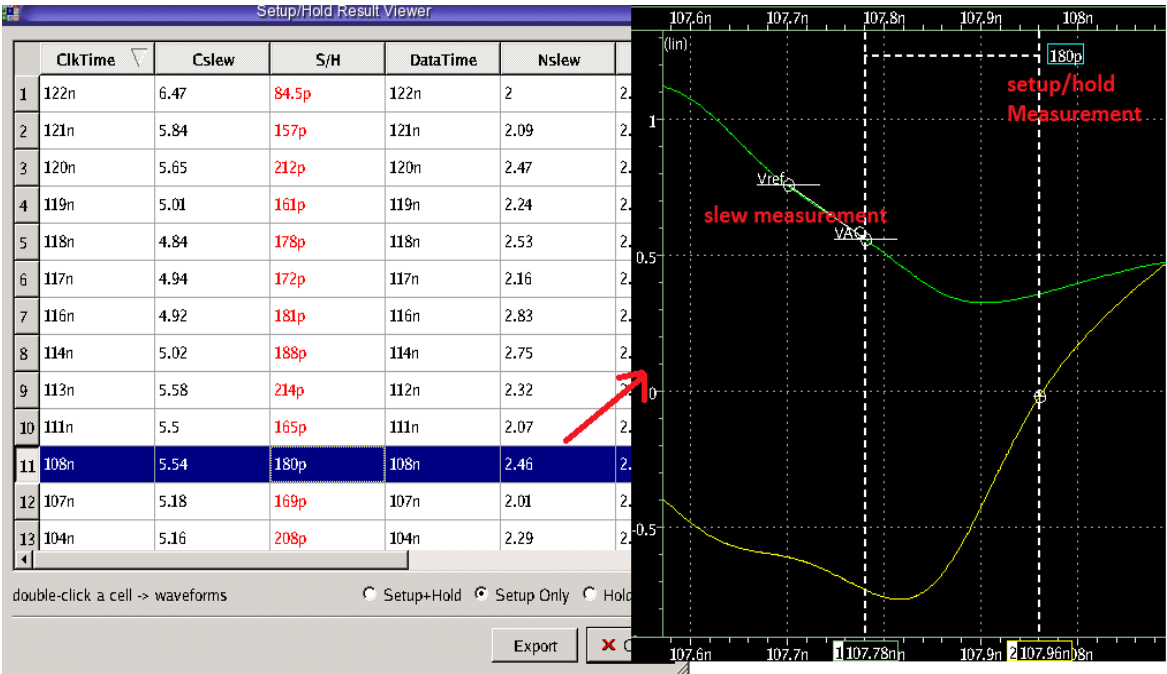
If the actual measured value is less than the required value, then it is a setup or hold violation. The output report displays the slew values of signals, as well as the required and actual measured values of setup or hold.

CLK slew Setup/Hold Measured value Normal/Tangent Slew of Data/Address Base(setup/Hold)+ Derating

	ClkTime	Cslew	S/H	DataTime	Nslew	Tslew	Derate	Derated
50	38.6n	5.18	70.3p	38.5n	2.24	2.29	128p	-57.7p
51	37.7n	6.12	68.4p	37.6n	1.67	1.71	115p	-46.3p
52	35.7n	5.11	66.3p	35.7n	2	2.04	128p	-61.7p
53	32.1n	5.28	82.1p	32n	1.91	2.06	128p	-45.9p
54	29.2n	5.32	69.3p	29.1n	2.31	2.31	128p	-58.7p
55	28.3n	6.37	63.8p	28.2n	1.64	1.66	112p	-48.5p
56	27.4n	5.56	77.5p	27.3n	1.75	1.75	116p	-38.8p
57	26.4n	4.91	71.6p	26.3n	2.01	2.02	128p	-56.4p
58	22.6n	5.83	94p	22.5n	1.65	1.65	112p	-18p
59	20.8n	6.73	74.6p	20.7n	1.94	2.06	128p	-53.4p
60	16.1n	6.13	61.9p	16n	1.6	1.61	110p	-47.9p
61	15.1n	5.45	82.2p	15.1n	1.7	1.77	118p	-35.3p
62	9.52n	5.36	75.7p	9.44n	1.74	1.74	116p	-40.4p

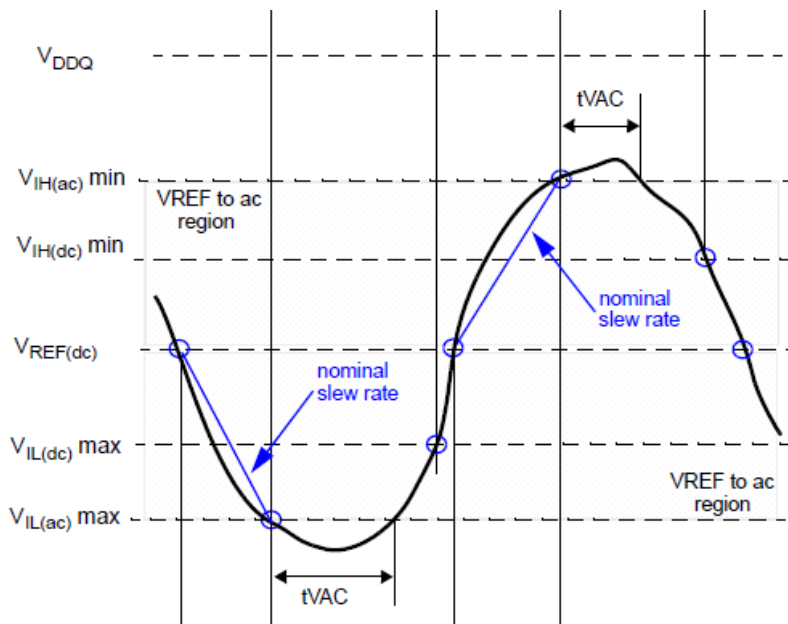
double-click a cell -> waveforms Setup+Hold Setup Only Hold Only Export Close

When you double-click a row, the measurements are displayed on a transient signal, which you can use for debugging the setup and hold violations. The transient portion of the signal that created the violation is displayed. The following figure shows the slew measurements and setup or hold measurements:



Using the tVAC Measurement

tVAC is the required time above $V_{IH(ac)}$ or $V_{IL(ac)}$ for a valid transition. For a valid transition to occur, a signal has to remain above or below an AC threshold for a period of time. This time span is defined in the JEDEC standard. The tVAC constraint is slew-rate dependent similar to the setup or hold derating.



Setting Up tVAC Measurements

The setup for a tVAC measurement is similar to setup or hold derating. To set up a tVAC measurement:

1. Before starting the WaveView tool, enter the following line to load a file that contains the value of tVAC:

```
setenv CX_DRTAB_FILE <tVAC_value_file>
```

The tVAC values can specify inside derating table file along with setup/hold derate values.

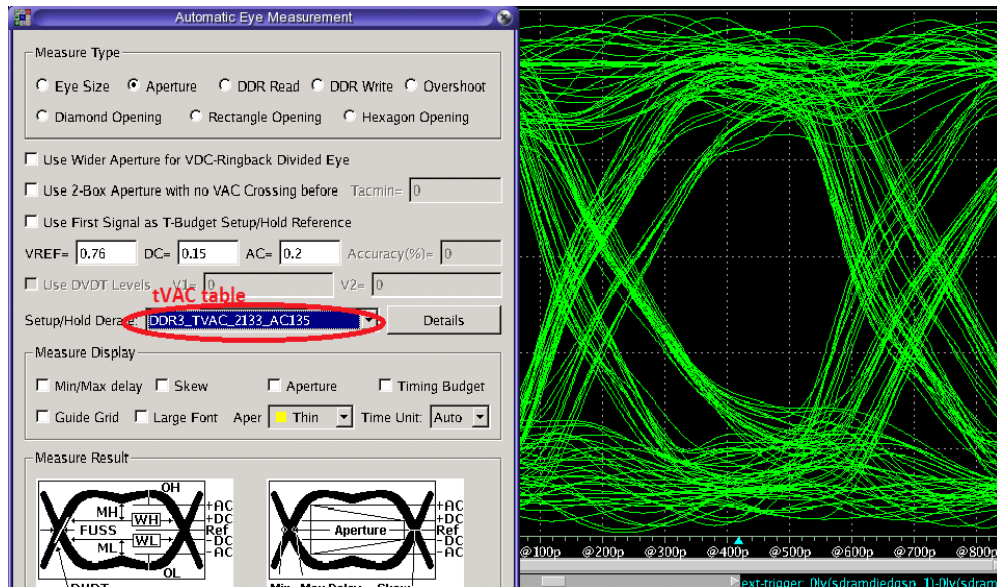
The measurement is performed after plotting the eye diagram with external trigger signal as in the case of setup or hold derating.

2. Open the WaveView tool.
3. Right-click the eye diagram, and choose **Measure Eye** from the menu that opens.

The **Automatic Eye Measurement** window opens.

4. Select atVAC table from the **Setup/Hold Derate** menu.
5. Click **Measure**.

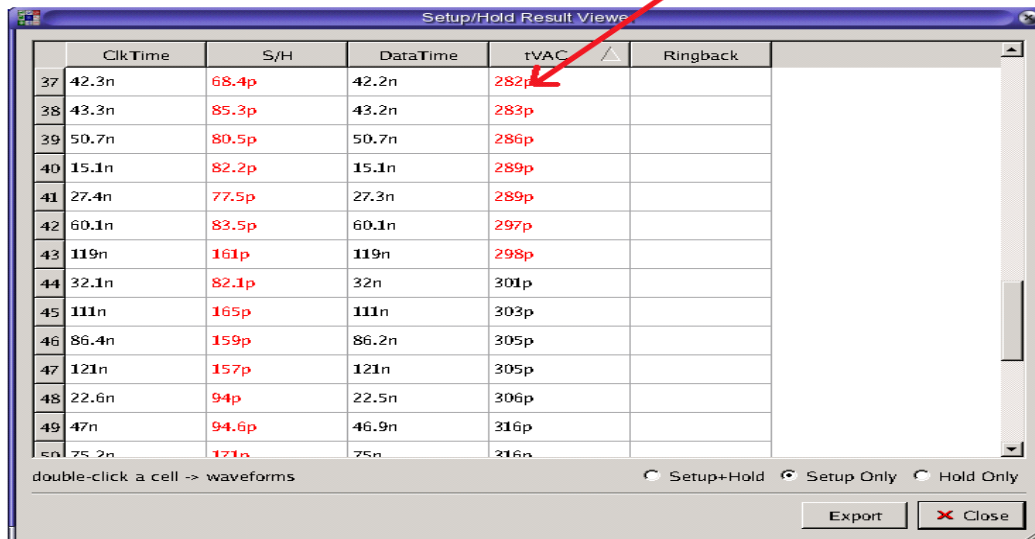
The eye diagram is measured using the tVAC value.



Viewing tVAC Measurement Output

If the measured tVAC values are smaller than the required value, violations are flagged red in the report that is created after the tVAC measurement is performed.

tVAC Violations



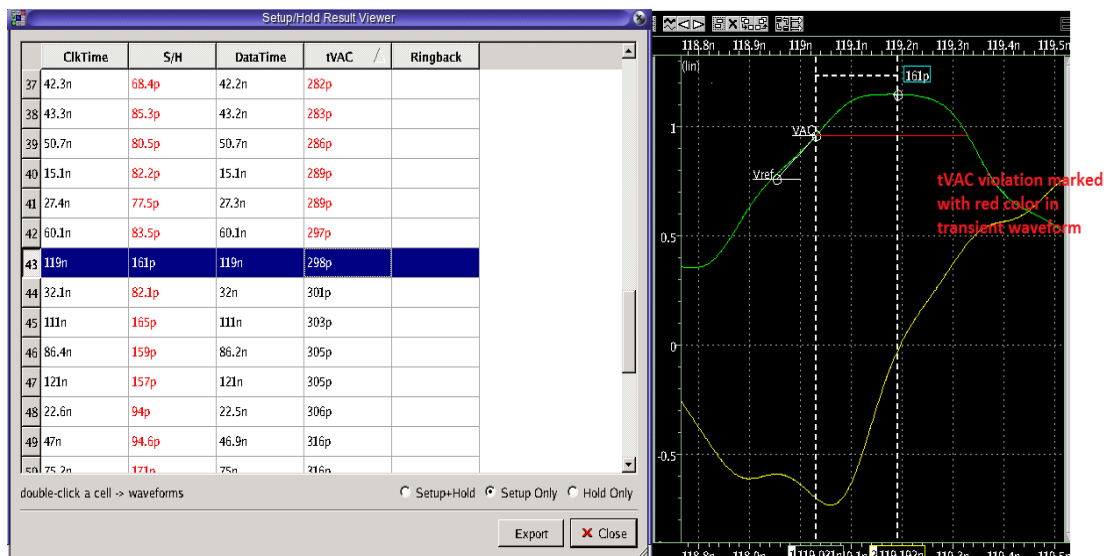
	ClkTime	S/H	DataTime	tVAC	Ringback
37	42.3n	68.4p	42.2n	282p	
38	43.3n	85.3p	43.2n	283p	
39	50.7n	80.5p	50.7n	286p	
40	15.1n	82.2p	15.1n	289p	
41	27.4n	77.5p	27.3n	289p	
42	60.1n	83.5p	60.1n	297p	
43	119n	161p	119n	298p	
44	32.1n	82.1p	32n	301p	
45	111n	165p	111n	303p	
46	86.4n	159p	86.2n	305p	
47	121n	157p	121n	305p	
48	22.6n	94p	22.5n	306p	
49	47n	94.6p	46.9n	316p	
50	75.2n	171n	75n	316n	

double-click a cell -> waveforms

Setup+Hold Setup Only Hold Only

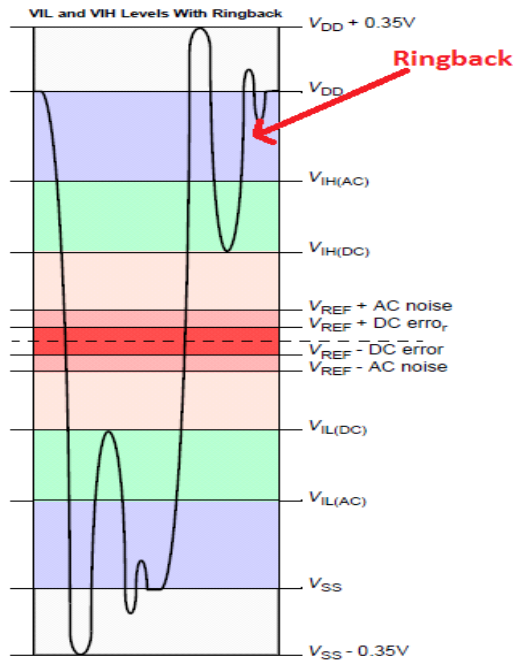
Export Close

You can click on any of the rows in the report that shows the transient signal and tVAC measurement. In the following figure, the tVAC violations are displayed on a transient waveform:



Measuring Ringback

The ringback measurement calculates how far the top side of a waveform dips below the AC threshold while not crossing the DC threshold. If the waveform moves below the DC threshold, then a ringback violation occurs.



The ringback measurement setup is similar to the tVAC measurement setup in that you also need to provide a tVAC table. The ringback is measured when a signal goes above AC threshold and then goes below the AC threshold values.

Viewing Ringback Measurement Output

Ringback results can occur with or without violations. The output report displays the ringback voltage value. A ringback value without a violation is a signal that goes above the AC threshold, then goes below AC threshold, and finally stays within the AC and DC thresholds. A ringback value with a violation is a signal that goes above the AC threshold and then goes below the DC threshold.

Ringback violations are highlighted in red in the report. When you select a row in the report, the ringback measurement is displayed on a transient signal. The following figure shows the ringback violation displayed on a transient signal in magenta:



Sample Derating Table File

A derating table file includes the derating lookup table values for different DDR3 standards, as well as any tVAC required values. The following file shows a sample derating table:

#table DDR3_800_ADDR_CMD_AC175	Header section shows the DDR3 standard, base value of setup/hold, slew values															
#base_setup 200																
#base_hold 275																
#clock_slew 4 3 2 1.8 1.6 1.4 1.2 1																
2	88	50	88	50	88	50	96	58	104	66	112	74	120	84	128	100
1.5	59	34	59	34	59	34	67	42	75	50	83	58	91	68	99	84
1.0	0	0	0	0	0	0	8	8	16	16	24	24	32	34	40	50
0.9	-2	-4	-2	-4	-2	-4	6	4	14	12	22	20	30	30	38	46
0.8	-6	-10	-6	-10	-6	-10	2	-2	10	6	18	14	26	24	34	40
0.7	-11	-16	-11	-16	-11	-16	-3	-8	5	0	13	8	21	18	29	34
0.6	-17	-26	-17	-26	-17	-26	-9	-18	-1	-10	7	-2	15	8	23	24
0.5	-35	-40	-35	-40	-35	-40	-27	-32	-19	-24	-11	-16	-2	-6	5	10
0.4	-62	-60	-62	-60	-62	-60	-54	-52	-46	-44	-38	-36	-30	-26	-22	-10
#table DDR3_1066_ADDR_CMD_AC175	Other standards sharing the same lookup table and tVAC table															
#base_setup 125																
#base_hold 200																
#use DDR3_800_ADDR_CMD_AC175																
#table DDR3_1333_ADDR_CMD_AC175	Table for required tVAC values															
#base_setup 65																
#base_hold 140																
#use DDR3_800_ADDR_CMD_AC175																
#use_tvac DDR3_1333_AC150																

The default units of derate table values are ps for time and V/ns for slew.

Measuring DDR4 Eye Diagrams

The following DDR4 measurements are supported in the WaveView tool:

- Vcent measurements from a Vref sweep
- Vcent_Dq (pin avg)
- TdiVW
- VIH_L_AC
- TdiPW
- Tr&Tf based on VdIVW_Total
- SRIN_diVW based on VdIVW_Total
- TdIVW & VdIVW margins
- Aperture alignment for LPDDR4

Please refer the *JEDEC Manual* (JEDEC standard no 79-4) for the details of these measurements.

This section contains information on the following topics:

- [Taking DDR4 Measurements](#)
- [Taking TdIVW and VdIVW Margin Measurements](#)
- [Aligning the Aperture for LPDDR4 Measurements](#)
- [Shifting the Trigger Clock Delay and Delay Clock to the Aperture Center](#)
- [Sample DDR4 Specification File](#)

Taking DDR4 Measurements

This section explains how to take the following DDR4 measurements:

- [Taking Vcent Measurements](#)
- [Taking TdIVW_Total and VIH_L_AC Measurements](#)
- [Taking TdiPW, Tr, Tf, and SRIN_diVW Measurements](#)

Taking Vcent Measurements

When a Vcent measurement is performed, the Vref value that corresponds to widest eye opening is calculated. If multiple eye diagrams exist on a single panel, the Vcent value of each eye is measured, and the average value is calculated to get the Vcent_DQ pin average.

To take a Vcent measurement:

1. Right-click the eye diagram, and choose **Measure Eye** from the menu that opens.

The **Automatic Eye Measurement** window opens.

2. Click the **DDR4** measurement.
3. Click the Input **Parameters** tab.
4. Click the **Enable** check box to enable a Vref sweep.

The Vcent_DQ pin average for all the bits is calculated when this option is enabled.

5. Enter an **AC** value ($V_{dIVW_Total}/2$), as well as the **Start**, **Stop**, and **Step** values.

Automatic Eye Measurement

Measure Type

☐ Eye Size ☐ Aperture ☐ DDR Read ☐ DDR Write ☐ Overshoot

☐ Diamond Opening ☐ Rectangle Opening ☐ Hexagon Opening ☒ DDR4

Input Parameters | Measure Options | Measure Diagram

VREF= 0 DC= 0 AC= 0.1 Accuracy(%)= 0

DDR4 : Find Vcent from VRef Sweeping

☒ Enable Start 0.23 Stop 0.26 Step 0.002

☐ Report Margins vs. Mask ☐ Align Bit Apertures ☐ Set VRef to Vcent

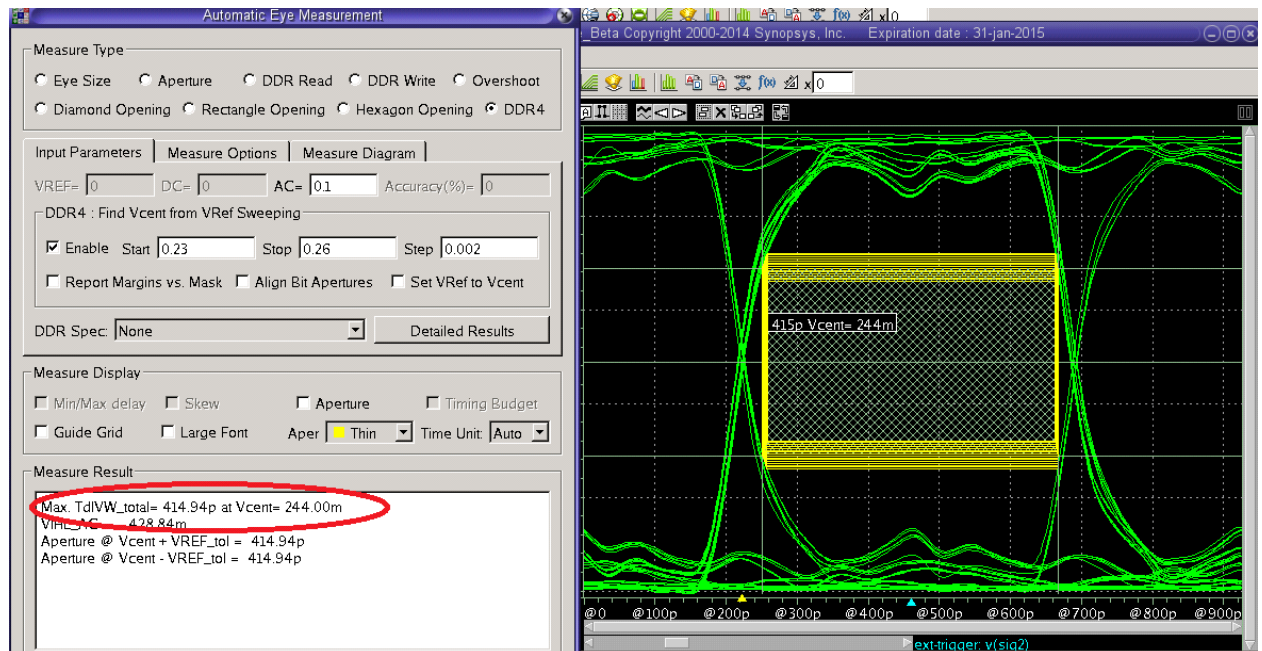
DDR Spec: None Detailed Results

Measure Display

☐ Min/Max delay ☐ Skew ☐ Aperture ☐ Timing Budget

☐ Guide Grid ☐ Large Font Aper Thin Time Unit: Auto

6. Click the **Measure** button.

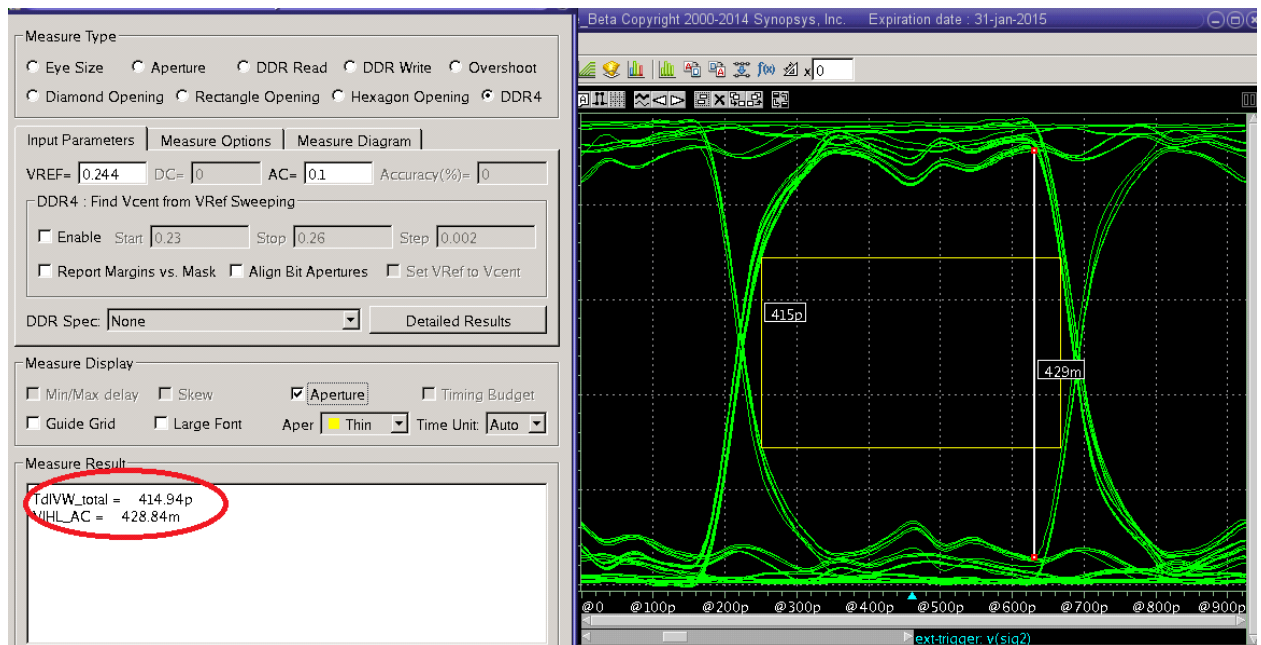


The values for Vcent are calculated and displayed in the Measure Result field, as well as on the eye diagram.

Taking TdIVW_Total and VIHL_AC Measurements

To take TdIVW_Total and VIHL_AC measurements:

1. Right-click the eye diagram, and choose **Measure Eye** from the menu that opens.
The **Automatic Eye Measurement** window opens.
2. Click the **DDR4** measurement.
3. Click the **Input Parameters** tab.
4. Enter values for **Vref** and **AC**, and click the **Measure** button.



The values for TdIVW_Total and VIHL_AC are calculated and displayed in the Measure Result field, as well as on the eye diagram.

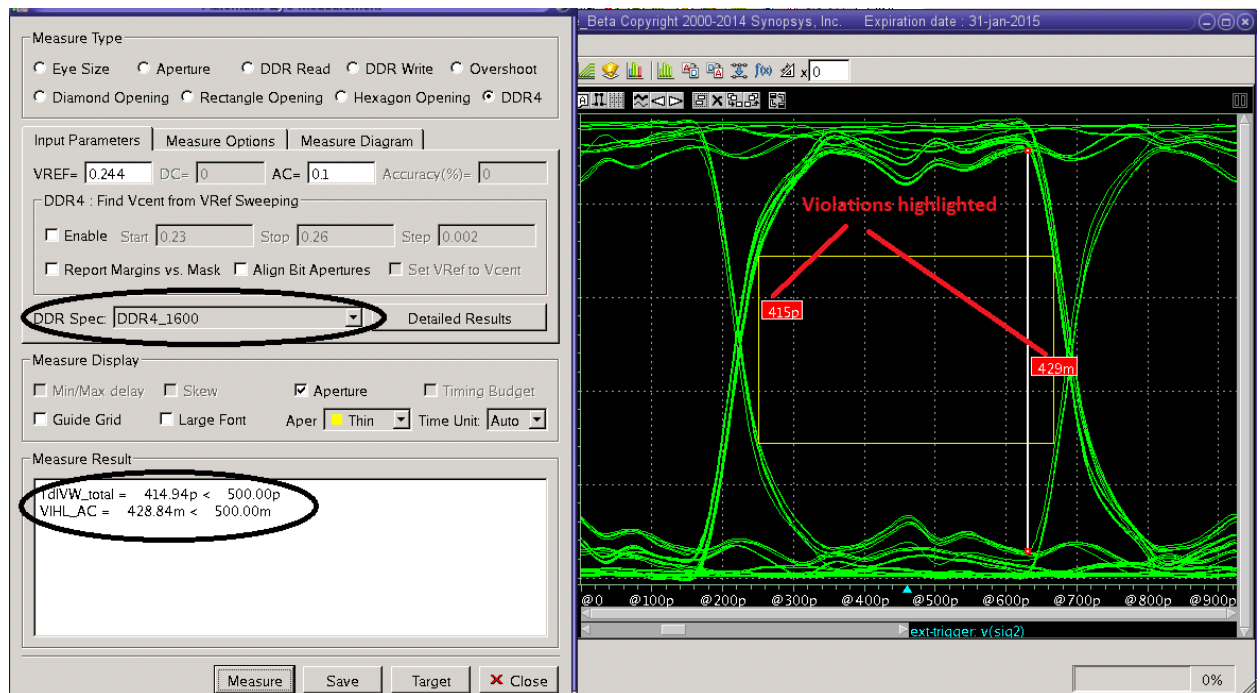
If you want to specify a target value for TdIVW_Total and VIHL_AC in a DDR4 specification file and load it into the WaveView tool, enter the `setenv CX_DRTAB_FILE <filename>` environment variable at the command line before invoking the WaveView tool. See the [Sample DDR4 Specification File](#) section for information on how to set up a DDR4 specification file.

Include the following commands in your DDR4 specification file to set these values:

```
#min_aperture <value> or #min_aperture_ui <value>
```

```
#min_vihl_ac <value>
```

If a measured value does not meet the target, the violations are highlighted in red.



Taking TdIPW, Tr, Tf, and SRIN_diVW Measurements

To take TdIPW, Tr, Tf, and SRIN_diVW measurements:

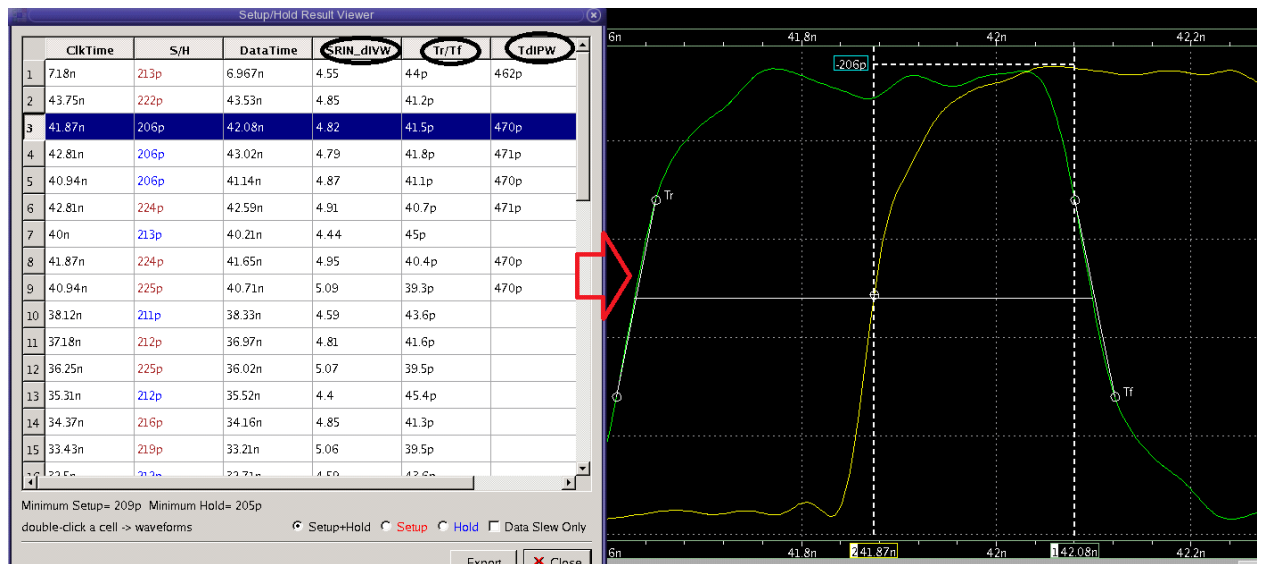
1. Right-click the eye diagram, and choose **Measure Eye** from the menu that opens.

The **Automatic Eye Measurement** window opens.

2. Click the **DDR4** measurement.
3. Click the **Input Parameters** tab.
4. Enter values for **Vref** and **AC**, and click the **Measure** button.

The values for TdIPW, Tr, Tf, and SRIN_diVW are calculated and displayed in the Measure Result field, as well as on the eye diagram.

5. Click the **Detailed Results** button.



The **Setup/Hold Result Viewer** window opens with the TdIPW, Tr, Tf, and SRIN_divW values displayed. You can double-click a row to show the corresponding measurements on a transient signal.

If you want to specify the target value for the TdIPW and slew, you can include the following commands in your DDR4 specification file:

```
#min_dataslew <value>
```

```
#min_tdipw_ui <value>
```

Measurements that do not meet the target value are highlighted.

Specifying VIH_{LAC}(min) for tr₂ Measurements

You can specify the VIH_{LAC}(min) value for calculating tr₂ measurements:

1. Select **DDR4** in the **Automatic Eye Measurement** window.
2. Check the VIH_{LAC}(min) box and specify a value.

If a VIH_{LAC}(min) value is not specified, by default the WaveView tool uses the calculated VIH_{LAC} value.

If the specified VIH_{LAC}(min) is too high or too low for certain clk period and tr₂ cannot be calculated for that period, the **Setup/Hold Result Viewer** shows a horizontal line at y=VIH_{LAC}(min).

Taking TdIVW and VdIVW Margin Measurements

The following calculations occur when you place a mask inside of an eye and measure the voltage and timing margins:

- The Vcent value of each bit is calculated.
- The Vcent average of all the bits (Vcent_Dq pin_avg) is calculated.
- A mask based on the Vcent_Dq and Triggering point is placed.
- The margin values are measured. The ringback effect is also considered during this measurement.

To measure the margin:

1. Right-click the eye diagram, and choose **Measure Eye** from the menu that opens.

The **Automatic Eye Measurement** window opens.

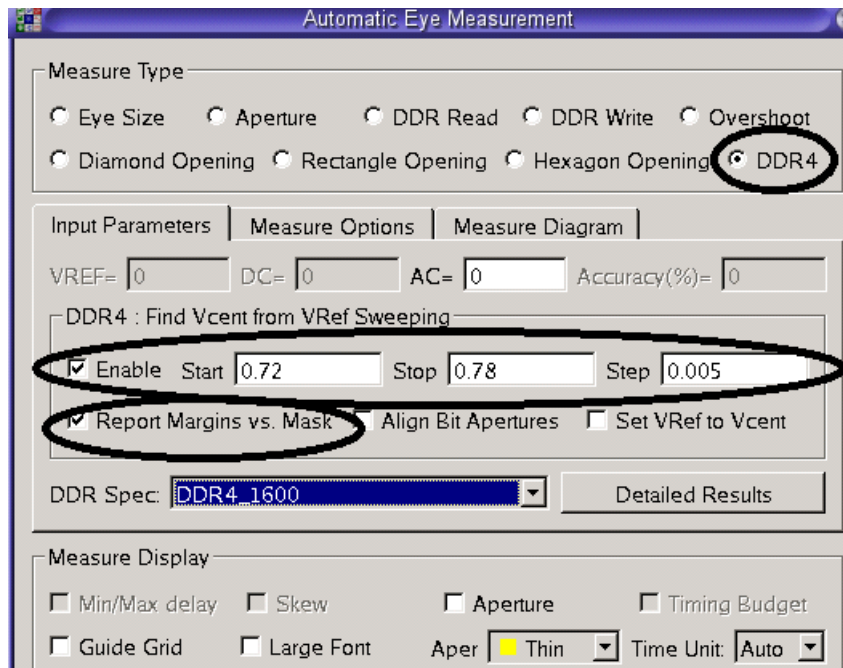
2. Click the **DDR4** measurement.
3. Click the **Input Parameters** tab.
4. Click the **Enable** check box to enable a Vref sweep.

The Vcent_Dq pin average for all the bits is calculated when this option is enabled.

5. Enter the **Start**, **Stop**, and **Step** values.

If you enter the Vref value directly, the measurement works without a Vref sweep.

6. Click the **Report Margins vs Mask** check box.
7. Choose a DDR4 spec from the **DDR4 Spec** menu that contains mask specifications.

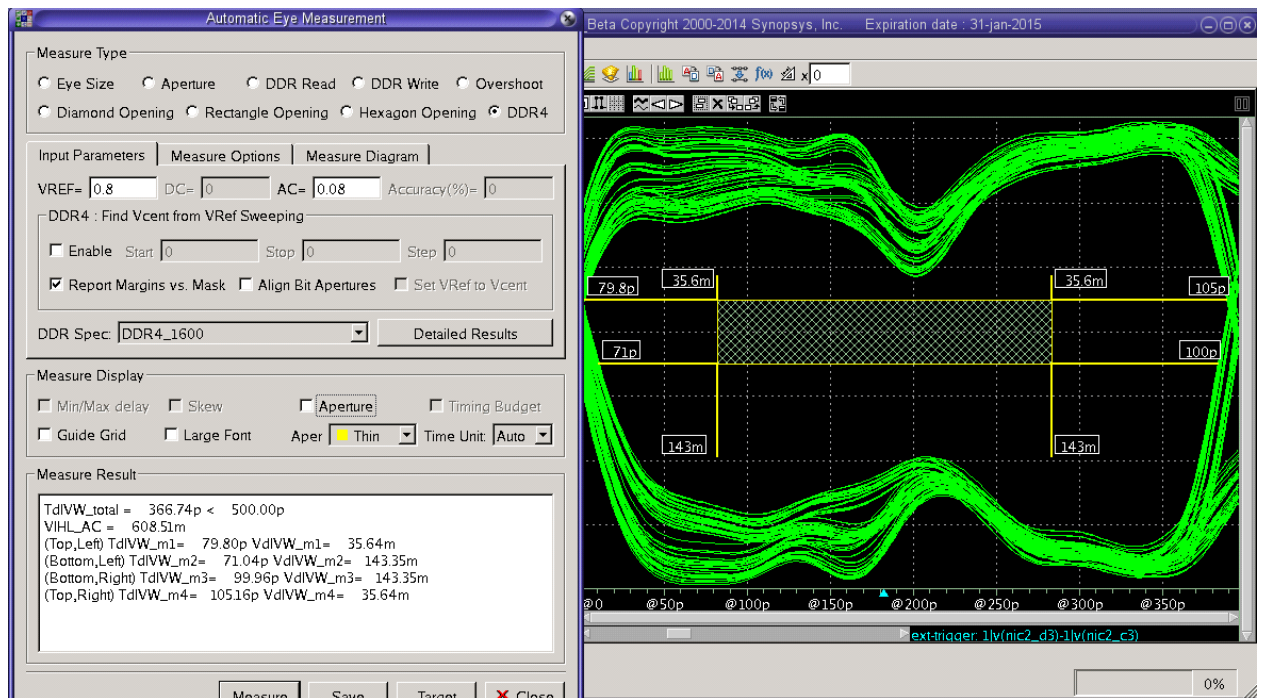


Use the following commands in your DDR4 specification file to set the width and height of mask:

```
#mask_tdivw_ui <width value>
```

```
#mask_vdivw <height value>
```

8. Click the **Measure** button.



The mask margin is calculated and displayed in the Measure Result field, as well as on the eye diagram.

Aligning the Aperture for LPDDR4 Measurements

Aperture alignment measures the mask offset values of each bit. You can plot the eye diagram of all the bits in a single panel and trigger it with a differential clock.

The first eye is used as the reference, and then the center point of the eye crossing is calculated. All other eye diagram center points are shifted so they overlap. Both the shift value of each bit and the aperture value are reported.

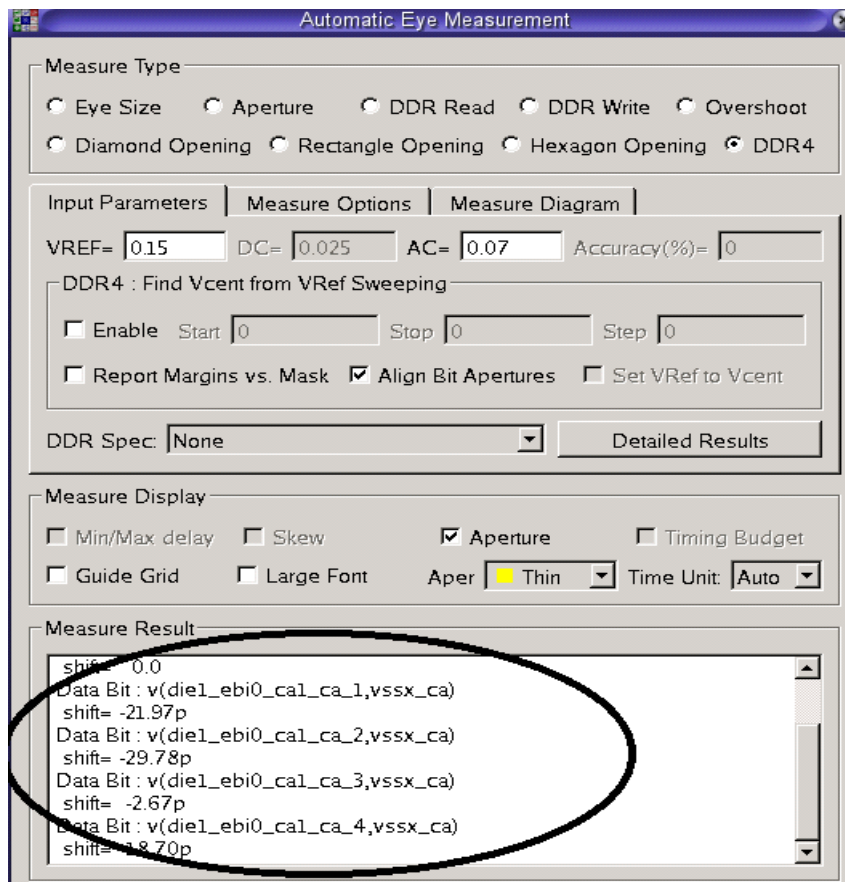
To align apertures:

1. Plot the eye diagrams of all the bits in a single panel, and use the same signal for triggering the eye.
2. Right-click the eye diagram, and choose **Measure Eye** from the menu that opens.

The **Automatic Eye Measurement** window opens.

3. Click the **DDR4** measurement.
4. Click the **Input Parameters** tab, and enter **Vref** and **AC** values for the eye.
5. Click the **Align Bit Apertures** option.

6. Click the **Measure** button.



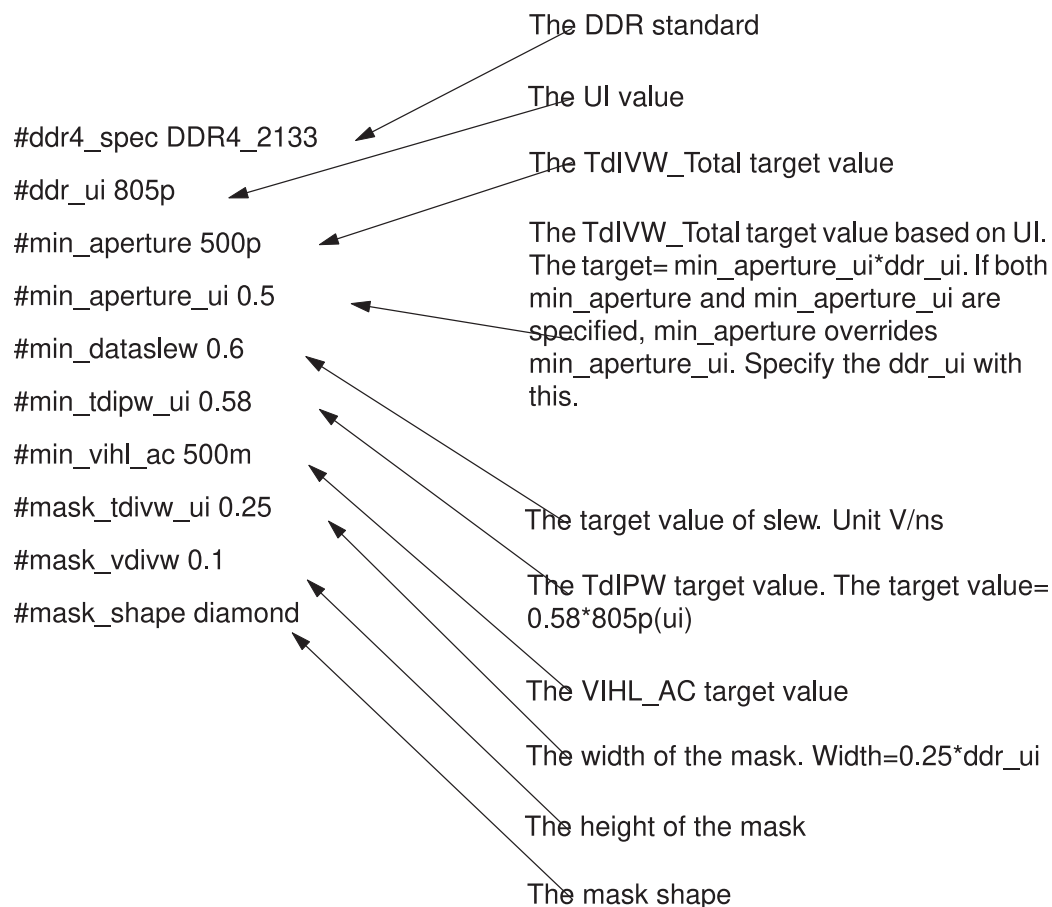
The shifted values of each bit are calculated and displayed in the **Measure Result** field.

Shifting the Trigger Clock Delay and Delay Clock to the Aperture Center

Sometimes the simulation data might not have the trigger signal well centered, and you might want to center the trigger signal during the DDR4 measurement. The **Trigger Clock Delay** option specifies the delay value for the triggering signal, and the **Delay Clock to Aperture Center** option aligns the trigger signal to the center of the eye.

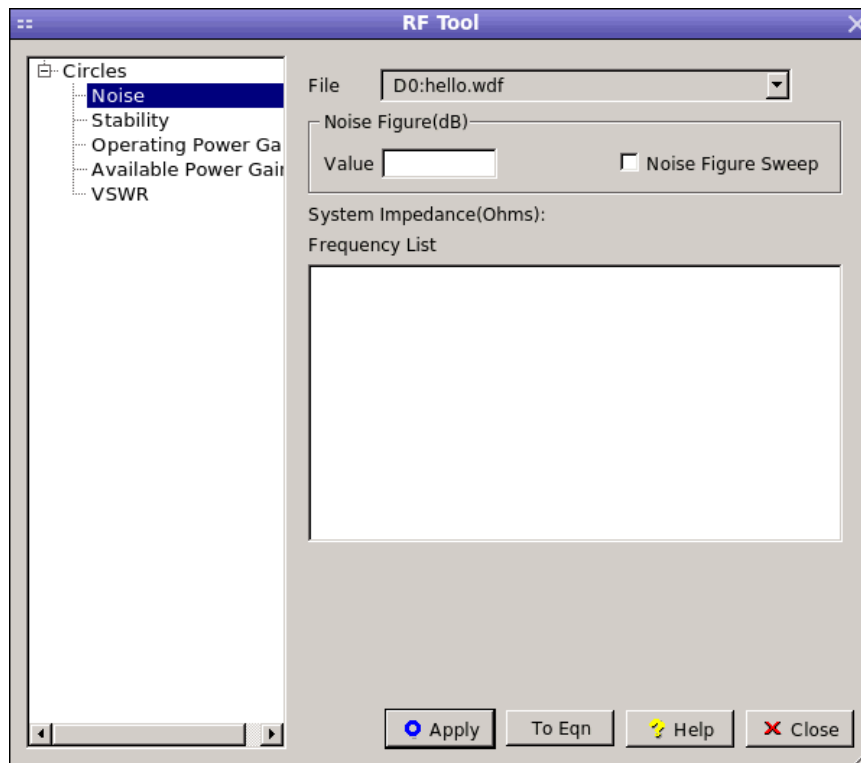
Sample DDR4 Specification File

You can specify the target values for different measurements in a DDR4 specification file. For example:



RF Measurements

To open the RF Tool, choose **RF Tool** from the menu located in the **Utilities** section of the **Waveform** tab. The **RF Tool** window opens.



The RF Tool takes special measurements or calculations for RF analyses. The RF analyses in the RF Tool are different from the measurements in WaveView Measurement Tool—they are not applied to a specific signal, but to a specific waveform file.

This section contains information on the following topics:

- [Measuring Circles](#)
- [Measuring Noise Circles](#)
- [Measuring Stability Circles](#)
- [Measuring Operating Power Gain Circles](#)
- [Measuring Available Power Gain Circles](#)
- [Measuring VSWR Circles](#)
- [Exporting Equations From RF Tool Setup](#)

Measuring Circles

An S-Parameter waveform file is required for all circle measurements. The **File** selection field at the top of the RF Tool window lists all currently open waveform files. If the specified

file is an S-Parameter waveform file, the frequency values are extracted from the specified file and listed in the Frequency List.

The center point and radius values for each circle are calculated. 360 data points are created to make a circle with the center and radius.

The result circles are plotted on a Smith Chart. If a Smith Chart does not yet exist in any waveview, a new one is created, and the circles are plotted on it.

This section contains information on the following topics:

Measuring Noise Circles

To measure a noise circle in the RF Tool:

1. Click the **Noise** category on the left side of the RF Tool window.
The Noise Circle measurement settings are displayed.
2. Choose a file from the **File** menu that contains the noise circle you want to measure.
3. Enter a dB noise figure value in the Noise Figure text box, or click the **Noise Figure Sweep** check box if you want to include a noise figure sweep as part of the measurement.

Note:

System Impedance value is not an input value, but is the value of signal $Z_o.Z_o(1)$ or $Z_o.Z_{od}(1)$ from the selected waveform file. The value is displayed for informational purposes.

4. Select one or more frequency values by holding the **Ctrl** key and clicking frequency values in the Frequency List.
If you want to select all frequency values, pres **Ctrl + A** to select all.
5. Click **Apply** to take the noise circle measurements and plot the results.

Measuring Stability Circles

To measure a stability circle in the RF Tool:

1. Click the **Stability** category on the left side of the RF Tool window.
The stability circle measurement settings are displayed.
2. Choose which S Parameter signals you want to include as part of the measurement in the S Parameters section.

The associated signals and parameters are defaults from the selected waveform file. You can change the defaults signals by choosing different signals from the menus in the S Parameter section.

3. Select one or more frequency values by holding the **Ctrl** key and clicking frequency values in the Frequency List.

If you want to select all frequency values, pres **Ctrl + A** to select all.

4. Choose the type of stability circle you want to plot. You can choose one or both types.

Click the **Output Stability Circle** check box to plot an output stability circle waveform.
Click the **Input Stability Circle** check box to plot an input stability circle waveform.

5. Click **Apply** to take the stability circle measurements and plot the results.

Measuring Operating Power Gain Circles

To measure an operating power gain circle in the RF Tool:

1. Click the **Operating Power Gain** category on the left side of the RF Tool window.

The operating power gain circle measurement settings are displayed.

2. Choose which S Parameter signals you want to include as part of the measurement in the S Parameters section.

The associated signals and parameters are defaults from the selected waveform file. You can change the defaults signals by choosing different signals from the menus in the S Parameter section.

3. Choose one of the following gain values:

- **Value**

Enter a gain value in the Value text box.

- **From/To Sweep**

Enter from and to values in the From and To text boxes, respectively. Enter a Step value as well.

- **Gmax-based Sweep**

Enter step and number values in the Step and Number text boxes, respectively.

4. Select one or more frequency values by holding the **Ctrl** key and clicking frequency values in the Frequency List.

If you want to select all frequency values, pres **Ctrl + A** to select all.

5. Click **Apply** to take the operating power gain circle measurements and plot the results.

Measuring Available Power Gain Circles

To measure an available power gain circle in the RF Tool:

1. Click the **Available Power Gain** category on the left side of the RF Tool window.

The available power gain circle measurement settings are displayed.

2. Choose which S Parameter signals you want to include as part of the measurement in the S Parameters section.

The associated signals and parameters are defaults from the selected waveform file. You can change the defaults signals by choosing different signals from the menus in the S Parameter section.

3. Choose one of the following gain values:

- **Value**

Enter a gain value in the Value text box.

- **From/To Sweep**

Enter from and to values in the From and To text boxes, respectively. Enter a Step value as well.

- **Gmax-based Sweep**

Enter step and number values in the Step and Number text boxes, respectively.

4. Select one or more frequency values by holding the **Ctrl** key and clicking frequency values in the Frequency List.

If you want to select all frequency values, pres **Ctrl + A** to select all.

5. Click **Apply** to take the available power gain circle measurements and plot the results.

Measuring VSWR Circles

To measure a VSWR circle in the RF Tool:

1. Click the **VSWR** category on the left side of the RF Tool window.

The VSWR circle measurement settings are displayed.

2. Choose which S Parameter signals you want to include as part of the measurement in the S Parameters section.

The associated signals and parameters are defaults from the selected waveform file. You can change the defaults signals by choosing different signals from the menus in the S Parameter section.

3. Choose the type of VSWR circle you want to plot. You can choose one or both types.

Click the **Output VSWR Circle** check box to plot an output VSWR circle waveform, and enter a value for VSWROut. Click the **Input VSWR Circle** check box to plot an input VSWR circle waveform, and enter a value for VSWRIn.

4. Select one or more frequency values by holding the **Ctrl** key and clicking frequency values in the Frequency List.

If you want to select all frequency values, press **Ctrl + A** to select all.

5. Click **Apply** to take the VSWR circle measurements and plot the results.

Exporting Equations From RF Tool Setup

To export an equation of the setup from an RF measurement to the Equation Builder:

1. Ensure you set up all the necessary settings for an RF measurement.
2. Click the **To Eqn** button at the bottom of the RF Tool window.

The Equation Builder opens with the created RF measurement equation. For example, the following equation is exported from the settings for a noise circle measurement:

```
nc("0|gsmlna.sc0","50 60 10",50,"100M 200M 300M")
```

PrimeSim HSPICE Measure Tool

To open the **PrimeSim HSPICE Measure** tool, choose **HSPICE Measure** from the menu located in the **Utilities** section of the **Waveform** tab.

Pareto Analysis

The Pareto Analysis tool can import your data mining results from HSPICE, performs sensitivity, R, and R² calculations, then produces results in one of the following formats:

- HistBar plot
- scatter plot

- text window plot
- text file

To perform a Pareto analysis:

1. Choose **Pareto Analysis** from the menu located in the **Utilities** section of the **Waveform** tab.

The **Pareto Analysis** window opens.

Pareto Analysis

Input Signal:

Param File:

	☒	Parameter name
1		
2		
3		
4		

Analysis Type

☒ Sensitivity ☐ R^2 ☐ R

Sensitivity normalized by(P: Parameter, M: Measure)

☒ P/M ☐ abs(P/M) ☐ P ☐ 1/M ☐ Un-normalized

Sorted by

☒ Sensitivity ☐ R^2 ☐ R

Filter minimum value

Sensitivity: R^2 : R:

Output Type

☒ HistBar Plot ☐ Scatter Plot ☐ Text Window ☐ Text File

Output File:

2. Select an Input Signal by dragging it from the signal list and dropping it in the text box.

The input signal can be from an *.mt or *.mc file (which are files that contain parameter names and values for each trace you can load by clicking **Browse**), or from a measurement you apply to a sweep tr0 signal. The number of points in this signal need to be equal to the number of sweeps. Signals are not editable.

3. Ensure a Param File is selected.

You can click the **Browse** button to browse to the location of a Param File, or enter the name in the Param File text box.

4. Choose which parameter signals you want to include by clicking the check box next to a signal name listed in the Parameter Name table.

By default all signals are selected. You can deselect all signals by unchecking the check box in the heading row.

5. Choose one or more analysis types: **Sensitivity**, **R^2**, or **R**.

Sensitivity analyses are calculated using the following function:

$$Sensitivity = \frac{\sum(x * y) - (\sum(x) * \sum(y) / n) * \sum(x)}{\sum(x^2) - (\sum(x))^2 / n} * \sum(y)$$

R^2 analyses are calculated using the following function:

$$Adjusted R - Square = 1 - (1 - R * R) * \frac{n - 1}{n - p - 1}$$

$$Adjusted R - Square = 1 - \left(1 - \frac{(\sum(x - \bar{x}) * (y - \bar{y}))^2}{\sum((x - \bar{x})^2) * \sum((y - \bar{y})^2)} \right) * \frac{n - 1}{n - p - 1}$$

R analyses are calculated using the following function:

$$R = \frac{\sum(x * y) - (\sum(x) * \sum(y) / n)}{\sqrt{(\sum(x^2) - (\sum(x))^2 / n) * (\sum(y^2) - (\sum(y))^2 / n)}}$$

6. If you choose **Sensitivity** as the analysis type, select one of the following normalization factors:

- **P/M**
- **abs(P/M)**
- **P**
- **1/M**
- **Un-normalized**

Otherwise, skip to the next step.

7. If you selected more than one analysis, choose how you want your results sorted in the Sorted By section.
8. Enter values for the minimum filter value.

If you select more than one analysis type, all the filter values are used together.

9. Choose which type of output you want.

If you choose **Text File**, enter a name for the text file.

10. Click **Apply** to run the Pareto analysis.

6

Using WaveView Tools

This chapter details how to use WaveView editors and tools.

This chapter contains information on the following topics:

- [Using the PWL Editor](#)
- [Using the ADC Toolbox](#)
- [Using the Coherent Sample Only \(CSO\) ADC Toolbox](#)

Using the PWL Editor

To open the PWL Editor, click **PWL Editor** in the **Tools** tab.

The PWL Source Editor is used to construct SPICE PWL input stimuli for circuit simulations. The editor supports interactive editing functions via direct point dragging. Multiple sources can be edited or overlapped simultaneously. You can store the edit result in the text PWL storage files (.pwl) or exported to SPICE netlist files.

The buttons are grouped into three groups:

- [File Operations](#)
- [Source Operations](#)
- [Point Operations](#)

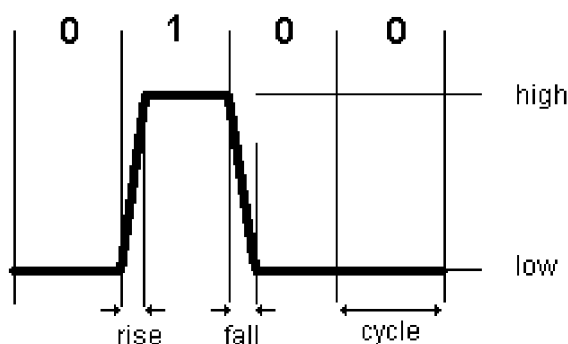
File Operations

Open a prestored PWL editing task with the **Open** button. Save an editing task with the **Save** button. The save function automatically appends file extension .pwl to the user-specified file name if the .pwl extension is not used. Click the **Check** button to check errors in the PWL sources such as overlapping time points.

Click **Close** to close the PWL editor window. All PWL sources are not removed when the PWL editor window is closed.

Source Operations

Click the **Add New Source** button to add a single PWL source or a bus pattern that contains multiple PWL sources. For bus pattern, the usage of high or low levels, cycle time, rise time, and fall time are indicated in the following figure:



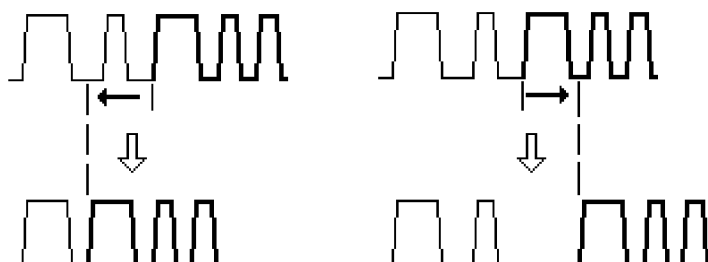
The source list window keeps a list of all PWL sources. The selected PWL source is the active source highlighted with thick line. Click **Modify a Source** to modify the name and parameters of the active source. Click **Display Source** to toggle the active source on and off.

Delete the active source with the **Delete a Source** button, or clear all sources with the **Delete All** button.

Point Operations

A newly added source has five points. Click on a point to select the point as the active point. You can insert a new point after the current active point using the **Insert a Point** button, or insert a pattern using the **Insert Pattern** button. A point can also be directly added with the Time and Value text fields and the **Add** button. The newly added point becomes the active point highlighted with a solid box. With multiple points in a source, click the **Next** or the **Prev** button to jump the active point of a source.

To shift the entire PWL pattern forward or backward after the active point, click the **Shift Pattern** button, and drag the active point horizontally to shift the pattern. For example:



The Time and Value fields are updated when a new active point is selected. To modify the values of the active point, enter new values and click the **Modify** button to make the change. You can also change the values of any point by dragging it directly. Dragging a point also selects it as the active point.

Click the **Delete a Point** button to delete the active point.

To assist point placement, snap functions can be enabled along the horizontal and vertical direction. Enter the preferred snap distance and check the corresponding option button to enable the snap function.

To zoom in an area, click and hold the mouse and move the mouse pointer to select the zoom area. Click the **Zoomout** button to zoom out or the **Full** button to reset to full viewing range.

Using the ADC Toolbox

To open the **ADC Toolbox**, choose **ADC Toolbox** from the menu located in the **Utilities** section of the **Waveform** tab. Simplified SNR/THD/SNDR/ENOB calculation functions, for sine input signals, are also provided in the FFT dialog box and the equation builder.

The ADC Toolbox is a built-in tool for detailed performance analysis of A-to-D converter designs. Based on simulation output waveform data or measured data, the ADC Toolbox reports the following performance merits:

- DC static characteristics—Histogram, INL/DNL min/max values, and their standard deviations.
- AC dynamic characteristics—Noise parameters SNR, THD, SNDR, SFDR and ENOB and the frequency-domain power spectrum.

This section contains information on the following topics:

- [Generic Versus Coherent ADC Toolbox](#)
- [Overview of the ADC ToolBox](#)
- [Input Signal of ADC Toolbox](#)
- [DC DNL/INL Analysis](#)
- [Sine Input](#)
- [Ramp Input](#)
- [Number of Sample Points and INL/DNL](#)
- [Sampling Frequency and INL/DNL](#)
- [AC Analysis: Coherent Sampling Versus Window Sampling](#)
- [Coherent Sampling](#)
- [Window Sampling](#)
- [Exporting the DNL/INL/FFT Results as Waveform Data](#)
- [Saving or Loading an Analysis Setup](#)
- [Dumping the Contents of the ADC Toolbox](#)
- [ADC Toolbox Display Controls](#)
- [Sampled Signal Window](#)
- [Histogram, DNL, and INL Windows](#)
- [Power Spectrum \(Sine Input Only\)](#)
- [AC Dynamic Characteristics](#)
- [DC Static Characteristics](#)
- [Running Batch-mode ADC Analyses](#)

Generic Versus Coherent ADC Toolbox

Before WaveView version 2004.4, the ADC Toolbox only supports the Coherent sampling method for sine input waveform and the input waveform have their signal amplitude in code values (signal values range from 0 to 2^N-1 , where N is the number of bits of the design).

From release 2004.4, a new generic ADC Toolbox is introduced. The new ADC toolbox accepts input signal in arbitrary amplitude, and a more generic Window sampling method is added to eliminate the need to sample the input sine waveform over integer cycles, which is a requirement of the Coherent sampling method.

To use the original Coherent-only ADC Toolbox, set the `SW_SX_ORG_ADC` environment variable to 1 in your shell:

```
setenv SW_SX_ORG_ADC 1
```

Without setting the `SW_SX_ORG_ADC` environment variable, the new generic ADC Toolbox is invoked as the default mode.

The new generic ADC Toolbox also allows the AC and the DC analysis to be carried out separately.

Overview of the ADC ToolBox

The ADC Toolbox window is divided into three tabs: the Input Parameters tab, DC Analysis Result tab, and AC Analysis Result tab.

The Input Parameter tab acquires input parameters from users for an ADC analysis task. The input parameters are divided into four categories: General input signal properties and data sampling parameters, DC analysis parameters, AC analysis parameters, and AC Coherent sampling parameters.

The DC Analysis Result tab displays sampled waveform and DC result including histogram, DNL and INL plots/distribution.

The AC Analysis Result tab displays the FFT spectrum of the sine input waveform and AC figure of merits including SNR/THD/SNDR/SFDR/ENOB.

Input Signal of ADC Toolbox

The ADC Toolbox accepts the following input signals:

- Sine: Both AC FFT analysis and INL/DNL analysis are available for the sine input type.
- Ramp: Only the INL/DNL analysis is available for the ramp input type.

The input signal of the ADC Toolbox can be an analog signal with arbitrary amplitude ranges, or a digital bus signal representing the digital output code of an ADC design. For digital bus input signal, the bus signal must be displayed in a waveview panel. Drag an analog signal from the Waveform Browser, or a bus signal from a panel to the Signal field in the ADC Toolbox.

The input signal is sampled based on one group of the following three sampling parameters: Start Time, Stop Time, and number of points, or Start Time, Sampling

Frequency (Fs), and number of points. When the stop time is used (in the first group), the last sample point is at:

```
stop_time - (stop_time - start_time)/number_of_points
```

For sine input type, the input frequency (Fin) also needs to be specified.

You can input analog bus signals directly by entering bus named delimited by semicolons (;). For example:

```
ADCOUT_3;ADCOUT_2;ADCOUT_1;ADCOUT_0
```

When you enter analog bus signals directly, the following requirements must be satisfied:

- Signals must be analog.
- The number of signals must be the same as the number of bits.
- The order bus signals must be MSB ; ; LSB.
- All signals in the bus must come from the same file.
- No more than 32 signals can be entered at a time.

Once the input signals are accepted, the median signal value among all signals is used as the A2D conversion threshold. For example:

```
0.5 * (ymax_of_all_bit + ymin_of_all_bits)
```

The analog bus signals are then converted to a logic bus whose code values are then used as the input to the ADC Toolbox. Analog bus signals are also supported during .INI file save/load operations.

For measured data that was sampled at known Fs, the data (x,y) value pair can use integer index as the x values. If this is the case, select the **Measured data** option and the ADC Toolbox processes the input as if the data is sampled at Fs. When the **Measured Data** option is selected, the Start T/P value is the starting index (0-based) of the measured data.

If you have a measurement data with only single column, (that is, the data is measured at fixed known sampling rate so the data file does not contain x-axis sample time information), edit the file header to add the following lines:

```
#format table data_only  
  
name_of_data  
  
measure data 1  
  
measure data 2  
  
....
```

With the keywords in the header line, the WaveView tool can load such a text data file as a regular waveform file and automatically assign integer index to the x-axis. You can then apply the **Measured Data** option and use the measurement data in the ADC Toolbox.

DC DNL/INL Analysis

To calculate DNL/INL, an analog input signal is first scaled into full code range based on the **Code Range** parameters in the DC Input Parameters section. If the **Number of Bits** value is used, the ADC Toolbox assumes that the input signal covers full code range from 0 to 2^N-1 . If the **User-defined Code Range** option is selected, the ADC Toolbox assumes that the input signal swings from the **Min** code value to the **Max** code value. Digital input bus signal is not subject to this amplitude scaling, instead the bus direct code values are used.

If the analog input signal value is already converted to the A/D converter output bus values, select the **Signal Value=Output Code** option.

Because the ADC Toolbox always automatically scales the analog input signal, PWL sampling is always used in sampling analog input signal.

After entering the necessary DC analysis parameters, click **Evaluate DC** to carry out a DC analysis.

Sine Input

For sine input signal, based on the sampled data and the F_{in} parameters, the ADC Toolbox needs to find the ideal sine wave to calculate INL and DNL. The ADC Toolbox uses two methods to calculate the ideal sine waveform:

- FFT fitting: If the input sine waveform data is not clipped. The amplitude and phase of the strongest FFT spectrum component of the input sine signal is used as the ideal sine waveform.
- LMS (least-mean-square) fitting: If the input sine waveform data is clipped. This is often seen with measure data. The ADC Toolbox uses the sample data within the Clip-Min and Clip-Max code values, and LMS fitting method to find the original ideal sine waveform. The offset and the amplitude of the fitted ideal sine waveform is displayed in the Sampled Data plot.

With the LMS fitting, since the clipped sampled data might be heavily distorted, FFT AC analysis is disabled.

Ramp Input

For the ramp input type, ADC Toolbox always assumes that the ideal ramp signal evenly covers the full code range. Based on this assumption, multiple complete cycles of ramp (up or/and down) are accepted.

Number of Sample Points and INL/DNL

In order to produce meaningful INL/DNL result, the number of sample points must be large enough so that there is at least one sample point in each code value. For example, for a 10-bit design, ramp input, at least 1024 points (210) sample points are needed. Sine input type needs even more points because the sample points do not distribute evenly over different code values.

Since the ADC Toolbox scales the original analog input signal and uses PWL method to sample the scaled input signal, it is the user's responsibility to make sure that the measurement/simulation generates enough true data points at each code value.

Sampling Frequency and INL/DNL

For sine input type, if multiple periods of the input sine waveform are sampled, the sampling frequency F_s and input frequency F_{in} must be carefully selected to avoid sampling the same code values in each period. For example, if a 5 MHz input sine waveform (6-bit) is sampled at 40 MHz over 128 cycles, the 8 sampled code values from each cycle are always the same. As a result, only 8 out of the 64 code values are available in the sampled data. If F_{in} is at 4.99 MHz, with the sampling frequency, the sampled data has different code values from cycle-to-cycle, and the INL/DNL result would be more meaningful.

AC Analysis: Coherent Sampling Versus Window Sampling

With the sine input type, the ADC Toolbox supports FFT AC analysis using two different types of sampling method:

- [Coherent Sampling](#)
- [Window Sampling](#)

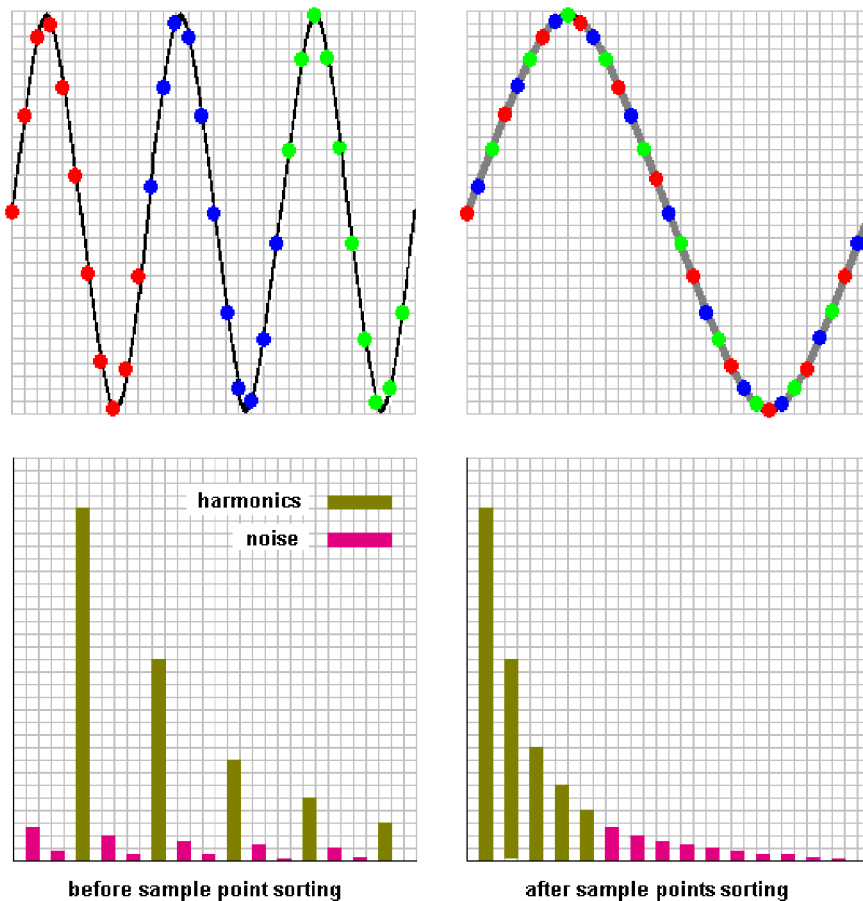
Coherent Sampling

With coherent sampling, exact integer multiple cycles (F_{inNUM}) of the input sine wave are sampled. The sampled data is then sorted into a single cycle of sine wave before performing the dynamic performance analysis using FFT.

Advantages of the coherent sampling method include:

- It allows an input signal at higher frequency to be sampled using a lower sampling rate. In the real world, physical measurement equipment might sometimes have limited sampling performance to meet the Nyquist criterion required by the bandwidth of the input signal.
- Harmonics and noise are readily separated in the power spectrum using coherent sampling.
- FFT windowing functions are not required because integer cycles of the input signal are sampled.

Figure 3 Coherent sine wave sampling



However, coherent sampling method does require the exact frequency (F_{in}) of the input sine wave to be known prior to running an ADC simulation. To use the coherent sampling method in the ADC Toolbox, you need to enter the preferred Precision, the estimated F_{in} and F_{clk} , and Sample Pts. Then, click **Evaluate AC** to find out the exact values for F_{in} .

Figure 3 is a coherent sampling example with $\text{FinNUM} = 3$, $N=32$. Coherent sampling requires that $N \times \text{Fclk} = \text{FinNUM} \times \text{Fin}$. In order to sort sampled points of multiple cycles into a single sine wave cycle, N and FinNUM cannot have any common factor. Since N is always power of 2, an odd number for FinNUM is always a valid choice. In the ADC Toolbox, the FinNUM value can be automatically determined based on user-defined Fin and Fclk values.

As depicted in the above graph, the power spectrum of the sorted sine wave is also a sorted version of the original multi-cycle sine waveform's power spectrum.

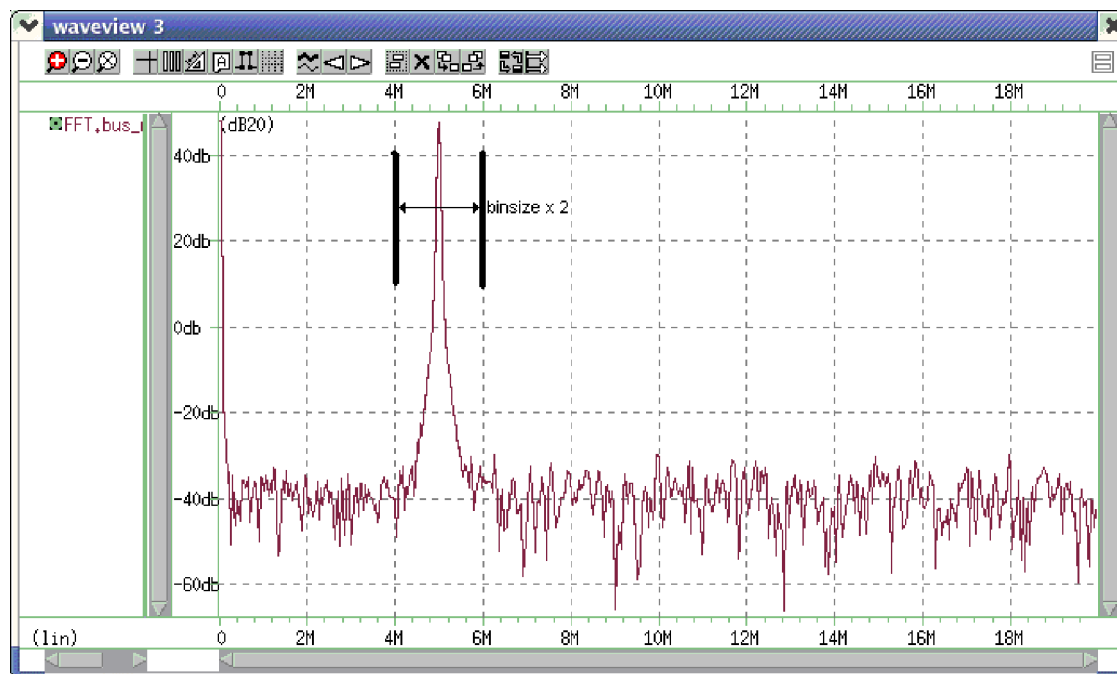
The sorted single-cycle sine wave is sometime also referred to as the unfolded sine wave.

Refer to [Using the Coherent Sample Only \(CSO\) ADC Toolbox](#) for details about entering parameters for Coherent Sample.

Window Sampling

The window sampling method does not require integer number of cycles of the input sine signal to be sampled. However, to overcome the spectral leakage problem from not sampling over an integer cycles of the input waveform, a FFT windowing function is required. **Figure 4** shows the FFT spectrum of a 4.99 MHz sine signal sampled over 127.75 cycles at 40 MHz.

Figure 4 *FFT Spectrum With Window Functions*



As depicted in the above spectrum, the power at the fundamental frequency (4.99 MHz) gets spread into neighboring spectral components due to spectral leakage. For this reason, a binsize parameter is required to include all neighboring components within +/- binsize spectral range as part of the original fundamental (or harmonic) components. The default of BinSize is 10.

After entering the necessary AC analysis parameters, click **Evaluate AC** to start an AC analysis.

Exporting the DNL/INL/FFT Results as Waveform Data

The ADC Toolbox result, including the sampled waveform data, histogram, INL/DNL and the FFT power spectrum, can be exported to a waveview window for further viewing purpose. Click **Waveform** next to the plot window to display the data into a new waveview window.

Saving or Loading an Analysis Setup

The ADC Toolbox setup, including the input parameters, sampled data, and the analysis result can be saved into data files. To restore a previously saved setup, load from the .INI (Input parameter) file.

Dumping the Contents of the ADC Toolbox

To output the contents of the ADC Toolbox to a graphic file (PNG, JPG, or BMP), click the **Dump** button on the ADC Toolbox window.

ADC Toolbox Display Controls

The following window displays are included in the ADC toolbox:

Sampled Signal Window

The Sampled Signal window displays the sampled data waveform based on the input parameters. The y-axis is in code value, with min-code value at the bottom and max-code value at the top.

Ideal/Acquired/Error/All switch:

- Ideal: Displays the ideal signal used for INL/DNL calculation.
- Acquired: Displays the sampled waveform data.

- Error: Displays the error between ideal signal and the sampled data.
- All: Displays all of the above.

Sin/Cos/Original (sine input only):

- Sin: Adjusts the phase of the displayed waveform into Sin phase (starts from zero).
- Cos: Adjusts the phase of the displayed waveform into Cos phase (starts from the maximum value).
- Original: No phase adjust for sampled waveform display.

Exp/Integer:

- Exp: Displays the y-axis value ranges using the engineering format.
- Integer: Displays the y-axis value ranges using the integer code values.

Histogram, DNL, and INL Windows

These windows display the histogram, DNLb, and INL plots of the sampled signal. Aligned with the y-axis of the Sampled signal plot, the y-axes of all three plots correspond to the code value. The x-axis is the histogram / DNL / INL value in LSB.

Power Spectrum (Sine Input Only)

The power spectrum window displays the noise spectrum of the sampled signal.

Line/Bar/Both

- Line: Plots the power spectrum using line chart.
- Bar: Plots the power spectrum using bar chart.
- Both: Plots the power spectrum using both line and bar chart.

Index/Frequency

- Index: Displays the spectrum x-axis value range using harmonic index.
- Frequency: Displays the spectrum x-axis value range using frequency.

AC Dynamic Characteristics

The AC Characteristics window displays the spectrum harmonic strength and the following values:

- SNR: The signal-to-noise ratio.
- THD: The total harmonic distortion.
- ENOB: The effective number of bits.
- SNDR: The signal-to-noise and distortion ratio.
- SFDR: The spurious free dynamic range. The harmonic index of the second strongest harmonic component is also displayed together with the SFDR value (for example, SFDR index_after_sorting (index_before_sorting)).

The strength of the power spectrum is in db relative to the strongest harmonic component (0 db for the strongest harmonic component). The AC characteristic window lists the value of each of the harmonic component. Use the **up** and **down** buttons to scroll up/down to view the spectrum values at different harmonic range.

Select the toggle check box next to a harmonic component to either exclude the harmonic for the THD calculation or include the harmonic for the SNR calculation.

Click **Evaluate AC** to update the AC noise analysis result (SNR/THD/ENOB/SNDR/SFDR) based on the new harmonic selection.

DC Static Characteristics

The DC Characteristics window displays the min/max values and the standard deviation of the INL/DNL distribution. The unit is 1 LSB.

Running Batch-mode ADC Analyses

ADC Toolbox analysis can be run in batch mode with the -adcin, -adcout,

-adcwdf, and -adccsv command line option. The usage is:

```
wv -adcin adc.INI -adcout out_sw -adcwdf wdf_sw
```

The adc.INI is a previously saved .INI file, out_sw is the _ (underscore) delimited type switch that controls the output text files; wdf_sw is the _ (underscore) delimited type switch that controls the output WDF waveform files. The -adccsv option sets the output text file in the CSV (comma separated values) format.

One of the -adcout and -adcwdf options must be specified or the batch mode operation is terminated. Supported type strings are INL, DNL, FFT, TLA, HIS, IND, ALL, and DAT (-adcout only). FFT and TLA have no effect on ramp input signals. For example:

```
wv -adcin adc.INI -adcout INL_DNL -adcwdf TLA_FFT -adccsv
```

Using the Coherent Sample Only (CSO) ADC Toolbox

To start the ADC Toolbox in the CSO mode, set the `SW_SX_ORG_ADC` shell environment variable.

This section contains information on the following topics:

- [Input Signal Requirements](#)
- [Preparing the Input Waveform](#)
- [Sampling the Input Waveform](#)
- [Preparing Test Benches for the CSO ADC Toolbox](#)
- [Input Signal Strength](#)
- [Input Sine Wave Frequency](#)
- [CSO ADC Toolbox User Interface](#)
- [Input Parameters](#)
- [Selecting Sampling Parameters](#)
- [Sine Input](#)
- [Ramp Input](#)
- [Exporting DNL/INL/FFT Results as Waveform Data](#)
- [Saving or Loading an Analysis Setup](#)
- [ADC Toolbox Display Controls](#)
- [Sampled Signal Window](#)
- [Histogram, DNL, and INL Windows](#)
- [Power Spectrum \(Sine Input Only\)](#)
- [AC Dynamic Characteristics](#)
- [DC Static Characteristics](#)

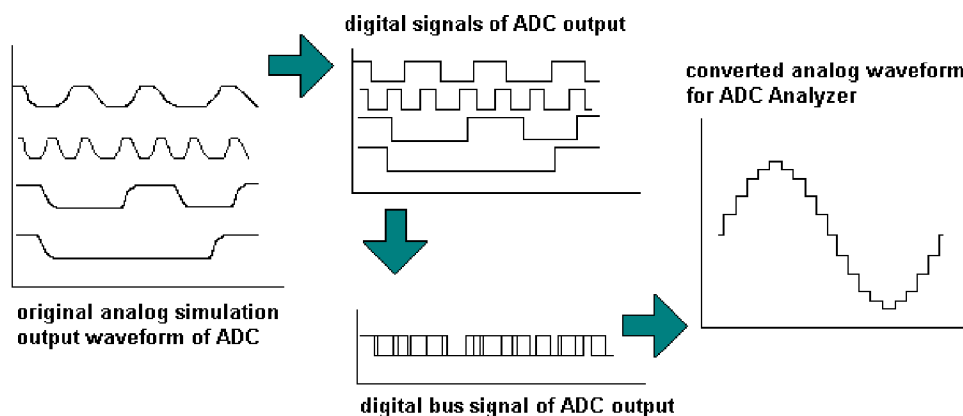
Input Signal Requirements

The input waveform for the CSO ADC Toolbox must have its signal strength in integer code values (for example, 0 to 63 for a 6-bit ADC). Run a simulation on an ADC design first and then convert the ADC output codes into an acceptable waveform.

Preparing the Input Waveform

As depicted in the flowchart below, the following steps demonstrate how to use the WaveView tool to convert the simulated ADC output code (in analog waveforms) to an acceptable input signal for the CSO ADC Toolbox.

Figure 5 Preparing an input signal for the ADC toolbox



To convert the simulated ADC output code to an acceptable input signal for the CSO ADC Toolbox:

1. Identify the analog output waveforms that represent the ADC output code, for example, "ad0" to "ad5" for a 6-bit design.
2. Display these signals in a waveview window and then highlight them by highlighting the panels these signals reside in.
3. Use the A/D converter tool in the WaveView single-bit mode to convert each of the highlighted analog waveforms to a digital waveform. Name the converted digital waveforms "AD0" - "AD5", and ensure proper threshold value is used for this A to D conversion step.
4. Group the converted logic signal "AD0" - "AD5" into a digital bus signal. Make sure that the signals are ordered (sorted) correctly so "AD5" is the MSB and "AD0" is the LSB.
5. Rename the grouped bus signal to "AD". You can use this logic bus signal directly as the input to the ADC Toolbox, or you can continue to [Step 6](#).

To use the logic bus signal as the input, the bus signal must be displayed in a waveview window, and you must drag and drop the logic bus signal from the display panel.

6. Use the D/A convertor tool in the WaveView tool to convert the grouped converted digital bus "AD", in multi-bit mode, into an analog waveform.
7. Ensure that the **Use logic bus value** option is selected so that the converted waveform has strength in code value.
8. Rename the final analog waveform "ad".

The analog waveform "ad" is now ready for use with the ADC Toolbox. Skip [Step 1](#) to [Step 3](#) if your simulator generates logic output waveforms directly. If your simulator directly generates a logic bus waveform, skip [Step 1](#) to [Step 5](#).

To specify the input signal, drag and drop the converted analog signal from the Waveform Browser to the Signal field in the ADC Toolbox.

Sampling the Input Waveform

The CSO ADC Toolbox samples the input signal waveform based on the following equation:

$$N \cdot F_{in} = F_{inNUM} \cdot F_{clk}$$

where:

- N is the number of sampling points (power of 2)
- F_{in} is the input frequency
- F_{inNUM} is the total number of cycles of a sine signal input used for coherent sampling.
- F_{clk} is the sampling frequency

Note:

N and F_{inNUM} must not have any common factors.

For sine wave input, the number of sample points N must to be power of 2 because FFT is used internally to calculate the ADC dynamic performance. It is important to choose a value for N , which is large enough to produce at least one sample point for each output code value. In the ADC Toolbox, PWC sampling is always used to sample the input signal waveform.

Preparing Test Benches for the CSO ADC Toolbox

To maximize the effectiveness of the CSO ADC Toolbox, the following guidelines can be used to prepare the input test tone for simulating an ADC design.

Input Signal Strength

For both ramp and sine input type, the ideal input signal strength drives the ADC output over a full swing output code from 0 to $2^{\text{BIT}}-1$, where BIT is the number of bits. On the other hand, the input signal cannot be too strong, because it can overdrive the ADC outside its operating range. Proper input signal strength results in more accurate INL/DNL analysis results.

Input Sine Wave Frequency

In general, depending on the speed performance of the ADC design, the input test tone frequency might be low enough so that the digital output does not skip codes. (All output codes can be exercised by the input test tone.) For coherent sine wave sampling, the exact input test tone frequency F_{in} must be determined before running the ADC simulation.

CSO ADC Toolbox User Interface

The CSO ADC Toolbox accepts user input parameters and generates graphic and textual analysis report.

After completing an analysis based on user input parameters, the CSO ADC Toolbox allows you to perform the following actions:

- Display analysis results in regular waveview windows
 - Export analysis result to text files
 - Save/load the analysis parameters setup
-

Input Parameters

The following table summarizes the input parameters required by the CSO ADC Toolbox.

Input Signal Type	Sine	Ramp
Start Time	mandatory	mandatory
Precision	mandatory	not required

Input Signal Type	Sine	Ramp
Decimation	mandatory	not required
Sample Points	mandatory	mandatory
Fclk User	mandatory	mandatory
Fin User	mandatory	not required
Fin #	optional	not required
Fclk	optional	not required
Fin	optional	not required
Bits	mandatory	mandatory
THD	mandatory	not required
SNR	mandatory	not required

- **Start Time:** The time at which the ADC Toolbox starts sampling the input signal.
- **Precision:** Number of digits after the decimal point for the internal Fclk, Fin and Fin# calculation. For the sine wave coherent sampling method, small error in Fclk and Fin can result in inaccurate analysis result. If the **Use user Fclk/Fin** option is not selected, the ADC Toolbox recalculates the precise values internally for Fclk, Fin, and Fin#.
- **Decimation:** Integer decimation-in-time. Default is 1. The decimation is used to mimic the sampling speed limitation of real measurement equipment by slowing down the sampling clock by the integer decimation factor. The actual sampling clock frequency is Fclk/decimation. When the ADC Toolbox is used on simulation data, set the value to 1.
- **Sample Points:** Number of sample points. Since the ADC Toolbox uses FFT for noise analysis, the number of sample points must be power of 2.
- **Fclk user:** The frequency of the sampling clock.
- **Fin user:** The frequency of the input sine wave. Ignored for ramp input.
- **FinNUM:** The total integer number of cycles of input sine signal for coherent sampling. Required only when the **Use user Fclk/Fin** option is selected. It is otherwise calculated internally based on other input parameters. Ignored for ramp input.

Note:

If the **Use user Fclk/Fin** option is selected, the Fclk-user value is used directly as the sampling frequency for acquiring data from the input signal. The input data is sampled for a total span of Sample-points / Fclk-user.

If the **Use user Fclk/Fin** option is not selected, the ADC Toolbox uses Precision, Fin-User, Fclk-User, Sample Points, and Decimation to calculate the precise internal Fclk, Fin, and FinNUM. In this case, you only need to provide approximate Fclk and Fin values. The final internally calculated Fclk, Fin, and FinNUM values are displayed in corresponding fields.

- **Bits/Min/Max:** Number of bits of the AD converter design, and the minimal and maximal code value (MIN/MAX). The MIN/MAX values, instead of the Bits value, are used if the **Use Code Value Range** option is selected. If the **Use Code Value Range** option is not selected, the Bits value is used and the code value is assumed to range from 0 to $2^{\text{Bits}} - 1$. Any sampled code values that are outside this range are clipped at the MIN/MAX values.
- **THD:** The highest harmonic order that is included as harmonic for the total harmonic distortion calculation.
- **SNR:** The highest harmonic order that is excluded from noise for the signal -to-noise ratio calculation.
- **Indexed Data:** If the input data has been sampled at a fixed rate without the x-axis time information for each of the data points (for example, measurement data), select this option and the ADC Toolbox assumes that the data is sampled at Fclk.

After entering all the required input parameters and the options, click **Evaluate** to complete the analysis.

Selecting Sampling Parameters

The following sampling parameters are available:

- [Sine Input](#)
- [Ramp Input](#)

Sine Input

To select a sine input sampling parameter:

1. Select the Sine option.
2. Enter a Sample Pts value.
3. If the **Use user Fclk/Fin** option is selected, enter proper Fclk, Fin, and FinNUM values. These values are used directly by the waveform sampling process.
4. If the **Use user Fclk/Fin** option is selected, enter proper Fclk, Fin, and FinNUM values. These values are used directly by the waveform sampling process.

5. If the **Use user Fclk/Fin** option is not selected, enter the preferred Precision value and estimated Fclk and Fin values. The FinNUM value and the exact Fclk and Fin values are calculated automatically.
6. Select the **Sort Sampled Points** option to view the unfolded power spectrum of the sorted sampled waveform, or de-select the **Sort Sampled Points** option to view the folded power spectrum of the unsorted sampled waveform. You need to click **Evaluate** every time the **Sort Sampled Points** option is changed.

Ramp Input

Select the **Ramp** option and enter the Start, Sample-Pts, and Fclk values.

Exporting DNL/INL/FFT Results as Waveform Data

The ADC Toolbox result, including the sampled waveform data, histogram, INL/DNL and the FFT power spectrum, can be exported to a waveview window for further viewing purpose. Select the preferred data type switches in the ADC Toolbox and click **Waveform** to display the data into a new waveview window.

Saving or Loading an Analysis Setup

The ADC Toolbox setup, including the input parameters, sampled data and the analysis result can be saved into data files. To restore a previously saved setup, load from the .INI (Input parameter) file.

ADC Toolbox Display Controls

The following ADC toolbox display controls are available:

- [Sampled Signal Window](#)
- [Histogram, DNL, and INL Windows](#)
- [Power Spectrum \(Sine Input Only\)](#)

Sampled Signal Window

The Sampled Signal window displays the sampled data waveform based on the input parameters. The y-axis is in code value, with min-code value at the bottom and max-code value at the top.

Ideal/Acquired/Error/All switch:

- Ideal: Display the ideal signal used for INL/DNL calculation.
- Acquired: Display the sampled waveform data.
- Error: Display the error between ideal signal and the sampled data.
- All: Display all of the above.

Sin/Cos/Original (sine input only)

- Sin: Adjust the phase of the displayed waveform into Sin phase (starts from zero).
- Cos: Adjust the phase of the displayed waveform into Cos phase (starts from the maximum value).
- Original: No phase adjust for sampled waveform display.

Exp/Integer

- Exp: Display the y-axis value ranges using the engineering format
- Integer: Display the y-axis value ranges using the integer code values

Histogram, DNL, and INL Windows

These windows display the histogram, DNL, and INL plots of the sampled signal. Aligned with the y-axis of the Sampled Signal plot, the y-axes of all three plots correspond to the code value. The x-axis is the histogram / DNL / INL value in LSB.

Power Spectrum (Sine Input Only)

The power spectrum window displays the noise spectrum of the sampled signal.

Line/Bar/Both

- Line: Plot the power spectrum using line chart.
- Bar: Plot the power spectrum using bar chart.
- Both: Plot the power spectrum using both line and bar chart.

Index/Frequency

- Index: Display the spectrum x-axis value range using harmonic index.
- Frequency: Display the spectrum x-axis value range using frequency.

AC Dynamic Characteristics

The AC Characteristics window displays the spectrum harmonic strength and the following values:

- SNR: The signal-to-noise ratio
- THD: The total harmonic distortion
- ENOB: The effective number of bits
- SNDR: The signal to noise and distortion ratio
- SFDR: The spurious free dynamic range. The harmonic index of the second strongest harmonic component is also displayed together with the SFDR value (for example, SFDR index_after_sorting (index_before_sorting)).

The strength of the power spectrum is in db relative to the strongest harmonic component (for example, 0 db for the strongest harmonic component). The AC characteristic window lists the value of each of the harmonic component. Use the **up** or **down** buttons to scroll up/down to view the spectrum values at different harmonic range. Select the toggle check box next to a harmonic component to exclude the harmonic for the THD calculation or include the harmonic for the SNR calculation.

Click **Evaluate** to update the AC noise analysis result (SNR/THD/ENOB/SNDR/SFDR) based on the new harmonic selection.

DC Static Characteristics

The DC Characteristics window displays the min/max values and the standard deviation of the INL/DNL distribution. The value is in LSB units.

7

Customizing the WaveView Tool

This chapter details how to customize options in the WaveView tool.

This chapter contains information on the following topics:

- [Setting Key Bindings \(Shortcut Keys\)](#)
- [Setting Preferences](#)
- [Mapping Bus Values](#)

Setting Key Bindings (Shortcut Keys)

To open the shortcut key settings, click **Settings** on the **Waveform** tab, and choose **Key Bindings** from the menu that opens. The BindKey Setting Finder window opens.

This section describes the following topics:

- [Customizing Shortcut Keys](#)
- [Top Menu-Operated Shortcut Key Functions](#)
- [WaveView Toolbar Shortcut Key Functions](#)
- [Other Shortcut Key Functions](#)

Customizing Shortcut Keys

Shortcut key settings can be customized for key action in the WaveView windows.

The following shortcut keys are allowed:

- A to Z
- 0 - 9
- Ctrl+A to Ctrl+Z
- Ctrl+0 to Ctrl+9

- Shift
- Esc
- Delete
- Backspace
- Up Arrow
- Down Arrow

Except for the five function keys (Esc, Delete, Backspace, Up, and Down), a key-down action and a key-up action can be associated with each key.

The available actions can be categorized into the following types:

- `menuFunctionName`: Equivalent to the top menu actions.
- `wctrlFunctionName`: Equivalent to the waveview control button actions.
- `wcmdFunctionName`: Special waveview command actions.

The Bind-Key dialog box automatically flags warnings if the selected action is not allowed for a key event. Select your auto-repeat and action preferences and click **Apply** to apply the change or click **Save** to apply and save the change.

Reading the Key Binding File

The key binding file must be named `.spxkey`. The WaveView tool can read this file from these different locations, with the following precedence:

- User-specified file name
- `-spxkey` option usage
- `current_working_directory/.spxkey`
- `$(SX_HOME)/.spxkey` (if `SX_HOME` is defined)
- `$(HOME)/.spxkey` (if `HOME` is defined)

The WaveView tool can also read the `.sae.spxkey` file with the following precedence:

- `$(SX_HOME)/sae.spxkey` (if `SX_HOME` is defined)
- `$(HOME)/sae.spxkey` (if `HOME` is defined)

`SX_HOME` is a user-defined environment variable; `HOME` is a shell variable which defaults to a user's home directory.

Settings read first are overridden by those read later in the sequence.

Top Menu-Operated Shortcut Key Functions

These functions can be also invoked from the top menu. The current shortcut key setting is indicated at the right end in the associated menu box.

`menuLoadWaveformFiles`

Opens the **Open: Waveform Files** dialog box.

`menuUpdateAll`

Updates all loaded waveform files.

`menuPrintWV`

Opens the **Print Setup** dialog box.

`menuExportWV`

Opens the **Waveform Export Parameters** dialog box.

`menuAbortLoading`

Aborts the current waveform file loading process.

`menuShowDirTable`

Opens the **Directory/File Table** dialog box.

`menuQuitApp`

Terminates the current WaveView process.

`menuLoadSession`

Opens the **Load Session** dialog box.

`menuSaveSession`

Opens the **Save Session** dialog box.

`menuNewWaveView`

Adds a new waveview window.

`menuUndoWaveView`

Undoes the previous waveview operation.

`menuDeleteWaveView`

Deletes the active waveview window.

`menuDeleteAllWaveView`

Deletes all waveview windows.

`menuRenameWaveView`

Renames the active waveview window.

`menuRefreshWaveView`

Refreshes the content of the active waveview.

`menuToggleWVDocking`

Toggles waveview between dock/undock modes.

`menuToggleBrowser`

Shows/hides the signal browser.

`menuAddCrossCursor`

Adds a cross-cursor. The default shortcut key is **CTRL+g.**

`menuAddPanelLinear` `menuAddPanelSmith` `menuAddPanelPolar`
`menuAddPanelEyeDiagram` `menuAddPanel2Dsweep` `menuAddPanel3Dsweep`
`menuAddPanelHistogram`

Adds an empty panel in various types.

`menuDeletePanels`

Deletes all selected panels.

`menuUngroupPanels` `menuGroupPanels`

Ungroups/groups all selected panels.

`menuSelectAllPanels` `menuUnselectAllPanels`

Selects/unselects all panels in the active waveview.

`menuFitAllPanels`

Fits all selected panels to waveview height.

`menuDefHAllPanels`

Resets all selected panels to default height.

`menuSetRadixBinary` `menuSetRadixOctal` `menuSetRadixHex`
`menuSetRadixDecimal` `menuSetRadixASCII`

Sets radix of all selected panels to various mode.

`menuShowShortVector` `menuShowFullVector`

Sets logic values of all selected panels to the shortest mode or the full bus width mode.

`MenuShowAlias menuShowOrgName`

Sets signal names in all selected panels to using alias or original signal name.

`menuSetStyleVCD menuSetStyleRTF`

Sets logic value style to the standard VCD logic or the Synopsys/Nassda HANEX RTF-style logic.

`menuUndoZoom menuRedoZoom`

Undoes/redoes the previous zoom/unzoom operation.

`menuSetLinearXAxis menuSetLogXAxis menuSetDBXAxis (db20)`
`menuSetLinearYAxis menuSetLogYAxis menuSetDBYAxis (db20)`

Sets the x/y axis type of selected panels in the active waveview.

`menuSetZoomRange`

Opens the **Zoom Settings** dialog box for manual zoom.

`menuResetZoom`

Fully unzooms selected panels in the active waveview.

`menuConfigSelectedYAxis`

Opens the **Setting Y Axis Full Range** dialog box.

`menuConfigYAxisDefault`

Opens the **Default Y Axis Range** dialog box.

`menuConfigXAxisVariable`

Opens the **Setting X Variable** dialog box.

`menuJumpCursorForward menuJumpCursorBackward`

Jumps the main cursor forward/backward.

`menuCallSignalFinder``menuCallFFT` `menuCallD2A` `menuCallA2D`
`menuCallDataReduction` `menuCallCalculator` (**calculator**) `menuCallDynamicMeter`
(**measurement tool**) `menuCallPWLEditor` (**PWL editor**)

Opens various analysis tools.

`menuShowSelectedSignals menuHideSelectedSignals`

Shows/hides all highlighted signals in selected panels of the active waveview.

WaveView Toolbar Shortcut Key Functions

These functions can be also invoked from the mini toolbar in each WaveView window.

`wbtnScanNext wbtnScanPrevious`

Scans signals in selected panels of the active waveview to the next/previous signal in the same waveform file.

`wbtnToggleDataPoints`

Toggles on/off the data point markers.

`wbtnToggleGrids`

Toggles on/off the X/Y grid lines.

`wbtnAddCursor`

Adds a new cursor in selected panels.

`wbtnAddCursorL`

Moves a left waveform cursor to the closest point on the waveform to the mouse cursor.

`wbtnAddCursorR`

Moves a right waveform cursor to the closest point on the waveform to the mouse cursor.

`wbtnDelete`

Deletes the selected panels.

`wbtnSwitchMainCursor`

Switches the main cursor.

`wbtnAddMonitor`

Adds a new monitor.

`wbtnAddTextLabel`

Adds a text label.

Other Shortcut Key Functions

These functions can be invoked using shortcut keys.

`wcmdAddDiffDMeter`

Adds a [Difference](#) measurement in the active waveform.

`wcmdAddToEquation`

Adds the selected signal to the Calculator.

`wcmdAddXYDMeter`

Adds a **Data(X,Y)** measurement in the active waveform.

`wcmdCopyLastDMeter`

Makes a copy of the last dynamic measurement added in a waveview.

`wcmdRepeatScript`

Repeats the last ACE script executed.

`wcmdTogglePanelFit`

Toggles all selected panels in the active waveview between fit/default panel height.

`wcmdToggleSyncMode`

Toggles the active WaveView between sync/unsync modes.

`wcmdArmBoxZoom wcmdUnarmBoxZoom`

Key-activated box-zoom operations. The Arm and the Unarm functions must be assigned to the key-down and the key-up actions respectively of the same bindkey.

`wcmdArmVertZoom wcmdUnarmVertZoom`

Key-activated vertical zoom operations. The Arm and the Unarm functions must be assigned to the key-down and the key-up actions, respectively, of the same shortcut key.

`wcmdArmHorzZoom wcmdUnarmHorzZoom`

Key-activated horizontal zoom operations. The Arm and the Unarm functions must be assigned to the key-down and the key-up actions, respectively, of the same shortcut key.

`wcmdArmSignalMode wcmdUnarmSignalMode`

Key-activated signal mode operations. The Arm and the Unarm functions must be assigned to the key-down and the key-up actions, respectively, of the same shortcut key.

`wcmdCopyPanels wcmdCutPanels wcmdPastePanels`

Copies, cuts, or pastes panels (or signals) in the active waveview.

`wcmdScrollWVUp wcmdScrollWVDown`

Scrolls the panel stack in the active waveview (stack-mode waveview only) up and down.

`wcmdXPanRight wcmdXPanLeft`

Pans the X-range in the active waveview (stack-mode waveview only) right and left.

`wcmdYPanUp wcmdYPanDown`

Pans the Y-range of all selected panels in the active waveview up and down.

`wcmdSetHorzZoom() wcmdSetVertZoom()`

Sets the zoom direction without zooming out the design.

`wcmdReloadSession`

Reloads the last session file used.

`wcmdQuitZoom`

Aborts a zoom operation during mouse drag.

`wcmdSortSignals`

Sorts signals order (by their names) in all selected panels.

`wcmdDumpScreen`

Dumps the content of the active waveview. Opens the **Save Screen Dump** dialog box.

`wcmdJumpCursorBackward wcmdJumpCursorForward wcmdResetZoom wcmdUndoZoom
wcmdRedoZoom`

These functions are duplicates of the same 'menu' shortcut key functions. They are supported only for backward compatibility.

`wcmdToggleLayoutMode`

Switches between Stack and Row/Column layout modes.

`wcmdSigAdd wcmdSigDiff wcmdSigMultiply wcmdSigDivide`

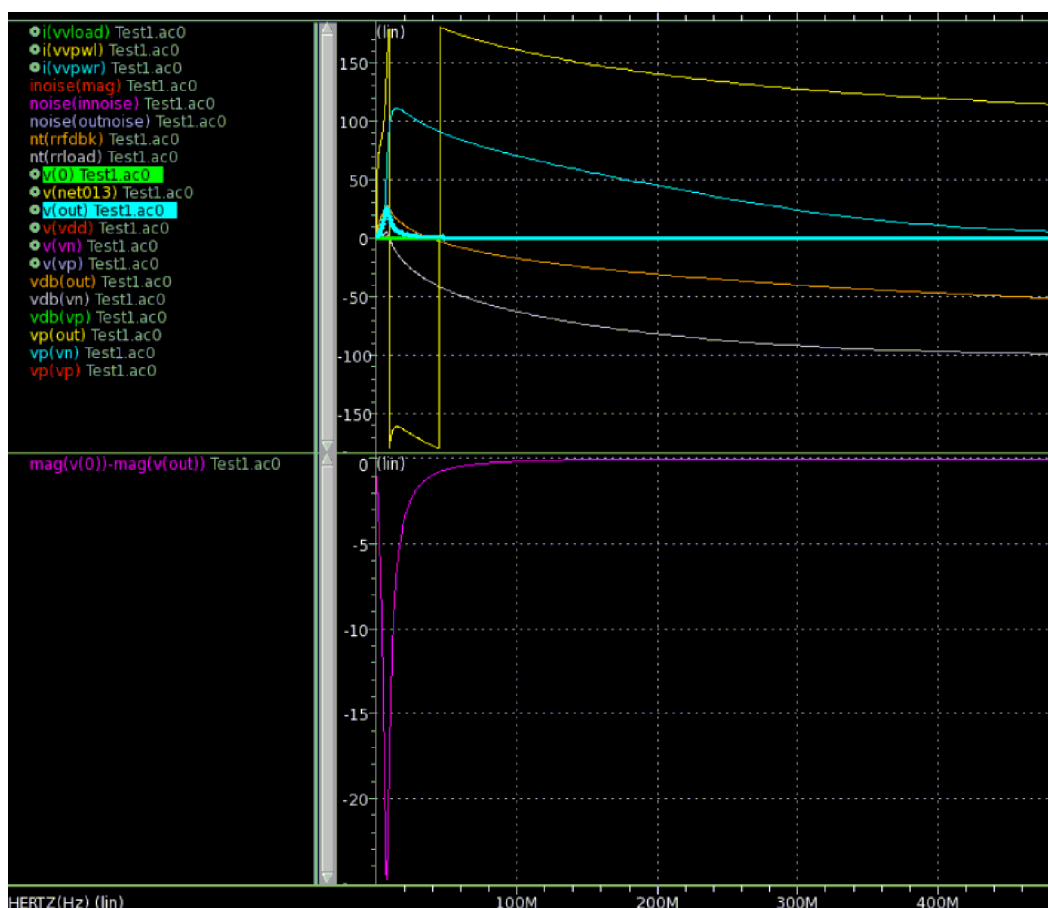
Performs Add (+), Differential (-), Multiply (* - represented as Shift+8 under KEY in the BindKey Settings Finder), and Divide (/) operations on selected single or multiple signals.

Behavior of operations is as follows:

- When a single signal is selected:
 - **Add, Multiply** — show the original signal.
 - **Differential** — show the data of `-signal`.
 - **Divide** — show the data of `/signal`.

- When multiple signals are selected:
 - **Add** — add the signals (order of signals does not matter).
 - **Multiply** — multiply the signals (order of signals does not matter).
 - **Differential** — perform the operation on selected signals in the order top-to-down as they appear in the Signal List.
 - **Divide** — perform the operation on selected signals in the order top-to-down as they appear in the Signal List.

For example, selecting $v(0)$ and $v(out)$ and selecting **Differential** (or the shortcut key -) calculates the differential operation " $v(0) - v(out)$ ".



Setting Preferences

This section contains information on the following topics:

- [General Settings](#)
- [Readers Settings](#)
- [Waveview Settings](#)
- [Panel Settings](#)
- [Signals Settings](#)
- [Axis Settings](#)
- [Color Settings](#)
- [Threshold Settings](#)
- [CDS Link Settings](#)
- [Saving Preference Files](#)
- [Loading Preference Files](#)

General Settings

The following general settings are available:

Option Name in GUI	Option Name	Value (Default)	Description
GUI Font	gui_font	large/(small)	Sets preferred GUI font size.
GUI Colors	gui_color	(bright)/dark	Select default GUI color tone.
Quick Bar Position		left/(right)	Select the position of the Quick Bar.
Canvas Tab Position	waveview_TabPos	(top)/bottom	Controls the position of waveview tabs.
Canvas Background	waveview_background	(black)/white	Select canvas background color.
Display Modeless Dialog	wv_iconizable	raise by default/(always on top)	Controls whether or not modeless windows are displayed on top.

Option Name in GUI	Option Name	Value (Default)	Description
Calculator Error/Warning Message	show_eqn_err_msg	0, (1), 2	0: Not show 1: Show in message window 2: Show in terminal
Use Native Qt File Dialog	use_qtfsb	true/(false)	Control the file loading dialog box style. When enabled, the native Qt dialog box is used, which is the native Windows dialog on Windows platforms.
Hide Ribbon Menu	hide_ribbon	true/(false)	Hides the tool ribbon menu on startup.
Show Console Window	show_console	true/(false)	Displays the Console window.
Suppress Canvas/Panel Delete Confirmation	delete_confirm	(true)/false	Set to false to disable confirm dialog box for waveview/panel delete action.
Disable Application Quit Confirm	exit_confirm	true/(false)	Set to false to disable confirm dialog when quitting.
Archive data files with waveform data files with logfiles on crash	archive_data_on_crash	true/(false)	Archives the <code>sx_crash.sx</code> , <code>sx_crash.log</code> , and <code>sxcmd.log</code> waveform data files to a zip file called <code>data_on_crash.tar.gz</code> . The archive file is created in the working directory
Enable Dynamic Filtering	dynamic_filter	(true)/false	Enables or disables dynamic filtering in the Signal List.
Keep Searching String and Type When Switching Hierarchy	keep_signalsearchstring	(true)/false	Keeps the filter string and type when changing hierarchy.
Include Sub-level Signals in Signal List	include_sub_sig	true/(false)	Includes sub-level signals in the signal list. When the signals have a different hierarchy, it includes signals with full hierarchy path at the top-level.
Suppress Function Completion for Building Equation	suppress_eqn_func_completion	true/(false)	Turns off auto-completion for functions in Equation Builder.

Option Name in GUI	Option Name	Value (Default)	Description
Disable Expanding Hierarchy Tree with Key-in '*'	disable_star_expand_tree	true/(false)	Turns off expanding the whole hierarchy of the sub-level signal when pressing the "shift+*" and selecting the top-level.
Disable Auto-Copy to Clipboard when Selecting Item	disable_autocopy_clipboard	true/(false)	Turns off automatically copying selected items to the clipboard for pasting.
Disable Auto Strip Off Zeros	disable_autostrip0	true/(false)	Turns off automatically removing zeroes.
Number of Significant Digits	precision_digits	integer_value (3)	Sets the number of valid digits for all numeric values. ([3,10]).
Replay Command Delay (ms)	replay_delay	integer_value (1)	Controls how fast the log is replayed.
Default Printout Title	print_title	string	Specifies the title displayed at the top of printouts.
Printer Devices	printer_device	printer_name (null)	Sets the default printer device in the Print Setup dialog box.
Linux Print Command	print_command	print command	Sets the user-defined print command (Linux only).
Waveform File Filter	wdf_fsb_filter	"label:pattern"	Adds a user-defined filter pattern to the WDF loading dialog box. This option can be used multiple times to add multiple filter patterns.
Netlist File Filter	spice_fsb_filter	Spice filename extension:*.sp, *.sp0, *.spi, *.scs (NULL)	Sets filter of file type when importing design from HSPICE Netlist.
Disable Auto Close Hint Message		true/(false)	Enables the pop-up hint window listing which signals could not be found in the FSDB due to changed names.
Default Browser Width	browser_width	integer_value (250)	Sets window width for the browsers. (All five left browsers).
Use width last saved in .spxlast	sx_tv_width	true/(false)	Uses the browser width saved in your .spxlast file.

Option Name in GUI	Option Name	Value (Default)	Description
Default WaveView Geometry	wv_geom	width*height+X+Y	Sets the size of the PrimeWave Design Environment GUI. The next time you open PrimeWave Design Environment, it will be of size width*height at (x,y). Note: Ensure the Use Geometry Last Saved in .spxlast preference is not selected. For example: 800x600+5+10

Readers Settings

The following readers settings are available:

Option Name in GUI	Option Name	Value (Default)	Description
Always Preload Waveforms to RAM	load_data_at_open	true/(false)	Sets WDF data loading schedule. If set to true, all waveform data are loaded when a file is initially open.
Sort Hierarchy Names by Length in OutputView	smart_hsort	true/(false)	Sorts the hierarchy names by length in the OutputView.
Automatically Connect to Subsequent Split Files	auto_connect_split_files	true/(false)	Automatically connects subsequent split files.
Disable Waveform Reloading Confirmation	disable_reload_wdf_confirmation	true/(false)	Disables the waveform reloading confirmation dialog box.
Merge Continuous Identical Logic Values	merge_digital_time	true/(false)	Merges continuous identical logic values.

Session File

Suppress Popup Warning Messages when Loading Session File	suppress_load_session_msg	true/(false)	Suppress warnings for missing signal when loading a session file.
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Option Name in GUI	Option Name	Value (Default)	Description
HSPICE Formats			
Always Remove Duplicated Signals in tr0 Files	remove_dupsig	true/(false)	Removes duplicated signal names in the signal browser.
Case Sensitive HSPICE Waveform Files	case_sens_tr0	(false)/true	Specifies case sensitivity for signal names in HSPICE tr0 files.
Convert S to Y/Z parameters in sc0 and TouchStone Files	convert_s_to_yz	(true)/false	Converts S parameters to Y/Z parameters in sc0 and TouchStone files.
Use S## naming convention when port number < 10 in sc0 and TouchStone files	use_snn	true/(false)	Uses the S## naming convention for files that have less than 10 port.
Use Signal Leaf Name for HSPICE and PSF Files	use_leaf_name	true/(false)	Uses the leaf name in the Signal List Browser.
FSDB Format			
Always Add toplevel for FSDB File	add_toplevel_for_fsdb	true/(false)	Adds toplevel to FSDB files. When disabled, the toplevel is added for FSDB files based on the signals it contains. If signal exists without a scope, a toplevel is added; otherwise, a toplevel is not added.
Add a point to extend waveform data to simulation max time for EPIC fsdb files	extend_waveform	(false)/true	Extends the EPIC-type waveforms in fsdb file to the maximum simulation time.
Filter Out Data Points Whose X Value Is Greater Than Simulation Max Time for FSDB File	filter_byxrange	true/(false)	When enabled, filters out the data points whose x value is out of range for FSDB files. When saving a .spxrc file, the preference uses the variable <code>filter_byxrange</code> , which can be set to false (default) or true.

Option Name in GUI	Option Name	Value (Default)	Description
Display Logic Real Type Signals in PWC by Default for FSDB File	logic_real_pwc_fsdb	(true)/false	Display logic type signals in PWC attribute by default.
Load Scope on Demand When Scope Counts Over	read_top_hier read_top_hier_threshold	(true)/false default is: 5 (million)	Controls FSDB file loading.
Display REAL/AMS WReal Type Signals in PWC by Default for FSDB File	amsreal_pwc_fsdb	true/(false)	When enabled, displays Real/AMS WReal Type signals in FSDB files as PWC.
Other Formats			
Flatten ai_pl File data segments when each segment has only one point	flatten_flags	(true)/false	Flattens the sweep signal in ai_pl files when each signal trace has only 1 point.
Turn off visibility of VHDL processes in VPD file in Signal Browser	not_show_vhdl_processes	(true)/false	Displays a confirmation dialog box when reloading. Otherwise, a confirmation dialog box is displayed first before reloading.
.meas File index X Values Start From 1 instead of 0		true/(false)	X index values start from 1 when reading measurement result files.

Waveview Settings

The following waveview settings are available:

Option Name in GUI	Option Name	Value (Default)	Description
Grid Brightness	grid_brightness	(normal)/low	Sets the background grid brightness for waveviews.

Option Name in GUI	Option Name	Value (Default)	Description
Cycle Color/Marker Per	cycle_color_by	panel/waveview/(panel waveview)/waveform_file	Selects color cycle mode. In the panel mode, the default line color restarts at color 0 for each panel. In the Waveform File mode, the waveform color is cycled when a waveform is from a different waveform file. Waveforms from the same waveform file are plotted in the same color.
Left Button Default	left_button_action	(zoom)/cursor	Sets default action for left mouse button (LMB). In zoom mode, box-zoom operation is the default action when LMB is pressed and dragged. If LMB is released without any drag action, the main cursor (if it exists) is moved to the mouse pointer location. In cursor mode, pressing LMB moves the main cursor to the pointer location. In both modes, if LMB is pressed near a cursor, the cursor is grabbed and becomes the main cursor.
XY-Panel Drop Mode	init_xy_mode	(hvsel)/vonly/honly	Selects the default orientation for XY-panels.
Inf Value Plotting	show_inf_data	(skip)/use value	Control the Inf value display and sets the value for Inf when plotting to WaveView.
Show Testbench Name in Waveview Tab Name	show_tbname_in_wvtab	true/(false)	Shows the testbench name in the waveview tab name. This is useful, for example, when two similarly named analyses are plotted from PrimeWave.
Cursor Operations			
Default Cursor Type	cursor_type	vertical/horizontal/cross	Set the cursor type for the cursor created by the Add Cursor toolbar button and shortcut keys [and].

Option Name in GUI	Option Name	Value (Default)	Description
Move Cursor to Mouse Click Position	click_move_cursor	(true)/false	Moves cursor to x-axis where canvas is clicked.
Enable Cursor Snap to Logic Value Change	logic_cursor_snap	(true)/false	Enables cursor snap function in logic panels if set to true.
Display Value Labels at Crossing Points by Default	show_cross_label	true/(false)	Shows crossing labels along cursor.
Hide X Values on the Value Labels at V-Cursor Crossing Points by Default	hide_cross_label_x_value	true/(false)	Hides the X values on the value labels at V-cursor crossing points by default if set to true.
Hide All Distances between Cursors	hide_cursor_distance	true/(false)	Hides distance lines and value boxes between reference and other cursors.
Show Reference Cursor Crossing Value	ref_cursor_value_indv	individually/Along with Target Cursor	Individually: P1 value is shown along reference cursor, P2 value and delta value are shown along target cursor. Along with Target Cursor: P1 value, P2 value and delta value are shown along with target cursor.
Canvas Operations			
Realtime Waveform Window Panning	real_time_scroll	(true)/false	Enables real-time scrolling of a zoomed panel if set to true.
Limit Side-Bar Grip to Bar Region	bar_grip_scroll	true/(false)	Limits slide-bar grip to the bar region if set to true.
Limit Phase of Complex Value Between (-180, 180)	clamp_ac_phase	(true)/false	Controls phase range of complex values. Phase is limited between -180/180 if set to true.
Extend Analog Signal Waveform to Fill X Range	auto_x_extend	true/(false)	Extends analog waveforms to fill the x-axis range.
Extend Logic Signal Waveform to Fill X Range	auto_x_extend_dig	(true)/false	Extends logic waveforms to fill the x-axis range.

Option Name in GUI	Option Name	Value (Default)	Description
Show Waveform X-Y Value Near Mouse Pointer	snap_marker	true/(false)	Enables waveform values mouse tracing cursor, which will snap to waveform data points and show the value box (will not be shown when not near a data point).
Display Logic Glitch (time-overlapping changes)	logic_glitch_highlight	(true)/false	Enables glitch highlight in logic panels if set to true.
Switch Mouse Left and Right Button Actions in Canvas	switch_mouse_button	true/(false)	When enabled, the left and right mouse button actions will be switched, meaning the left mouse button will bring up the context menu and the right mouse button is for zoom and selection.
Enable Mouse Right Button Drag to Zoom in	right_mouse_zoom	true/(false)	When enabled, you can use right click + mouse drag to zoom-in in the waveform canvas.
Enable VNC Display Acceleration Mode	fast_vnc_mode	true/(false)	Shows performance enhancement in the VNC display.
Enable Line Drag-n-Drop on Canvas		true/(false)	Enables drag and drop of a waveform or a group of waveforms to a new window or sub-window.
Unit Scale Factor For Monitor	unit_scale	(auto)/f/p/n/u/m/none/k/M/G/T	Sets the scale factor for Monitor values.
Disable Waveform Marching Reload Confirmation	reload_confirm	true/(false)	Disables the waveform marching reloading confirmation dialog box.
Do Not Update Loaded Signals When Plotting New Signal	disable_update_signal	(true)/false	When enabled, the WaveView tool does not update loaded signals automatically; to update loaded signals, you must click the Update button.

Option Name in GUI	Option Name	Value (Default)	Description
Disable Equation Update when Reloading Signals		true/(false)	When selected, equations are not updated automatically when signals are updated; to reevaluate an equation, you must click the Update button and then click Evaluate .
Synchronize Signal List with Equation Builder	sync_eqnbuilder	true/(false)	If an equation is deleted in the Calculator, WaveView deletes the equation result on the canvas and in the browser tree.

Panel Settings

The following panel settings are available:

Option Name in GUI	Option Name	Value (Default)	Description
Set Grid Default to Off	grid_default	true/(false)	Sets default grid on/off mode.
Automatically Switch to Row/Column Mode when Plotting Signals with Different X Ranges	auto_switch_horz	true/(false)	Switches modes when plotting two different signals with different X ranges (units) in the same waveview, to help in telling them apart.
Limit Zoom Out Range to Axis Full Scale Range	limit_zoom_range	true/(false)	Limits zooming out to, but not beyond, the full scale range.
Zoom Direction Default	def_zoom_dir	(xy_zoom)/x_zoom/y_zoom	Specifies the default zoom mode for panel.
Double-Click to plot signal at	plotsig_bottompanel	(new_panel)/bottom_panel	Specifies where a signal is plotted when double-clicked in the signal list.
Stack-Mode Panel Height			
Automatically Fit Analog Panels to Waveview Height	stack_auto_fit	(true)/false	Enables panel height auto-adjustment for non-logic panels in vertical waveviews.

Option Name in GUI	Option Name	Value (Default)	Description
Hide Dividers Between Stacked Panels	hide_panel_div	true/(false)	Hides the dividing lines between panels in stack mode.
Max. # of Visible Panels	max_visible_panels	integer_value (0)	Sets the maximum number of visible XY panels of vertical stack waveviews.
Minimum Analog Panel Height	min_panel_height	integer_value (20)	Sets minimum height for non-logic panels (in screen pixels) in vertical waveviews. [20,1000].
Analog Panel Height	analog_panel_height	integer_value (120)	Sets analog panel height for non-auto-height-fit mode.
Logic Panel Height	logic_panel_height	integer_value (24)	Sets logic panel height.
General Panel Settings			
Row/Column Mode Default Panels Per Row	default_panels_perrow	integer_value (4)	Specifies the default number of panels per row in row/column mode.
Row/Column Mode Minimum Panel Height	horz_panel_height	integer_value (80)	Specifies the minimum panel height in row/column mode.
Displaying Margin for Y Axis of Full Scale Panels	auto_y_margin panel_fs_margin_xy	on/(off) integer_value(0)	Allows WaveView to determine the good Y axis display margin. Sets the display margin for y-axis when Fixed Size is selected.
Displaying Margin for X Axis of Full Scale Panels	auto_x_margin panel_fs_margin_x	on/(off) integer_value(0)	Allows WaveView to determine the good X axis display margin. Sets the display margin for x-axis when the Fixed Size is selected.
Default Histogram Bin Size	default_hist_binsize	integer_value (10)	Sets the default bin size of histograms.
Show Single Histogram of All Signals by Default	default_hist_one	true/(false)	Drags multiple singles and shows histogram as one signal by default.

Option Name in GUI	Option Name	Value (Default)	Description
Signals Per Panel (d-n-d)	lines_per_panel	integer_value (0)	Sets number of lines per panel for signal allocation after the initial signal drag-and-drop operations. The default setting 0 causes all signals to be placed in one new panel.
Maximum Log Scale Span in Decades	max_log_span	(12)	Defines maximum log scale span in decades.
Zoom In Ratio (%)	zoom_in_ratio	(50)	Sets ratio for zoom.
Smith Chart Characteristic Impedance	smith_char_impedance	real_value(50.0)	Sets the default characteristic impedance of Smith charts.
Use Fixed Y Full Range	user_fixed_yrange	true/(false)	Uses the user-defined values for vmin (default 0), vmax (default 3.3), imin (default 0), and imax (default 1m).
Displaying Margin for Y Axis of Full Scale Panel		true/(false)	Displays the Y-axis start and end values.
Displaying Margin for X Axis of Full Scale Panel		true/(false)	Displays the X-axis start and end values.

Signals Settings

The following signals settings are available:

Option Name in GUI	Option Name	Value (Default)	Description
Synchronize Signal Alias Between Browser and Display	sync_alias	(true)/false	Synchronizes the signal name alias between the signal browser and the name column in the canvas.
Add Brackets [] for Signal Alias Names	alias_bracket	(true)/false	Adds bracket to user-defined signal alias name.
Alias Signal Name Only	alias_signal_name_only	true/(false)	Sets the alias to replace leaf node names only.

Option Name in GUI	Option Name	Value (Default)	Description
Always Name Sweep Family Signals as Individual Traces	sweep_expand	true/(false)	Treats the sweep signal lines as an individual or a family of lines.
Plot Single Point Sweep Family Signals on 2D Panel by Default	plot_single_point_sweep_on_2d	true/(false)	View abstract line when enabled, and detailed line when disabled. If the y value is too small to show, disable to see the detail of the signal.
Flatten Single Point Sweep Signals on 2D Panel	flatten_single_point_sweep	(true)/false	Flattens sweep signals that have only 1 point in each trace when plotting on a 2D panel.
Display Signal Name in Tooltip Popup	signame_popup	true/(false)	Enables signal name popup tooltip.
Display Sweep Trace Parameters in Tooltip Popup		true/(false)	Enables sweep trace parameters popup tooltip.
Display Complex Plotting Type as Modifier in Signal Name	show_MP_prefix	true/(false)	Sets the plotting type as the signal name modifier when plotting complex signals.
Use Simplified PWL Algorithm in rmsval Calculation	simplified_rms_calc	true/(false)	Uses simplified RMS calculation algorithm.
Use little-endian format to display bus value	use_little_endian	true/(false)	Specifies little endian or big endian for bus value display.
Show Toggle Button in Toolbar for Signal Selection Mode and Panel Selection Mode	show_SM_button	true/(false)	Displays the Signal/Panel Mode Selection button in the WaveView toolbar
Set Panel Mode as Default	show_panel_mode	true/(false)	Sets Panel mode as the default mode.
Highlighting Signal when Mouse Hovers on its Name	enable_wv_highlight	(true)/false	Controls the automatic highlighting of signals when hovering the mouse over it.
Support Multiple Selection for Signals and Panels with Single Click	enable_mul_click_items	true/(false)	Supports selecting multiple signals with a single click.

Option Name in GUI	Option Name	Value (Default)	Description
Signal Search Options	signal_filter_option	(Glob pattern)/Regular expression	Specifies the string pattern used to search against waveform names is a glob-style or a regular expression pattern in the Signal Finder.
Support Substring Search	support_substr_search	true/(false)	Matches the substrings of waveform names with the specified searching pattern.
Signal Name Width	signame_width	widest/(fixed size)	Specifies the width of signal names.
Signal Name Alignment	signame_align	(left)/right	Sets signal name alignment.
Signal Name Content	signal_name	netname/leafname/fullname/(fullname+file)/(full name+path+file)/(netname+file)/(netname+path+file)file	Sets signal name mode in signal name column.
Signal Hierarchy/Name Order	signal_path	(hierachy first)/name first	Sets signal hierarchy path style.
Order Signals from Grouped Files	group_display_order	(per file)/per signal	Sets the signals (from grouped files) order during the drag-and-drop display operations.
Signal Name Letter Spacing	signame_spacing	(normal)/large	Controls the letter spacing in signal names.
Signal Mode Drag-n-Drop	signal_dnd_mode	(move)/copy	Moves or copies a signal when dragged and dropped.
Default Logic Value Type	logic_style_default	(vcd)/rtf	Specifies the text style for logic signal values.
Default AC Signal Display	def_ac_stack	(complex)/magphase/realimag/mag+phase	Sets the default display mode for AC signals.
Signal Name Font	signal_font	(normal)/large	Controls the font size of signal names.
Default AC Signal	def_acx_type	(linear)/log/db20	Sets the default X-axis type for AC signals.

Option Name in GUI	Option Name	Value (Default)	Description
Default AC Signal	def_acy_type	(linear)/log/db20/db10	Sets the default Y-axis type for AC signals.
Default TR Signal	def_trx_type	(linear)/log/db20	Sets the default X-axis type for real type signals.
Default TR Signal	det_try_type	(linear)/log/db20/db10	Sets the default Y-axis type for real type signals.
Default FFT Signal	def_fft_xscale	linear/(log)/db20	Sets the default X-axis type for FFT signals.
Default FFT Signal	def_fft_yscale	linear/log/(db20)/db10	Sets the default Y-axis type for FFT signals.

Axis Settings

The following axis settings are available:

Option Name in GUI	Option Name	Value (Default)	Description
Automatic Dual Y-Axes (X-Y Panels)	auto_y_axes	on/(off)	Sets automatic dual Y-axes for mixed I/V signals.
Display Y-Axis Unit Label	show_ytype	true/(false)	Displays the y-axis unit labels on a WaveView window.
Display Prefix in Axis Unit Labels	show_unit_prefix	true/(false)	Displays the unit prefix axis labels on a WaveView window.
Place Label in the Middle and Parallel to Axes	center_axis_label	true/(false)	When enabled, places units and labels in the middle of and parallel to the x/y-axis, which can enhance clarity.
Show Abstract Line for Small Y Range (Difference at >9th Digit)	adjust_small_y_range	(true)/false	When Y Range is very small, the difference at or after the 9th significant digits, the tool shows a straight abstract line.

Option Name in GUI	Option Name	Value (Default)	Description
Use Default X Variable while Setting X Variable		(true)/false	The default X-variable as the x-axis of the plot. This is useful when multiple parameter sweeps are present and enables user to plot across original x-axis instead of the first sweep parameter.
Y-Axis Label Width	auto_y_width	automatic/(fixed)	Automatically sets the y-axis width based on the length of labels, or set to a fixed width.
X-Axis Pan Step (%)	horizontal_slider_step	(50)	Controls how fast the display pans left or right via the left or right keys or the WaveView slider arrows.
Font Size for Axis	axis_font	(Normal)/Large/Gothic/Times/Roman/Courier	Sets the default font size for axis labels.
X-Axis Auto Grid	xaxis_auto_grid	(true)/false	Justify and show the grid in x-axis direction.
Left Y-Axis Auto Grid	yaxis_auto_grid	(true)/false	Justify and show the grid in y-axis direction on the left side.
Right-Y-Axis Auto Grid	y2axis_auto_grid	(true)/false	Justify and show the grid in y-axis direction on the right side.

Color Settings

The following color settings are available:

Option Name in GUI	Option Name	Value (Default)	Description
Waveform Colors	user_colors	(Pre-Defined)User-Defined/Predefined and User-Defined	Selects pre-defined colors, user-defined colors, or a combination of both.
Lock Auto Selected Line Color	lock_color_sweep	true/(false)	Retains the color of a plotted signal when moving signals (for example, dragging and dropping a signal).

Option Name in GUI	Option Name	Value (Default)	Description
Display Sweep Signal Traces With Same Color	same_color_sweep	true/(false)	Plots all sweep signal traces with the same color.
Display Sweep Signal Traces With Looped Line Styles by Default	loop_linestyles_sweep	true/(false)	Plots sweep signals with looped line styles by default.
Display Signal Labels with White Color	white_label_color	true/(false)	When enabled, Waveview will always show the signal label with white color. If disabled, Waveview will show the signal label with same color as the trace color.
Default Waveform Line Thickness	line_thickness	1 to 7	Selects thick line as the default waveform line width.
User-Defined Waveform Colors	user_rgb	R:G:B	A user-defined color. Select Clear Colors to remove old colors. Select Add Color or Change Color to bring up a color dialog box where you can select or set colors (including true colors).

Threshold Settings

The following threshold settings are available:

Option Name in GUI	Option Name	Value (Default)	Description
D2A Convert X State	d2a_xtol	to high/(to low)	Selects D/A conversion method for the X/Z logic states.
D2A X-Axis Range	d2a_fullxt	line range/(waveview range)	Selects the converted signals' x-axis range for D/A conversion.
D2A Threshold Levels	d2a_high_value	real_value(3.300E+00)	Defines the analog level of the high logic state for D/A conversion.

Option Name in GUI	Option Name	Value (Default)	Description
D2A Threshold Levels	d2a_low_value	real_value(0)	Defines the analog level of the low logic state for D/A conversion.
D2A Rise/Fall Time	d2a_rise_slew	real_value(10ps)	Defines the rise slew for the D/A conversion process.
D2A Rise/Fall Time	d2a_fall_slew	real_value(10ps)	Defines the fall slew for the D/A conversion process.
A2D Threshold	a2d_single_threshold	high-low/(single)	Selects default A/D conversion method.
A2D Center Threshold	a2d_center_threshold	real_value(2.5)	Defines center threshold for the A/D conversion process.
A2D Threshold Levels	a2d_high_threshold	Real_value (4.0)	Defines high threshold for the A/D conversion process.
A2D Threshold Levels	a2d_low_threshold	real_value (1.0)	Defines low threshold for the A/D conversion process.
Meter level/margin scale changes according to panel scale	SX_DM_DMScale	true/(false)	Specifies if the level and margin values for dynamic measurements depend on the axis scale (for example, dB, log, linear).
Meter Results Use Panel Scale	SX_DM_ResultPanelScale	true/(false)	Uses the panel scale for measurement results.
Move label Independently on Difference meter	dm_diff_movelabel	true/(false)	Moves label Independently on Difference meter
Display Unit in Value Labels	show_value_unit	true/(false)	Displays unit in value labels
Count floor noise components in SNR/SNDR calculation	SNR_inc_floor_noise	true/(false)	Counts the floor noise components in SNR and SNDR calculations.
Scalar Result Format	calc_eng	true/(false)	Sets scalar result format in the Equation Builder. True sets to engineering notation. Default is scientific notation.

CDS Link Settings

The following CDS Link settings are available:

Option Name in GUI	Option Name	Value (Default)	Description
CDS-Link Plot Mode	cds_link_plot_mode	(replace)/append	Replaces or appends a plotted signal. This preference is only available when the WaveView tool is started in CDS-Link mode.
CDS-Link Update Mode	cds_link_update_mode	replace/(append)	Replaces or appends a plotted signal when updating waveform files. This preference is only available when the WaveView tool is started in CDS-Link mode.
Signal Crossing Probing Color Scheme	x-probing_color_schem	(Use Schematic Color Probe)/Use WaveView Signal Color	Specify to use Schematic Probe Color or to use WaveView Signal Color as doing cross probing between WaveView and another program.
Use Different Color for Bus Signal	bus_diff_color	true/(false)	WaveView will plot all bus signals in the same color by default. If the user selects this preference, WaveView will plot signal in different color.

Saving Preference Files

To save preferences into a file:

1. Ensure the **Preference Settings** window is open (click **Preferences** on the **Configuration** tab).
2. Click the **Apply-Save** button.

The **Preference Save Location** dialog box opens.

3. Choose a location, either in your home directory, the working directory, or in a path you enter by typing or choosing from the file browser.

Note:

Select **Save as PrimeWave Design Environment Preference Name** to save the preference settings in PrimeWave preference format.

For example, if you have **Number of Significant Digits** set to 13 in the **Preference Settings** window, this will be saved as `precision_digits 5` by default, but if you select **Save as PrimeWave Design Environment Preference Name**, it will be saved as `wfWVPrecisionDigits 5`.

4. Click **OK** to save the file.

Loading Preference Files

To load an existing preference file:

1. Ensure the **Preference Settings** window is open (click **Preferences** on the **Configuration** tab).
2. Click the **Load** button.

The **Load Preference File** dialog box opens.

3. Browse to and select your preference file, and then click **OK**.

Note:

The WaveView tool can read WaveView-related preferences from the `synopsys_custom.spxrc` file.

Your preference file is now loaded into the WaveView tool.

Mapping Bus Values

To open a bus mnemonic map file, click **Settings** on the **Waveform** tab, and choose **Bus Mnemonic Map** from the menu that opens.

Bus values can be mapped to predefined text strings. You can provide a translation table in an external file using the following syntax:

Syntax

```
# comment lines start with '#'
value1 name1
value2 name2
...
```

The values are the bus values to be mapped.

The following formats are supported:

- 2400
- 0xAF32
- 'hAF32
- 'b1001_0100_0101
- 'd2400
- 'b110110111100
- 'o371

The names are mapped to text strings. A text string must be quoted if the name contains spaces.

Wildcards are supported in Bus Mnemonic expressions. The ? character is a single bit for wildcards. It works only on 'b, 'o, and 'h.

For example:

```
2'b1?  
3'o1?1  
4'h1?1?
```